Inherent Case Features and Case Assignment in Double Object Constructions in Kazym Khanty

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Abstract

This dissertation contributes to the research on case by addressing questions about default case-marking and the mechanism of case assignment. This study is based on novel fieldwork data from the Kazym dialect of Northern Khanty (Ob-Ugric, Uralic) and makes three central claims. First, it makes an empirical and theoretical generalization that there exist two default cases in the language. I argue that the default case in argument positions is distinct from the default case in adjunct positions, e.g. on hanging topics (cf. Schütze 2001). Apart from the unmarked default, discussed in multiple previous studies (e.g. Schütze 2001; Legate 2008; Caha 2024 etc.), there exists a marked default case. Nominals in argument positions can bear the most marked case in the absence of syntactic case licensing, i.e. as default marking. To derive marked default, I argue that case features are inherently overspecified on arguments, which leads to an exceptional full spell-out in the absence of syntactic case checking. In particular, I suggest that arguments have an inherent [uCase]-feature, which is overspecified, i.e. valued for all case values available in the language. If a noun is not assigned case by a syntactic mechanism, all inherent case values are spelled out, which gives rise to oblique case morphology being used as the default.

The second claim of this study addresses the mechanism of case assignment. As a baseline, I adopt a model where case is assigned under Agree with a valued Case feature on a functional projection. Consequently, case assignment happens under Agree between two valued features, which is considered impossible in all current theories of Agree (cf. both Preminger 2014 and Deal 2022). This problem is resolved by proposing a modified definition of Agree. I follow the theory that divides Agree into two parts: Agree-Link, which creates a pointer from a functional head to a nearby DP, and Agree-Copy, which transfers features from the goal to the probe (e.g., Arregi & Nevins 2012; Bhatt & Walkow 2013; Marušič, Nevins, & Badecker 2015; Baker & Camargo Souza 2020; Lyskawa 2021). However, the original definitions of Agree-Link and Agree-Copy must be modified to account for Agree between two valued features. Under the modified definition, two features can undergo Agree-Link if at least one of the features is uninterpretable and/or unvalued. This allows me to account both for ϕ-agreement between a valued and an unvalued feature and for case assignment, seen as agreement between two valued features. Furthermore, Agree-Copy is replaced by a more general Agree-Match that modifies the value of the uninterpretable feature so that it matches the value of the interpretable feature. Agree-Match either copies the value to an unvalued feature or deletes the redundant, “unmatching” values of the valued uninterpretable feature.

The third claim addresses the nature of DP-restriction on unlicensed arguments. This definition of Agree provides a motivation for the DP-restriction on secundative themes. An explanation is proposed as to why DPs must be case-licensed in Khanty, while structurally reduced nominals can survive unlicensed (cf. “derivational time-bombs” in Kalin 2018). I propose that D^{o} in Khanty has an unvalued [uCase]-feature, which can enter into an Agree-Link relation with the valued [uCase]-feature on K^{o}. Crucially, both features are uninterpretable, and Agree-Match cannot prefer impoverishment over copying the feature values. Such derivation crashes at PF. However, if a second Agree-Link with an interpretable Case feature takes place, Agree-Match can modify both uninterpretable features on K^{o} and on D^{o}, and the case is successfully assigned.

The proposed approach is able to account for the inherent case overspecification on arguments, to model case assignment as Agree between two valued features, and to describe a mechanism behind the DP-restriction on unlicensed arguments.

Acknowledgments

My connection with Khanty goes back to the first years of my life, when my mother worked with the Kazym Khanty people, helping them to develop a culture-oriented school program. Although neither of us can speak Khanty, I grew up with stories about life in Kazym and the woods around it. So I did not think twice when the Moscow research group invited new people to join their fieldwork trip to Kazym in 2019. I am deeply grateful for this opportunity to work with the language and to learn more about Khanty and linguistic fieldwork.

The work on this dissertation has been a long and exciting journey. As in every long journey, it had its ups and downs, but there were more good days, full of admiring the beauty of the language and the elegance of the linguistic theory. As with every long journey, many people have been part of it. Without them, this journey would never have come to an end, and it would definitely have been much less enjoyable.

First, I would like to thank Maria Kouneli, Gereon Müller, and András Bárány for their interest in my work and for many hours of consultations. Thank you for the fruitful discussions of the data and my small and big ideas. Thank you for teaching me to be very careful about every piece of the analysis and to work out the details before applying my ideas to more and more new data. Thank you for your feedback on the many abstracts and drafts of this dissertation.

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To my parents
and to the loving memory of Marina

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List of symbols and abbreviations

Glosses

1	First person
2	Second person
3	Third person
1SG>NSG	First person singular subject and non-singular object
ABL	Ablative case
ABS	Absolutive case
ACC	Accusative case
ADD	Additive particle
ADJ	Adjective
ALL	Allative case
AP	Antipassive
APPL	Applicative
ATTR	Attributive
AUX	Auxuliary
CAR	Caritive
CAUS	Causative
COMIT	Comitative case
CVB	Converb
DAT	Dative case
DEF	Definite
DEM	Demonstrative
DETR	Detransitivizer
DIM	Diminutive
DU	Dual
EMPH	Focus particle
ERG	Ergative case
F	Feminine
FREQ	Frequentative
FUT	Future
GEN	Genitive case
GER	Gerund
IMP	Imperative

IMPF	Imperfective
IND	Indicative
INF	Infinitive
INSTR	Instrumental
INSFIN	Instrumental-final
INTER	Interrogative
INTR	Intransitive
LAT	Lative case
LOC	Locative case
M	Masculine
MOD	Modalis case
MOM	Momentative
NEG	Negation
NOM	Nominative
NFIN.NPST	Non-finite form in non-past tense
NFIN.PST	Non-finite form in past tense
NPST	Non-past tense
NSG	Non-singular
O	Object
PART	Particle
PTCP	Participle
PASS	Passive
PEJ	Pejorative
PERF	Perfective
PL	Plural
POSS	Possessive
PREV	Preverb
PROH	Prohibitive
PROP	Proprietive
PRS	Present tense
PST	Past tense
REL	Relitivizer
REM.PST	Remote Past tense
SG	Singular
S	Subject
TR	Transitive
TRANS	Translative case

Other abbreviations

DEF	Definite
DO	Direct object
DOM	Differential Object Marking
INDEF	Indefinite
IO	Indirect object
NI	Noun incorporation
NON-SPEC	Non-specific
PNI	Pseudo noun incorporation
PRON	Pronoun
SPEC	Specific
UQ	Universal quantifier

Chapter 1

Introduction

1.1 General introduction

This dissertation addresses three aspects of Case theory. First, it discusses the mechanism of case assignment and provides arguments in favor of the checking approach and against the valuation approach. Second, it models how case assignment is connected to the abstract licensing of nominal arguments and size restrictions on unlicensed arguments. Third, it investigates default case marking, especially the default case in argument positions.

Case is one of the core linguistic phenomena and has always been at the center of linguistic research. At the same time, it is a very complex phenomenon, so it is not surprising that it is still a topic that needs further research. In the early minimalist model, case was seen as a simplex abstract feature without semantic interpretation. This [uCase]-feature must be checked during the derivation, i.e. two uninterpretable Case features with the same value must enter into an Agree relation and be checked against each other. Crucially, in the initial model (Chomsky 2000, 2001), two [uCase]-features could not agree independently, and case checking was seen as a side effect of the agreement between the finite verb and a noun for ϕ-features. Subsequent research has shown that case assignment is an independent operation that does not always coinside with ϕ-agreement.

A crucial question for a theory of Case concerns the mode of assignment of abstract Case. In Chomsky's model, the [uCase]-features on a noun and on a verbal projection have identical inherent values, so case assignment is in fact checking of the case values against each other. In recent studies, an alternative approach has been adopted, where the Case feature on a noun is unvalued and case values are copied from a verbal projection under Agree or assigned by the Dependent Case mechanism (e.g. Legate 2008; Baker & Vinokurova 2010; McFadden & Sundaresan 2011; Bárány 2017; Preminger 2020; Bárány & Sheehan 2021; Clem & Deal 2024 etc.). The valuation approach is adopted in the most current literature on Case (but e.g. Bošković 2006b explicitely argues against valuation approach).

Another line of research concerns the connection between abstract nominal licensing and syntactic case assignment (e.g. Kalin 2018; van Urk 2020; van der Wal 2022; Irimia

2022), though see Nie 2020, who links licensing to ϕ-agreement). Many studies have shown that DPs in argument positions require additional syntactic licensing, which can be achieved via case assignment or through (pseudo-)noun incorporation (e.g., van Urk 2020; Driemel 2020, 2023). Importantly, van der Wal (2022) and Sheehan & van der Wal (2018) argue that abstract Case exists even in languages that lack morphological case, and that arguments must be licensed via the assignment of an abstract Case. Importantly, multiple studies have discussed the asymmetry between full DPs and structurally reduced nominals. Lyutikova & Pereltsvaig (2015), Kalin (2018), and Irimia (2022) argue that in Tatar, Amharic, and Romance, respectively, NPs can survive derivation unlicensed, while DPs must be obligatorily licensed via assignment of case in Syntax. There is no consensus among linguists on how this asymmetry should be explained. One line of research (Lyutikova & Pereltsvaig 2015 a.o.) argues that the Case feature is located on D^0 and requires valuation. If the DP-projection is absent, the Case feature is absent too, and the nominal does not need case-licensing. Crucially, both Kalin (2018) and Irimia (2022) point out that NPs can be assigned case, even though they do not need it. Kalin (2018) suggests that all functional projection inside DP have a Case feature, i.e. are able to receive case value. Additionally, some of these Case features have an abstract diacritic, a "derivational time-bomb". Case features without "time-bomb" can remain unlicensed, i.e. do not require case assignment. Projections with a "time-bomb" must be licensed via case assignment; otherwise, the derivation crashes. However, Kalin (2018) does not discuss the nature of the "derivational time-bomb" in detail.

Another aspect of Case theory relevant for this study is the theory of the default case. Schütze (2001: p.206) defines default case as a morphological case marking used "to spell out nominal expressions (e.g., DPs) that are not associated with any case value assigned or otherwise determined by syntactic mechanisms." There are two main questions about the default case that must be answered by a satisfactory analysis.

The first question is whether any morphological case can be used as a default, i.e. whether different languages have different default cases. If no, a theory of the default case must provide a motivation for why a particular case is used as the default. Schütze (2001) argues that the value of the default case is language-specific. In particular, nominative is used as the default in German and accusative in English and Italian. Following this line, Pesetsky (2013) suggests that genitive is the default case in Russian. Most modern studies assume that the least marked case, i.e. nominative or absolutive (or absence of a morphological case marker), is the morphological default (Legate 2008; McFadden & Sundaresan 2011; Lyutikova & Pereltsvaig 2015; Weisser 2020; Caha 2024 etc.). This trivially follows in case-valuation approaches, where nominals lack inherent case value. If the case is not assigned by a syntactic mechanism, the Case feature on a nominal remains unvalued in Syntax and only the least specified morphology can be inserted at PF. Furthermore, Bošković (2006a), Weisser (2020), and Caha (2024) argue that accusative and genitive are not true default cases and require alternative analyses.

The second question is whether the default case is restricted, i.e. whether it can surface in all or only some syntactic positions. Under Schütze's 2001 definition, the default case is unrestricted, i.e. it is used whenever a syntactic case has not been assigned. Importantly, Fanselow (1986, 1991) noticed for German that not every nominal is able to survive without a syntactically assigned case. This means, that default nominative is not freely assigned, i.e. there are some syntactic restrictions. This raises a problem for the current definition of the default case, and it is still an open question how the distribution of the default case is restricted.

This dissertation contributes to the theory of Case by addressing three questions. First, I argue that there are two types of default case marking, attested on adjuncts and arguments. Second, I argue that case is assigned via checking rather than valuation. I also propose a definition of Agree that can account for both checking and valuation. Finally, I address the asymmetry between full DPs and structurally reduced nominals with respect to nominal licensing. The proposed analysis motivates why argument DPs must be licensed via assignment of case in Syntax, while structurally reduced nominals can survive with an unchecked Case feature. The remainder of this section discusses each of these aspects in detail.

The first claim is an empirical and theoretical generalization that language makes use of two default cases. I argue that the default case in argument positions is different from the default case in adjunct positions, e.g. hanging topics (cf. Schütze 2001). Apart from the unmarked default discussed in previous studies (e.g. Schütze 2001; Legate 2008; Caha 2024 etc.), there exists a "marked" default case. Nominals in argument positions can have oblique morphology in the absence of syntactic case licensing, i.e. bear a "marked" default case. The marked default can be found e.g. on oblique themes in double object constructions, as illustrated in 1 for the Kazym dialect of Khanty (Ob-Ugric, Uralic). An important property of the "marked default" is that it is accompanied by DP-restriction. Unlicensed arguments bearing the marked default case obligatorily lack DP-projection.

(1)	Kašəŋ χujat	λөχs-əλ	**lipət-ən**	mă-s-λe
	Every person.[NOM]	friend-POSS.3SG.[ACC]	flower-LOC	give-PST-3SG>SG

'Everyone gave a flower/flowers to his friend.'

In order to derive the marked default, I adopt the strong version of the Case Contiguity Hypothesis in 2 and assume that (a) case is decomposed into primitive features, and (b) more marked cases include all features of the least marked cases plus an additional feature.

(2) Case Contiguity hypothesis (Caha 2009: p.21,23)

NOM	>	ACC	>	GEN	>	DAT	>	INSTR	>	COMIT…
[A]		[A,B]		[A,B,C]		[A,B,C,D]		[A,B,C,D,E]		[A,B,C,D,E,F]

I argue that arguments have an inherent overspecified Case feature, i.e. are valued for all available case values. In the absence of syntactic case checking, all inherent case values survive at PF, and the noun bears the marked default case. In particular, I suggest that arguments have a KP-projection with an inherent uninterpretable [uCase]-feature, which is overspecified, i.e. valued for all case values available in the language. If a noun is not assigned case under Agree with an interpretable [iCase]-feature, all inherent case values are spelled out, which gives rise to oblique morphology being used as the default case marking.

The second claim of this study addresses the mechanism of case assignment as a subtype of Agree. As a baseline, I adopt a model where case is assigned under Agree with a valued interpretable Case feature on a functional projection in the clausal spine, e.g. Voice^{o} or T^{o} (cf. Pesetsky & Torrego 2007). Recall that arguments have valued Case features. Consequently, case assignment occurs under Agree between two valued features, i.e. case is assigned via checking and not via valuation. This goes against current theories of Agree (both Preminger 2014 and Deal 2022), which attempt to treat all instances of Agree as valuation. In particular, Preminger (2014) explicitly defines a probe as an unvalued feature. In this dissertation, I argue that Agree can be realized as valuation or checking. I propose a modified definition of Agree that can account for agreement between a valued and an unvalued feature, as well as for agreement between two valued features.

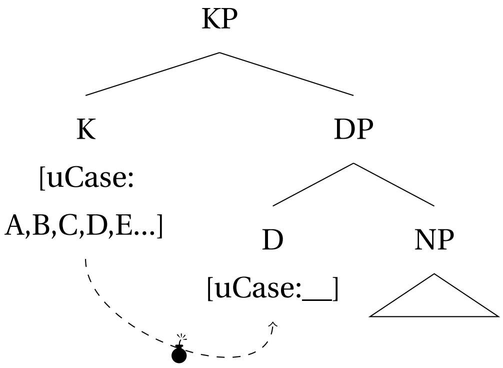
Several recent studies divide Agree into two parts: Agree-Link, which creates a pointer from a functional head to a nearby DP, and Agree-Copy, which transfers features from the goal to the probe and deletes the pointer (e.g., Arregi & Nevins 2012; Bhatt & Walkow 2013; Marušič et al. 2015; Himmelreich 2017; Baker & Camargo Souza 2020; Lyskawa 2021). I suggest modifying the original definition as in 3. Under the modified definition, two features can enter Agree-Link if at least one of the features is uninterpretable and/or unvalued. Hence, it captures both ϕ-agreement and case assignment. In ϕ-agreement, one of the features (probe) is unvalued, so Agree-Link can take place. This is similar to the current approach to ϕ-agreement. Case is assigned under Agree between two valued features. Because the Case feature on the argument is uninterpretable, Agree-Link can successfully take place. On the PF side, Agree-Copy is replaced by a more general Agree-Match that modifies the value of the uninterpretable [uF], so that it matches the values of the interpretable [iF]. There are two possible scenarios for this. Under ϕ-agreement, ϕ-values are copied to the unvalued feature on the verbal projection. Under case assignment, Agree-Match compares the set of values of the uninterpretable [uCase] on an argument to the value of the interpretable [iCase] on the case assigner (e.g. T^{o} or Voice^{o}). Then, Agree-Match deletes the "unmatching" values of the uninterpretable [uCase].

(3) a. AGREE-LINK:
Feature A is linked to feature B with a pointer, iff (i) A is a probe on the head of a functional projection, and (ii) B is the closest available goal in the maximal projection of A, and (iii) either A or B is defective (lacks value and/or interpretation);

b. AGREE-MATCH:
The value of a defective feature must match the value of the interpretable feature. The pointer is deleted.

The third claim addresses the nature of nominal licensing. The definition in 3 allows us to define "derivational time-bombs" on DPs (Kalin 2018). An explanation is proposed as to why DPs must be case-licensed in Khanty, while structurally reduced nominals can survive unlicensed. I propose that D^{o} in Khanty has an unvalued [uCase]-feature, which is able to enter an Agree-Link relation with the valued [uCase]-feature on K^{o}. The derivation is illustrated in 4a. Crucially, both features are uninterpretable, and Agree-Match cannot prefer impoverishment over copying of the feature values. Such derivations crash at PF unless a subsequent Agree-Link with an interpretable Case feature takes place. When this happens, Agree-Match has one "stable" interpretable feature in the Agree-chain and is able to modify both uninterpretable features on K^{o} and on D^{o}, and a case is successfully assigned. An abstract derivation of a successful case assignment is shown in 4b.

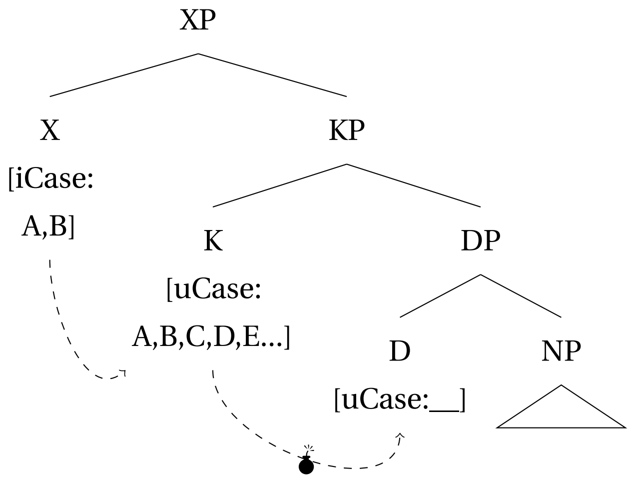
(4) a. Agree-Link for Case inside KP



→ Agree-Match crashes

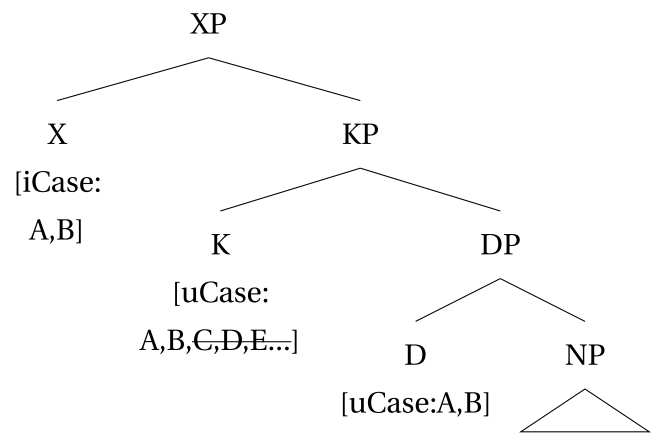
b. Case assignment by an external head

Agree-Link



→

Agree-Match



Summing up, this dissertation provides empirical evidence for the existence of two default cases. In addition to the unmarked default (nominative/absolutive), there exists a marked default or oblique default that is found on nominals in argument positions. On the theoretical side, case is argued to be assigned via checking rather than valuation (cf. Chomsky 2000, 2001; Bošković 2006b). Furthermore, checking is formally defined as a subtype of Agree. Additionally, the requirement for DPs to be licensed via case assignment is motivated and accounted for by the same mechanism of case assignment. The argumentation is primarily based on data from Kazym Khanty; however, a brief discussion of typological parallels in Eastern Khanty, Eastern Mansi (Ob-Ugric), and Kalaallisut (Inuit) provides additional support for the presented analysis.

1.2 Design of the study

The present study is based on data from the Kazym dialect of Northern Khanty (Kazym Khanty). Data from Kazym Khanty were analyzed independently and compared to existing analyses of other varieties and dialects of Khanty. The most well-researched varieties are the Obdorsk dialect of Northern Khanty and the Surgut (Eastern) Khanty variety. The existing literature on Kazym Khanty does not address the question discussed in this study. This study introduces a significant amount of new fieldwork data from the Kazym dialect, primarily on the syntax of double object clauses and the inner structure of the secundative theme, as well as a theoretical analysis of case alignment alternation.

Unless otherwise indicated, the Kazym Khanty data come from my own fieldwork. Some of the data come a corpus of Northern (aka. Western) Khanty (Kaškin & Muraviev 2014) and fieldwork handouts by the Moscow research group. Data for other dialects of Khanty, as well as Mansi, and Inuit languages come from the existing linguistic literature. The fieldwork data for Kazym Khanty were collected by me in multiple online and onsite fieldwork sessions with 11 native speakers of Kazym Khanty during 2022-2024. The consultants form a relatively homogeneous group, with one notable exception. There are 4 male and 6 female consultants aged 50-64, who currently live in Kazym village. Kazym Khanty is their first native language, spoken in the families of their parents. Almost all of them learned Russian at school between the age of 6-8. Some can speak or understand a bit of Forest Nenets, and four learned German at school. In everyday life, they mostly use Russian but speak Khanty with friends who also know the language. In their families, they only speak Russian, since they are married to Russian spouses who do not know Khanty. All consultants have a school education, where classes were taught in Russian. One consultant had only completed 9 classes of school education. Eight people have a technical college education, also in Russian. Finally, three people have a university education; for two of them, it is a second education after a technical college. I have also consulted Irina Moldanova, a native speaker linguist working at the Ob-Ugric Institute of Applied Research and Development in Khanty-Mansiysk, Russia.

All main types of elicitation were used to acquire the data. This includes translation tasks from Russian to Khanty, grammaticality, and felicity/preference judgements of Khanty stimuli. Importantly, all consultants are bilingual, and all but one speak Russian at a native speaker level. On the one hand, this is an advantage, as it minimizes the chance of misunderstanding the task because of a language barrier. On the other hand, it increases the risk of Russian grammar influencing their judgements about Khanty. This is particularly important because Russian does not have a regular indirective-secundative alignment alternation, which could potentially create a bias towards indirective alignment. Furthermore, most tests used in this study required data to be organized into minimal pairs, and a translation necessarily provides only one member of the pair. Therefore, I attempted to minimize the presence of the Russian language and used translation tasks as rarely as possible. Since I do not speak Kazym Khanty, Russian was necessarily used during the session to communicate with the consultants, explain the task, and discuss the judgements and intuitions. The duration of one elicitation session was 1-1,5 hour, depending on the consultants' availability and how tired they were.

During the translation sessions, native speakers were presented with stimuli in Russian and asked to translate them into Khanty. Subsequently, the initial translation was modified to create a minimal pair for the respective test, and the consultant was asked for a grammaticality or preference judgement.

For felicity judgements, stimuli were presented with a context in Khanty. Minimal pairs were presented to the consultants together or directly after each other. The consultants were asked to read the context and stimulus aloud and judge its general grammaticality. If the stimulus was judged to be grammatical, further questions were asked depending on the research question. For instance, consultants were asked about the felicity of a given expression in a given context, or which member of a minimal pair is preferred in the given context.

1.3 Overview of the dissertation

The remainder of this dissertation is organized as follows. **Chapter 2** comprises two bigger parts. The first part introduces the Khanty language, its dialectal classification, and sociolinguistic situation, and provides a general overview of the grammar of the Kazym dialect of Khanty. The second part presents a detailed discussion of the alternation between indirective and secundative alignment (cf. Malchukov, Haspelmath, & Comrie 2010 for terminology). This alternation, illustrated in 5, is fully productive and attested in low applicative and causative clauses, but not in high applicative clauses.

(5) a. Indirective alignment

Kašəŋ χujat	λɵχs-əλ-a	lipət	mă-s-(λe)
Every person.[NOM]	friend-POSS.3SG-**DAT**	flower.**[ACC]**	give-PST-(3SG>SG)

'Everyone gave a flower/flowers to his friend.'

b. Secundative alignment

Kašəŋ χujat	λɵχs-əλ	lipət-ən	mă-s-λe
Every person.[NOM]	friend-POSS.3SG.**[ACC]**	flower-**LOC**	give-PST-3SG>SG

'Everyone gave a flower/flowers to his friend.'

The current research tradition of Uralic languages links the alternation in 5 to the information-structural properties of the indirect object. In particular, it is argued that indirect objects must be a secondary topic when promoted to accusative (e.g. Dalrymple & Nikolaeva 2011; Virtanen 2012; Sosa 2017 etc.). This study shows that information structure is insufficient to capture the distribution of indirective and secundative alignments in Kazym Khanty. Although promoted indirect objects tend to be topics, there are notable exceptions that prevent one from adopting an analysis based solely on information structure. Similarly, prominence hierarchies based on definiteness, specificity, animacy, and the count-mass distinction cannot capture the distribution of indirective and secundative alignments in Kazym Khanty.

Chapter 3 presents a syntactic analysis of the derivation behind indirective alignment. First, the base-generation order of the two objects in lexical ditrastive and applicative clauses is discussed. Several tests on asymmetric c-command, available in the language, show that the base-generation order of the two objects is IO>DO in both alignments.

The analysis of the case alignment alternation is developed within an approach where case is assigned by the head of a functional projection under Agree (e.g. Chomsky 2000, 2001; Pesetsky & Torrego 2007; Bárány 2017 etc.). I argue that dative in indirective alignment is assigned by HighAppl^{o}. An applied argument base-generated in SpecHighApplP is correctly predicted to be obligatory marked with (inherent) dative. In low applicative and causative clauses, an a-thematic flavor of HighApplP is merged optionally, which gives rise to the alignment alternation in 5. If HighApplP is present, the indirect object receives the dative case, and the whole clause has indirective alignment. This analysis is supported by the data showing that the dative-marked indirect object raises to a position outside the VP, as well as by the fact that secundative alignment is unavailable in clauses with high applied arguments.

Indirective case alignment in causative clauses is discussed separately. Causatives in Kazym Khanty are argued to be verb selecting (cf. Pylkkänen 2008). Crucially, they alternate between indirective and secundative alignments, similar to low applicative structures. I argue that an a-thematic HighApplP can be merged on top of CausP, deriving indirective alignment by assigning dative to the causee.

In **Chapter 4**, the syntax of secundative alignment is discussed in detail. The discussion is mostly concerned with the properties of oblique secundative themes. Despite being an articleless language, Kazym Khanty clearly shows structural restrictions on theme arguments in secundative alignment. All DP-level modifiers, i.e. a universal quantifier, demonstratives, and possessive suffixes, are ungrammatical on oblique-marked themes. I argue that this is a structural restriction and that secundative themes lack DP-projection. This generalization is based on novel fieldwork data that has not been discussed in the literature before. To account for this restriction, I suggest that HighApplP-projection is missing from the structure in secundative alignment. This allows Voice^0 to assign ACC to the indirect object (recipient/causee), but the theme remains unlicensed for case. The size restriction on the theme follows from the lack of licensing (cf. a similar proposal by Kalin 2018), but the oblique case marking requires an independent explanation.

Chapter 5 concerns with the origin of the marked default case and the nature of the DP-restriction on unlicensed arguments. This chapter presents the main theoretical proposal of this dissertation. First, it defines the mechanism of case assignment and Agree between two valued features. Second, it motivates DP-restriction as a consequence of Agree between two uninterpretable features.

Finally, **Chapter 6** compares the Kazym Khanty data to related phenomena in West Greenlandic (aka. Kalaallisut, Inuit, Eskimo-Aleut) and Eastern Mansi (Ob-Ugric, Uralic). The core properties of indirective-secundative alignment alternation are common to all three languages. West Greenlandic provides support for the analysis developed in this dissertation. First, morphological evidence supports the analysis that HighApplP is a source of the dative case. Second, West Greenlandic has an additional context in which the marked default case can be found. In this chapter, I also argue that secundative alignment requires an analysis distinct from that of objects in antipassive and unmarked objects in DOM. A systematic comparison of these phenomena in Inuit and Eastern Mansi provides evidence of their different natures. Similar to antipassive objects, secundative themes are marked with an oblique case, but in contrast to the antipassive, secundative alignment is restricted to ditransitive clauses. The West Greenlandic data makes it possible to distinguish between antipassive and secundative derivations and analyze them differently. Eastern Mansi, on the other hand, allows secundative alignment to be distinguished from differential object marking. This is particularly important in light of multiple analyses of DOM, which treat unmarked objects as missing case values (e.g. Ormazabal & Romero 2010; López 2012; Bárány 2017, 2018). Eastern Mansi data provide evidence that DOM should not be universally analyzed as an opposition between nominals with and without syntactic case licensing (cf. e.g. Keine & Müller 2015; Irimia 2023). Rather, marked and unmarked objects in DOM languages can bear two different allomorphs of the accusative case.

Chapter 2

Background

Khanty is an endangered Ob-Ugric language in the Uralic language family spoken in North West Siberia. This dissertation deals primarily with the Kazym dialect of Northern Khanty, spoken in the region around the Kazym River. Although there are several scholars who are native speakers of the Kazym dialect of Khanty, there are not many theoretically oriented studies on this dialect. There is a descriptive grammar by Kaksin (2010) and a comparative study of different Khanty dialects by Solovar, Nakhracheva, & Shijanova (2016), written by native speakers. Both studies provide a non-formal description of the language data, while Solovar et al. (2016) provide a functionalist analysis of certain aspects of Khanty syntax. Functionalist analyses on certain aspects of the Kazym Khanty grammar have been made e.g. by Koškareva (2002), Solovar (2016), and Moldanova (2018). A grammar by Nikolaeva (1999) is written in formal terms but describes another dialect of Northern Khanty, namely, Obdorsk Khanty. The situation around the Kazym dialect has been changing in the last few years, when formal analyses of some aspects of Kazym Khanty have been presented at conferences and published in linguistic journals (e.g. Bikina, Rakhman, Potseluev, Starchenko, & Toldova 2022; Mikhailov 2023 etc.). However, there is still no comprehensive grammatical description of Kazym Khanty in formal (minimalist) terms. This chapter does not attempt to fill this gap and replace such grammar. Rather, it provides an overview of the basic grammatical properties of the Kazym dialect of Northern Khanty and sets the ground for further discussion of one particular phenomenon of the language system, namely, the indirective-secundative alignment alternation. The grammatical description below leaves phonology aside and addresses only the core morphological and syntactic features of Kazym Khanty. When necessary, data from other varieties, especially the Obdorsk dialect of Northern Khanty and Surgut Khanty, are introduced to complete the description. If not explicitly mentioned, all examples are given for the Kazym dialect.

Section 2.1 describes the dialectal classification of Khanty varieties, points out the place of the Kazym dialect in it, and provides general sociolinguistic information on Kazym Khanty. Section 2.2 introduces the core features of nominal and verbal morphology in the Kazym dialect. In Section 2.3, the core syntactic properties of Kazym Khanty are considered, such as passive and object agreement, *pro*-drop, word order, and the position of wh-elements. This is by no means intended as a comprehensive description of Kazym

Khanty grammar, but it provides the necessary background for the subsequent discussion. The grammatical sketch is mainly focused on the Kazym dialect of Northern Khanty. Comments on other Northern Khanty dialects are made when necessary and explicitly mentioned. Data from the Obdorsk dialect are given after Nikolaeva (1999) and occasionally from Solovar et al. (2016). Data for Kazym Khanty come from my own fieldwork, as well as from the Western Khanty Corpus (Kaškin & Muraviev 2014) and fieldwork handouts of the Moscow research group, which has conducted a series of field trips to Kazym village over the last several years. Finally, Section 2.4 contains a discussion of indirective-secundative alternation in Kazym Khanty, based exclusively on my own fieldwork data. This alternation has been researched in the Obdorsk dialect of Northern Khanty, Surgut Khanty, and different dialects of Mansi (Ob-Ugric, Uralic), but not in the Kazym dialect of Northern Khanty. It is generally assumed that the indirective-secundative alternation is regulated by information structure, i.e. secundative alignment is used when the indirect object is topical. In Section 2.4, this generalization, made for other Ob-Ugric languages and dialects, is tested on Kazym Khanty data. It is argued that dependency of the alignment alternation on information structure is present in Kazym Khanty as well, but only as a tendency, and cannot capture all restrictions on secundative alignment. Therefore, information structure is insufficient for modelling the indirective-secundative alternation and requires further investigation.

2.1 Khanty language, dialectal classification and sociolinguistic situation

Khanty, also known as Ostyak, is an endangered language or group of languages spoken in North-West Siberia, along the river Ob and its tributaries. Modern Khanty people live in Khanty-Mansi Autonomous Okrug, Yamalo-Nenets Autonomous Okrug, and parts of Tomsk Oblast in Russia (Nikolaeva 1999, Solovar et al. 2016).

Khanty is a language in the Ob-Ugric branch of the Uralic language family. Its closest relatives are Mansi (also known as Vogul) and Hungarian. The exact classification of the different Khanty varieties, as well as their status as dialects or languages, remains a matter of discussion. A detailed history of the classification can be found in Kaksin (2010) and Fedotova (2022).

Karjalainen (1902) proposed two-part division of Khanty varieties into Western and Eastern Khanty. This classification is adopted by e.g. Tereškin (1974, 1981) and Solovar et al. (2016: p.13). On the other hand, Steinitz (1937) proposed a more detailed classification that distinguishes between three Khanty dialects:

1. Eastern Khanty dialects (dialects of Vakh-Vasyugan, Surgut and Salym)
2. Southern Khanty dialects (dialects of Irtysh and Demyanka)
3. Northern Khanty dialects (dialects of middle Ob, Kazym, Shuryshkary and Obdorsk)

This classification is adopted by Nikolaeva (1999) and accepted in most modern studies, e.g. Filchenko (2007); Forsberg (2024). Eastern and Northern varieties are not mutually intelligible and differ in phonology, morphology, and syntactic characteristics (e.g. Nikolaeva 1999 and Solovar et al. 2016 for details). These radical differences among certain varieties led Balandin (1955) to conclude that Khanty is more correctly described as a group of closely related languages. Vakh, Vasyugan and Surgut Khanty form a group of Eastern Khanty langauges. Northern and Southern Khanty correspond to the respective dialects in Steinitz's classification.

1. Vakh Khanty + Vasyugan Khanty
2. Surgut Khanty
3. Northern Khanty (includes Kazym, Shuryshkary and Obdorsk dialects)
4. Southern Khanty

Fedotova (2022) provides further arguments in favor of this view. Based on lexical statistics, she argues that not only grammatical but also lexical differences among these varieties are sufficient to treat them as separate languages. The map in Figure 2.1 illustrates the geographical distribution of the different Khanty languages.

The sociolinguistic situation is different for all four languages and even between dialects in one language. The Southern Khanty dialects became extinct in the second half of the 20$^{\text{th}}$ century, when native speakers changed to Siberian Tatar and Russian (Eberhard, Simons, & Fennig (eds.) 2024). Eastern varieties, especially Surgut Khanty, are seen as more stable than Northern dialects. However, both the Eastern and Northern dialects of Khanty are endangered, with limited intergenerational transfer. Khanty is spoken at home, especially in reindeer herder families, but school education is only available in Russian, and children change to Russian once they reach school age. In general, most children can only understand Khanty and do not speak it (http://jazykirf.iling-ran.ru/groups/Ob-Ugric.shtml).

Aristova (2023) conducted an interview-based study on the linguistic situation in the Kazym Khanty community and reported a gradual loss of the language. The older generation knows and actively uses the language in the community. The middle generation knows the language but does not use it in everyday life. The younger generation understands Khanty but cannot speak it, and children have completely lost the language.

Similar tendencies are reflected in official sources, which do not distinguish between different varieties and dialects. The population census from 1989 reported that 61% of the ethnic Khanty population spoke Khanty, while only 37% knew the language in 2010. The number of the ethnic Khanty who reported Russian being their first language has grown from 36% in 1989 to 62,5% in 2010 (Koškareva 2016).

Khanty has its own alphabet, based on the Russian Cyrillic script, with the addition of several special characters. Throughout the dissertation, a transcription developed by

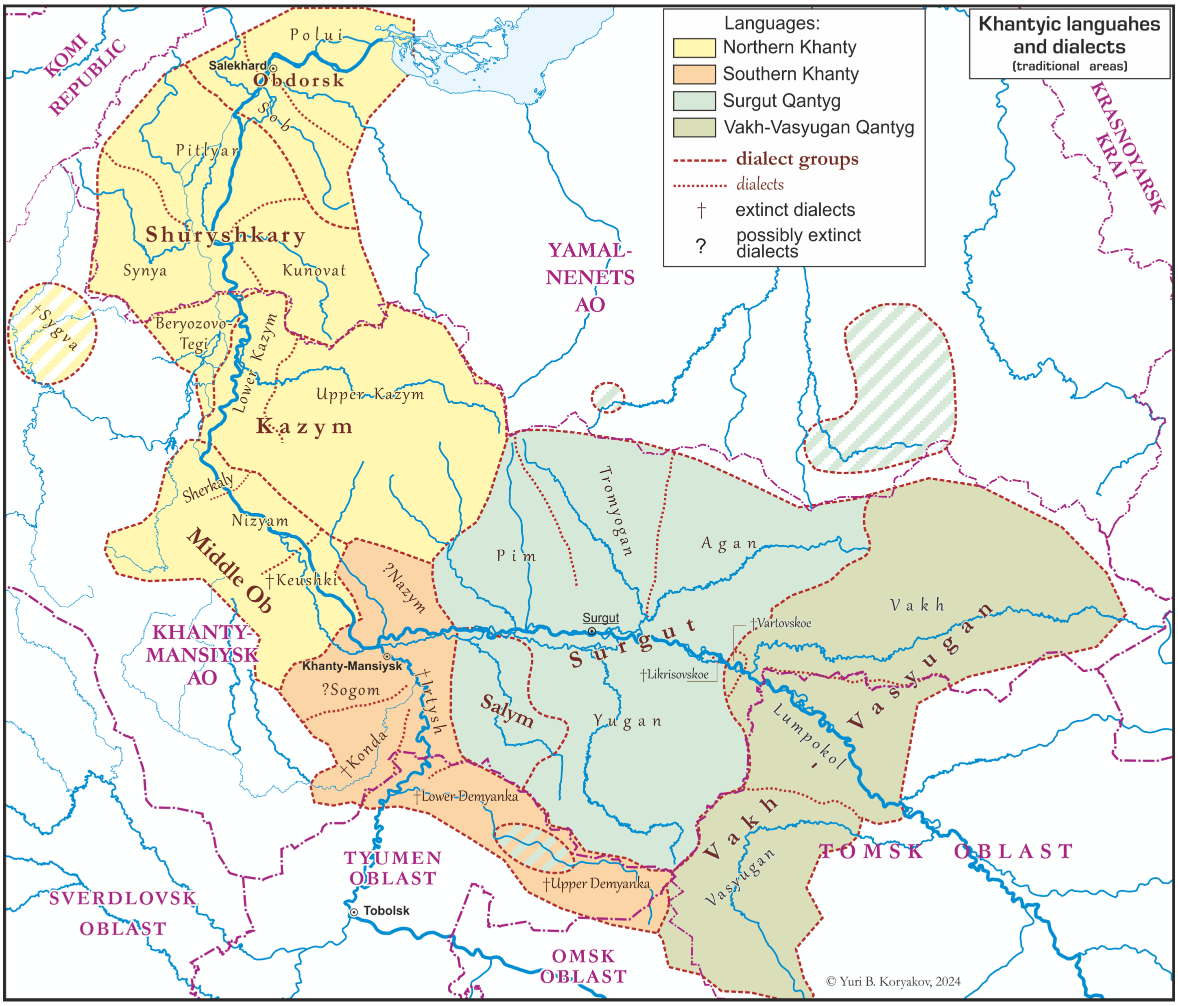


Figure 2.1: Khanty languages and dialects (Koryakov 2024)

the Moscow research group is used for Kazym Khanty examples. There are two other transcription traditions based on Latin script: one used by Kaksin (2010) and another used by Koškareva (2013) and in the Western Khanty Corpus (Kaškin & Muraviev 2014). Correspondences across the three systems are summarized in Table 2.1 for vowels and in Table 2.2 for consonants.

IPA	Current transcription	Kaksin (2010)	Koškareva (2013)
ɑ	a	a	a
a	ă	ă	ă
e	e	e	e
ɛ	ɛ	ɛ	ɛ
i, ɨ	i	i	i, i̬
ɔ	o	o	ɔ
ɵ	ɵ	ŏ	ǫ
u	u	u	ŏ
ʉ	ʉ	ŭ	ŭ
ə	ə	ə	ə

Table 2.1: Vowels orthography

IPA	Current transcription	Kaksin (2010)	Koškareva (2013)
j	j	j	j
k	k	k	k
l	l	l	l
ɬ	λ	ɬ	λ
ɬ’	λ’	ɬ’	λ’
m	m	m	m
n	n	n	n
ɲ	ń	ń	ń
ŋ	ŋ	ŋ	ŋ
p	p	p	p
r	r	r	r
s	s	s	s
ʂ	š	š	š
ɕ	ś	ś	ś
t	t	t	t
c	t’	-(t)	t’
v	w	w	w
χ	χ	χ	χ

Table 2.2: Consonants orthography

2.2 Morphology

Khanty is a predominantly agglutinative language. Derivational and inflectional morphology is suffixal in both the verbal and nominal domains. This subsection provides an overview of inflectional morphology relevant to the rest of this dissertation. The first part of this subsection describes nominal morphology, namely number marking, possessive markers, and case markers. The second part provides an overview of the verbal domain and introduces tense morphology and ϕ-agreement markers.

2.2.1 Nominal morphology

In Kazym Khanty, similar to other varieties of Khanty, nouns are inflected for number and case and can bear a suffixal possessive marker.

(6) stem - (derivational morphology) - (number) - (possessor) - (case)

There is no nominal concord in number, possession, or case inside the NP, i.e. these features are only spelled out on the phrase-final noun.

Khanty personal pronouns differentiate between three persons and three numbers, as summarized in Table 2.3.

Like other Uralic languages, Kazym Khanty has no grammatical gender distinction; for instance, the third-person pronoun *λʉw* can refer to both male and female referents. Similarly, nouns do not show any syntactic correlates of the grammatical gender.

	SG	DU	PL
1	ma	min	mʉŋ
2	năŋ	nin	nin
3	λʉw	λin	λiw

Table 2.3: Personal pronouns

Number marking

As already mentioned for pronouns, Khanty has a three-way number distinction, differentiating between singular, dual, and plural forms. The number exponent is directly adjoined to the stem. When derivational suffixes are present, e.g. diminutive morphology, the number exponent adjoins to it. Each number marker has two allomorphs. The choice of a particular allomorph depends on the presence of a possessive suffix that directly follows the number morpheme. Both paradigms are presented in Table 2.4. Singular is zero-marked in both declension types. If possessive morphology is absent, the dual and plural exponents take the forms in 7.

	bare declension	possessive declension
SG	-∅	-∅
DU	-ŋan	-ŋaλ-
PL	-ət	-(ə)λ-

Table 2.4: Number exponents

(7) a. χop-ŋan
boat-DU
'(two) boats'

b. χop-ət
boat-PL
'(many) boats'

The allomorphs in the second column are used in the context of possessive markers, which reflect the person and number of the possessor. Some examples are provided in 8.

(8) a. χop-λ-am
boat-PL-POSS.1SG
'my boats'

b. χop-λ-aw
boat-PL-POSS.1PL 'our boats'

c. χop-aw
boat.[SG]-POSS.1PL
'our boat'

Note that the distribution of number marking is not straightforward. Number marking is obligatory on all referential and human DPs, regardless of their syntactic position. However, non-human and non-referential direct objects mark number optionally, as e.g. in 9.

Furthermore, subjects can also be optionally marked for number, but only if the subject DP is inanimate and non-referential, as in 10 (Pisarenko 2020).

(9) λʉw wʉλi / wʉλe-t λapt-əs
(s)he.[NOM] reindeer[SG.ACC] / reindeer-PL.[ACC] feed-PST.[3SG]
'He fed reindeers.' (Pisarenko 2020: p.184, ex.10)

(10) pʉt-ət / pʉt oməs-λ-ət / oməs-λ păsan ɵχti-jən
pot-PL pot sit-NPST-3PL sit-NPST.[3SG] table on-LOC
'Pots (usually) are on the table.' (Pisarenko 2020: ex.12)

Finally, the number exponent is always absent when the noun is modified with a numeral. As the example in 11 illustrates, the presence of a numeral blocks the overt realization of the plural marker.

(11) wet pʉt / *pʉt-ət oməs-λ păsan ɵχti-jən
five pot pot-PL sit-NPST.[3SG] table on-LOC
'There are five pots on the table.' (Pisarenko 2022: p.1, ex.4)

As 12 illustrates, a subject with a numeral modifier can trigger both singular and plural agreement on the verb. Presence of plural agreement seems to correlate with definiteness, since plural agreement on the finite verb is obligatory, when the subject has a demonstrative (Burov, Kulikova, Pisarenko, Starčenko, Tyutyunnikova, & Čeremisinova 2024).

(12) wet kinška iλ pit-s-ət / pit-əs.
five book.[NOM] down become-PST-3PL become-PST.[3SG]
'Five books fell down.' (Burov et al. 2024)

Possessive markers

Possessors in Kazym Khanty are merged as DPs-modifiers and optionally trigger agreement on the head noun. The suffixal possessive marker reflects the person and number values of the possessor, e.g. 13.

(13) (mʉŋ) amp-ew
we dog-POSS.1PL
'our dog'

Note that pronominal possessors obligatorily trigger possessive agreement on nouns and are often non-overt (Mikhailov 2023). Overt nominal possessors, on the other hand, are only optionally doubled by a possessive suffix, as illustrated in 14.

(14) Tɵχtər-en λʉweλ [[mɵšən ut-λ]
doctor-POSS.2SG.[NOM] (s)he.DAT sick something-POSS.3SG
purtɵŋ-(əλ)] mă-s-λe
medicine-(POSS.3SG).[ACC] give-PST-3SG>SG
'The doctor gave him patient's medicine.'

Mikhailov (2023) distinguishes between four main functions of possessive markers in

Kazym Khanty. Apart from the proper possessive semantics, possessive markers in Kazym Khanty have several discourse-related functions.

First, possessive morphology can express canonical possessive relations. In this case, the exponent spells out the person and number features of an overt or covert possessor. Note that a proper possessive marker does not require the possessum to be unique or definite. The possessum 'cup' in 15 is modified by an existential quantifier 'some' and the possessive marker *-en* is fully grammatical.

(15)

muλsər	an-en	mij-a
some	cup-POSS.2SG.[ACC]	give-IMP.[SG]

'Give me some of your cups.' (Mikhailov 2023: p.20, ex.7)

Second, there is an associative use of possessive morphology. The associative possessive marker reflects temporal and context-dependent relations between two items. In 16, the cup on the table is not owned by the addressee, but the addressee stands closer to the cup and is, therefore, associated with it.

(16) [The speaker has a friend as a guest. There is only one cup on the table. The speaker says to their friend:]

an-*(en)	mi-je
cup-POSS.2SG	give-IMP.SG>SG

'Give me the cup.' (Mikhailov 2023: p.22, ex.9)

This function of possessive markers presupposes the uniqueness of the referent. In particular, the sentence in 16 cannot be uttered if there are several cups on the table. Furthermore, the associative possessive presupposes existing and known possessums, making it similar to the English definite article.

In the two functions above, the possessive morphology references the ϕ-features of a possessor in a wide sense. In addition, two possessive exponents can be used in article-like functions.

The marker -əλ (POSS.3SG) can be used as a partitive article referring to a part of a contextually defined group. In 17, the dog is not necessarily owned by someone or stands close to someone. Rather, it is a part of the group of dogs mentioned in the previous sentence. Note that there is no overt or covert noun that can control the person and number values of the possessive exponent. The 'two dogs' in the preceding sentence cannot be seen as an antecedent because the number value does not match.

(17) [I walked down the street and saw two dogs.]

i	amp-*(əλ)	ma	pɛλ-am-a	χurət-ti	pit-əs
one	dog-POSS.3SG	I	at-POSS.1SG-DAT	bark-NFIN.NPST	turn-PST.[3SG]

'One of the dogs started to bark at me.' (Mikhailov 2023: p.26, ex.16)

The possessive marker *-en* (-POSS.2SG) can be also used as a topic marker, signaling known and contextually unique referents. In 18, the two dogs are mentioned in the immediate context, which makes them known and contextually unique. In this function, possessive marker is similar to a definite article (see e.g. Schwarz 2013).

(18) [I walked down the street and saw two dogs.]

amp-ŋəλ-an	ma	pɛλ-am-a	χurət-ti	pit-əs-əŋən
dog-DU-POSS.2SG	I	at-POSS.1SG-DAT	bark-NFIN.NPST	turn-PST-3DU

‘The two dogs started to bark at me.’ (Mikhailov 2023: p.28, ex.21a)

Despite Mikhailov’s terminology, this marker is not restricted to (aboutness) topic. It marks the most salient DP in the clause, and in certain cases, can mark a (contrastive) focus, as is illustrated in 19.

(19) [“Who hit you?"]

ma	śi	aj	ik-en-ən	χǎtś-əs-ij-əm
I	DEM	small	man-POSS.2SG-LOC	hit-PST-PASS-1SG

‘That boy hit me.’ (lit. ‘I was hit by that boy’) (Mikhailov 2023: p.31, ex.24)

Importantly, the topic marker can be used only once per clause. A second known and unique noun is marked with an associative possessive marker, as 20 shows.

(20) [I went down the street and saw a boy and a girl.]

ew-en	aj	ik-eλ/*-en	χǎtś-əs-λe
girl-POSS.2SG.[NOM]	small	man-POSS.3SG/POSS.2SG.[ACC]	hit-PST-3SG>SG

‘The girl hit the boy.’ (Mikhailov 2023: p.32, ex.25)

Finally, the marker *-en* (-POSS.2SG) is obligatory on proper names in all argument positions. It does not depend on salience or topicality and is present on every proper name in the clause, e.g. 21.

(21)

Maša-jen	Petr-en	šiwaλə-s-λe
Masha-POSS.2SG.[NOM]	Petr-POSS.2SG.[ACC]	see-PST-3SG>SG

‘Masha saw Peter.’

In summary, possessive markers in Khanty have various functions. In addition to conveying information about a possessor, they can also have discourse- and definiteness-related functions. In the latter case, possessive markers are similar to definite articles.

Case

Case is the category where Eastern and Northern Khanty languages and even their dialects differ the most. The three dialects of Northern Khanty have a relatively small case system with only three morphological case forms in each dialect. Interestingly, even with such a small number of cases, there are significant differences among the dialects. Moreover, the pronominal case system is never identical to the case system of nouns and is also different in all three dialects.

In this section, I describe the case system of the Kazym dialect of Northern Khanty.

Case system of Eastern Khanty is described in Chapter 5, Section 5.4. Case systems of Odborsk and Shuryshkary dialects of Northern Khanty are discussed in Appendix A.

As already mentioned, Khanty has different sets of exponents for nouns and personal pronouns (e.g. Kaksin 2010; Solovar et al. 2016). As the paradigm in Table 2.5 shows, nouns have three case exponents. Nominative and accusative are zero-marked, dative is marked with *-a*, and locative-instrumental (usually labeled as locative, e.g. Kaksin 2010) is marked with *-ən*.

NOM/ACC	-∅
DAT	-a
LOC-INSTR	-ən

Table 2.5: Case exponents (nominal declension)

Across all Khanty varieties, personal pronouns and nouns have different sets of case exponents (Solovar et al. 2016). In Kazym Khanty, personal pronouns have three separate morphological case forms: nominative, accusative, and dative. Recall that nouns have one exponent for nominative and accusative and an additional exponent for locative-instrumental. The full declension paradigm of personal pronouns is presented in Table 2.6. Crucially, nominative is distinct from accusative, which suggests that the syncretism in the nominal paradigm does not indicate a reduced case system, and Kazym Khanty has both nominative and accusative as separate cases. Note also that accusative and dative have two allomorphs. Koškareva (2002) and Klumpp (2012) argue that distribution of these allomorphs depends on information structure. In particular, they argue that non-reduced forms, i.e. *a*-forms in dative and *ti*-forms in accusative are used, whenever the pronoun is a narrow focus.[1]

	NOM	ACC	DAT
1SG	ma	mănət/mănti	mănɛm/mănɛma
2SG	ŋăn	ŋănət/ŋănti	ŋănen/ŋănena
3SG	λʉw	λʉwət/λʉwti	λʉweλ/λʉweλa
1DU	min	minət/minti	minɛmən/minɛmna
2DU	nin	ninət/ninti	ninan/ninana
3DU	λin	λinət/λinti	λinan/λinana
1PL	mʉŋ	mʉŋət/mʉŋti	mʉŋew/mʉŋewa
2PL	niŋ	niŋət/niŋti	niŋan/niŋana
3PL	λiw	λiwət/λiwətti	λiweλ/λiweλa

Table 2.6: Case exponents (pronominal declension) (Solovar et al. 2016: p.110)

Overall, Kazym Khanty has a system of four cases, which have separate morphological exponents either in nominal or pronominal declension.

Nominative case marks subjects and possessors, in 22 and 23 respectively. Complements of postpositions are also always zero-marked, e.g. 24.

[1] But note that in an earlier paper, Koškareva (2001) argues that the two different forms of accusative are "regional variants" within the Kazym dialect, while Kaksin (2010) does not mention the reduced accusative form.

(22) a. ńawrɛm χoλλ-əλ.
child.[NOM] cry-NPST.[3SG]
'A/the child cries.'

b. λʉw χoλλ-əλ.
(s)he.NOM cry-NPST.[3SG]
'He/she cries.'

(23) a. tăm ɛw-en jɛrnas-λ-aλ
this girl-POSS.2SG.[NOM] dress-PL-POSS.3SG
'This girl's dresses'

b. ma jɛrnas-λ-am
I.NOM dress-PL-POSS.1SG
'My dresses'

(24) păsan ɵχti-jən
table on-LOC
'on the table'

Direct objects are marked with accusative, which is syncretic with nominative on nouns (25a) and has a separate exponent in pronominal declension (25b).

(25) a. tʉm moləpś-en wʉj-e.
That fur_coat-POSS.2SG.[ACC] take-IMP
'Take that coat.'

b. Ma λʉwt(i) woš-ən wan-s-əm.
I.NOM (s)he.ACC town-LOC see-PST-1SG
'I saw him/her in town.'

I assume that Kazym Khanty differentiates between nominative and accusative in syntax, despite the fact that both cases are marked with a ∅-exponent on nouns. Two different forms of pronouns cannot be derived if no nominative-accusative distinction is present in the syntax.

Dative case is used to mark experiencers (26a), recipients (26b), and directions (26c-26d). Dative can also be used as a lexical case on complements of certain verbs, e.g. 'touch' in 27.

(26) a. Petja-jen-a χot kăλ
Petya-POSS.2SG-DAT house seen.NPST.[3SG]
'Petja can see a/the house.' (Ryžowa 2021: ex.6)

b. Ma năŋena juλən măw mă-λ-əm
I.NOM you.DAT at_home candy.[ACC] give-NPST-1SG
'I give you a candy at home.'

c. Petja-jen χot-a λoŋ-əs
Petya-POSS.2SG.[NOM] house-DAT enter-PST.[3SG]
'Petya entered the house.' (Ryžowa 2021: ex.7)

d. Vasja-jen woš-a posilka kit-s.
Vasja-POSS.2SG.[NOM] town-DAT parcel-[ACC] send-PST.[3SG]
'Vasya sent a parcel to the town.'

(27) Maša-jen λɛpət λońś-a ketəm-s
Masha-POSS.2SG.[NOM] soft snow-DAT touch-NPST.[3SG]
'Masha touched the soft snow.'

Note that the dative form of pronouns is never used for directions and goals of movement (Kaksin 2010). A postposition with a possessive marker and an optional overt pronoun is used instead, as illustrated in 28 for the directional postposition *χuśa* 'to'.

(28) (năŋ) χuś-en-a
you.NOM to-POSS.2SG-DAT
'to you, in your direction'

Locative-instrumental case (also locative) is used to mark instruments and locations, as well as temporal adjuncts. All three uses are illustrated in 29.

(29) a. Instrument

Kartλʉŋk săŋkəp-ən śiw sɛŋk-e.
nail.[ACC] I.[NOM] hammer-LOC there_PREV hit-IMP
'Hit (the/a) nail with a hammer!'

b. Location

Ar joχ **tăm** woš-ən wɵλ-λ-ət.
many people.[NOM] this town-LOC be-NPST-3PL
'There live many people in this village/town.'

c. Temporal adjunct

towi-jən
spring-LOC
'in spring'

Demoted arguments are also marked with locative, e.g. the agent in the passive clause in 30 and the theme in secundative alignment in 31.

(30) Passive

Maw-λ-am Maša-**jen**-ən ńawrɛm-ɛm-a
candy-PL-POSS.1SG.[NOM] Masha-POSS.2SG-LOC child-POSS.1SG-DAT
mă-s-i-jət.
give-PST-PASS-3PL

'My candy was given by Masha to my kid.' (Colley & Privoznov 2020)

(31) Secundative alignment

Kašəŋ χujat λɵχs-əλ lipət-ən mă-s-λe
Every person.[NOM] friend-POSS.3SG.[ACC] flower-LOC give-PST-3SG>SG
'Everyone gave a flower/flowers to his friend.'

Recall from Table 2.6 that personal pronouns do not have locative-instrumental forms.

They are never used as demoted agents in the passive or themes in secundative construction. As for instrumental and locative adjuncts, pronouns are merged inside a PP in this positions (Kaksin (2010); Solovar et al. 2016: p.112). The fact that pronouns cannot be used as passive agents or secundative themes cannot be attributed to the absence of morphological form alone. Passive agents in Khanty must be (i) focal, (ii) new information, and (iii) less animate than the internal argument (Muravyev 2023). Pronouns do not satisfy these requirements and hence cannot be used as passive agents. As for secundative themes, I show in Chapter 4 that they have structural restrictions, which ban pronouns in this position, irrespective of the availability of a morphological form.

It can be concluded that the case system of the Kazym dialect of Khanty distinguishes four cases: nominative, accusative, dative, and locative-instrumental. I am not aware of any morphological or syntactic data that would allow the locative-instrumental to be divided into two distinct cases. Similarly, there is no reason to postulate the existence of a separate genitive case, since it would be syncretic with the nominative in both the nominal and pronominal paradigms.

2.2.2 Verbal morphology

(32) stem-(actionality)-(causative)-tense-(passive)-agreement

There is no applicative morphology on the verb, but an indirect object can be added to any transitive verb, as illustrated in 33-34 for two lexically monotransitive verbs.

(33) Poχatʉr (χon iki-ja) păsti woj weλəmt-əs
hero.[NOM] king man-DAT wolf.[ACC] kill-PST.[3SG]
'The hero killed a wolf/wolves for the king.'

(34) Ma (năŋena) kartɵpka lottə-s-əm
I.[NOM] you.DAT potato.[ACC] bury-PST-1SG
'I buried potatoes for you.'

Causatives can be derived from nominal and adverbial roots, as well as from intransitive and transitive verbs. Some examples of causative verbs derived from nouns, adjectives, and adverbs are given in 35. Deverbal causatives are illustrated in 36. Causativization is marked with various suffixes with unclear distribution (see, e.g., the overview by Moldanova 2018). Causative derivation is unproductive and lexically restricted (see Chapter 3, Section 3.3 for details).

(35) a. noun
χor 'ox' > χorj-aλə-ti 'rob, take by force'
b. adjective
λiλəŋ 'alive' > λiλŋ-ət-ti 'revive'
c. adverb
măšja 'silently' > măš-at-ti 'embarrass, put in a tricky position, oppose'

(36) a. unaccusative
pot-ti 'freeze, be cold' > pot-əλtə-ti 'cool, to make cold'
b. unergative
măn-ti 'go' > măn-əλt-ti ´make go'
c. transitive
want-ti ´see, look' > wan-(ə)λtə-ti 'show'

Kazym Khanty differentiates between two morphological tenses form. The past tense is marked with the exponent *-(ə)s* and the non-past with *-(ə)λ*, as illustrated in 37.

(37) a. Maša-jen arij-əλ
Masha-POSS.2SG.[NOM] sing-NPST.[3SG]
'Masha sings.'
b. Maša-jen ari-s
Masha-POSS.2SG.[NOM] sing-PST.[3SG]
'Masha sang.'

The passive exponent follows the tense exponent. There are two allomorphs: *-a*, as in 38, and a more restricted *-i*.

(38) upe-λ-ən ar-en ari-s-a
sister-POSS.3SG-LOC song-POSS.2SG.[NOM] sing-PST-PASS
'His sister sang him a song.' [Lit.: 'The song was sung by his sister.'] (Kaškin & Muraviev 2014)

Northern Khanty dialects have a straightforward nominative-accusative alignment in both ϕ-agreement and case marking. The finite verb obligatorily agrees with the nominative argument (subject) in person and number, and optionally agrees with the direct (ACC-marked) object in number. Some examples are provided in 39. The phi-agreement marker is always the last exponent in the morphological word, i.e. it follows both tense and passive exponents.

(39) a. Ma kašəŋ ewi-ja juntut(-əλ) mă-s-əm
I.NOM every girl-DAT toy-POSS.3SG.[ACC] give-PST-1SG
'I gave every girl her toy.'
b. Ma kašəŋ ewi-ja juntut(-əλ) mă-s-ɛm
I.NOM every girl-DAT toy-POSS.3SG.[ACC] give-PST-1SG>SG
'I gave every girl her toy.'
c. Ma kašəŋ ewi-ja juntut-ət mă-s-λam
I.NOM every girl-DAT toy-PL.[ACC] give-PST-1SG>NSG
'I gave toys to every girl.'

Subject and object agreement are expressed with a single portmanteau morpheme in Kazym Khanty, as the glossing in (39b-c) suggests.[2] In Obdorsk Khanty, on the other

[2]The -λ--part in *-λam* in (39c) can be argued to be a separate marker for non-singular object, which makes *-λam* a partially superfluous extended exponent, as the features of the object a expressed both with -λ- and *-am*.

hand, two separate morphemes can be clearly distinguished (Nikolaeva 1999; Dalrymple & Nikolaeva 2011).

The factors that regulate the presence of object agreement in Khanty are a matter of ongoing debate. The existing proposals are discussed in the next section.

2.3 Syntax

2.3.1 Passive

It is widely assumed in the research tradition on Ob-Ugric languages that both Northern and Eastern Khanty are information-structure configurational languages (see e.g. Nikolaeva 1999, 2001; Dalrymple & Nikolaeva 2011; Sosa 2017; É. Kiss 2019, 2021; Muravyev 2023, among many others). Passive is used when the internal argument (direct object, theme) is a primary topic and the agent (external argument) is unknown/unimportant, as in 40, or a new information focus, as in 41.

(40) Waśa-jen λapət-s-a
Vasya-POSS.2SG.[NOM] feed-PST-PASS.[3SG]
What about Vasya? 'Vasya was fed.' ((Muravyev 2023: p.50, ex. 3))

(41) Waśa-jen petja-jen-ən λapət-s-a
Vasya-POSS.2SG.[NOM] Petya-POSS.2SG-LOC feed-PST- PASS.[3SG]
Who fed Vasya? 'Vasya was fed by Petya.' ((Muravyev 2023: p.50, ex. 4))

For Kazym Khanty, Muravyev (2023) argues that the distribution of the passive is regulated by both the information structure and prominence scales, such as animacy and definiteness/givenness. The passive is used whenever the internal argument is more topical, given, and/or more animate than the external argument of the clause. In 42, the patient 'Vasya' is higher on the animacy scale than the agent 'wasp'; hence, the passive voice is preferred.

(42) a. Waśa-jen śi pos-ən tɵχəm-s-a
Vasya-POSS.2SG.[NOM] DEM wasp-LOC sting-PST-PASS.[3SG]
'Vasya was stung by the bee.'

b. %śi pos-en waśa-jəλ tɵχəm-s-əλλe
DEM wasp-POSS.2SG.[NOM] Vasya-POSS.3SG.[ACC] sting-PST-3SG>SG
Exp.: 'The bee stung Vasya.' ((Muravyev 2023: p.58, ex. 30))

Muravyev (2023) suggests the following generalization. When both participants are new information, the choice of voice is regulated by animacy. When the external argument is more animate, the active voice is used. When the internal argument is more animate, the passive voice is preferred. When both participant are given, the choice of voice depends on which one of them is a primary topic of the clause.[3]

[3] For the definition of primary vs. secondary topic, see Nikolaeva (2001) and Dalrymple & Nikolaeva (2011).

2.3.2 Object agreement

Object agreement is one of the central topics of discussion in present-day research on Khanty. Nikolaeva (1999) and Dalrymple & Nikolaeva (2005, 2011) argue that in the Obdorsk dialect of Northern Khanty, verbs agree only with non-focal objects (secondary topics in their terminology). This analysis is followed in most studies on various Khanty dialects (e.g. Kulonen 1989; Nikolaeva 2001; Virtanen 2013, 2014, 2015; Virtanen & Sosa 2018; É. Kiss 2019, 2021; Muravyev 2023).

On the other hand, Nikolaeva (2001) and especially Däbritz (2020) argue that the givenness of the direct object might be a more important factor than topicality. Given objects are cross-referenced on the verb, while new-information objects never are. Filchenko (2012) for Eastern Khanty and Koškareva (2002) for Northern Khanty combine the two approaches, suggesting that only objects that are continuous topics control the agreement on the verb. Muravyev (2024) shows in a corpus study that all these analyses yield unsatisfactory statistical results. Therefore, neither the givenness nor the topicality of the object can provide a satisfactory explanation for the distribution of object agreement in Khanty.

From a syntactic point of view, there are two factors regulating the presence of object agreement.

Nikolaeva (1999) shows that in the Obdorsk dialect of Northern Khanty, non-agreeing referential objects are strictly required to stay VP-internal (verb-adjacent) and are syntactically inert. This is supported by four syntactic tests. First, non-agreeing referential objects cannot be scrambled. This prevents the possibility of quantifier stranding, as shown in 43.

(43) a.

luw	(asa)	a:n-ət	(asa)	il	pa:jət-s-əlli
he	all	cup-PL.[ACC]	all	down	drop-PST-3SG>NSG

‘He dropped all the cups.’ (Nikolaeva 1999: ex.180b)

b.

luw	(asa)	a:n-ət	(*asa)	il	pa:jət-əs
he	all	cup-PL.[ACC]	all	down	drop-PST.[3SG]

‘He dropped all the cups.’ (Nikolaeva 1999: ex.181)

Second, non-agreeing referential objects cannot bind a possessive marker. In 44a, the agreeing direct object can bind a possessor on the locative adjunct ‘his house’. The non-agreeing direct object in 44b cannot be an antecedent, and the possessive marker can only be bound by the subject.

(44) (Nikolaeva 1999: ex.178)

a.

a:śi	pox-əl	xo:t-əl-na	wa:n-s-əlli
father.[nom]	son-POSS.3SG.[ACC]	house-POSS.3SG-LOC	see-PST-3SG>SG

‘The father$_{i}$ saw his$_{i}$ son$_{j}$ in his$_{i/j}$ house.’

b.

a:śi	xo:t-əl-na	pox-əl	wa:nt-əs
father.[nom]	house-POSS.3SG-LOC	son-POSS.3SG.[ACC]	see-PST.[3SG]

‘The father$_{i}$ saw his$_{i}$ son$_{j}$ in his$_{i/*j}$ house.’

Third, there is an asymmetry in possessor topicalization, as illustrated in 45. A possessor can be topicalized out of an agreeing direct object. Non-agreeing direct objects do not allow possessor topicalization.

(45) a. juwan motta xo:t-əl kaś́a:l-əs-e:m
Ivan before house-POSS.3SG.[ACC] see-PST-1SG>SG
'I saw Ivan's house before.' (Nikolaeva 1999: ex.182)

b. *juwan motta xo:t-əl kaś́a:l-əs-əm
Ivan before house-POSS.3SG.[ACC] see-PST-1SG
'I saw Ivan's house before.' (Nikolaeva 1999: ex.184a)

Finally, object control into participial embedded clauses is unavailable for non-agreeing direct objects. The non-agreeing direct object 'fish' in 46 cannot bind a pro in a participial clause.

(46) [**pro* / u:n u:l-m-(al) u:rəŋna] xul nox an t:al-səm
fish large be-PP-3SG fish up NEG carry-PST-1SG
'I didn't take out the fish because it was large.' (Nikolaeva 1999: ex.182)

Similarly, Smith (2020) shows for Surgut Khanty that agreeing objects occur ex situ (and are, presumably, A-scrambled). Hence, referential non-agreeing objects do not allow any overt or covert scrambling. This relates them to pseudo-incorporated nouns, which also tend to stay in situ and be syntactically inert (e.g. Massam 2009; Dayal 2011; Borik & Gehrke 2015; Driemel 2020). A pseudo-incorporation analysis has indeed been suggested for direct objects that do not control agreement on the verb and are number netral, i.e. bare singulars (Tiutiunnikova, Mikhailov, & Golosov 2024, and especially Tiutiunnikova 2024). Syntactic properties and the status of non-agreeing non-bare objects, i.e. those overtly marked for (non-singular) number and/or modified with possessive markers or demonstratives, have not been discussed in the literature and are less clear.

Non-referential objects in Odrosk Khanty are less restricted, e.g. they do not trigger object agreement on the verb even when scrambled. For example, the non-referential object in 47 precedes the wh-in-situ adverb 'where', but object agreement is not possible on the verb.

(47) li-ti pil xal'sa kas-l-əm / *kas-l-e:m
eat-NFIN.NPST companion.[ACC] where find-NPST-1SG find-NPST-1SG>SG
'Where shall I find a companion to eat with?' (Dalrymple & Nikolaeva 2011: p.149 ex.6)

Nikolaeva (1999) and Dalrymple & Nikolaeva (2011) argue that scrambled direct objects are topical and therefore trigger object agreement. Objects in situ, on the other hand, are focal, and the verb does not agree with them.

This is not a unique feature of Khanty that scrambling, and short object scrambling in particular, is related to topicality (e.g. Diesing 1992; Choi 1998, 1999 and references therein for German and Korean, É. Kiss 2003 and references therein for Hungarian, Biskup 2006 for Czech, Bailyn 2001 for Russian, Hinterhölzl 2012 for German, Neeleman & Van de Koot 2008; Schoenmakers, Poortvliet, & Schaeffer 2022 and references therein for Dutch, among others).

The Kazym dialect has additional factors that regulate the presence of object agreement. Similar to the Obdorsk dialect, the topicality of the object has been argued to trigger object agreement (e.g. Koškareva 2002). However, Kozlov (2022) shows that the aspectual value of the verb also influences the presence of object agreement. For example, perfective contexts favor object agreement, so that the verb can agree even with focal objects 48. In contrast, the imperfective aspect can block agreement, even with topical objects. If the verb is imperfective and the direct object is focal, as in 48b, object agreement is unavailable.

(48) λʉw muj wɛrs?
'What did he do?'

a. λʉw năŋ pɵšas-en λeśit-s-əλλe
he.[NOM] you fence-POSS.2SG.[ACC] fix-PST-3SG>SG
'He has fixed your fence' (perfective)

b. λʉw năŋ pɵšas-en λeśit-s
he-[NOM] you fence-POSS.2SG.[ACC] fix-PST.[3SG]
'He was fixing your fence' (imperfective) (Kozlov 2022)

Moreover, my fieldwork data show that object agreement is unavailable in clauses with a syntactically complex aspect, e.g. in habitual and perfect contexts. This is most clearly visible in ditransitive clauses with secundative alignment, where the indirect object is promoted to a primary object (terminology after Malchukov et al. 2010, sometimes referred to as Dative Shift, e.g., Sipos 2022). As 49 shows, the finite verb obligatory agrees with accusative-marked indirect objects when the aspect value is neutral.

(49) a. Ma komnata-ja χɵn λun-s-əm, λʉw
I.[NOM] room-DAT when come-PST-1SG (s)he.[NOM]
ew-eλ moś-ən mońś-s-*(əλλe)
daughter-POSS.3SG.[ACC] story-LOC tell-PST-3SG>SG
'When I came into the room, (s)he was telling a story to his/her daughter.'

b. λiw aškola-jew rɵpatnək-ən tɵ-s-*(eλ) pa
They.[NOM] school-POSS.1PL.[ACC] worker-LOC bring-PST-3PL>SG ADD
juχi Beloyarskij-ja măn-s-ət.
home Beloyarsky-DAT go-PST-3PL
'They brought workers to our school and went home to Beloyarky.'

In the aspect-neutral clause in 49, object agreement is obligatory present. On the other hand, object agreement is ungrammatical in clauses with sematically complex aspect. In the complex clause with a sequence of events in 50, the high aspect is valued as perfect, and object agreement is unavailable. High imperfectivization, e.g. the habitual aspect in 51, blocks object agreement as well.

(50) Maša-jen λant jiŋk-əλ lopša-jən
Masha-POSS.2SG.[NOM] flour water-POSS.3SG.[ACC] noodles-LOC

ɛsʎ-əs / *ɛsʎ-s-əʎʎe , omsəmt-əs pa
put_down-PST.[3SG] / put_down-PST-3SG>SG sit_down-PST.[3SG] ADD
kińška ʎʉnət-ti wʉ-s.
book.[ACC] read-NPST.NFIN take-PST.[3SG]
'Masha put noodles in the soup, then sat down, then took a book to read <then read a bit, then checked the soup, then went out of the kitchen...>'

(51) ʎiw aškola-jew rɵpatnək-ən tɵ-s-ət /
they.PL.[NOM] school-POSS.1PL.[ACC] worker-LOC bring-PST-1PL /
*tɵ-s-eʎ pa juχi Beloyarskij-ja măn-s-ət.
bring-PST-3PL>SG ADD home Beloyarsky-DAT go-PST-1PL
'They were bringing workers to our school and going home to Belojarsky (every day, as a usual route).'
Alternatively: 'The company <did X, Y, Z,> brought workers to school and went home to Beloyarsky.'

From a crosslinguistic perspective, it is not surprising that a structurally complex aspect blocks ϕ-agreement. Further examples are discussed by Coon (2013); Coon & Preminger (2017); Kalin & van Urk (2015); Kalin (2018); Levin (2015) (for a formal analysis, that captures Kazym Khanty data straightforwardly see Csirmaz 2004; Kardos & Farkas 2021).

2.3.3 pro-drop

Khanty allows pro-drop in both subject and object positions. The subject pro-drop is illustrated in 52. An example of pro-drop in the object position is provided in 53.

(52) *pro* ʎajm-ɛm jiŋk-a wɵtšə-s-əm
axe-POSS.1SG water-DAT lose-PST-1SG
'I've dropped my axe into the water.' (Kaškin & Muraviev 2014)

(53) – χot-ɛm jɵrən nuχ wu-s-i.
house-POSS.1SG by_force PREV_up take-PST-PASS

– Wuχsar-en ki *pro* jɵrən nuχ wu-s-ʎe ...
fox-POSS.2SG if by_force PREV_up take-PST-3SG>SG

'(– What happend with your house?) – My house was taken by force. – If the fox has taken *it* by force, [we kick him out now].'

Pro-dropped objects obligatory trigger object agreement on the finite verb, as in 53. This is expected, also because pro-dropped objects tend to be topics, and topical objects trigger agreement on the verb, even when they are realized overtly.

2.3.4 Word order

Khanty, and the Kazym dialect in particular, is consistently head-final. It has only postpositions, with one example given in 54. All modifiers are strictly prenominal, as illustrated in

55 for demonstratives, adjectives, nominal modifiers, and relative clauses.

(54) păsan χuśa
table at
'at the table'

(55) a. śit piti χoŋra
that.one black woodpecker
'That black woodpecker' (Kaškin & Muraviev 2014)

b. juχan tum pɛlka
river that side
'That side of the river'

c. [ma aś-ɛm ari-ti ar] jis
I father-POSS.1SG.[NOM] sing-NFIN.NPST song.[NOM] ancient
'The song that my father is singing/will sing is old.' (Bikina et al. 2022: ex.11)

Khanty is an SOV language, where the finite verb stays clause-finally in the majority of the clauses, e.g. 56. The neutral and the most common order of the arguments is [subject > indirect object > direct object], but Khanty is a free word-order language. Preverbal elements can be freely scrambled, as illustrated in 57. In 57a, the order of the two objects is reversed with respect to the unmarked order in 56. Adverbs can also be scrambled with arguments in the middle field, as (57b) illustrates for a low adverb and indirect object.

(56) Kašəŋ ewi(-jen) kat'-əλ-a juntut mă-s
Every girl-POSS.2SG.[NOM] cat-POSS.3SG-DAT toy.[ACC] give-PST.[3SG]
'Every girl gave her cat a toy.'

(57) Scrambling

a. Kašəŋ ewi(-jen) juntut kat'-əλ-a mă-s
every girl-POSS.2SG.[NOM] toy.[ACC] cat-POSS.3SG-DAT give-PST.[3SG]
'Every girl gave her cat a toy.'

b. Vasja-jen (sora) χujat-λ-a (sora) jəš
Vasya-POSS.2SG quickly someone-POSS.3SG-DAT quickly way
wanλta-s
show-PST.[3SG]
'Vasya quickly showed someone the way...'

Focus fronting is also present in Modern Kazym Khanty. My consultants readily allowed examples in 58, where an indirect and a direct object precede an overt subject.

(58) Topicalization

a. Kat'-əλ-a kašəŋ ewi-jen juntut-əλ
cat-POSS.3SG-DAT every girl-POSS.2SG.[NOM] toy-POSS.3SG.[ACC]
mă-s-λe
give-PST-3SG>SG
'It was the cat, that every girl gave its/her toy.'

b. Juntut-əλ kašəŋ ewi-jen kat'-əλ-a
toy-POSS.3SG.[ACC] every girl-POSS.2SG.[NOM] cat-POSS.3SG-DAT
mă-s-λe
give-PST-3SG>SG
'It was a toy, that every girl gave to a cat.'

The exact syntactic position of the fronted elements is a matter of further research, but such fronting is clearly different from the short scrambling in 57.

Subordinate clauses – both finite and non-finite – obligatorily precede the matrix clause. The two examples in 59 illustrate this for a finite conditional clause and a non-finite clausal adjunct.[4]

(59) a. [Wet sorńi wuχ ki ma-λ-ən], naŋ-ti ɛsλəmt-λ-ɛm.
five gold money if give-NPST-2SG you-ACC let_go-NPST-1SG>SG
'If you give me five golden coins, I'll let you go.' (Belkind 2023: ex.21)

b. [Wewλi juw-m-ew-ən] pʉt wɛr-s-əw
tired come-NFIN.PST-1PL-LOC pot.[ACC] make-PST-1PL
'When we got tired we cooked soup' (Muravyev 2021: ex.1)

Despite general verb-finality, extraposition is allowed (e.g. Nikolaeva 1999; Solovar et al. 2016; Asztalos, Borise, Gugan, Mus, Schmidt, & Surányi 2022). For Surgut Khanty, Asztalos et al. (2022) argue that both given and new constituents are allowed to occur postverbally. Extraposition in Kazym Khanty has not been sufficiently studied, but corpus data show that the same generalization seems to hold for this dialect as well. In 60a, the element in extraposition is a background topic, as it was mentioned in the immediate context. In 60b, the element in extraposition introduces a new character into the story and is a new information focus.

(60) a. Mʉŋ joχ-λ-əw piλa torn χăr-a ăkt-əś-s-əw.
we people-PL-POSS.1PL with grass place-DAT collect-DETR-PST-1PL
χop-ən măn-λ-əw torn χăr-a
boat-LOC go-NPST-1PL grass place-DAT
'My family and I are going to a meadow. We are going to the meadow with a boat.' (Kaškin & Muraviev 2014)

b. Tăm at i woš-ən i imi-lɛ xuśa sɛm-a pit-əs
this night one town-LOC one woman-PEJ.[NOM] at eye-DAT fall-PST.[3SG]
puχ
son.[NOM]
'This night, a woman in one town gave birth to a son.' (Kaškin & Muraviev 2014)

Solovar et al. (2016), as native speakers of Northern Khanty, report that extraposition of an element is allowed only under "expressive rhematisation" (p.111), i.e. extraposition

[4] Kiss (2023) reports that dialects of Eastern Khanty have finite subordinate clauses that follow the matrix clause. Importantly, such clauses are introduced by a complementizer borrowed from Russian. Moreover, Kiss argues that such subordinate clauses have only emerged in the language because extraposition and SVO order have become increasingly widespread over the years.

is reserved for given elements that are not part of a background. In formal terms, only contrastive topics or contrastive foci can be postverbal. Although Solovar et al. (2016) argue that subjects and direct objects are never allowed in extraposition, 60b shows that native speakers produce such structures. It remains unclear whether argument extraposition is used under Russian influence, as Nikolaeva (1999) argues, or whether Solovar et al. (2016) are too restrictive in their judgment. The latter view is supported by the fact that some of my native speaker consultants strongly forbade extraposition, while others did not. Tentatively, this may be a result of more prescriptive grammar being taught at teachers' colleges or universities (e.g. Sipos 2022). Regardless of the restrictions, it can be safely assumed that extraposition is a part of modern Khanty grammar (cf. Kiss 2023).

2.3.5 Position of wh-elements

Wh-elements in Kazym Khanty can remain in situ, as illustrated by 61.

(61) ma jaj-um mujt-əλ wʉ-s?
I brother-POSS.1SG.[NOM] what-POSS.3SG.[ACC] take-PST.[3SG]
'What did my brother take?' (Kasjanova 2019: p.3, ex. 11d)

If wh-element is agent in a transitive clause, passive voice is preffered, as e.g. in 62

(62) ńuχ-εŋ ńań χujt-en-ən / χuj-ən λε-s-i
meat-PROP bread.[NOM] who-POSS.2SG-LOC / who-LOC eat-EAT-PASS
'Who ate the meat pie?' (Kasjanova 2019: p.3, ex. 11b)

Nevertheless, wh-elements do not have to remain in situ and can be fronted. In 63, a wh-adverbial is fronted over the subject. In 64, the wh-theme 'what' is fronted over the subject and the indirect object.

(63) χuta naŋ wɵnλt-ijλ-λ-ən
Where you.NOM learn-IMPF-NPST-2SG
'Where do you study?' (Kaškin & Muraviev 2014)

(64) muj-ən aŋk-εm jaj-em tɵ-s-λe
what-LOC mother-POSS.1SG.[NOM] brother-POSS.1SG.[ACC] bring-PST-3SG>SG
'What did my mother bring to my brother?' (Kasjanova 2019: p.3, ex. 15b)

Kaksin (2010) argues that wh-fronting is the default option. When the wh-element is not clause-initial, the preceding element must be the topic. If this is true, wh-words must always be fronted, while a topical element can be fronted across them to the clause-initial position. This predicts that a wh-word can only be the first or second element in a clause. As 65 shows, this is not borne out.

(65) [Maša-jen] [naŋ saχ-en] **χuλta** tɵ-s-λe?
Masha-POSS.2SG.[NOM] you fur_coat-POSS.2SG.[ACC] where bring-PST-3SG>SG
'Where did Masha bring your coat?'

Therefore, I assume that Khanty is a wh-in-situ language. While wh-elements can be optionally fronted, this process is not triggered by the wh-feature and seems more likely to be an optional focus-fronting rather than wh-fronting.

2.4 Alignment alternation in Kazym Khanty

Crosslinguistically, applicative and causative constructions can have different alignments in ϕ-agreement and case marking. In parallel to the nominative-accusative and ergative-absolutive systems, Malchukov et al. (2010) differentiate between three main alignment types:

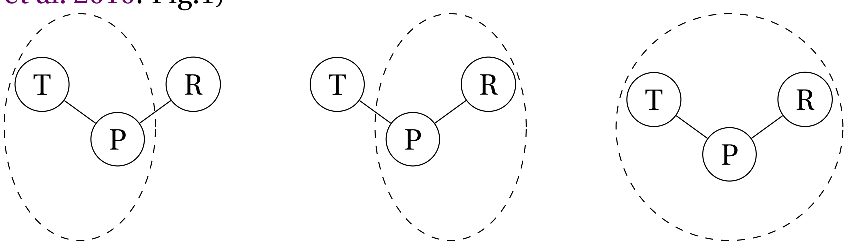
(i) indirective alignment: the theme (T) is marked the same way as the object in a monotransitive clause (P), and the recipient argument (R) bears a special marking

(ii) secundative alignment: R is marked the same way as the object in a monotransitive clause (P), and T bears a special marking

(iii) neutral alignment: both R and T are marked the same way as the object in a monotransitive clause (P)

These three alignments, schematically represented in 66, are the most common across languages. The other two theoretical possibilities – (i) when R is marked the same way as P, and (ii) when all three thematic roles have different markings – are almost unattested (Malchukov et al. 2010).

(66) indirective alignment secundative alignment neutral alignment (Malchukov et al. 2010: Fig.1)



There is significant crosslinguistic variation in the alignments used in applicative and causative clauses, i.e. clauses with two objects. Certain languages use only indirective alignment, e.g. Hungarian, German, and Russian. On the other hand, Nez Perce uses only secundative alignment, and English uses neutral alignment. There also exist languages where two alignment types are present simultaneously. The current study primarily concerns with the latter types of languages.

The central case study is Kazym Khanty, which has an alternation between indirective and secundative alignments. The alternation is illustrated in 67-68 for a lexically ditransitive verb.

(67) Indirective alignment

a. Sub-[NOM] IO-**DAT** DO-**[ACC]** V

b. Kašəŋ χujat λɵχs-əλ-a lipət mă-s
Every person.[NOM] friend-POSS.3SG-**DAT** flower.**[ACC]** give-PST.[3SG]
'Everyone gave a flower/flowers to his friend.'

(68) Secundative alignment

a. Sub-[NOM] IO-**[ACC]** DO-**LOC** V

b. Kašəŋ χujat λɵχs-əλ lipət-ən mă-s-λe
Every person.[NOM] friend-POSS.3SG.**[ACC]** flower-**LOC** give-PST-3SG>SG
'Everyone gave a flower/flowers to his friend.'

The indirective-secundative alternation is not restricted to lexically ditransitive verbs and is attested for applicativized monotransitive verbs, e.g. 'buy' in 69, and in causative clauses, e.g. 70.

(69) a. χăλewət woš-a măn-t-aλ sa, ɛwi-je-λ-a
tomorrow town-DAT go-NFIN.NPST-3SG while daughter-DIM-POSS.3SG-DAT
akań λɵt-λ
doll.[ACC] buy-NPST.[3SG]
'When he tomorrow goes to the city, he buys a doll for his daughter.'

b. χăλewət woš-a măn-t-aλ sa, ɛwi-jeλ-əλ
tomorrow town-DAT go-NFIN.NPST-3SG while daughter-DIM-POSS.3SG.[ACC]
akań-ən λɵt-λ-əλλe
doll-LOC buy-NPST-3SG>SG
'When he tomorrow goes to the city, he buys a doll for his daughter.'

(70) a. λawərt χir mănɛm aλm-əλt-s-əλe
heavy sack.[ACC] I.DAT lift-CAUS-PST-3SG>SG
'(S)he loaded me with a/the heavy sack.' [Moldanova 2018]

b. Mănti λawərt χir-ən aλm-əλt-s-əλe
I.ACC heavy sack-LOC lift-CAUS-PST-3SG>SG
'(S)he loaded me with a heavy sack.' [I.Moldanova p.c.]

This alternation is not a unique feature of the Kazym dialect of Northern Khanty. It is also present in the Obdorsk dialect, as well as in Surgut Khanty and all dialects of the closely related Mansi language (Ob-Ugric, Uralic). There is a consensus in the field that this alternation is regulated by similar factors in all Ob-Ugric languages and dialects. Indeed, the reported data for all of them are very similar, if not identical. The consensus is that secundative alignment is triggered by the topicality of the indirect object, which is then promoted to a direct object. In Subsection 2.4.1, I compare this claim with my own fieldwork data from the Kazym dialect of Northern Khanty and show that topicality alone cannot explain the alignment alternation. Subsection 2.4.2 introduces novel data on how prominence of both objects affects the availability of secundative alignment. I argue that secundative alignment in dispreferred if the IO does not exceed the DO in prominence. Nevertheless, a stronger claim about a triggering factor for secundative alignment cannot be made, since in most cases, both alignments seem to be in free variation.

2.4.1 Information structure

Nikolaeva (1999, 2001) and Dalrymple & Nikolaeva (2011), followed by Sipőcz (2015), argue at length that alignment choice is sensitive to the information structure in the Obdorsk dialect of Northern Khanty. This view is adopted in most of the literature on other Ob-Ugric languages (e.g. Sosa 2017; Filchenko 2007, 2012; É. Kiss 2019, 2021 for Eastern Khanty and Skribnik 2001; Virtanen 2012; Bíró & Sipőcz 2017; Bárány 2018; É. Kiss 2019, 2021 for different dialects of Mansi).

Nikolaeva (2001) proposes that secundative alignment is a marked option and is present when the indirect object is a secondary topic in the clause. She argues that topic is not unique in a clause. When two elements are already mentioned in the previous discourse, they are both topical, but their status is not identical. There can be only one aboutness topic, and the second element acts as a secondary topic.

(71) (Dalrymple & Nikolaeva 2011: p.55, ex.9)

a. Whatever became of John?
b. He married Rosa,
c. but he didn't really love her.

In 71b, only the subject is topical, since it is mentioned in the immediate context and is what the proposition is about. The object referent (Rosa) is part of the assertion, i.e. part of the (predicate) focus. In 71c, the object is already given in the immediate context and therefore cannot be part of the focus. Similarly, it cannot be considered (a part of) the primary topic, which stays the same throughout the discourse. The proposition is still primarily about the subject (John). Nikolaeva (2001) suggests that multiple topics can exist in one clause and uses the notion of a secondary topic. Secondary topic is defined as "an entity such that the utterance is construed to be ABOUT the relationship between it and the primary topic" (Nikolaeva 2001: p.26).[5] Importantly for the discussion in this subsection, both primary and secondary topics are parts of a presupposition and should be given in the immediate context.

A careful investigation of the Kazym Khanyt data shows that information structure and topicality of the indirect object indeed play a role, but cannot be seen as the only factor affecting the choice of alignment. A more accurate generalization for Kazym Khanty would be that narrowly focused R-participants strongly disfavor secundative alignment. In 72, the recipient 'sister' is a narrow focus, and secundative alignment is indeed dispreferred, though not completely ruled out by native speakers.

(72) {Who did Vasya bring the paint for?}

a. Indirective alignment: 5 ok

ok Vasja-jen	oʎəp	up-eʎ-a	tɵ-s.
Vasja-POSS.2SG.[NOM]	paint.[ACC]	sister-POSS.3SG-DAT	bring-PST.[3SG]

[5] Note that É. Kiss (2019, 2021) uses a different definition of topichood. For her, topic = [definite, specific], but see more on that in the next subsection.

b. Secundative alignment: 2 ok(?), 3 (?)?/*

?/?? Vasja-jen up-eλ oλəp-ən
Vasja-POSS.2SG.[NOM] sister-POSS.3SG.[ACC] paint-LOC
tɵ-s-λe.
bring-PST-3SG>SG

'Vasja brought paint to his sister.'

However, as 73 and 74 show, even when the R-participant is a secondary topic in the clause, i.e. mentioned/activated in the immediately preceding context, there is no clear preference for secundative alignment. Both indirective and secundative alignments are possible options.

(73) { Masha and I are walking together and see children paying. I say:}

a. ok Want-e, i ńawrɛm porχśɵp ăn tă-jəλ. Ma tăm
look-IMP one child.[NOM] coat.[ACC] NEG have-NPST.[3SG] I.[NOM] this
ńawrɛm-a porχśɵp mă-λ-əm. Tăta iśki.
child-DAT coat.[ACC] give-NPST-1SG here cold

b. ok Want-e, i ńawrɛm porχśɵp ăn tă-jəλ. Ma tăm
look-IMP one child.[NOM] coat.[ACC] NEG have-NPST.[3SG] I.[NOM] this
ńawrɛm porχśɵp-ən mă-λ-ɛm. Tăta iśki.
child.[ACC] coat-LOC give-NPST-1SG>SG here cold
'Look, one child does not have a coat. I'm giving a coat to this child. It's cold.'

(74) {'Why our dog is barking?'}

a. ok/? Pet'a-jen amp-əλ-a λɛtut ănt mă-s
Petja-POSS.2SG.[NOM] dog-POSS.3SG-DAT food.[ACC] NEG give-PST.[3SG]

b. ok Pet'a-jen amp-əλ λɛtut-ən ănt
Petja-POSS.2SG.[NOM] dog-POSS.3SG.[ACC] food-LOC NEG
mă-s-λe
give-PST-3SG>SG
'Patya hasn't given the dog (lit. his dog) any food.'

Interestingly, secundative alignment is slightly preferred in 74, even though the degree of topicality should be identical in 74 and 73. I argue that the choice of alignment – at least in Kazym Khanty – depends not only on the information structural status of the R-participant but also on certain characteristics of the theme. In particular, when the theme is a mass noun, secundative alignment is clearly preferred in most examples. Moreover, 75 shows that secundative alignment is preferred even when the recipient is a narrow focus.

(75) {Where does Masha sew the blue glass beads?}

a. Indirective alignment: 2 ??/*, 1 ?, 2 ok

*/? λʉw ɛtərχări sak akań-a jont-λ-əλλe
(s)he.[NOM] blue glass_beads.[ACC] doll-DAT sew-NPST-3SG>SG

b. Secundative alignment: 4 ok, 1 ?

ok λʉw ɛtərχări sak-ən akań jont-λ-əλλe
(s)he.[NOM] blue glass_beads-LOC doll.[ACC] sew-NPST-3SG>SG

'She is sewing them onto a doll.'

At the same time, both alignments are equally acceptable in other clauses, where the theme is a mass noun, as in 76.

(76) {Where is Masha?}

a. ok Maša-jen śăta oməs-λ, molśij-a
Masha-POSS.2SG.[NOM] there sit-NPST.[3SG] fur_coat-DAT
sak jont-λ.
glass_beads.[ACC] sew-NPST.[3SG]

b. ok Maša-jen śăta oməs-λ, molśij-əλ
Masha-POSS.2SG.[NOM] there sit-NPST.[3SG] fur_coat-POSS.3SG.[ACC]
sak-ən jont-λ-əλλe.
glass_beads-LOC sew-NPST-3SG>SG
'Masha sits there, sews glass beads to a/her fur coat.'

These new data show that previous analyses based solely on information structure are insufficient, at least for capturing the alignment alternation in the Kazym dialect. Indirect alignment is far less restricted than secundative alignment. Therefore, it is impossible to argue that they are in complementary distribution, regulated by the topicality of the indirect object. In the next subsection, I review the role of prominence scales in the choice of alignment.

2.4.2 Prominence hierarchies

An immediate alternative to information structure is prominence hierarchies in 77, as suggested by Aissen (1999, 2003) for Differential Object Marking (see also Keine & Müller 2015). In this model, the more prominent an argument is on these scales, the more likely it is to be overtly marked for a case.

(77) Aissen (2003: p.437, ex. 4)

a. Animacy scale:
Human > Animate > Inanimate

b. Definiteness scale:
PRON (1,2) > PRON (3) > Proper name > DEF > INDEF SPEC > INDEF NON-SPEC

The hypothesis is that secundative alignment is preferred in clauses with a more prominent IO and/or a less prominent DO. Conversely, indirective alignment is expected in clauses with a less prominent IO and/or a more prominent DO. This is not an unsubstantiated hypothesis, since animacy and definiteness have been argued to play a role in the choice of voice in Kazym Khanty (Muravyev 2023). However, neither the animacy scale, nor the definiteness scale are sufficient to predict the distribution of indirective and secundative alignments in Kazym Khanty.

Animacy alone does not play a role in the choice of alignment: the alternation is present in clauses with two inanimate objects in 78, as well as with two animate objects in 79. Native speakers dis not report any significant preference for either alignment.

(78) ńuχi aj-a ewət-λ-ew. kawərt-λ-əw λεpta
meat.[ACC] small-DAT cut-NPST-1PL>SG cook-NPST-1PL soft
ji-t-aλ wonta.
go-NFIN.NPST-3SG until
'We cut the meat into small pieces, cook it until it gets soft.'

a. śăλta %ńuχ-en-a / okput-en-a kurpa
then meat-POSS.2SG-DAT / pot-POSS.2SG-DAT groat.[ACC]
εsəλ-λ-ew.
put-NPST-1PL>SG

b. śăλta okńuχ-en / okput-en kurpa-jən εsəλ-λ-ew.
then meat-POSS.2SG / pot-POSS.2SG groat-LOC put-NPST-1PL>SG
'Then we put groats to it.'

(79) a. Maša-jen iki-jəλ-a χoλəm puχ
Masha-POSS.2SG.[NOM] man-POSS.3SG-DAT three son.[ACC]
to-s.
bring-PST.[3SG]

b. Maša-jen iki-jəλ χoλəm puχ-ən
Masha-POSS.2SG.[NOM] man-POSS.3SG.[ACC] three son-LOC
to-s-λe.
bring-PST-3SG>SG
'Masha gave her husband three sons.'

Moreover, both alignments are equally available if the indirect object is higher on the animacy scale than the direct object. In 80, the recipient is a human and the theme is an inanimate noun. If prominence on the animacy scale plays a role, secundative alignment would be preferred, but native speakers judge both alignments as equally acceptable.

(80) a. Lena-jen-a pasilka kit-s-əw.
Lena-POSS.2SG-DAT parcel.[ACC] send-PST-1PL

b. Lena-jen pasilka-jən kit-s-ew
Lena-POSS.2SG.[ACC] parcel-LOC send-PST-1PL>SG
'We sent Lena a parcel.'

It has long been noted that the two prominence scales are not completely independent. For example, the presence of the [PARTICIPANT] feature implies animacy, and the presence of person specification implies definiteness. Therefore, 1/2-person is always [+animate] and [+definite], while [-animate] and [-definite] elements are always 3-person. (For more detailed discussion see Richards 2014; Driemel 2023.) Indeed, specificity and definiteness, combined with animacy yield better results in predicting the availability of secundative alignment in Kazym Khanty than topicality or animacy alone. The combined scale in 81 predicts that secundative alignment is more likely to be used, if IO is higher on the scale, e.g. animate and definite, and the theme (DO) lower on the scale, i.e. inanimate, indefinite and non-specific.

(81) Animacy-definiteness scale in secundative alignment:

ANIMATE > INANIMATE

PRON (1,2) > PRON (3) > PROPER NAME > DEF SPEC > INDEF SPEC > INDEF NON-SPEC

← $\mathrm{IO}_{\mathrm{ACC}}$ $\mathrm{DO}_{\mathrm{LOC}}$ →

At first glance, this prediction appears to be correct. In part of the data, native speakers' judgements confirm that inanimate IOs must be specific and definite in secundative alignment. In 82, only indirective alignment can be used if the (inanimate) IO is indefinite and non-specific, i.e. if the statement holds for any river. In secundative alignment, the indirect object is reinterpreted as definite and specific, e.g. the river the speaker stands by, thus being higher on the prominence scale.

(82) a. okJuχan-a tăpər ăλ tăχ-a
river-DAT trash.[ACC] PROH trow-IMP
'Don't throw trash into a river.' (in general)

b. #Juχan tăpər-ən ăλ tăχ-a
river.[ACC] trash-LOC PROH trow-IMP
Intend.: 'Don't throw trash into a river.' (in general)
Actual meaning: 'Don't throw trash in THE river.'

When the IO is animate, as in 83, it is already high enough on the animacy scale and can be indefinite non-specific in both indirective and secundative alignments.

(83) 'Vasja wants to help people. He wants to be a doctor.'

a. Joχ-a purtɵŋ măt-λ
people-DAT medicine.[ACC] give-NPST.[3SG]

b. Joχ purtɵŋ-ən mă-λ-λe
people.[ACC] medicine-LOC give-NPST-3SG>SG
'He will give people medicine.'

The prediction for themes (direct objects) is that more prominent themes are less likely to appear in secundative alignment. Indeed, pronouns and proper names are illicit DOs in secundative alignment. Note that pronouns do not have a morphological LOC/INSTR form, but 84 shows that a silent *pro* is also ungrammatical as secundative theme. Hence, the restriction on pronouns is not due to a paradigmatic gap. Moreover, when the theme is a proper name, as in 85, it can be locative-marked, e.g. in the passive, but secundative alignment is equally unavailable.

(84) *śit wɛr aj *copula.* ma năŋ-ti *pro* nuχ wɛr-λ-ɛm
this business small is I you.ACC PREV do-NPST.1SG>SG

'This business is small. I'll do it for you.'

(85) *Maša-jen λɵχs-əλ Ira-jəλ-ən
Masha-POSS.2SG.[NOM] friend-POSS.3SG.[ACC] Ira-POSS.3SG-LOC
kit-s-əλλe
send-PST-3SG>SG
'Masha sent Ira to her friend.'

Definite themes are also illicit in secundative alignment. Schwarz (2013) shows that languages distinguish between two types of definiteness. Anaphoric use is a strong definiteness, whereas uniqueness and bridging are weak definiteness. However, the type of definiteness does not play a role in secundative alignment. Both strong and weak definites cannot be used as secundative themes. 86 shows the ungrammaticality of secundative alignment in clauses, where the theme is a strong definite, i.e. given in the immediate context.

(86) In kińška-jen χuti? – *Ma Vasja-jen
this book-POSS.2SG.[NOM] how I.NOM Vasya-POSS.2SG.[ACC]
kińška-jen-ən mă-s-ɛm.
book-POSS.2SG.LOC give-PST-1SG>SG
Intend.: 'What about the book? – I gave it to Vasya.'

Weak definite themes, i.e. situationally or generally unique items, are equally ungrammatical in secundative alignment. The museum in 87 is generally unique because there is only one museum in the village of Kazym. As a unique item, it is definite, and secundative alignment is ungrammatical. Similarly, glasses in 88 are situationally unique, leading to the same result.

(87) *Olja-jen kašəŋ χujat Kasəm musej-ən
Olya-POSS.2SG.[NOM] every someone.[ACC] Kazym museum-LOC
wanλtə-s-λe.
show-PST-3SG>SG
'Olya showed everyone the museum in Kazym.'

(88) *Iśńi χop sɛm karti-jən ăλ pun-a.
window boat.[ACC] eye iron-LOC PROH put-IMP
'Don't put (your) glasses on a window.'

Possessive markers and demontratives are unexceptionally ungrammatical on secundative themes, as shown in 89b-90. Hence, all strong definites, which are marked with 2SG and 3SG possessive suffixes are automatically ruled out from the set of potential secundative themes.

(89) a. okToχtər-en λʉwti [mošən ut-λ
Doctor-POSS.2SG.[NOM] (s)he.ACC ill someone-POSS.3SG
purtəŋ-ən] mă-s-λe
medicine-LOC give-PST-3SG>SG
'Doctor gave him$_i$ the medicine of his$_i$ patient.'

b. *Toχtər-en λʉwti [mošən ut-λ
Doctor-POSS.2SG.[NOM] (s)he.ACC ill someone-POSS.3SG
purtəŋ-**λ**-ən] mǎ-s-λe
medicine-POSS.3SG-LOC give-PST-3SG>SG
'Doctor gave him$_i$ the medicine of his$_i$ patient.'

(90) *Ma Vasja-jen **śi** kińśka-ən mǎ-s-ɛm.
I.[NOM] Vasya-POSS.2SG.[ACC] DEM book-LOC give-PST-1SG>SG
Intend.: 'I gave that book to Vasya.'

However, this piece of data is better described as a restriction on DP-level modification (i.e. a structural restriction on DPs) than a semantic definiteness restriction. In particular, modification with the superlative adjective in 91 and the *non-agreeing* lexical possessor in 89a do not block secundative alignment.

(91) [ok]Maša-jen λɵχs-əλ mɛt χʉw kina-jən wanλta-s-λe.
Masha-POSS.2SG friend-POSS.3SG.[ACC] most long film-LOC show-PST-3SG>SG
'Masha showed her friend the longest film.'

Note also that possessive markers are ungrammatical even on indefinite themes. As 92 shows, an indefinite theme, modified by the existential quantifier, is still ungrammatical in secundative alignment if it bears a possessive suffix. Note that a proper possessive marker is compatible with an indefinite nonspecific interpretation (Mikhailov 2023). The theme in 92 is non-specific and inanimate, i.e. located very low on the prominence scale. Hence, if the use of secundative alignment is regulated by prominence scales, it is expected to be licit as a secundative theme, contrary to fact.

(92) *Vasja-jen aŋk-eλ muλsər an-əλ-ən
Vasya-POSS.2SG.[NOM] mother-POSS.3SG.[ACC] some cup-POSS.3SG-LOC
mǎ-s-λe
give-PST-3SG>SG
'Vasya gave his mother one of his cups.'

This variation cannot be properly captured by prominence scales because the examples in 89-91 and 91-90 are located at the same level in the prominence hierarchy. In Chapter 4, I argue that reference to syntactic structure makes better predictions for restrictions on secundative theme than the prominence scales. In other words, I suggest that secundative themes obey a syntactic restriction on DP-level modification (cf. Richards's 2014 proposal that all prominence-sensitive restrictions should be reanalyzed in terms of a syntactic DP-NP asymmetry). This approach not only captures the data in 89-91 in a uniform manner, but also correctly captures the role of semantic definiteness. In particular, it explains, why 87-88 are ungrammatical, even though the theme in both examples is bare, i.e. does not have any modifiers.

Moreover, prominence scales fail to motivate the choice of the alignment in scenarios, when both indirective and secundative alignments are grammatical. The prominence scale in 81 suggests that secundative alignment must be preferred with the least prominent themes, i.e. inanimate and indefinite non-specific, and dispreferred with animate specific themes. This is not the case: both alignments are judged as equally acceptable with animate (indefinite) specific DO 93.

(93) Ma piλ măn-λ-əw ajλtijewa.
I.NOM with go-NPST-1PL carefully.
'Come with me quietly.'

a. Ma năŋena wɵntər wanλta-λ-əm
I.NOM you.DAT otter.[ACC] show-NPST-1SG

b. Ma năŋti wɵntər-ən wanλta-λ-ɛm
I.NOM you.ACC otter-LOC show-NPST-1SG>SG
'I'll show you an otter.'

Moreover, the prominence scales cannot explain the unexpected contrast between 94 and 95. The two stimuli are identical with respect to the relative prominence of IO and DO: they are both inanimate, indefinite, and non-specific. Nevertheless, 94 clearly prefers indirective alignment, whereas both alignments are equally grammatical in 95.

(94) (repeated from 82)

a. okJuχan-a tăpər ăλ tăχ-a
river-DAT trash.[ACC] PROH trow-IMP
'Don't throw trash into a river.' (in general)

b. #Juχan tăpər-ən ăλ tăχ-a
river.[ACC] trash-LOC PROH trow-IMP
Intend.: 'Don't throw trash into a river.' in general)

(95) a. okMaša-jen wońśumutəŋ ńań-a suλ i puša ăn
Masha-POSS.2SG.[NOM] berry.ATTR bread-DAT salt.[ACC] at_all NEG
punij-λ.
put-NPST.[3SG]

b. okMaša-jen wońśumutəŋ ńań i puša suλ-ən ăn
Masha-POSS.2SG.[NOM] berry.ATTR bread.[ACC] at_all salt-LOC NEG
punij-λ-əλλe.
put-NPST-3SG>SG
'Masha never put salt in a berry pie.'

As mentioned in the previous subsection, secundative alignment is always available, if not preferred, with mass noun themes.

2.4.3 Conclusion

A careful investigation of how prominence interacts with availability and choice of alignment in ditransitive clauses leads to the following generalization.

(96) GENERALIZATION 1: AVAILABILITY OF SECUNDATIVE ALIGNMENT

a. Secundative alignment is available if the IO exceeds the DO in at least one of the following features: topicality, definiteness and specificity, animacy, or countability.

b. Secundative alignment is ungrammatical in clauses with the most prominent themes (personal pronouns, proper names, and definite nouns).

This generalization explains cases where secundative alignment is degraded or ungrammatical. Importantly, it does not explain why possessive markers are ungrammatical even on indefinite themes in secundative alignment. Furthermore, it cannot account for the majority of contexts where both alignments seem to be equally grammatical, and native speakers do not show any clear preferences towards either of them.

(97) GENERALIZATION 2: AVAILABILITY OF INDIRECTIVE ALIGNMENT
Indirective alignment is always grammatical

Note that there is no complementary distribution of the two alignment types, and indirective alignment is always available and grammatical. The only scenario in which indirective alignment can be dispreferred is a clause with a mass noun theme and an inanimate IO. Importantly, however, this is not a proper restriction, but rather a tendency, since there are always some native speakers who allow indirective alignment in this scenario. It can be safely concluded that indirective alignment is never truly ungrammatical.

In contrast, secundative alignment is more restricted. Importantly, at least a subset of restrictions on secundative themes cannot be accounted for purely by means of prominence scales. At this point, it seems promising to follow Richards' (2014) suggestion that the explanatory power of prominence scales is an epiphenomenon of the syntactic contrast between full DPs and structurally reduced nominals (NPs). This allows to reformulate the GENERALIZATION 1.

(98) GENERALIZATION 1': AVAILABILITY OF SECUNDATIVE ALIGNMENT
Secundative alignment is ungrammatical in clauses where the theme argument is full DP.

In Chapter 4, I follow Richards' (2014) approach and introduce a syntactic analysis of the restrictions on the theme in secundative alignment. This analysis has greater explanatory power than the approach based on prominence scales.

Chapter 3

Indirective alignment

Of the two ditransitive alignments present in Kazym Khanty, indirective alignment is the default option. As discussed in the previous chapter, its availability seems to be unrestricted. The indirect object in the indirective alignment is marked with the dative case, while the theme is marked with the accusative.[6] Only the accusative-marked theme can optionally control agreement (in number) on the finite verb, while the dative-marked indirect object never agrees with the verb, as illustrated in (99).

(99) a.

λiw	ew-eλ	arsər	tɵχtər-ət-a	wan-λta-s-eλ
3.PL.NOM	girl-POSS.3PL.[ACC]	many	doctor-PL-DAT	see-CAUS-PST-3PL>SG

‘They showed their daughter to different doctors.’

b.

*λiw	ew-eλ	arsər	tɵχtər-ət-a	wan-λta-s-λaλ
3.PL.NOM	girl-POSS.3PL.[ACC]	many	doctor-PL-DAT	see-CAUS-PST-3PL>NSG

The remainder of this chapter is organized as follows. In the first section, I discuss the base generation order of the two applied object (IO) and the theme (DO). The available tests show that the base-generation order is IO>DO both in indirective and secundative alignments. The second section presents an analysis of indirective alignment. The proposed derivation is based on the suggestion that high applicative projection is the source of dative case marking. Therefore, HighApplP is present in the structure even in low applicative clauses, including lexical ditransitives, and causative clauses. Furthermore, two flavors of HighAppl° are suggested: (i) a thematic flavor that introduces a new (high applied) argument and assigns dative case to it, and (ii) an athematic flavor that is merged when there are two caseless arguments inside its future c-command domain. The sole purpose of the athematic HighAppl° is to assign case to the higher argument, allowing Voice° to assign accusative to the lower argument.

3.1 Base-generation structure in double object clauses

The base-generation order of the two objects in ditransitive clauses is a matter of debate and possibly, cross-linguistic variation. The alternation between the prepositional dat-

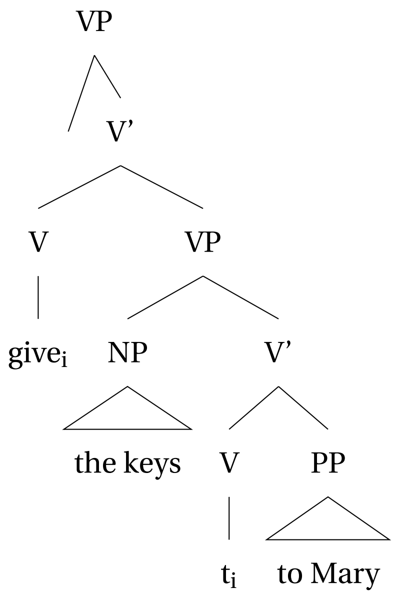
[6] Recall that accusative is zero-marked on nouns but is visible on pronouns.

ive construction in (100a) and the double object construction in (100b) in English has determined the debate.

(100) a. John gave the keys to Mary
b. John gave Mary the keys

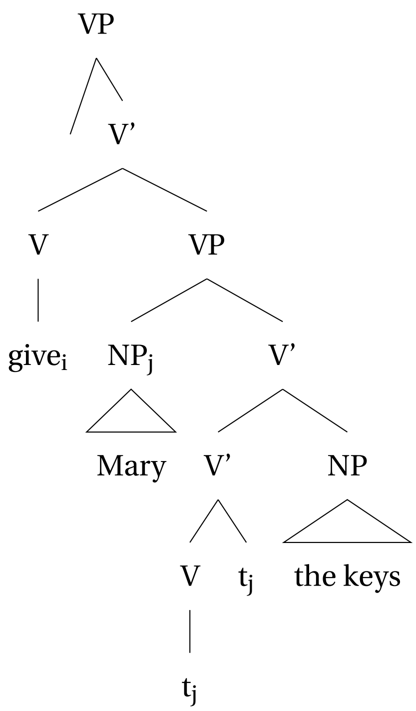
Larson (1988, 2014) (followed by, e.g., Ormazabal & Romero 2010; Michelioudakis 2012) suggests a VP-shell analysis, where the direct object asymmetrically c-commands the indirect object. If the indirect object is merged as a PP, the base-generation order is preserved (101).

(101)



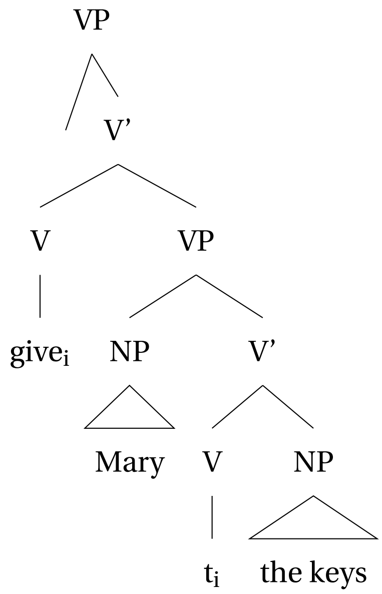
The Double object frame is derived by demoting the theme to a V'-adjunct and merging the recipient as an NP (102).

(102)



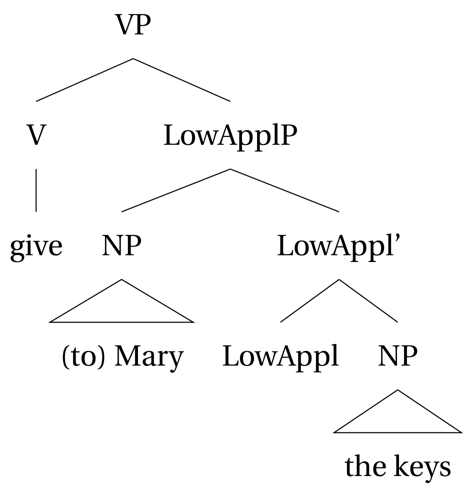
Another group of approaches argues for two independent base-generated structures (Harley 1997, 1995,Harley 2002,Harley & Jung 2015, Hale & Keyser 1993, den Dikken 1995, Bruening 2001, 2010a,b, Anagnostopoulou 2003, Holmberg, Sheehan, & van der Wal 2019, Newman 2021; Overfelt 2022). The prepositional dative construction has the structure in (101), and in the double object construction, IO c-commands DO, as in (103).

(103)



A third group of theories argues for only one base-generation structure, where IO asymmetrically c-commands DO, and the prepositional dative construction is derived from it (Bowers 1981; Dryer 1986; Aoun & Li 1989; Pylkkänen 2008; Georgala 2011, 2012; Roberts 2010; Hallman 2015; Collins 2020; Hallman 2024, etc.). The tree in (104) illustrates the base-generation structure proposed by Pylkkänen (2008) for a low applicative (ditransitive) clause.

(104)

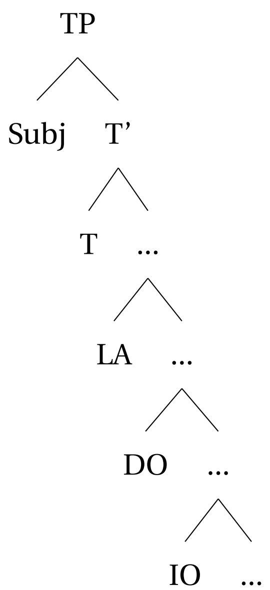


Hallman (2024) specifically argues in favor of a single base-generation structure for Syrian Arabic, where the alternation of the prepositional dative and double object construction is identical to English. He argues that the theme precedes the dative PP (IO) on the surface because the theme agrees with Voice and raises over the IO to SpecVoiceP. Therefore, the linear order of the two objects does not reflect their base-generation order. In double object constructions, the IO raises to SpecVoiceP, and the linear order of the two objects is the same as their base-generation order.

The indirective-secundative alternation in Kazym Khanty is not identical to the dative shift in English on the surface, e.g. the unmarked surface order of the two objects is IO>DO.[7] However, it is theoretically possible that the alternation in Khanty results from two different underlying base structures. For instance, in indirective alignment, it can be assumed that the DO asymmetrically c-commands the IO in the base position. For example, Müller (1999) argues that DO c-commands IO in the base-generated position in German, where indirective alignment is the only available alternative and the unmarked surface word order is IO>DO (but see e.g. Georgala 2011, 2012, for an alternative view).

For Kazym Khanty, Colley & Privoznov (2018, 2020) suggest that there are different base-generation structures for lexical ditransitives and derived low applicative structures. In their view, the DO c-commands the IO in lexical ditransitives. In applicativized monotransitive clauses, on the other hand, the applied argument (LA = low applied object) c-commands DO. An abstract structure with both types of indirect objects is presented in (105). Their argumentation comes from variable binding, which is discussed in detail in Section 3.1.4.

(105)



In the rest of this section, I argue against Colley & Privoznov (2018, 2020) and show that the base-generation order of the two objects is the same for both lexical ditransitives and applicativized monotransitives. The indirect object (recipient) in Kazym Khanty asymmetrically c-commands the direct object (theme) in the base position (contra e.g. Larson 1988, 2014, and crucially contra Colley & Privoznov 2018, 2020's analysis of Kazym Khanty). Note also that I follow the research tradition based on Pylkkänen (2008), where lexical ditransitives and low applicatives have identical syntactic structures.

[7] Here, I abstract from the optionality of scrambling of the objects in the middle field. A pragmatically neutral word order in Kazym Khanty is IO>DO.

3.1.1 Short scrambling in Kazym Khanty

Most tests for asymmetric c-command are sensitive to the type of movement that an argument can undergo. The difference between A- and A'-scrambling is a long-standing problem in generative research (Chomsky 1981; Saito 1985; Déprez 1989; Mahajan 1990; Heim & Kratzer 1998; van Urk 2015; Keine 2018; Bhatt & Keine 2019; Lohninger, Kovač, & Wurmbrand 2022, etc.). The earlier approaches assumed that there are two types of positions: A-positions related to θ-role assignment and A'-positions, where a θ-role cannot be assigned to an argument (Chomsky 1981). Consequently, two types of scrambling are distinguished based on whether the landing site is an A- or an A'-position. Alternatively, some more recent approaches (e.g. van Urk 2015; Bhatt & Keine 2019; Lohninger et al. 2022; Lohninger & Yip 2023) advocate for a feature-based A/A' distinction. This allows to account for mixed A/A'-properties of scrambling in certain languages. It also allows one to explain why one type of scrambling, e.g. short scrambling, shows A-properties in one group of languages and A'-properties in another.

Irrespective of the adopted analysis of the opposition, A- and A'-movement show different syntactic properties. First, a bound variable pronoun can only be bound by an A-scrambled antecedent (e.g. Bhatt & Keine 2019). Wh-fronting in 106a does not allow the variable on *his mother* to be bound by the fronted object. In contrast, raising to subject in 106b is an instance of A-movement, and the variable *his* is bound by the derived subject.

(106) a. *Which boy$_1$ did his$_1$ mother say [t$_1$ is intelligent]?
b. Every boy$_1$ seems to his$_1$ mother [t$_1$ to be intelligent].

Second, only A-movement can strand quantifiers, while A'-movement cannot (Déprez 1989). The contrast is illustrated in (107b) vs. (107a), where wh-movement cannot strand quantifiers, but subject movement to SpecTP can.

(107) a. *Which drug dealers$_1$ did the mayor say that the police will [all t$_1$] arrest?
b. The drug dealers$_1$ have [all t$_1$] been arrested.

Finally, A'-movement obligatorily reconstructs for Principle A and Principle C, which ensures the ungrammaticality of 108a. In contrast, A-movement in English can resist reconstruction, and 108b is grammatical.

(108) a. *[Which friend of John's$_1$]$_2$ did he$_1$ visit t$_2$?
b. [A picture of John$_1$]$_2$ seems to him$_1$ to be t$_2$ on sale.

However, in some languages, reconstruction for Principle C is obligatory, even for A-movement (see e.g. Bhatt & Keine 2019).[8]

[8] Note that the initial approach (Chomsky 1981) defined A-movement as a movement to or through a case position. More recent approaches have abandoned this definition. Bhatt & Keine (2019) argue that reconstruction for Priciple C is independent from A/A'-type of movement (see also Keine 2018). Instead, a copy in a case-assignment position is interpreted for Principle C. Hence, if a case is assigned in situ, the lower copy will be interpreted, even if the argument is subsequently A-moved. The Wholesale Late Merger

Kazym Khanty is a language with a free word order, where both IO>DO and DO>IO orders are attested (109a vs. 109b, respectively).

(109) a. Kašəŋ ewi(-jen) kat'-əλ-a juntut mă-s-(λe)
Every girl-POSS.2SG.[NOM] cat-POSS.3SG-DAT toy.[ACC] give-PST-3SG>SG
'Every girl gave her cat a toy.'

b. Kašəŋ ewi(-jen) juntut kat'-əλ-a mă-s-(λe)
every girl-POSS.2SG.[NOM] toy.[ACC] cat-POSS.3SG-DAT give-PST-3SG>SG
'Every girl gave her cat a toy.'

Note that the scrambling of the two objects is equally available in clauses with and without object agreement. Importantly, however, the short scrambling of agreeing and non-agreeing direct objects in Kazym Khanty is not uniform. In particular, only agreeing direct objects can bind a variable on the indirect object (110), and non-agreeing direct objects cannot.

(110) a. Maša-jen$_k$ kašəŋ χujat$_i$ λɵχs-əλ-a$_{k/i}$
Masha-POSS.2SG.[NOM] every someone.[ACC] friend-POSS.3SG-DAT
χur-ən want-λta-s-λe
photo-LOC see-CAUS-PST-3SG>SG
'Masha$_k$ showed everyone$_i$ to his$_i$/her$_k$ friend on the photo.'

b. Maša-jen$_k$ kašəŋ χujat$_i$ λɵχs-əλ-a$_{k/i}$
Masha-POSS.2SG.[NOM] every someone.[ACC] friend-POSS.3SG-DAT
χur-ən want-λta-s
photo-LOC see-CAUS-PST.[3SG]
'Masha$_k$ showed everyone$_i$ to his$_{*i}$/her$_k$ friend on the photo.'

Similarly, only agreeing objects can strand quantifiers (111). Non-agreeing direct objects cannot be coreferent with the quantifier, so the only available interpretation in (111b) is 'all children', and not 'all books'.

(111) a. ńawrɛm-ət piśma-λ-aλ tɛrmat [χuλ]$_F$ (śi)
child-PL.[NOM] letter-PL-POSS.3PL.[ACC] hasty all EMPH
šɵnχit-s-əλλe
burn-PST-3SG>SG
'Children burned all letters in haste.' (Solovieva 2019: p.3, ex.13a)

b. ńawrɛm-ət kinška-ət χot jit-ən [xuλ]$_F$ (śi) wu-s-ət
child-PL.[NOM] book-PL.[ACC] house part-LOC all EMPH take-PST-3PL
'Children$_i$ books$_j$ in the room [all t$_{i/*j}$]$_F$ took.' (Solovieva 2019: p.3, ex.9)

In the following discussion, I assume that agreeing direct objects are A-scrambled when they precede an indirect object and non-agreeing direct objects are A'-scrambled. I furthermore argue that both agreeing and non-agreeing direct objects are generated below indirect objects and the order DO>IO is derived via movement of the DO.[9]

approach (Takahashi & Hulsey 2009; Gong 2023) also argues that reconstruction for Principle C is sensitive to case-assignment position and not to the type of scrambling.

[9]Whether agreeing and non-agreeing objects are scrambled to the same position or land in two different projections is a subject for further research. It is also orthogonal to the question of the base-generation order

3.1.2 Causatives

It is useful to compare ditransitive and applicative clauses to a structure with a less ambiguous base-generation order of the two objects, such as causative clauses.

Causative clauses participate in indirective-secundative alternation, similar to applicative clauses. The causative derivation in Kazym Khanty is unproductive, but there is a significant number of lexicalized causatives derived from transitive verbs. Such causatives allow indirective-secundative alternation, similar to applicative verbs, 112-113.[10]

(112) a. ʎawərt χir mănɛm aʎm-əʎt-s-əʎe
heavy sack.[ACC] I.DAT lift-CAUS-PST-3SG>SG
'(S)he loaded the heavy sack on me.' (Moldanova 2018)

b. Mănti ʎawərt χir-ən aʎm-əʎt-s-əʎe
I.ACC heavy sack-LOC lift-CAUS-PST-3SG>SG
'(S)he loaded me with a heavy sack.' (I.Moldanova p.c.)

(113) a. Ma năŋena wɵntər want-ʎta-ʎ-əm
I.NOM you.DAT otter.[ACC] see-CAUS-NPST-1SG

b. Ma năŋti wɵntər-ən want-ʎta-ʎ-ɛm
I.NOM you.ACC otter-LOC see-CAUS-NPST-1SG>SG
'I'll show your an otter.'

In all approaches to causatives, the causee c-commands the theme in the base-generation order (Zubizarreta 1985; Ippolito 2000; Folli & Harley 2007; Pylkkänen 2008; Jerro 2016, 2017; Nie 2020; Akkuş 2021, 2022; Hallman 2024, etc.). There is no doubt that the causee in both alignments has the same thematic role. Under the UTAH hypothesis (Baker 1988), it is undesirable to argue that an argument with the same thematic role can be externally merged in different positions (see e.g., this kind of argumentation used in Hallman 2024). Even if the causee theta-role is not tied to a singular projection in a clause, there is no reason to adopt a highly unconventional assumption that the causee is c-commanded by the theme in either of the alignments. Therefore, indirective-secundative alternation cannot be explained by postulating two different base-generation structures with two different orders of objects, as has been proposed for the alternation between Double Object Construction and Preposition Dative Construction in English (see the discussion above). Without evidence to the contrary, it should be assumed that both indirective and secundative alignments are derived from a single syntactic structure, where the IO (recipient or causee) is base-generated higher than the theme. Moreover, the remainder of this section shows that syntactic tests for asymmetric c-command give the same result in both alignments in causative and applicative clauses. Therefore, I assume that the

of the two objects and it will not be discussed here.
For the same reason, I abstract away from the discussion of the nature of the A/A'-distinction. Whether it is feature- or position-driven is irrelevant for the diagnostics of the base-generation order of the two objects.

[10] Note that the verb 'to show' *want-ʎta-ti* is often treated in the literature as a lexical ditransitive. This is not entirely correct. Even though (Kazym) Khanty has no productive causative derivation and the existing causative forms are completely frozen, the verb *want-ʎta-ti* 'to show' is clearly morphologically derived from WANT-TI 'to see' (see also Moldanova 2018). In the following discussion, I treat this verb as a derived causative.

DAT/ACC-maked argument c-commands the ACC/LOC-marked theme argument in all cases of indirective-secundative alternation, i.e. both in applicative and causative clauses.

3.1.3 Condition C

One of the widely used tests for asymmetric c-command is Condition C and its violations. Condition C states that an R-expression must be free, i.e it cannot be co-indexed with a c-commanding pronoun (Chomsky 1981). Indeed, in 114, a pronominal subject cannot be co-referent with a possessor in the object DP because the subject c-commands the direct object. Similarly, in 115, a pronominal indirect object (recipient) cannot be co-referent with an R-expression inside the direct object DP because the IO c-commands DO.

(114) a. *He$_i$ loves John's$_i$ mother.
b. ^{ok}John$_i$ loves his$_i$ mother.

(115) a. *I gave him$_i$ John's$_i$ laptop.
b. ^{ok}I gave John$_i$ his$_i$ laptop.

Saito (1985) shows that movement can affect the possibility of coreference and application of Condition C. Once an R-expression object is moved across the pronominal subject, they can be co-referent in Japanese, while in the base-order Sub > Obj, the co-reference violates Condition C and is ungrammatical 116.

(116) a. *Kare$_i$-ga [$_{NP}$ Mary-ga John$_i$-ni okutta tegami]-o mada yonde inai (koto)
he-NOM Mary-NOM John-to sent letter-ACC yet read have-not fact
*'He$_i$ has not read the letter that Mary sent to John$_i$.' (Saito 1985: ex.20)
b. [$_{NP}$ Mary-ga John$_i$-ni okutta tegami]-o kare$_i$-ga mada yonde inai (koto)
Mary-NOM John-to sent letter-ACC he-NOM yet read have-not fact
'The letter that Mary sent to John$_i$, he$_i$ has not read.' (Saito 1985: ex.20)

Wh-movement does not help avoid Condition C violation, as 117 shows for English. Hence, the wh-moved element reconstructs into its base position and is interpreted there with respect to Condition C.

(117) *[Whose examination of John$_i$]$_k$ did he$_i$ fear k_k? (Sportiche 2006: ex.18a)

Importantly, reconstruction for Condition C is obligatory for A'-movement, e.g. for wh-movement in 117, but not for A-movement in 118 (see, e.g., Sportiche 2006 for an overview).

(118) [This picture of John$_i$]$_k$ seems to him$_i$ [t$_k$ to be fuzzy]. (Sportiche 2006: 87a)

However, the reconstruction effects for Condition C do not behave uniformly cross-linguistically. As discussed in Bhatt & Keine (2019), intermediate scrambling in Hindi obligatorily reconstructs for Condition C, despite having other properties of A-movement. As illustrated by 119a, a pronoun in the subject position cannot be coreferential with an R-expression inside the direct object because the R-expression is c-commanded by the pronoun. Crucially, Condition C is still violated in 119b, where the object is A-moved across the subject. The R-expression in the landing position is not c-commanded by the pronoun, so Condition C can only be violated if the object DP is interpreted in its base position, where the pronoun c-commands the R-expression.

(119) (Bhatt & Keine 2019: p.129, ex.8)

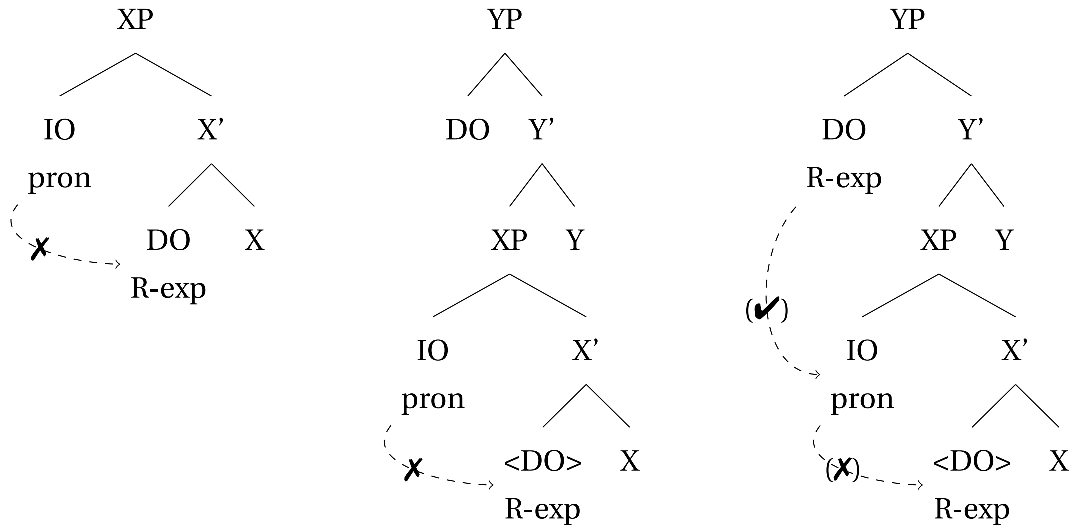
a. *us-ne$_1$ [Sita-ke$_1$ bhaaii-ko] ḍããṭaa
she-ERG Sita-GEN brother-ACC scolded
'She$_1$ scolded Sita's$_1$ brother.'

b. *[Sita-ke$_1$ bhaaii-ko]$_2$ us-ne$_1$ __$_2$ ḍããṭaa
Sita-GEN brother-ACC she-ERG scolded
'Sita's$_1$ brother, she$_1$ scolded.'

In a language where IO c-commands DO in the base position, as in 120a, and the overt order of the objects is free, the following pattern is expected for Condition C:

- A pronominal IO in its base position cannot be co-referent with an R-expression inside an in situ DO (120a);
- obligatory reconstruction under A'-scrambling: a pronoun in the IO position cannot be co-referent with an R-expression inside the preceding DO (120b);
- reconstruction / no reconstruction under A-scrambling: a pronoun in the IO position may or may not be co-referential with an R-expression inside the DO if the DO precedes the IO (120c);

(120) a. base position b. A'-movement c. A-movement



DO obligatory reconstructs DO may or may not reconstruct

Recall that intermediate scrambling in Kazym Khanty can be divided into two operations. THe scrambling of the direct object, accompanied by object agreement on the verb, is classified as A-scrambling. Non-agreeing direct objects undergo A'-scrambling.[11] As expected, A'-moved (non-agreeing) direct objects obligatory reconstruct for Condition C. Furthermore, A-moved (agreeing) direct objects also reconstruct for Condition C in Kazym Khanty. Recall, that intermediate (A-)scrambling in Hindi also reconstructs for Condition C (Bhatt & Keine 2019). Keine (2018) and Bhatt & Keine (2019) argue that a copy in a Case position is relevant for Condition C. This suggests that direct object in Khanty receives accusative case in situ. Importantly, reconstruction for Condition C in Kazym Khanty follows the pattern in 120 and provides evidence for the IO>DO base-generation order.

If the direct object does not agree with the verb, Condition C is violated, irrespective of linear order, as illustrated in 121. This is expected if non-agreeing direct objects undergo A'-scrambling.

(121) a. Jaj-en λʉweλ$_{k/*i}$ [puχ-en$_i$ moλəpśi]
elder_brother-POSS.2SG.[NOM] (s)he.DAT son-POSS.2SG fur_coat.[ACC]
kit-s
send-PST.[3SG]

b. Jaj-en [puχ-en$_i$ moλəpśi] λʉweλ$_{k/*i}$
elder_brother-POSS.2SG.[NOM] son-POSS.2SG fur_coat.[ACC] (s)he.DAT
kit-s
send-PST.[3SG]
'Brother send him$_{k/*i}$ boy's$_i$ fur coat.'

Agreeing direct objects also reconstruct to a position below the IO, as 122 shows. The possessor 'patient' inside the DO cannot be coreferent with the pronominal IO, even though the DO moves over the IO and c-commands it in the landing position. Since Condition C is violated even under A-scrambling of an (agreeing) direct object, it can be safely concluded that IO asymmetrically c-commands DO in the base-generation order.[12]

[11] Presence of object agreement can be used as a test for A- vs. A'-scrambling only in clauses with unvalued (neutral) high aspect. Recall that (overt) object agreement is blocked in clauses with the valued high Asp-projection, i.e. in habitual and perfect clauses. I assume that the direct object can be A-scrambled in habitual and perfect clauses as well, even in the absence of an overt object agreement on the finite verb. For the sake of clarity, only clauses with neutral high aspect are considered in this section.

[12] One of my informants allowed co-reference of an R-expression *puχ* 'boy, son' inside the DO *puχ-en$_i$* *moλəpśi* 'boy's fur coat' with the pronominal IO in (i), even when the DO was A-scrambled. Importantly, however, she disallowed this relation in two other examples. Note also that the pronoun *λʉweλ* is not uniquely coreferent with the possessor inside the DO and can refer to a third person. It is not clear what differs (i) from other tested examples, and I preliminarily conclude that this might have been a false judgement.

(i) a. Jaj-en λʉweλ$_{k/*i}$ [puχ-en$_i$ moλəpśi] kit-s-əλλe
elder_brother-POSS.2SG.[NOM] (s)he.DAT son-POSS.2SG fur_coat.[ACC] send-PST-3SG>SG
'Brother send him$_{k/*i}$ boy's$_i$ fur coat.'

b. Jaj-en [puχ-en$_i$ moλəpśi] λʉweλ$_{k/i}$ kit-s-əλλe
elder_brother-POSS.2SG.[NOM] son-POSS.2SG fur_coat.[ACC] (s)he.DAT send-PST-3SG>SG
'Brother send boy's$_i$ fur coat to him$_{k/i}$.'

(122) Tөχtər-en [[møšən ut-λ]$_{*i}$ purtөn] λʉweλ$_i$
doctor-POSS.2SG.[NOM] ill someone-POSS.3SG medicine.ACC (s)he.DAT
mǎ-s-λe
give-PST-3SG>SG
'The doctor gave him$_i$ patient's$_{k/*i}$ medicine.'

Causative clauses always require obligatory reconstruction, and Condition C is violated irrespective of the presence or absence of object agreement on the verb, i.e. irrespective of A- or A'-scrambling of the DO 123.

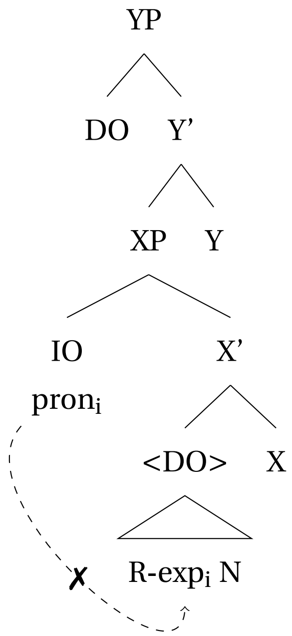
(123) a. Maša-jen ([Ira-jen$_i$ χur]) λʉweλ$_{k/*i}$ ([Ira-jen$_i$
Masha-POSS.2SG.[NOM] Ira-POSS.2SG picture.[ACC] (s)he.DAT Ira-POSS.2SG
χur]) want-λta-s-λe
picture.[ACC] see-CAUS-PST-3SG>SG

b. Maša-jen ([Ira-jen$_i$ χur]) λʉweλ$_{k/*i}$ ([Ira-jen$_i$
Masha-POSS.2SG.[NOM] Ira-POSS.2SG picture.[ACC] (s)he.DAT Ira-POSS.2SG
χur]) want-λta-s
picture.[ACC] see-CAUS-PST.[3SG]
'Masha has shown him/her$_{*i}$ Ira's$_i$ photo.'

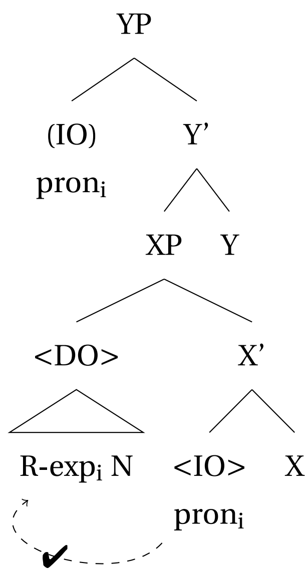
The pattern in which a pronominal IO can never be coreferential with an R-expression in the DO can only be predicted if the IO c-commands the DO in the base position, as in 124a. The scenario in which A-movement fixes the violation of Condition C can also be easily captured once it is assumed that A-movement can resist reconstruction. Then, the higher copy of the DO is active for Condition C in 124a, and the R-expression does not bind its antecedent.

If the base-generation order is reversed and the DO c-commands the IO in the base position, Khanty data cannot be captured. There are two possible sub-scenarios. (i) If reconstruction is obligatory, as in 124b, the R-expression inside DO should be always free and coreferential with IO, contra to the Khanty data. (ii) If there is no reconstruction, as in 124c, then every clause with the surface DO>IO order should not violate Condition C, which also contradicts the Khanty data. Note that Condition C can be violated under scrambling of non-agreeing as well as agreeing direct objects. Hence, differentiating between A- and A'-scrambling cannot help to fix the incorrect prediction.

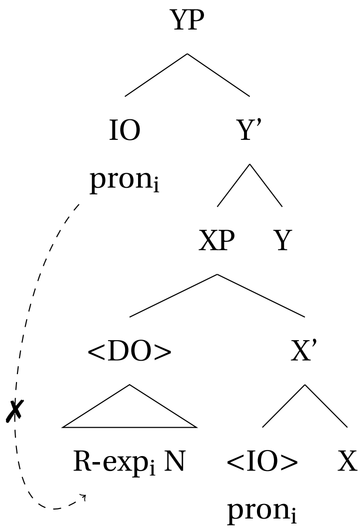
(124) a. IO c-commands DO in the base position; reconstruction for Condition C is obligatory;
=> Condition C is violated; the Kazym Khanty data captured;

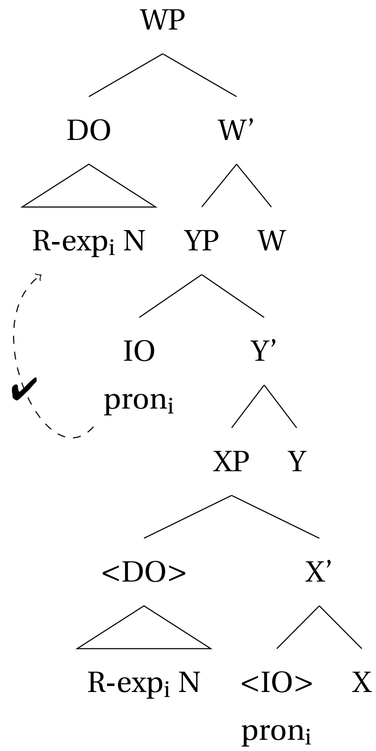


b. DO c-commands IO in the base position; reconstruction for Condition C is obligatory;
=> Condition C is never violated; the Kazym Khanty data not captured;



c. DO c-commands IO in the base position; IO obligatory raises; no reconstruction for Condition C;
=> Condition C is violated in the IO>DO surface order (i);
=> Condition C is never violated in the DO>IO surface order (ii);
the Kazym Khanty data not captured;

(i)


(ii)


In the secundative alignment, the picture is similar: irrespective of the linear order, the R-expression inside the oblique DO can never be coreferential with the pronominal IO$_{ACC}$ in applicative 125 and causative 126 clauses.

(125) Jaj-en ([puχ-en moλəpśi-jən]) λʉwti
elder_brother-POSS.2SG.[NOM] son-POSS.2SG fur_coat-LOC (s)he.ACC
([puχ-en moλəpśi-jən]) kit-s-əλλe
son-POSS.2SG fur_coat-LOC send-PST-3SG>SG
'Brother send him$_i$ boy's$_{k/*i}$ fur coat.'

(126) Maša-jen ([Ira-jen$_i$ χur-ən]) λʉwti$_{k/*i}$ ([Ira-jen$_i$
Masha-POSS.2SG.[NOM] Ira-POSS.2SG picture-LOC (s)he.ACC Ira-POSS.2SG
χur-ən]) want-λta-s-λe.
picture-LOC see-CAUS-PST-3SG>SG
'Masha has shown him/her$_{*i}$ Ira's$_i$ photo.'

As shown in this subsection, the data from Condition C violations provide evidence that the base-generation structures in indirective and secundative alignments are identical. In other words, the indirect object (both ACC- and DAT-marked) always c-commands the theme (LOC- or ACC-marked) in the base positions.

3.1.4 Weak crossover and Variable binding

Another test for determining asymmetric c-command relations between two arguments is variable binding, in particular, the binding of a possessor pronoun. If the indirect object can bind a variable inside the direct object, but not vice versa, as in 127, the IO asymmetrically c-commands the DO.

(127) a. I showed every friend$_i$ of mine his$_{i/k}$ photograph.
b. I showed its$_{k/*i}$ trainer every lion$_i$.
(Barss & Lasnik 1986: ex.6b,7b)

The availability of scrambling induces weak crossover effects. In particular, only A-moved items can bind a variable, even if their base-generation position is below the variable. For example, the subject 'every boy' in the raising construction in (128a), can bind a variable in the DP 'his mother', even though the subject does not c-commands the variable in the base-generation position.On the other hand, A'-movement in (128b) has no effecton the binding relation.

(128) a. Every boy$_i$ seems to his$_i$ mother t$_i$ to be intelligent.
b. *Who(m)$_i$ does his$_i$ mother like t$_i$?

As discussed in Chapter 4, secundative themes are ungrammatical with possessive markers and cannot host a variable. Therefore, variable binding can be used in (Kazym) Khanty as a test for the base-generation order of the two objects only in indirective alignment.

In contrast to reconstruction for Principle C, variable binding is sensitive to the presence of object agreement on the verb, i.e. to A- vs. A'-scrambling. It has been noted that agreeing direct objects move from their base position (see e.g. Smith 2020; É. Kiss 2021). Whether this movement is obligatory has not been extensively discussed in the literature. As a baseline for further discussion, I will assume that agreeing direct objects can stay in situ. See, in particular, 129. If IO linearly precedes DO, a low adverbial cannot follow the (agreeing) DO.[13] The adverbial *sora* is left-adjoined to the VP and cannot linearly follow the VP-internal material. Hence, the agreeing DO in (129) stays in situ inside the VP.

(129) Vasja-jen (sora) χujat-λ-a (sora)
Vasya-POSS.2SG.[NOM] quickly someone-POSS.3SG-DAT quickly
jɵš-əλ (*sora) want-λta-s-λe pa jeλλi šɵšm-əs
road-POSS.3SG.[ACC] quickly see-CAUS-PST-3SG>SG ADD further go-PST.[3SG]
'Vasya quickly showed someone the way and went on.'

The pattern for variable binding in Khanty is rather complex, since it is sensitive not only to the surface order of the elements but also to the presence of object agreement on the verb.

[13] This judgement was received from one informant, and this restriction must be further tested. Nevertheless, there is no doubt that the order [low adv>DO] is grammatical. Hence, agreeing objects can stay in situ.

In causative clauses, variable binding works as follows. When DO does not control agreement on the verb, as in 130, IO binds a variable on DO, irrespective of the linear order. A non-agreeing DO, in turn, can never bind a variable on IO, even when linearly preceding it, as 131 illustrates.[14]

(130) IO can always bind a non-agreeing DO

a. Maša-jen$_k$ kašəŋ amp$_i$(-əλ)-a χot-əλ$_{i/k}$
M.-POSS.2SG.[NOM] every dog-POSS.3SG-DAT house-POSS.3SG
want-λta-s
see-CAUS-PST.[3SG]

b. Maša-jen$_k$ χot-əλ$_{i/k}$ kašəŋ amp$_i$(-əλ)-a
M.-POSS.2SG.[NOM] house-POSS.3SG.[ACC] every dog-POSS.3SG-DAT
want-λta-s
see-CAUS-PST.[3SG]
'Masha$_k$ showed each dog$_i$ its$_i$/her$_k$ house'

(131) Non-agreeing DO can never bind IO

a. Maša-jen$_k$ λɵχs-əλ$_{k/*i}$-a kašəŋ χujat$_i$
Masha-POSS.2SG.[NOM] friend-POSS.3SG-DAT every someone.[ACC]
χur-ən want-λta-s
photo-LOC see-CAUS-PST.[3SG]

b. Maša-jen$_k$ kašəŋ χujat$_i$ λɵχs-əλ$_{k/*i}$-a
Masha-POSS.2SG.[NOM] every someone.[ACC] friend-POSS.3SG-DAT
χur-ən want-λta-s
photo-LOC see-CAUS-PST.[3SG]
'Masha$_k$ showed everyone$_i$ on the photo to *his$_i$/her$_k$ friend.'

This is an expected result if non-agreeing direct objects can only undergo A'-scrambling, which does not affect binding relations (e.g. Saito 1985; Déprez 1989; McGinnis 1999). A'-scrambled DPs are interpreted in their base position both as binders and bound variables. If the base-generation order is IO>DO, the data above follow straightforwardly.

In clauses with agreeing direct objects, the availability of binding depends on the linear order of the objects. When an IO linearly precedes an agreeing DO, the IO can bind a variable on the DO, as illustrated in (132a). As expected, a DO cannot bind a variable on a preceding IO (132b).

(132) IO>DO => IO binds DO / DO does not bind IO

a. Maša-jen$_k$ kašəŋ amp(-əλ)-a$_i$ χot-əλ$_{i/k}$
M.-POSS.2SG.[NOM] every dog-POSS.3SG-DAT house-POSS.3SG.[ACC]
want-λta-s-λe
see-CAUS-PST-3SG>SG
'Masha showed every dog its/her house.'

[14] Note that habitual and perfect clauses, i.e. clauses with a valued high aspect projection, behave differently. High aspect blocks object agreement, even with topical (referential and definite) objects. An analysis of the interaction between aspect and object agreement requires further study, but A-scrambling of the object is still possible, since variable binding patterns together with clauses with agreeing (A-scrambled) objects. Habitual and perfect clauses are excluded from the following discussion for the sake of clarity.

b. Maša-jen$_k$ λɵχs-əλ-a$_{k/*i}$ kašəŋ χujat$_i$
Masha-POSS.2SG.[NOM] friend-POSS.3SG-DAT every someone.[ACC]
χur-ən want-λta-s-λe
photo-LOC see-CAUS-PST-3SG>SG
'Masha$_k$ showed everyone$_i$ on the photo to *his$_i$/her$_k$ friend.'

When the order of the objects is reversed and the agreeing DO linearly precedes the IO, the binding relation is also reversed and the DO can bind a variable on the IO (133b).[15] Agreeing DOs do not reconstruct for variable binding; therefore, the IO cannot bind a variable on the preceding agreeing DO, as in (133a).

(133) DO>IO => DO binds IO / IO does not bind DO

a. Maša-jen$_k$ χot-əλ$_{k/*i}$ kašəŋ amp(-əλ)-a$_i$
M.-POSS.2SG.[NOM] house-POSS.3SG.[ACC] every dog-POSS.3SG-DAT
want-λta-s-λe
see-CAUS-PST-3SG>SG
'Masha showed every dog her/*its house.'

b. Maša-jen$_k$ kašəŋ χujat$_i$ λɵχs-əλ-a$_{k/i}$
Masha-POSS.2SG.[NOM] every someone.[ACC] friend-POSS.3SG-DAT
χur-ən want-λta-s-λe
photo-LOC see-CAUS-PST-3SG>SG
'Masha$_k$ showed everyone$_i$ to his$_i$/her$_k$ friend on the photo.'

These data suggest that agreeing objects can either remain in situ or undergo A-scrambling. An A-scrambled element is interpreted in its landing position for binding relations and does not reconstruct into the base-generation position; hence, only agreeing DOs can bind a variable on IO (cf. Bhatt & Keine 2019 for a recent analysis of the interaction between A-movement and WCO).

The data in the applicative clauses are similar. If the direct object does not control agreement on the verb, the surface word order does not affect the binding relation. As (134) demonstrates, a non-agreeing DO can never bind a variable on the IO. In clauses with an agreeing DO, as in (135b), only a scrambled DO can bind the IO. When the DO stays in a low position, it cannot bind a variable on the IO.

(134) Non-agreeing DO never binds IO

χon ik-en (jaŋ χujat$_i$) kɵrt-əλ$_{*i}$-a (jaŋ
king man-POSS.2SG.[NOM] 10 someone.[ACC] village-POSS.3SG-DAT 10
χujat$_i$) part-əs.
someone.[ACC] send-PST.[3SG]

'The king$_k$ sent (each of) ten men$_i$ to his$_{k/*i}$ village.'

[15] It should be noted that some clauses with IO>DO surface order behave similar to 133b with respect to variable binding. There are two ways to reconcile these facts with the existence of 132a: (i) covert movement of the DO over the IO, or (ii) movement of the IO over the scrambled DO. I will return to this problem at the end of this subsection and in Appendix B when discussing quantifier scope.

(135) Ageeing DO can bind IO only in DO>IO order

a. Vasja-jen$_k$ χot-əλ$_{k/*i}$-a kašəŋ amp-əλ$_i$
Vasya-POSS.2SG.[NOM] house-POSS.3SG-DAT every dog-POSS.3SG.[ACC]
par-s-əλλe
send-PST-3SG>SG
'Vasya$_k$ sent [every one of his dogs]$_i$ to his$_k$/*its$_i$ house.'

b. ?Vasja-jen$_k$ kašəŋ amp-əλ$_i$ χot-əλ$_{i/k}$-a
Vasya-POSS.2SG.[NOM] every dog-POSS.3SG.[ACC] house-POSS.3SG-DAT
par-s-əλλe
send-PST-3SG>SG
'Vasya$_k$ sent [every one of his dogs]$_i$ to his$_{?k}$/its$_i$ house.'

On the other hand, an IO can bind a variable inside an agreeing DO only if the IO linearly precedes the DO, as in 136a. In 136b, the variable on the DO can only be bound by the subject.[16]

(136) IO can bind agreeing DO only in IO>DO order

a. Vasja-jen$_k$ kašəŋ rөpatńek-a$_i$ tiλəś wuχ-əλ$_i$
Vasya-POSS.2SG.[NOM] every worker-DAT month money-POSS.3SG.[ACC]
mă-s-λe
give-PST-3SG>SG
'Vasya$_k$ gave every worker$_i$ his$_{i/(k)}$ paycheck.'

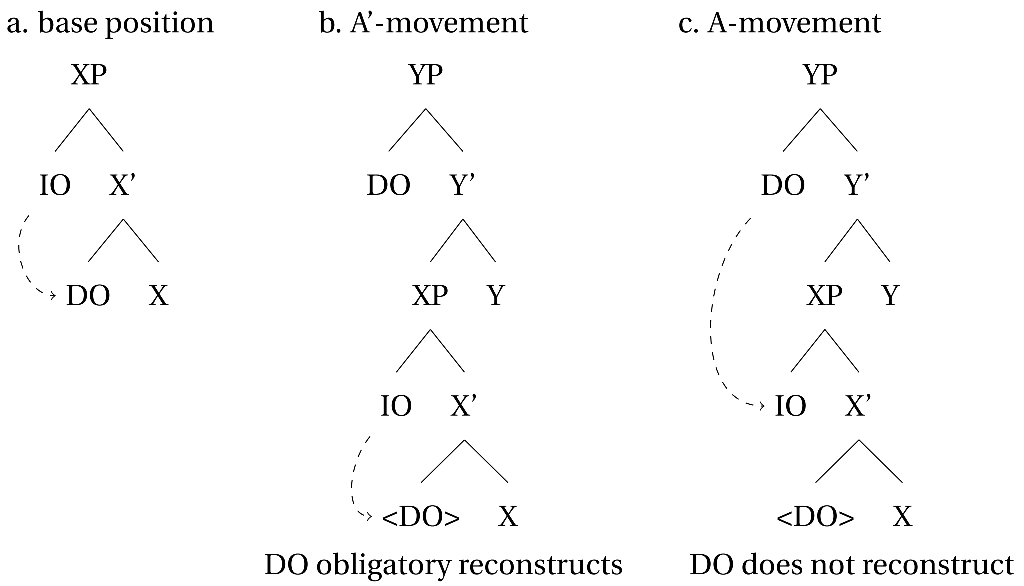
b. Vasja-jen$_k$ tiλəś wuχ-əλ$_{k/*i}$ kašəŋ rөpatńek-a$_i$
Vasya-POSS.2SG.[NOM] month money-POSS.3SG.[ACC] every worker-DAT
??mă-s-λe / or-s-əλλe
give-PST-3SG>SG divide-PST-3SG>SG
'Vasya$_k$ gave every worker$_i$ his$_{k/*i}$ paycheck.'
Actual meaning: 'Vasya$_k$ divided his$_k$ paycheck to every worker.'

In summary, an IO in Kazym Khanty can always bind a variable on a DO unless the latter is A-moved. A DO can only bind a variable on an IO if the DO is A-moved. This pattern can be easily derived if the IO asymmetrically c-commands the DO in the base position. Agreeing direct objects undergo A-scrambling, while non-agreeing direct objects can only undergo A'-scrambling (see e.g. Nikolaeva 1999; Smith 2020; É. Kiss 2021 for discussion of the relation between object agreement and object raising). A-scrambling is known to be able to create new binding relations and resist reconstruction (e.g. Saito 1985; Déprez 1989; Sportiche 2006; Bhatt & Keine 2019). In contrast, non-agreeing direct objects can only be A'-scrambled and always reconstruct for binding relations.

The derivations in (137) illustrate three possible scenarios: (i) no scrambling, (ii) A'-scrambling, and (iii) A-scrambling of the theme over the IO.

[16]Note that the verb *măti* 'to give' is highly dispreferred and must be replaced with *orti* 'to divide', since each of the multiple recipients cannot receive one a the same object (in this case, the boss's paycheck).

(137)



As mentioned earlier, Colley & Privoznov (2018, 2020) argue for the existence of two base structures in Kazym Khanty. In applicativized monotransitives, the base-generation order is IO>DO. For lexical ditransitive verbs, they claim that the DO is base-generated above the IO. They base their conclusion on the data in 138-139. If the linear order of the objects is DO>IO, as in 138, a negative pronoun DO can bind a variable inside the IO. Note that this is not sufficient evidence that DO is generated above IO. My analysis can capture 138 equally well, since the direct object triggers agreement on the verb and is therefore A-scrambled.

(138) CONTEXT: Kids were at school, writing an exam. Their parents were worried and really wanted to see them. But Masha (the teacher) didn't let anyone in to see their kid (exam rules).

Maša-jen [nɛmχujăt]$_k$ [aŋke-λ-aλ$_k$-a –
Masha-POSS.2SG.[NOM] nobody.[ACC] mother-PL-POSS.3SG-DAT
aśe-λ-aλ$_k$-a] ăn wantλta-s-λe
father-PL-POSS.3SG-DAT NEG show-PST-3SG>SG

'Masha did not show anybody$_k$ to their$_k$ parents.' (Colley & Privoznov 2018: 20a)[17]

Crucially, in their data, the IO cannot bind a variable in the DO. In 139, a dative-marked negative pronoun cannot bind a variable in 'X's toys', even when the order is IO>DO. This directly contradicts the data discussed so far and is unexpected if IO c-commands DO in the base-generation position.

(139) CONTEXT: Masha is a teacher, she has all the toys. Kids are asking for their toys. She didn't give anybody their toy.

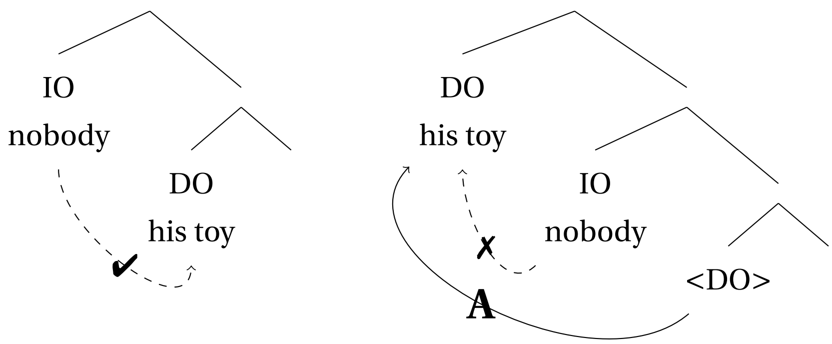
[17] The IO *aŋke-λ-aλ-a – aśe-λ-aλ-a* 'parents' is a co-compound. Borise & Halm (2024) suggest that such co-compounds result from the merger of two nominal roots into a complex N-head. All modifiers, including possessive suffixes, are merged on top of the complex N^0 and lowered onto the roots in the morphology. Such co-compounds refer to a single entity, the 'parents' in this case. Crucially, there is only one possessive marker in syntax, which is doubled later at PF.

Maša-jen$_n$ [nɛmχujăt-a]$_k$ [juntut-əλ$_{n/*k}$] ăn
Masha-POSS.2SG.[NOM] nobody-DAT toy-POSS.3SG.[ACC] NEG
mă-s-λe
give-PST-3SG>SG

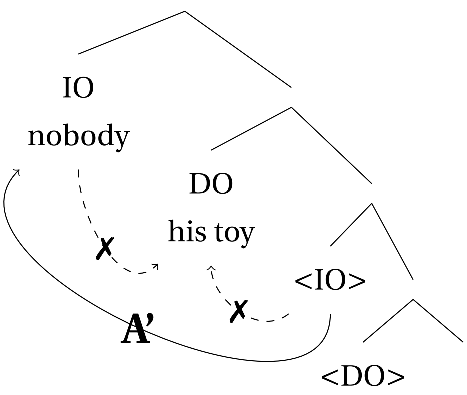
1. 'Masha$_n$ did not give her$_n$ toy to anybody.'
2. '*Masha did not give their$_k$ toy to anybody$_k$.' (Colley & Privoznov 2018: 20b)

It is not clear why the IO cannot bind a variable on the DO in 139. At first glance, it provides an argument in favor of the DO>IO base-generation order. Nevertheless, the picture is not straightforward, since IO can bind a variable on agreeing DO in other tested examples, e.g. in 136a.[18] A possible explanation for the absence of binding in 139 is double scrambling. A possible abstract derivation is provided in 140. First, the DO 'toy' is A-scrambled above the IO 'nobody', which changes the binding configuration, and the IO cannot be a binder anymore. In the second step, IO is A'-moved above DO (possibly to SpecNegP, cf. Kayne 1998 for negative pronouns in Norwegian). Crucially, A'-scrambling does not affect binding, and IO is still unable to bind the DO.[19]

(140) step 1: merge step 2: A-scrambling of DO



step 3: A'-scrambling of IO



[18]Gereon Müller suggested that this asymmetry can be due to the different nature of the quantifiers $\forall x$ and $\neg\exists x$. The available data do not allow to prove this hypothesis.

[19]There is one potential problem with the analysis in 140. The A-scrambling of DO in step 2 has to be obligatorily; otherwise, the binding relation in base-generation positions (step 1) would be further available. However, this does not seem to be a fatal problem. The A-scrambling of agreeing direct objects seems to be a default scenario, while the in situ position is a marked option. Factors preventing A-scrambling of an agreeing direct object must be absent in 139, or even generally unavailable in clauses with negative pronouns.

Even though the absence of binding by the IO in (139) does not straightforwardly follow from the IO>DO base-generation order, it cannot be used as an argument for the DO>IO base-generation order without further discussion. Other collected examples show that the IO can bind a variable on the DO. The binding pattern discussed earlier in this subsection can only be explained under the IO>DO base-generation order. In contrast, several factors can prevent binding by the IO in 139.

Finally, the verb *want-λta-ti* 'to show' in 138 is not a lexical ditransitive but a causative derivation from *want-ti* 'to see'. As a causative, it can only merge the two objects so that the causee (IO) c-commands the theme (DO). More importantly, in light of the previous discussion, the presence of object agreement suggests that DO in 138 is (or at least can be) A-scrambled, which would create a new binding relation. Therefore, the data in 138 do not provide sufficient evidence for the base-generation order of the objects being DO>IO.

3.1.5 Floating quantifiers

In the current minimalist theory, there are two main analyses of floating quantifiers. Under the first account, free standing quantifiers are base-generated inside the DP, that they associate with (Sportiche 1988; Giusti 1990; Koopman & Sportiche 1991; Shlonsky 1991; Merchant 1996; McCloskey 2000; Bošković 2004; Henry 2012; Ott 2015; Zyman 2018). In 141, only part of the DP is moved, and the quantifier is stranded in its original position.

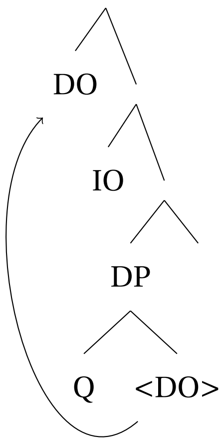
(141) The students$_i$ seem [all t$_i$] to know French. (Bošković 2004: ex.2)

In an alternative approach, the quantifier and the associated noun never form a constituent. Instead, the quantifier is merged as an adverbial, adjoined to the vP or some other projection in the clausal spine 142 (Doetjes 1992; Baltin 1995; Torrego 1996; Brisson 1998; Benmamoun 1999; Bobaljik 2003; Heck & Himmelreich 2017 for German *alles*).

(142) The walruses$_i$ are [$_{vP}$ all [$_{vP}$ t$_i$ painting murals]]. (Zyman 2018: ex.3)

In light of the ongoing discussion, there is a possibility that different languages use different strategies in clauses with floating quantifiers. Note that the first analysis allows us to define the base-generation position of a moved (scrambled) element if the quantifier is stranded in the in situ position. Therefore, it can be used to diagnose the base-generation order of the two objects in ditransitive, applicative, and causative clauses. If a DO can strand a quantifier below the IO, then the base-generation order is most likely IO>DO, as illustrated in 143.

(143)



Kazym Khanty has two universal quantifiers: the collective *χuλ* and the distributive *χuλijewa*.[20] Both of them can be separated from the head noun, e.g. by a locative adverbial in 144.

(144) ńawrɛm-ət kinška-ət χot jit-ən xuλ / xuλijewa wu-s-ət
child-PL.[NOM] book-PL.[ACC] house part-LOC all each take-PST-3PL
'Children$_i$ took all/each$_i$ books$_j$ in the room.' (Solovieva 2019: p.3, ex.9)

For the collective *χuλ*, Solovieva (2020) argues in favor of the stranding analysis and against the adverbial one. The strongest argument comes from the fact that *χuλ* cannot modify pronouns both in the linearly preceding and in the stranded position, as illustrated in 145. However, adverbial analysis cannot capture this restriction. Under modifier analysis, the universal quantifier is base-generated inside the DP and thus can be sensitive to the DP-internal structure.

(145) a. *χuλ mʉŋ roχpi-λ-əw
all we.PL lie-NPST-1PL

b. *mʉŋ *χuλ roχpi-λ-əw
we.PL all lie-NPST-1PL
Intend.: 'We all lie.' (Solovieva 2020: ex.15)

The universal quantifier *χuλ* is base-generated inside the argument DP. Hence, it marks the base position of the matrix DP, and quantifier stranding can be used to diagnose the base-position and the base-generation order of the two objects.

Recall that only A-moved elements can strand quantifiers, and A'-moved elements cannot (cf. Déprez 1989).[21] Indeed, only external arguments (144) and agreeing direct objects (146) allow quantifier stranding in Kazym Khanty. In the absence of object agreement, as in 147, a direct object cannot strand a quantifier (Solovieva 2019).

[20] Solovieva (2020) proposes that *χuλijewa* is derived by raising of the universal quantifier *χuλ* to the specifier of FocP. The head Foc is spelled out with the exponent *-ijewa*. This analysis captures the fact that *χuλijewa* is obligatory focused and can be stranded more easily than the non-derived *χuλ*.

[21] This contradicts Bošković's (2004) generalization that quantifiers cannot float in θ-positions. However, this is a matter of variation even in different dialects of English (Henry 2012).

(146) ńawrɛm-ət piśma-λ-aλ$_i$ tɛrmat [χuλ t$_i$] (śi)
child-PL.[NOM] letter-PL-POSS.3PL.[ACC] hasty all EMPH
šɵnχit-s-əλaλ
burn-PST-3PL>NSG
'Children burned all letters in haste.' (Solovieva 2019: p.3, ex. 13a)

(147) ńawrɛm-ət$_i$ kinška-ət$_j$ χot jit-ən [xuλ t$_{i/*j}$] (śi) wu-s-ət
child-PL.[NOM] book-PL.[ACC] house part-LOC all EMPH take-PST-3PL
1. 'All children took books in the room.'
2. '*Children took all books in the room.' (Solovieva 2019: p.2, ex.8b)

Note also that DAT-marked indirect objects can never have stranded quantifiers. An attempt to have a floating quantifier associated with IO$_{DAT}$ in 148 leads to ungrammaticality.

(148) *ma kɵr-λ-əw-a$_i$ [χuλ t$_i$] ńań oməs-s-əm.
I stove-PL-POSS.1PL-DAT all bread put-PST-1SG
Intend.: 'I put bread into all stoves.'

One way to derive this would be to assume that DAT-marked objects are case-licensed in situ and never undergo A-movement. As will be shown in Section 3.2.3, this is not the case. In contrast, DAT-marked IOs are obligatorily raised to a higher position in order to receive a case. Therefore, I suggest that quantifiers are stranded in the Case position and only under A-movement (see Bhatt & Keine 2019 for dissociation of A-movement from case assignment). Direct objects always receive case in their base position, as shown by reconstruction for Condition C (cf. Bhatt & Keine 2019) and can, therefore, strand a quatifier. Indirect objects receive dative case ex situ (in SpecHighApplP, see Section 3.2.3, cf. similar proposals by Müller 1995 and Georgala 2011) and cannot strand quantifiers in the base-generation position.

Nevertheless, stranded quantifiers can be used as a test for the base-generation order of the two objects. In indirective alignment, the quantifier can be either right-adjacent to a scrambled object or stranded below the indirect object. [22] Examples of an applicative and a causative clause are given in 149 and 150, respectively.

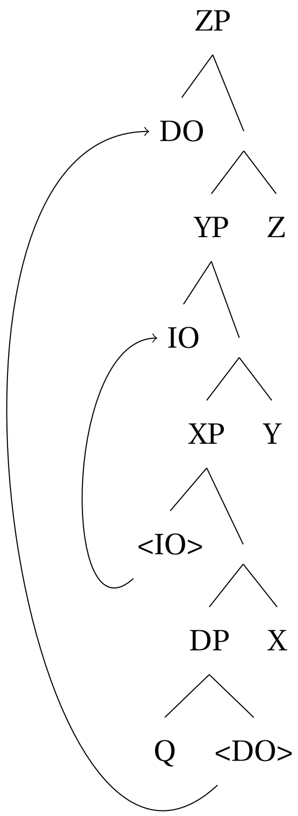
(149) *pro* ma wuχ-λ-am (?χuλ) năŋena (okχuλ) mă-s-λe
I money-PL-POSS.1SG all you.DAT all give-PST-3SG>NSG
'(S)he gave you all my money.'

(150) Ma χot-λ-am (χuλ) năŋeŋa (χuλ) want-λta-s-λam
I.NOM house-PL-POSS.1SG all you.DAT all see-CAUS-PST-1SG>NSG
'I showed you all my houses.'

This data is easily captured if the IO is base-generated higher than the DO. As discussed later in Section 3.2, IO moves string-vacuously to a higher position for case assignment. In the derivation in 151, the A-movement of the DO leaves a stranded quantifier in the position below both copies of the IO.

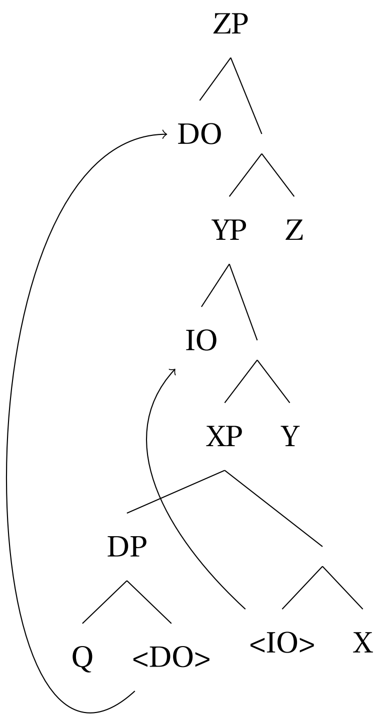
[22] Only the lower position of the floating quantifier is relevant for my purposes. The higher position requires some further research. In particular, it is unclear whether the quantifier is stranded in the scrambled position, while the noun is raised to an even higher projection. An alternative is the DP-internal raising of N^o to SpecDP.

(151)



An alternative scenario, in which the IO is base-generated below the DO, violates the Minimal Link Condition (Rizzi 1990). As 152 illustrates, the first movement step would cross a caseless DO reach the IO. The violation cannot be avioded, because the first movement step is movement for case assignment and must probe for the first caseless DP in its c-command domain (see more on that in Section 3.2.3).

(152) *

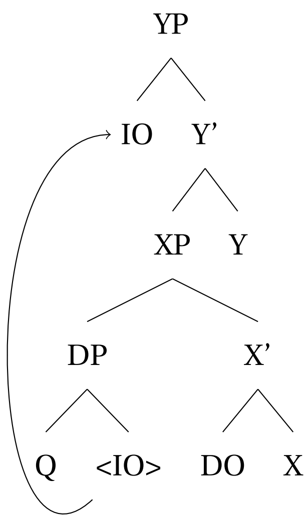


In secundative alignment, the indirect object is ACC-marked and can have stranded quantifiers. The prediction is that the stranded quantifier can never follow an in situ DO if the IO is base-generated higher than the DO. As 153 shows, this is indeed borne out by the data. In 153a, the standed quantifier linearly precedes the oblique theme. If the quantifier is stranded below the secundative theme, as in 153b, the clause is ungrammatical. Therefore, the base-generation order of the objects in secundative alignment is IO>DO, identical to indirective alignment.

(153) a. $^{\text{ok}}$Maša-jen nurum-ət$_i$ [χuλ t$_i$] kińška-jən pun-s-əλλe
Masha-POSS.2SG shelf-PL all book-LOC put-PST-3SG>NSG
'Masha put books on all shelves.'

b. *Maša-jen nurum-ət$_i$ kińška-jən [χuλ t$_i$] pun-s-əλλe
Masha-POSS.2SG shelf-PL book-LOC all put-PST-3SG>NSG

The derivation with a stranded quantifier in secundative alignment is illustrated in 154.

(154)


The data on quantifier stranding provide further evidence for a single base-generation structure in both indirective and secundative alignments. The base-generation order of two objects is IO>DO for lexical ditransitives, causative and low applicative clauses.

3.1.6 Further tests

Further tests for asymmetric c-command include reciprocal binding, quantifier scope, superiority effects in wh-movement, and NPI licensing by a c-commanding DP (Barss & Lasnik 1986). All these tests are unavailable in (Kazym) Khanty.

According to Condition A, a reciprocal pronoun must be locally bound by a c-commanding R-expression. If an IO can bind a reciprocal pronoun in the DO position, then the base-genration order is IO>DO. If this is not possible, but DO can bind a reciprocal in the IO position, the base-generation order is DO>IO. However, this test is unavailable in Kazym Khanty. There are no reciprocal pronouns in the language and the semantics is expressed by other means, e.g. the postposition *kʉt-ən* 'together' (see Volkova & Toldova 2023 for details).

Quantifier scope, and especially the availability of inverse scope, has also been used as a test for base-generation order in some languages (Frey 1993; Aoun & Li 1993; Bruening 2001; Harley & Jung 2015; Collins 2020 among others). For instance, if two quantified objects can have inverse scope, when the surface order is DO>IO, it can be concluded that the DO is base-generated below the IO and can reconstruct for scope. The data in Kazym Khanty are discussed in detail in Appendix B. Crucially, wide or narrow scope of the DO relative to IO is independent from the surface word order and correlates only with presence or absence of object agreement on the verb. Hence, these data cannot be used as a test for the base-generation order.

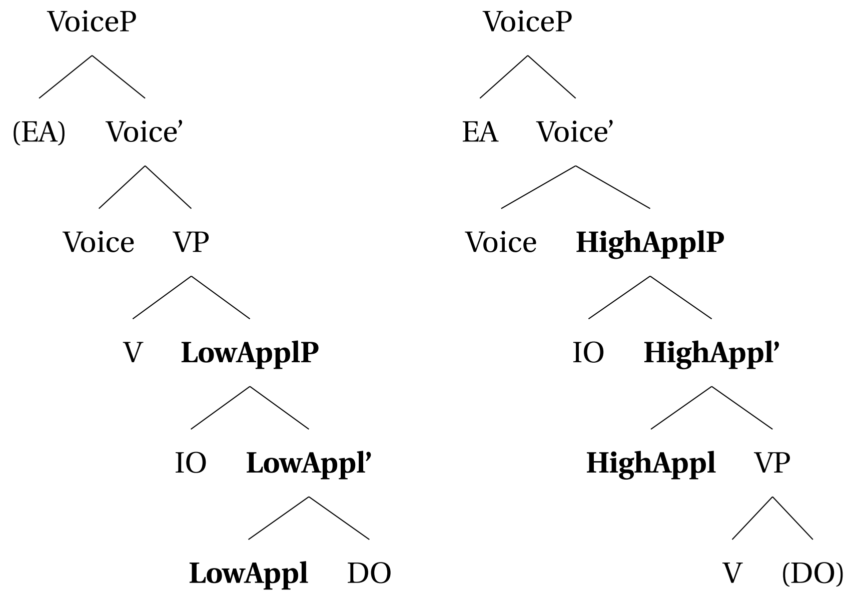
Two other existing tests are also unavailable in Kazym Khanty. As discussed in Chapter 2, Kazym Khanty is a wh-in-situ language. Therefore, the superiority effects in wh-movement cannot be tested. The NPI licensing test suggests that a negation on the higher argument would license an NPI on a lower argument, but not vice versa. If an NPI in the DO position is licensed by a negative pronoun in the IO position, the base-generation order is IO>DO. If a negative pronoun on the DO can license an NPI on the IO, the base-generation order is DO>IO. Crucially, Kazym Khanty is a negative concord language. Furthermore, a negative pronoun in an argument position cannot be used without sentential negation being present in the clause. Sentential negation is expected to be merged above VoiceP, i.e. above both IO and DO. Hence, NPI (i.e., negative concord) on any of them would always be licensed by the sentential negation, irrespective of their base-generation order. Therefore, this test is also unavailable for diagnosing asymmetric c-command between the two arguments.

3.1.7 Summary

There are three tests for asymmetric c-command available in Kazym Khanty: reconstruction for Condition C, asymmetries in variable binding and Weak Crossover Effect, and quantifier float. All three tests show that IO c-commands DO in their base-generation positions in lexically ditransitive, applicativized, and causative clauses. Moreover, these tests give identical results for indirective and secundative alignments and provide a baseline for an analysis of alignment alternation in the next Section and in Chapter 4.

For the analysis below, I follow Pylkkänen (2008) and distinguish between high and low applicative projections. The two structures are presented in 155. The low applicative structure corresponds primarily to lexically ditransitive and recipient clauses. Both the indirect object (recipient) and the theme are selected by LowAppl^{o}. Then, LowApplP is merged with the verb (verbalized root). In the high applicative derivation, on the other hand, the lexical verb (verbalized root) selects the internal argument (theme). HighApplP is merged only afterwards, and the applied argument is introduced in its specifier. Crucially, the theme and the applied argument are introduced in separate projections in the high applicative structure and do not form a constituent.

(155) a. low applicative b. high applicative



One further difference between the two types of applicatives concerns the presence of the internal argument. Low applicative structures are obligatorily ditransitive, as LowAppl^{o} always selects for two DPs. In contrast, HighApplP does not require the presence of an internal argument. Hence, unergative verbs can have only high applied arguments.

Pylkkänen (2008) argues that languages differ in which one of the applicative projections introduces which θ-role. Recipients tend to be cross-linguistically associated with LowApplP, but benefactors can be introduced either as high or as low applied arguments. Furthermore, in contrast to Baker (1988), this approach to applicative construction does not require Appl-projection to be marked with special morphology. Applicative heads can remain silent at the PF if a language does not have a corresponding vocabulary item.

3.2 Indirective alignment

3.2.1 Dative as a structural case

The nature of the dative case is a matter of discussion. Quite often, dative shows mixed properties of an inherent and structural case. Furthermore, one morphological exponent can correspond to two different syntactic case values. For instance, Bárány (2018) and Irimia (2023) suggest that the Romance DOM marker is distinct from the proper dative case, even though they are morphologically syncretic. In this subsection, I argue that dative in Kazym Khanty can be used as a lexical, structural, and inherent case. I do not follow Bárány (2018) and Irimia (2023) and do not suggest that dative in Kazym Khanty should be split into three abstract cases. Rather, I assume that one abstract case can be assigned in syntax by three different types of heads. The existence of structural dative is crucial for the subsequent discussion.

Woolford (2006b) differentiates among three types of cases: lexical, inherent, and structural. Lexical case is invariable and is randomly determined by a specific verbal root or an adposition. According to Woolford (2006b)' definition, lexical case can only be assigned to themes/internal arguments or complements of adposition. For example, the Icelandic verb 'capsize' in 156 assigns dative to its complement. The lexical dative is idiosyncratic and lexically selected by this particular verb.

(156) Bátnum hvolfdi.
boat.DAT capsized
'The boat capsized.' (Woolford 2006b: p.112, ex.2)

An inherent case is assigned to a DP in a particular θ-position. In contrast to lexical case, inherent case is not random. It is associated with a particular thematic role of the argument and is assigned by a functional head to the argument it introduces (Woolford 2006b; Legate 2012). It does not change if the argument moves to a different structural position in a clause, i.e. cannot be overridden by another functional head. The inherent dative case is often used to mark applied arguments and experiences. For instance, 157 illustrates this for experiencer subjects in Icelandic. As 157b shows, dative marking is preserved under passivization. This means that dative is not a structural case. Neither is it a lexical case, since all experiencers are dative-marked, i.e. case marking depends on the θ-role, and not on the verbal root.

(157) (Woolford 2006b: p.118, ex.16)

a. þeir skiluðu Maríu bókinni
They returned Mary-DAT book-the-DAT
'They returned the book to Mary.'

b. Maríu var skilað þessari bók
Mary-DAT was returned this book-DAT

All non-structural cases are preserved under A-movement (e.g., passivization or raising) or in a changed syntactic environment (e.g., in ECM configurations). The non-structural cases do not compete for case with other arguments and are associated with specific thematic roles (Woolford 2006b). In contrast, structural cases are determined by and sensitive to the configurational syntactic environment in which a DP appears and to nothing else. A canonical example of a structural case is accusative. The direct object in the active transitive clause in 158a is marked with accusative. However, in the passive clause in 158b, accusative is overridden by nominative.

(158) a. Meine Schwester hat einen Kuchen gebacken
my.F.NOM sister AUX a.M.ACC pie bake.PTCP
'My sister baked a pie'

b. Der Kuchen ist gebacken.
the.M.NOM pie is bake.PTCP
'The pie is ready (lit.: baked).'

Anagnostopoulou & Sevdali (2020) show that dative is not uniform across languages. In some languages, dative has all the properties of an inherent case (cf. e.g. Woolford 2006b for Icelandic). In other languages, dative can act as a structural case (cf. Anagnostopoulou & Sevdali 2020 for Greek). As the rest of this subsection shows, dative can be non-uniform even within one language. Dative in Kazym Khanty surfaces on the complements of certain verbs, i.e. shows the properties of a lexical case. On the other hand, experiencers and some applied arguments are marked with inherent dative. And most crucially, dative can also be used as a structural case, e.g. on low applicatives and causees. Two examples of lexical datives in Kazym Khanty are provided in 159. Indeed, some verbs mark their internal arguments with dative morphology. These uses are idiosyncratic and cannot be attributed to syntactic properties of a certain verb class.

(159) a. Vasja-jen tөχtər-a ji-s
Vasya-POSS.2SG.[NOM] doctor-DAT become-PST.[3SG]
'Vasya became a doctor.'

b. Maša-jen λɛpət λońś-a ketəm-s
Masha-POSS.2SG soft snow-DAT touch-NPST-[3SG]
'Masha touched the soft snow.'

The inherent dative is used in Kazym Khanty to marke experiencers (in 160), benefactives (in 161), and unaffected goals (directions) (in 162). Dative is assigned to these arguments based on their θ-role and cannot be changed to another case value. As 161b and 162b demonstrate, inherent datives do not participate in indirective-secundative alignment alternation.

(160) Experiencer

Pet'a-jen-a χot kă-λ.
Petja-POSS.2SG-DAT house seen-NPST.[3SG]

'Petya sees a house.' [Ryžowa 2021: ex.6]

(161) Benefactive

a. Ar χujat λʉweλa rөpat-λ-ət / rөpata wɛr-λ-ət
many someone.[NOM] (s)he.DAT work-NPST-3PL work do-NPST-3PL
'Many people work for her.'

b. *Ar χujat λʉwti rөpat-λ-eλ / rөpata-jən wɛr-λ-eλ
many someone.[NOM] (s)he.ACC work-NPST-3PL work-LOC do-NPST-3PL

(162) "Unaffected" goal

a. Tөrəm λor-a pasilka kit-s-əw a Polnowat-a ănt kit-s-əw.
Numto-DAT parcel.[ACC] send-PST-1PL but Polnowat-DAT NEG send-PST-1PL
'We've sent a parcel to Numto, and haven't sent any parsel to Polnowat.'

b. *Tөrəm λor pasilka-jən kit-s-ew a Polnowat ănt kit-s-ew.
Numto-[ACC] parcel-LOC send-PST-1PL but Polnowat-[ACC] NEG send-PST-1PL

Furthermore, some instances of dative in (Kazym) Khanty must be analyzed as structural, and not as an inherent case (see also Baker & Vinokurova 2010; Baker 2015; den Dikken 2023 for dative as a dependent case in Sakha and Shipibo). In particular, DAT marking on low applied arguments and causees in Kazym Khanty should be seen as a structural case. Crucial for this conclusion is the fact that low applied arguments and causees participate in case-alignment alternations, i.e. they can be marked both with dative, as in 163, and with accusative, as in 164. There is no evidence for different θ-roles of *λeχs-əλ* 'friend', which is a recipient in both clauses. According to Woolford (2006b), inherent case is assigned based on the θ-role and cannot be overridden. Therefore, dative in 163 must be treated as a structural case.

(163) Indirective alignment

a. Sub-[NOM] IO-**DAT** DO-**[ACC]** V

b. Kašəŋ χujat λeχs-əλ-a lipət mă-s
Every person.[NOM] friend-POSS.3SG-**DAT** flower.**[ACC]** give-PST.[3SG]
'Everyone gave a flower/flowers to his friend.'

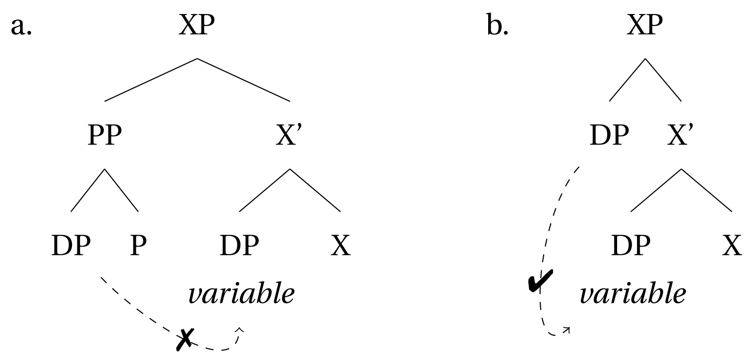
(164) Secundative alignment

a. Sub-[NOM] IO-**[ACC]** DO-**LOC** V

b. Kašəŋ χujat λeχs-əλ lipət-ən mă-s-λe
Every person.[NOM] friend-POSS.3SG.**[ACC]** flower-**LOC** give-PST-3SG>SG
'Everyone gave a flower/flowers to his friend.'

Note also that den Dikken (1995); McFadden (2004); Rezac (2008); Alexiadou, Anagnostopoulou, & Sevdali (2014); Anagnostopoulou & Sevdali (2020) argue that the inherent dative corresponds to a PP-structure. If this is cross-linguistically true for all inherent cases, then dative in Khanty cannot be an inherent case. A complement of P cannot participate in c-command-dependent relations, as it does not c-command anything outside the PP. A complement of P does not c-command a direct object; hence, it cannot act as an antecedent for the direct object. By contrast, a (case-marked) DP c-commands a lower DP and can therefore bind it. The contrast is illustrated in 165.

(165)



Since DAT-marked nominals in Khanty can be antecedents for both possessive markers, as in 166, and reflexive pronouns, as in 167, one can conclude that they are DPs rather than PPs (cf. e.g. Woolford 2006b and Michelioudakis 2012 for a similar view).

(166) Vasja-jen$_k$ kašəŋ rəpatńek-a$_i$ tiλəś wuχ-əλ$_i$
Vasya-POSS.2SG.[NOM] every worker-DAT month money-POSS.3SG.[ACC]
mă-s-λe
give-PST-3SG>SG
'Vasya$_k$ gave every worker$_i$ his$_{i/(k)}$ paycheck.'

(167) a. Ma χur-ən Pet'a-jen-a λʉwti want-λta-s-ɛm.
I.[NOM] image-LOC Petja-POSS.2SG-DAT (s)he.[ACC] see-CAUS-PST-1SG>SG
'I showed to Petja himself on the photo.' (Volkova & Toldova 2023: ex.25)

b. Maša-jen ńawrɛm-λ-aλ-a$_i$ λiw$_i$ oλŋ-eλ-ən
Masha-POSS.2SG.[NOM] child-PL-POSS.3PL-DAT they about-POSS.3PL-LOC
putərt-əs.
tell-PST.[3SG]
'Masha told the children about them.' (Volkova & Toldova 2023: ex.33)

This is not the strongest argument, however, because some languages allow binding outside of a PP for quantificational variables and even for anaphors under certain circumstances (e.g. Van Riemsdijk & Williams 1986; Baltin, Déchaine, & Wiltschko 2015 for English). I do not have relevant data for Kazym Khanty. The only example mentioned by Volkova & Toldova (2023: p.22, ex.60a) illustrates co-reference rather than anaphoric or quantificational binding. Despite the potential weaknesses of this argument, it adds to the evidence for the existence of the structural dative in Kazym Khanty, provided by the case alternation in 163 and 164.

For the analysis below, I assume that dative in (Kazym) Khanty is a structural case, assigned on the basis of the syntactic structure and the syntactic position of the argument.

3.2.2 Indirective alignment: derivation

This subsection introduces an analysis of indirective alignment in clauses with low applicative projection in Kazym Khanty. The next subsection discusses the three pieces of evidence for this analysis. These include (i) high applied arguments, (ii) evidence for raising of the low applied argument to a higher projection, and (iii) (in)availability of double-dative constructions in the language.

For the current analysis, I follow theories where case is assigned under Agree between a DP and a functional projection with a valued Case feature (e.g. Chomsky 2000, 2001; Müller 2009; Bárány 2017; Bárány & Sheehan 2021; Clem & Deal 2024 etc.).[23] An important difference from the Chomsky's original proposal (cf. Kalin 2018; Nie 2020; Clem & Deal 2024) is that in the current model, ϕ-agreement and case assignment are not treated as two sides of one Agree operation, but are dissociated into two independent operations. There are both crosslinguistic/theoretical reasons and Khanty-internal reasons for this assumption. (i) First, Khanty does not show any reflexes of ϕ-agreement with objects marked with structural dative. Since object agreement is generally possible in the language, the

[23]The analysis suggested in this section for indirective alignment can be reformulated in terms of Dependent Case Theory. However, secundative alignment is more problematic for the Dependent Case approach. I will return to this issue in Chapter 5, Section 5.5.

absence of ϕ-agreement with dative-marked objects should be explained independently. A model where HighAppl° has only a Case probe and no ϕ-probe gives the desired result: a DP can receive a case under Agree with HighAppl but can never agree for ϕ-features with it. (ii) Second, ACC-marked objects trigger object agreement only in a subset of clauses (see discussion in Chapter 2, Section 2.3). If the assignment of the accusative case can succeed without simultaneous object agreement, then these operations should be two separate instances of Agree and should be able to take place independently from one another. (iii) Third, Chomsky's system predicts that whatever argument receives a structural case from head A should also agree for ϕ-features with this head. Although this is often the case, this pattern is not universal. Bárány (2021b) shows that ϕ-agreement alignment can be independent of case alignment. For example, in Amharic, only the recipient agrees with the verb, both when it is marked with accusative (168a) and dative case (168b). If ϕ-agreement and case assignment were two sides of one operation, they would always coincide, and 168b would be impossible.

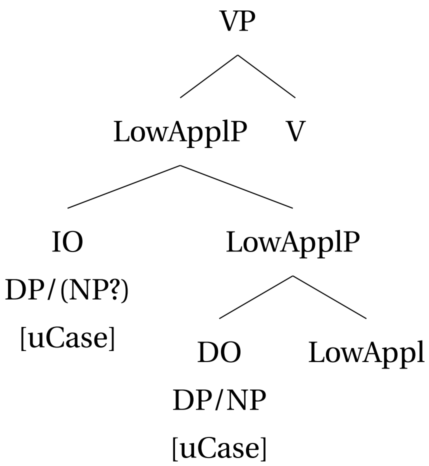
(168) Amharic (Baker 2012: 261, 258, 261)

a. Ləmma [$_R$ Aster-ɨn] [$_T$ hɨs'an-u-n] asaj-at.
Lemma.M Aster.F-ACC baby-DEF-ACC show.3.M.S-3.F.O
'Lemma showed Aster the baby.'

b. Ləmma [$_R$ l-Almaz] [$_T$ tarik-u-n] nəggər-at.
Lemma.M DAT-Almaz.F story.M-DEF-ACC tell.3.M.S-3.F.O
'Lemma told Almaz the story.'

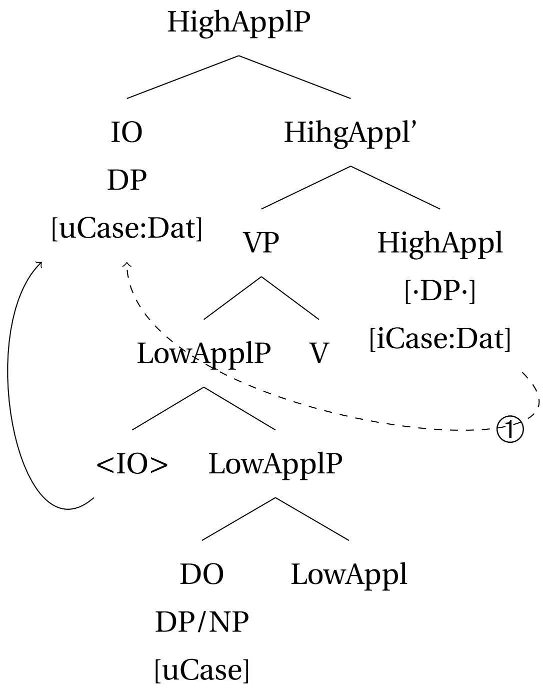
For indirective alignment, I suggest the following derivation. At the first step, the indirect object and the theme are merged in the low applicative projection (cf. Pylkkänen 2008). Then, LowApplP is merged with the lexical verb. At this stage of the derivation, neither object can receive case.

(169) Step 1: Argument DPs are merged



In the second step of the derivation, the high applicative projection is merged. I suggest that that there are two flavors of HighApplP: (i) one with [iCase:Dat]- and [$\cdot$DP$\cdot_\theta$]-feature, that externally merges an argument in its specifier and simultaneously assigns inherent case to it; and (ii) an "a-thematic" HighApplP with [iCase:Dat]- and [·DP·]-feature, that moves a DP to its specifier without assigning a θ-role to it.[24] The moved DP receives dative. The second type of HighAppl° is semantically empty. Its only function is to ensure that a DP in its c-command domain is assigned a case and is properly licensed/incorporated into the clause. I suggest that the "a-thematic" HighApplP is always merged in clauses with indirective alignment and is responsible for the assignment of the dative case.[25]

(170) Step 2: HighAppl° assigns dative case to the highest caseless argument in its c-command domain (recipient)

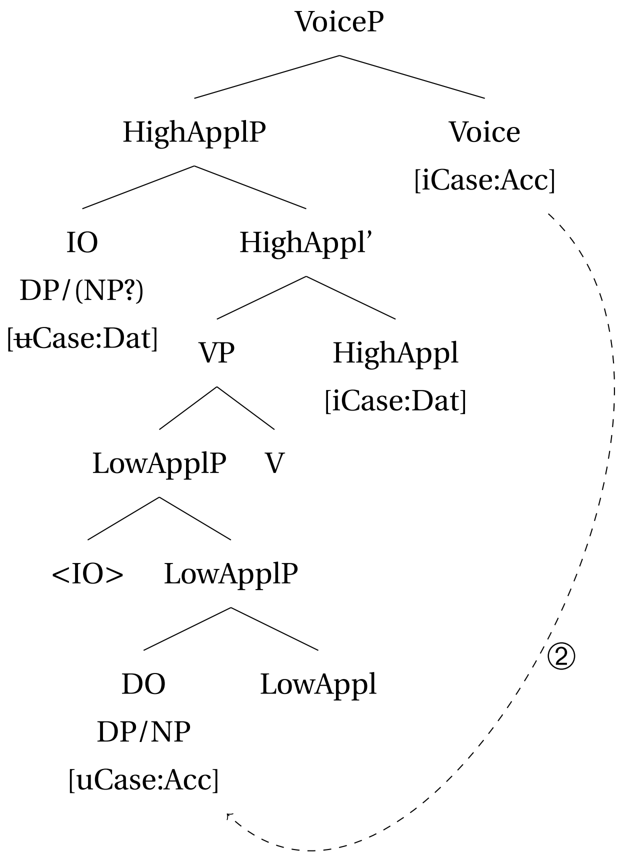


In the third step, VoiceP is merged. Transitive Voice has a [·DP·]-feature that is responsible for the merger of the external argument and a [iCase:Acc]-feature that assigns case to the closest caseless DP/NP in its c-command domain. Since the indirect object has already received a case, Voice can reach the complement of LowApplP and assign accusative to the theme.

[24]Notation for Merge is used after Heck & Müller (2007). I do not use their notation for probes ([*F*]) because syntactic features in the current model are not inherently marked as probes or goals. Furthermore, a single feature can act as a probe in one Agree operation and be a goal in another Agree operation (see Chapter 5 for details).

[25]Note that in the current model, a valued feature can act as a probe ([iCase:Dat] on HighAppl°). Even though this goes against the approach that defines probehood as the lack of value (cf. Preminger 2014), suggestions along this line have been made (e.g. Pesetsky & Torrego 2007; Heck & Himmelreich 2017). I will return to this problem and discuss the notion of probehood in Chapter 5, Section 5.2.3.

(171) Step 3: Voice is merged and assigns accusative case to the theme



The current proposal states that HighAppl° is the only source of the dative case in the clause, or at least in the lower phase (i.e., VoiceP). Hence, indirective alignment requires the presence of HighApplP in the structure. When merged, HighAppl° assigns dative to the closest caseless DP in its c-command domain, i.e. to the indirect object. The next section reviews three pieces of evidence supporting this assumption.

3.2.3 High Applicative as the source of Dative

In recent years, multiple proposals have been introduced in the linguistic literature, suggesting that a functional projection above vP is responsible for the presence of dative case in the clause (e.g. μP in Müller 1995, high applicative in Georgala 2012, den Dikken 2023, or raising applicative in Nie 2019, 2020). In this section, I review the evidence for this assumption in Kazym Khanty and argue that this projection is HighApplP. There are three arguments for treating HighAppl° as a source of the dative marking. First, I show that HighAppl° assigns the inherent dative case, i.e. at least its thematic flavor must have the [iCase:Dat]-feature. Furthermore, there is a general ban on double datives in Kazym Khanty. This restriction cannot be modeled if high and low applicative projections are completely unconnected in the clausal spine. However, if HighApplP assigns a case to an argument inside LowApplP, it cannot assign it a second time to its own specifier, and the ban on double datives can be easily predicted. Second, evidence is presented that low applied arguments are obligatory raised outside of the LowApplP and VP in order to receive case. Given that HighAppl^{o} assigns inherent dative case, it is logically plausible that low applied arguments raise to HighApplP and are assigned dative there. Third, some examples of clauses with two dative-marked arguments are considered. It is argued that only one of them receives the dative case from HighApplP^{o}, and the second one is merged as a PP. Crucially, such clauses with dative PPs are only grammatical in indirective alignment. This can be accounted for if PP-arguments are merged in SpecHighApplP. Hence, HighAppl^{o} is present in the structure and obligatory assigns dative to first available goal, i.e. to the low applied argument.

I. High applied arguments

In Kazym Khanty, high applied arguments (e.g., benefactives of unergatives) are merged as PPs or DPs. When merged as DPs, benefactives can only be DAT-marked and do not allow for case alignment alternation. As 172 shows, the unergative verb 'to work' allows merging a benefactor 'for her', but requires it to be dative-marked.

(172) a. Ar χujat λʉweλa rɵpat-λ-ət / rɵpata wɛr-λ-ət
many someone.[NOM] (s)he.DAT work-NPST-3PL work.[ACC] do-NPST-3PL
'Many people work for her.'

b. *Ar χujat λʉwti rɵpat-λ-eλ / rɵpata-jən wɛr-λ-eλ
many someone.[NOM] (s)he.ACC work-NPST-3PL work-LOC do-NPST-3PL

When the high applied argument is marked with any other case, e.g. accusative, the clause becomes ungrammatical, as in 172, or the clause structure is reinterpreted. In 173, the dative marked *λoχs-əλ* 'friend' is interpreted as a benefactor of the whole event of learning the dance, i.e. a girl is learning a dance in order to dance for her friend later. When *λoχs-əλ* is marked with accusative, the clause meaning changes, and 'her friend' is interpreted as a recipient, receiving the knowledge of how to perform this dance.

(173) a. λʉw λoχs-əλ-a jak wonλta-λ
(s)he.NOM friend-POSS.3SG-DAT dance.[ACC] learn-NPST.[3SG]
'She learns a dance for her friend.'

b. λʉw λoχs-əλ kăša jak wonλta-λ
(s)he.NOM friend-POSS.3SG for dance.[ACC] learn-NPST.[3SG]
'She learns a dance for her friend.'

c. # λʉw λoχs-əλ jak-ən wonλta-λ-λe
(s)he.NOM friend-POSS.3SG.[ACC] dance-LOC learn-NPST-3SG>SG
Intend.: 'She learns a dance for her friend.'

Actual meaning: 'She teaches her friend to dance this dance.'

In 174, the dative-marked argument is a benefactor of the singing event. When marked with accusative, it is reinterpreted as the topic of the song, i.e. the non-external argument is reinterpreted as the direct object of a transitive verb.

(174) a. Im-əλ Vasja-en-a isašək ari-jəλ
wife-POSS.3SG.[NOM] Vasya-POSS.2SG-DAT often sing-NPST.[3SG]
'His wife often sings for Vasya.'

b. # Im-əλ Vasja-en isašək
wife-POSS.3SG.[NOM/ACC] Vasya-POSS.2SG.[ACC/NOM] often
ari-jəλ-λe
sing-NPST-3SG>SG
Intend.: 'His wife often sings for Vasya.'
Actual meaning: 'It's his wife, Vasya often praises.' or ' His wife often praises Vasya.'

Following Woolford (2006b)'s classification, inherent case is connected to the θ-role of the argument, i.e. it is assigned by a functional head to the DP that this head selects for. As inherent case cannot be overridden, high applied arguments (benefactives of unergatives) must have an inherent dative case. Since they are merged in HighApplP, HighAppl^{o} must be seen as a source of the dative case.

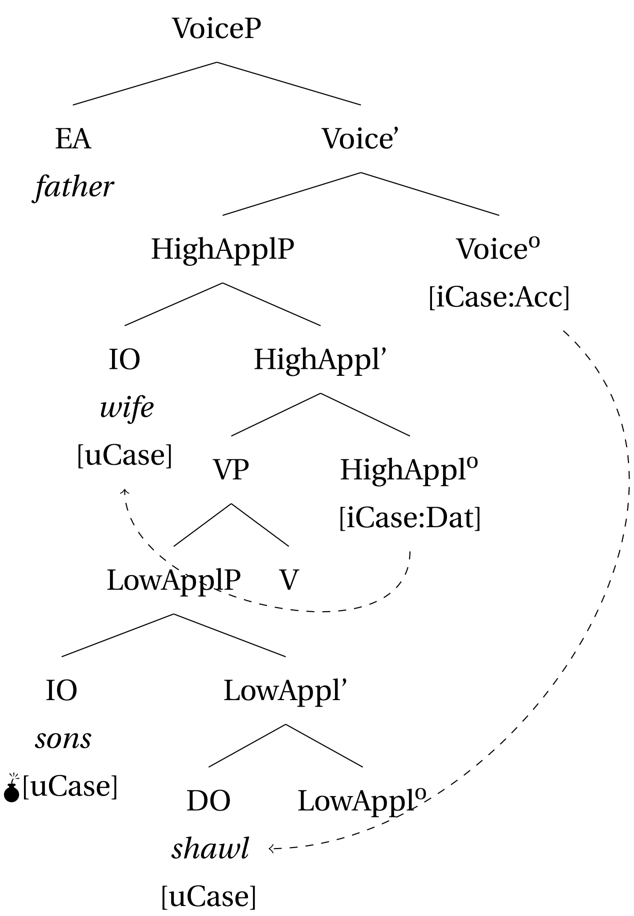
Furthermore, I argue that HighAppl^{o} is the only head in the θ-domain (i.e., inside the VoiceP) that can assign structural dative.[26] Nie (2019, 2020) derives the ban on recursive applicatives in languages by assuming that only some of the functional heads are able to license nominals. The same logic can be used to explain the ban on two DAT-marked applied arguments in Khanty. 175 illustrates that an attempt to introduce both a recipient and a general benefactor of the action into a clause leads to ungrammaticality.

(175) *Aś-əλ kašəŋ puχ-əλ-a im-əλ-a
Father-POSS.3SG.[NOM] every son-POSS.3SG-DAT woman-POSS.3SG-DAT
uχšam mă-s
shawl.[ACC] give-PST.[3SG]
'The father gave each of his sons a shawl for their wives.'

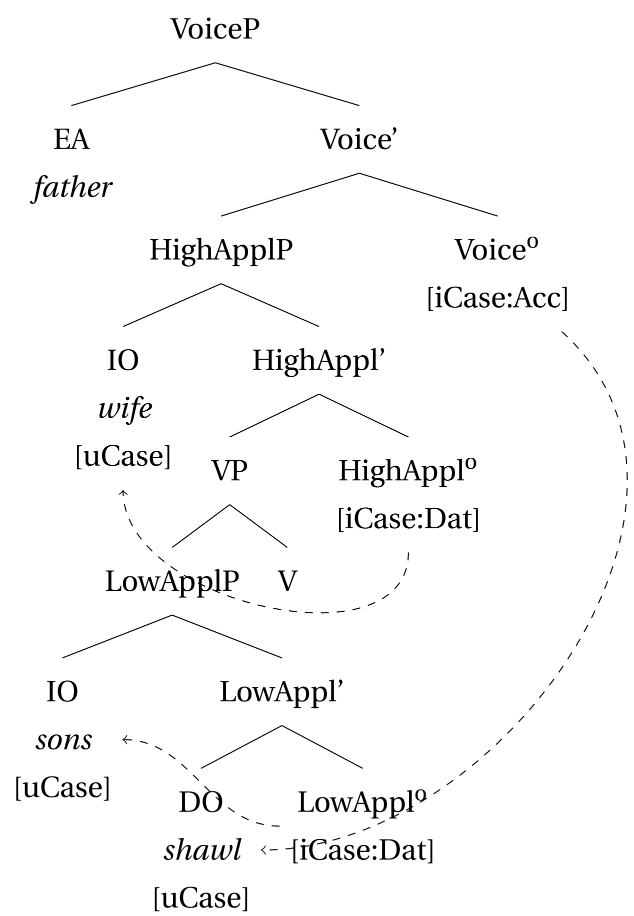
If only HighAppl^{o} is able to assign DAT and LowAppl^{o} is not a case assigner, the ungrammaticality of 175 can be easily captured. There are four arguments in need of a case: the external argument, the recipient *kašəŋ puχ-əλ* 'every son', the benefactor *im-əλ* 'his wife', and the theme *uχšam* 'a shawl'. The external argument is assigned case by T^{o} and can be omitted from the discussion. Crucially, there are only two case-assigning heads inside VoiceP: Voice^{o} and HighAppl^{o}. In 175, Voice^{o} assigns accusative to the theme. Therefore, two applied arguments are competing over a single case assigning head. As HighAppl^{o} can agree for Case with just one DP, either the low or the high applied argument will not receive case value, and the derivation will crash. The first option is illustrated in 176a. However, if LowAppl^{o} could assign DAT, all three arguments would be assigned a case, as illustrated in 176b. This scenario cannot predict the ungrammaticality of 175.

[26]There is an alternative possibility, where neither HighAppl^{o} nor LowAppl^{o} can assign dative. Instead, an additional applicative head (raising applicative) is merged immediately below VoiceP and assigns dative (see Nie 2019, 2020 for this suggestion). Khanty does not allow to choose between these two analyses, but Kalaallisut (aka. West Greenlandic, Inuit) provides additional morphological evidence, that HighAppl^{o} is the source of dative case. The Kalaallisut data are discussed in Chapter 6, Section 6.1.1.

(176) a.


VoiceP
EA
father
Voice'
HighApplP
Voiceº
[iCase:Acc]
IO
wife
[uCase]
HighAppl'
VP
HighApplº
[iCase:Dat]
LowApplP
V
IO
sons
[uCase]
LowAppl'
DO
shawl
[uCase]
LowApplº

b. *


VoiceP
EA
father
Voice'
HighApplP
Voiceº
[iCase:Acc]
IO
wife
[uCase]
HighAppl'
VP
HighApplº
[iCase:Dat]
LowApplP
V
IO
sons
[uCase]
LowAppl'
DO
shawl
[uCase]
LowApplº
[iCase:Dat]

The restriction on double datives suggests that only one of the applicative heads can assign a case. HighAppl° undoubtedly assigns dative to the benefactives of unergatives. Therefore, there is independent evidence of the [iCase:Dat]-feature on this head. However, no such evidence exists for LowAppl°. Therefore, I conclude that HighAppl° must be seen as the only source of the dative case inside VoiceP.

II. Raising to Dative

Georgala (2012) argues that applied arguments in German raise out of their base position inside the VP to a higher applicative projection, where they receive the dative case. Word order data in Kazym Khanty suggest that there is indeed a projection outside the VP associated with the dative case. As 177 demonstrates, DAT-marked IO precedes circumstantial adverbials in unmarked word order. The reversed order in 177b is possible, but only when the adverbial is focused.

(177) a. ɛwi-je aŋk-eʎ-a [wʉrti χăńś-i
daughter-DIM.[NOM] mother-POSS.3SG-DAT red write-NFIN.NPST
jʉχ-ən] piśma χăńś-əs
tree-LOC letter.[ACC] write-PST.[3SG]
'A girl wrote a letter to her mother with a red pen.' (unmarked word order)

b. ɛwi-je [wʉrti χăńś-i jʉχ-ən]
daughter-DIM.[NOM] red write-NFIN.NPST tree-LOC
aŋk-eʎ-a piśma χăńś-əs
mother-POSS.3SG-DAT letter.[ACC] write-PST.[3SG]
'It was with a red pen, that a girl wrote a letter to her mother.' (narrow focus on the instrument)

Since circumstantial adverbials are adjoined to VP (e.g. Cinque 1999), the DAT-marked IO is either base-generated outside VP or moved out of it. The second option appears to be more plausible. It is uncontroversial that the recipient IO is base-generated in the same phrase as the theme (Larson 1988; Pylkkänen 2008; Harley & Jung 2015, etc.). As Khanty is a head-final, left-branching language, nothing can intervene between a specifier and a complement in the linearized structure unless one of them is moved out of its base position. Moreover, a stranded quantifier in secundative alignment can follow the adverbial, as shown in 178. As argued in Section 3.1.5, floating quantifiers in Kazym Khanty are stranded in the base position. This provides evidence that the IO is base-generated inside the VP (in LowApplP; Pylkkänen 2008) and moves to a higher position outside of the VP.

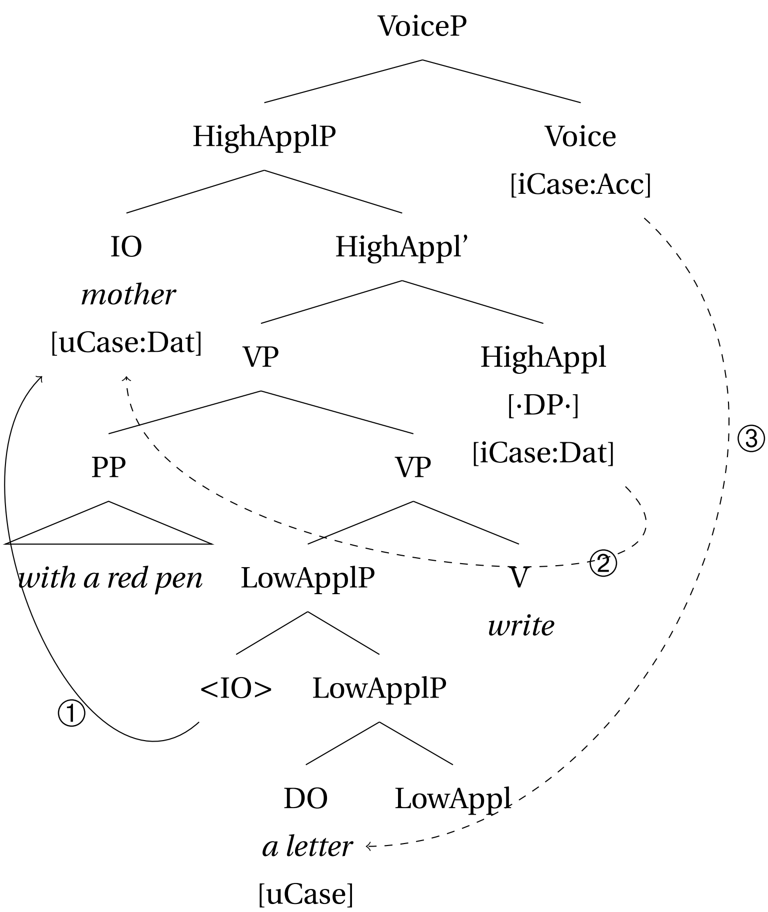
(178) [ok]Aŋk-eʎ ńawrɛm-ʎ-aʎ woš-ən χʉʎ măw-ən
Mother-POSS.3SG.[NOM] child-PL-POSS.3SG.[ACC] town-LOC all candy-LOC
ʎɵt-s-əʎʎe
buy-PST-3SG>NSG
'Mother bought all her children candy in the city.'

The landing position of the DAT-marked IO should be above the VP. At the same time, 179 shows that IO_{DAT} follows agent-oriented adverbials in unmarked word order. Since agent-oriented adverbials are adjoined to VoiceP (Cinque 1999), the landing site for IO_{DAT} is between VP and VoiceP.

(179) Maša-jen amtəŋ suχ-əλ-ən εw-eλ-a
Masha-POSS.2SG.[NOM] happy skin-POSS.3SG-LOC daughter-POSS.3SG-DAT
tăm nuwi wuλ-eλ mă-s-λe.
this white reindeer-POSS.3SG.[ACC] give-PST-3SG>SG
'Masha gladly gave her daughter this white reindeer.'

HighApplP occupies the required position. I therefore suggest that IO raises from its base position to SpecHighApplP, where it receives the dative case. The syntactic structure of 177a is illustrated in 180. Note that the Case probe on HighAppl^{o} is directed upwards, i.e. it agrees with a nominal in its specifier. Müller 2009; Assmann, Georgi, Heck, Müller, & Weisser 2015; Georgi 2014; Himmelreich 2017 propose that direction of Agree is pre-defined for each probe. According to the definition of Agree adopted in this dissertation (see Chapter 5), a probe on X^{o} agrees with the highest possible goal in the maximal projection of XP. Therefore, the direction of Agree is not randomly predefined for each head. the ordering of Marge and Agree features allows for anambiguously defining the direction of Agree in each particular case. If a Merge feature is ordered before Agree, the probing is directed upwards. In Kazym Khanty, features on HighAppl^{o} are ordered Merge>Agree. Hence, HighAppl^{o} agrees with its specifier. If Agree is ordered before Merge, as it is on Voice^{o}, the probe is directed downwards.

(180)



After both internal arguments are introduced in LowApplP, VP and HighApplP are merged. HighAppl° has both [·DP·] and [iCase:Dat] features. The first feature attracts the highest DP in its c-command domain (IO), and the second assigns dative to it. In the next stage of the derivation, [iCase:Acc] on Voice° can successfully skip the IO and assign ACC to the lower DP.

The word order in 177b, where the adjunct precedes the IO, can be derived if the adjunct is subsequently scrambled to a position higher than SpecHighApplP. This is supported by the fact that the adverbial in 177b is narrowly focused, and a narrow focus in Kazym Khanty can trigger scrambling.

III. Double Datives and ban on alignment alternation

There are two main scenarios in Kazym Khanty in which two DAT-marked nominals can co-occur in a clause. However, I want to argue that this is only possible if only one of them is a DP argument in need of a structural case.

The first scenario is illustrated in 181. Here, only *mănema* 'to me' is an argument, which needs to be assigned case. The second DAT-marked nominal, *imi-ja* 'as a wife', is a manner adverbial and not an argument of the verb *mă-ti* 'to give'. As an adverbial, it is inherently case-marked and does not compete with the recipient for the structural dative case.

(181) ɛwi-jen mănema imi-ja mij-e
daughter-POSS.2SG.[ACC] I.DAT woman-DAT give-IMP.SG>SG
'Give me your daughter to marry.'

The second scenario is more complex. Some native speakers of Kazym Khanty allow clauses with two datives if one of them is a recipient and the second one is a direction of movement. This scenario is illustrated in 182. There is a significant variation among native speakers in this regard. Some speakers allow for such double datives. Others clearly disprefer them and have complications processing such stimuli in Khanty.[27] As a repair strategy, the second dative can be separated prosodically from the rest of the clause and reinterpreted as a comment that does not belong to the main clause.

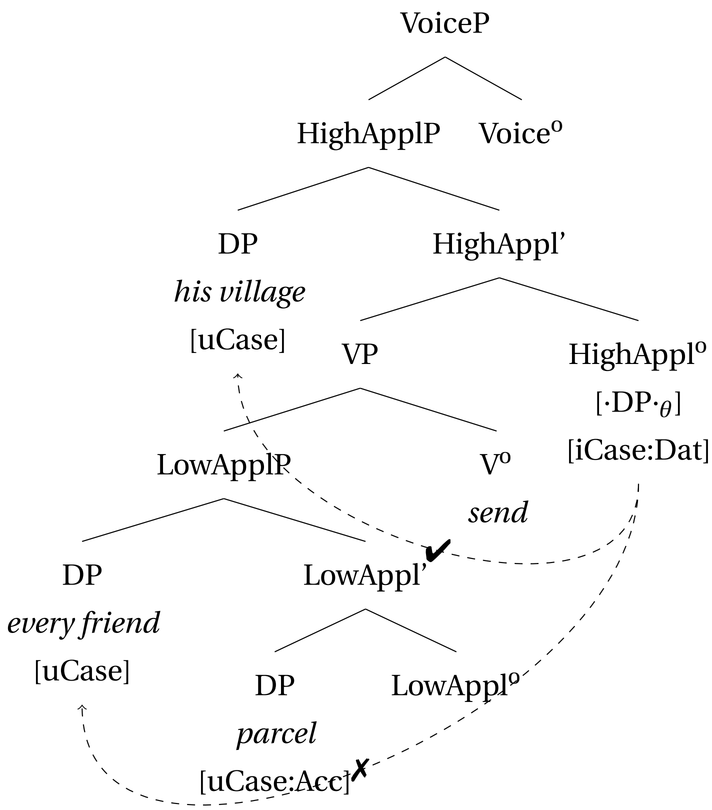
(182) %Maša-jen kašəŋ λɵχəs-a λʉw kɵrt-əλ-a
Masha-POSS.2SG.[NOM] every friend-DAT (s)he village-POSS.3SG-DAT
pasilka kit-s
parcel.[ACC] send-PST.[3SG]
Intend.: 'Masha sent every friend a parcel to his village.'

(182) ?Maša-jen kašəŋ λɵχəs-a, λʉw kɵrt-əλ-a,
Masha-POSS.2SG.[NOM] every friend-DAT (s)he village-POSS.3SG-DAT
pasilka kit-s
parcel.[ACC] send-PST.[3SG]
'Masha sent a parcel to every friend, to his village.' ('λʉw kɵrt-əλ-a' is a comment)

[27] They start to be more allowing in translation tasks from Russian. In Russian, a goal argument/adjunct is realized as a PP with an overt P-head. In Khanty, both the goal and the recipient are realized as DAT-marked DPs. It may be the case that more restrictive native speakers allow recursive datives under the influence of Russian, especially during a word-by-word translation from Russian stimuli.

Those native speakers who do not allow double datives merge the direction argument as a DP in SpecHighApplP, where it is assigned an inherent case by HighAppl^{o}. For such speakers, HighAppl^{o} can have either a thematic [·DP·$_{\theta}$] or an a-thematic [·DP·] feature, but not both simultaneously. Therefore, there can only be one DP in the specifier of HighApplP. The derivation is illustrated in 183. HighAppl^{o} has only one [·DP·$_{\theta}$] feature. Hence, there is only one DP in SpecHighApplP. The Case feature probes up, is satisfied with a DP in SpecHighApplP, and cannot assign case to the second DP in SpecLowApplP.

(183)

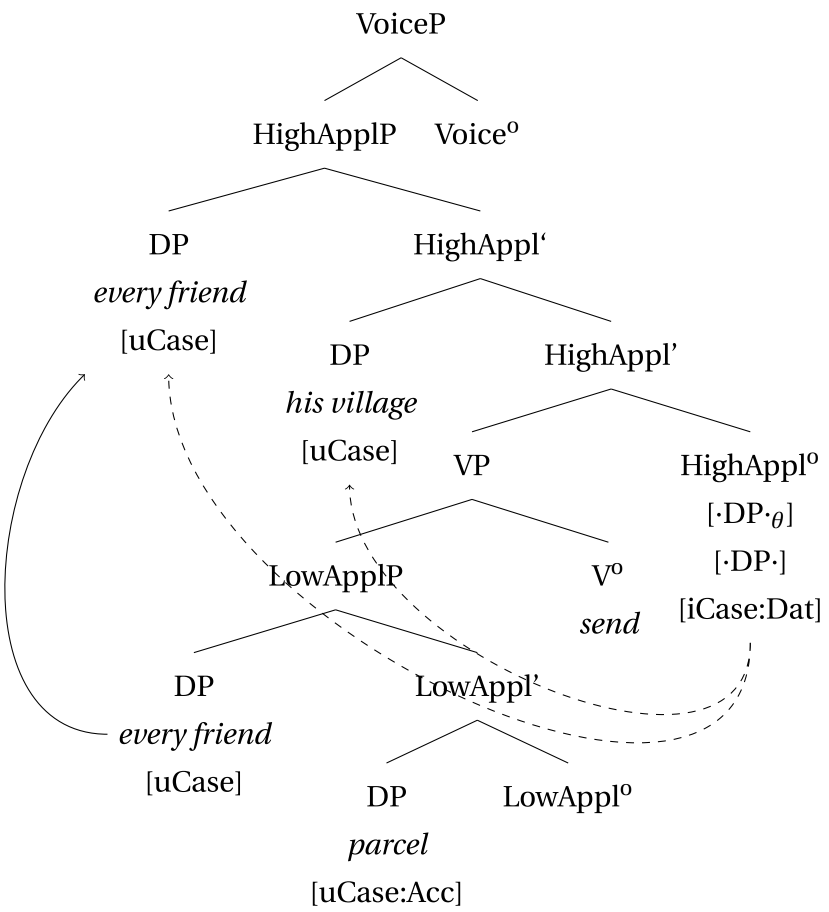


Those native speakers who allow double datives can have HighAppl^{o} with two Merge-features: a thematic [·DP·$_{\theta}$] and an a-thematic [·DP·].[28] Hence, HighAppl^{o} can attract the low applied argument to its specifier after merging a direction DP. The case probe assigns dative to both specifiers under Spec-Head Agree.[29] The derivation is illustrated in 184.

[28]A. Bárány expresses concern that such manipulations with the features of a particular head make the whole system more "brittle". It seems to me that the inter-speaker variation does not differ in this respect from the inter-linguistics variation. To account for the latter type of variation, Richards (2008) suggests that Universal Grammar (UG) is maximally underspecified, and that a child collects external evidence for the inventory of projections and feature specification for each of them. "If the universal inventory F defines the range of features that a child can postulate in constructing its lexicon, then variation in the actual features (and perhaps projections) present in a given language is an automatic consequence of a conservative learning algorithm <...>, whereby only those functional categories are postulated for which there is positive evidence in the input." (Richards 2008: p.136). It can be assumed that inter-speaker variation in a single language arises for the same reason, when children receive insufficient data and some fail to derive a possibility of a particular (rarely attested) feature specification.

[29]Alternatively, one of the dative-marked nominals can be merged as a PP-argument with a silent P-head, as illustrated in (i). For the sake of uniform treatment of the dative case, I assume that all (non-adjunct)

(184)



However, this analysis wrongly predicts that if a direction DP is merged in a monotransitive clause, the theme should be able to receive DAT-marking. As illustrated in 185, the a-thematic [·DP·] should be able to attract the theme to SpecHighApplP if the direction DP is the only indirect object. However, this is not what happens in Khanty, and 185a is ungrammatical.

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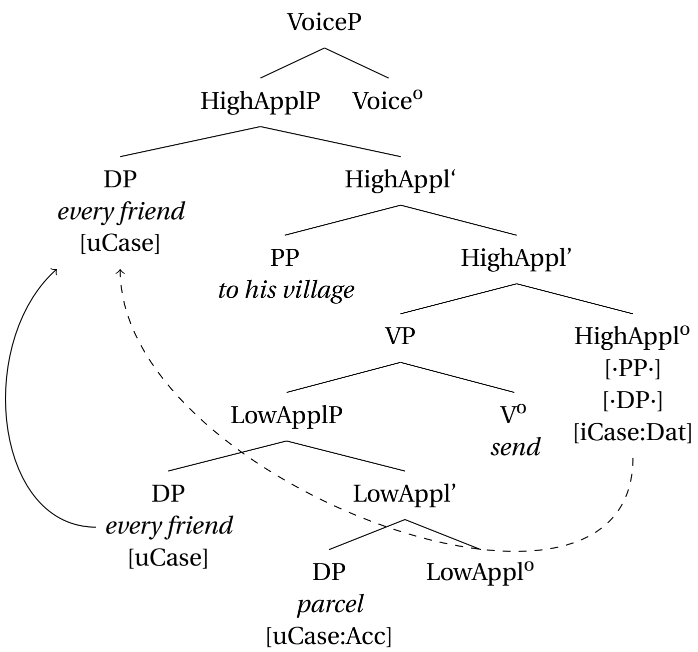
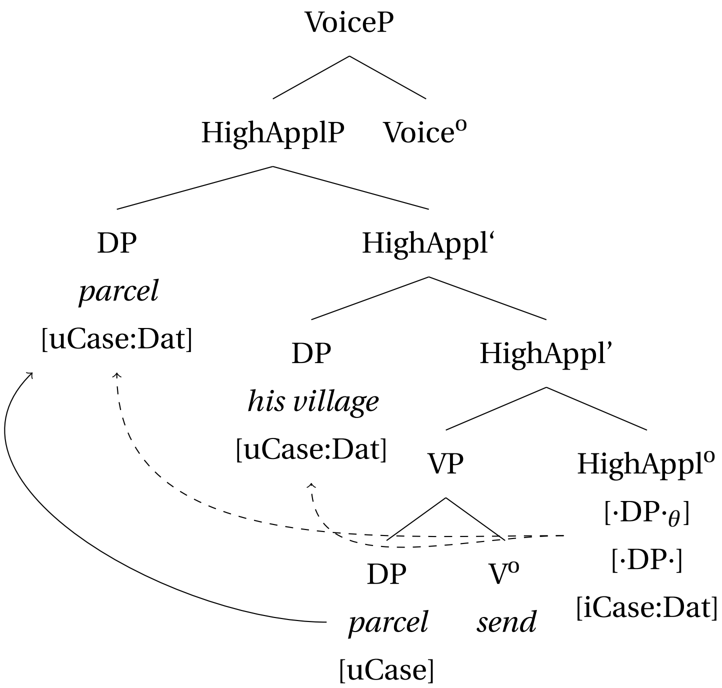
datives are DPs. However, the PP-based analysis does not change the reasoning below.

(i)



(185) a. *Maša-jen pasilka-ja woš-a kit-s
Masha-POSS.2SG.[NOM] parcel-DAT town-DAT send-PST.[3SG]
Intend.: 'Masha sent a parcel to the city.'

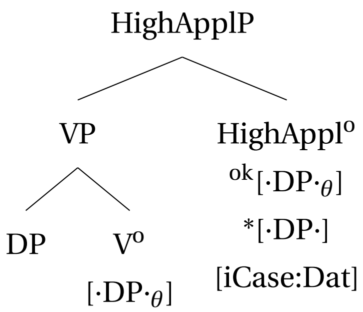
b.



To avoid redundant assignment of dative, as in 185, I suggest that the inventory of features on HighAppl° is sensitive to the structure below. Crucially, the features of the head must be visible at the maximal projection level (López 2001). More precisely, I suggest that HighAppl° can have an a-thematic [·DP·]-feature only if two DPs have already been merged inside its complement. Compare three following scenarios.

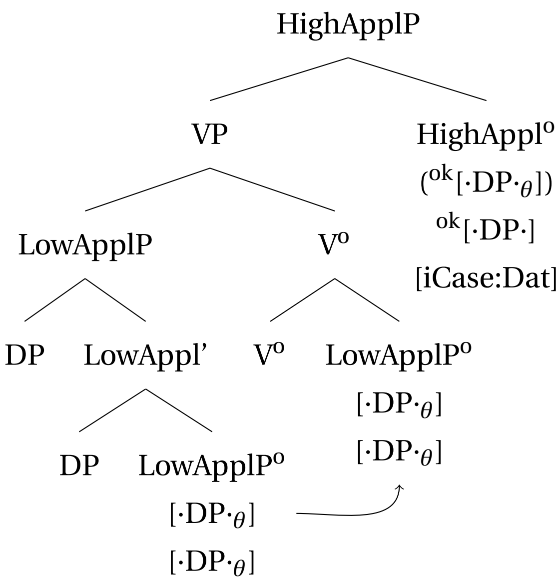
In the first scenario in 186, HighApplP is merged with VP, containing only one argument. Because HighAppl° in 186 is merged to a projection that has just one [·DP·$_\theta$]-feature, the a-thematic [·DP·]-feature cannot be present on HighAppl°.

(186)



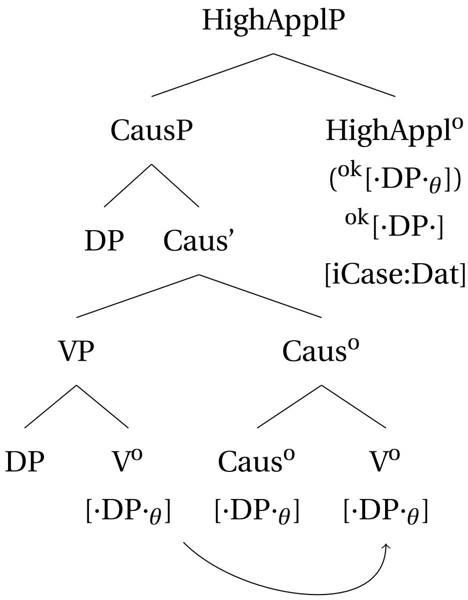
In the low applicative clause, two DPs are merged in LowApplP. Due to head movement, LowAppl° raises to V (see Belkind 2023 for arguments in favor of verb raising in Kazym Khanty). Consequently, two [·DP·$_\theta$]-features are present on V°. A VP with two [·DP·$_\theta$]-features can only merge with HighAppl°, which contains an additional a-thematic [·DP·]-feature.

(187)



Causative clauses are similar to the low applicative clauses. Both Vº and Causº have one [·DP·$_\theta$]-feature. Due to head movement, these two features are simultaneously present at the higher head (Causº). Hence, a HighApplº with an additional a-thematic [·DP·]-feature can be merged. It attracts the highest DP (causee) to its specifier and assigns the dative case to it.

(188)



The Case probe on HighApplº is directed upwards, i.e. it can only see a nominal in SpecHighApplP (cf. e.g. the proposal in Müller 2009; Assmann et al. 2015; Georgi 2014 that different probes can have a pre-defined different direction of Agree). Therefore, in the absence of a-thematic [·DP·$_\theta$]-feature, it cannot probe downwards and assign dative to the theme.

In a nutshell, I propose that only those speakers allow 182, who allows the Case probe on HighApplº to agree with multiple specifiers of HighApplP.

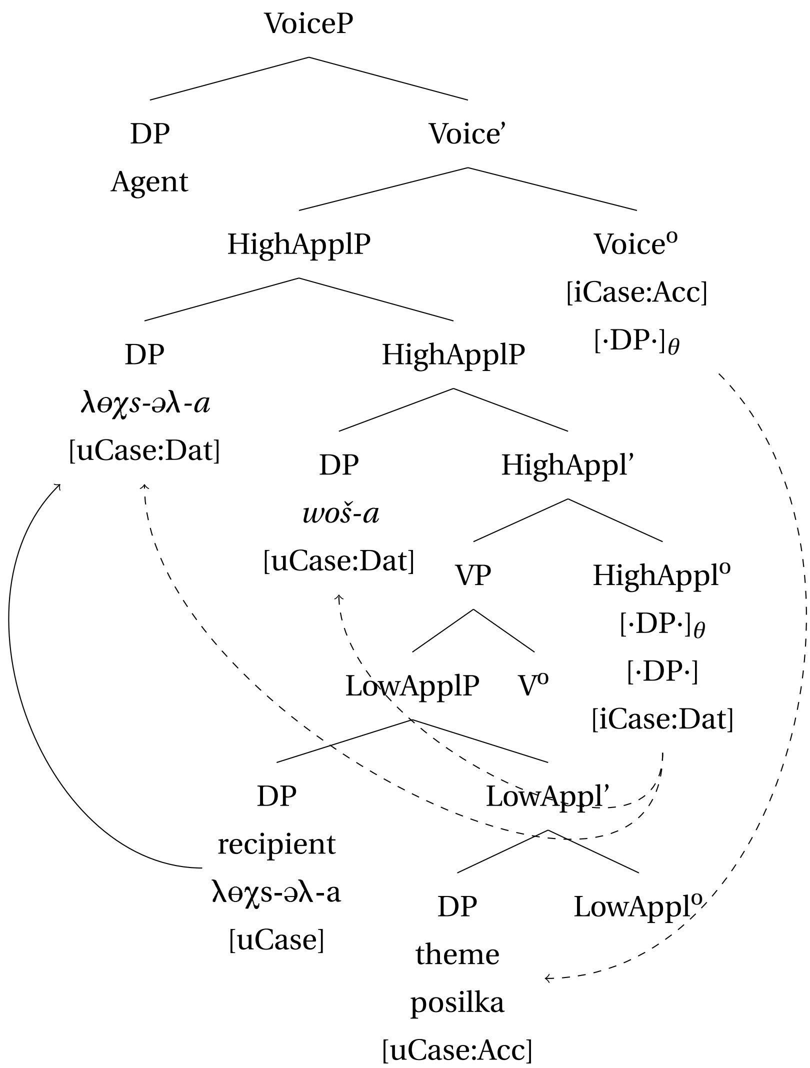
More importantly for the discussion in this section, double-dative clauses provide an additional argument in favor of HighAppl° as a source of the dative case-marking. As 189 demonstrates, double-dative clauses do not allow alignment alternation, i.e. whenever direction DP is present in the clause, the recipient IO is obligatory DAT-marked.

(189) a. %Vasja-jen λɵχs-əλ-a woš-a posilka kit-s
Vasya-POSS.2SG.[NOM] friend-POSS.3SG-DAT town-DAT parcel send-PST.[3SG]
'Vasya sent a parcel to his friend to the town.'

b. *Vasja-jen λɵχs-əλ woš-a posilka-jən kit-s-əλλe
Vasya-POSS.2SG.[NOM] friend-POSS.3SG.[ACC] town-DAT parcel-LOC send-PST-3SG>SG

The derivation in 190 captures this restriction. After HighAppl° is introduced into the structure, it externally merges the DP *woš-a* 'to the town' and attracts the closest DP in its c-command domain (the recipient) to the second specifier of HighApplP. Then, [iCase:Dat] assigns dative to both specifiers. Finally, Voice is merged and assigns accusative to the theme, as to the last caseless DP in its c-command domain.

(190)



Note that HighAppl^{0} introduces the direction DP and is, therefore, obligatory merged in 190. Its presence in the structure makes the DAT marking on the recipient obligatory and triggers indirective alignment in the clause. It can again be concluded that indirective alignment is dependent on the presence of HighAppP in the clausal spine.

3.2.4 Summary

This section presents an analysis of indirective alignment. The core proposal concerns the nature of the dative case. I argue that only HighAppl^{0} can assign dative inside the lower phase (VoiceP). Consequently, HighApplP is always merged in indirective alignment. Similar to what Georgala (2012) suggests for German, I assume that the dative-marked indirect object raises ex situ to SpecHighApplP, where it receives dative case. This analysis is supported by several facts. First, it explains why arguments base-generated in SpecHighApplP are invariably dative-marked. Second, it straightforwardly derives the ban on clauses with two dative arguments, which is attested by most of the native speakers. Third, it can derive double-dative clauses (direction+recipient) that are allowed by some speakers. And most crucially, it can explain why the presence of one dative argument (direction) enforces the obligatory dative marking of the second applied argument (recipient) and blocks indirective-secundative alignment alternation.

3.3 Causatives and HighApplP

The alternation between indirective and secundative alignments is present not only in clauses with applicativized verbs but also in causative clauses. The two exmaples in 191-192 show the alternation for two different causative verbs.

(191) a. ʎawərt χir mănεm aʎm-əʎt-s-əʎe
heavy sack-[ACC] I.DAT lift-CAUS-PST-3SG>SG
'(S)he loaded me with a/the heavy sack.' [Moldanova 2018]

b. Mănti ʎawərt χir-ən aʎm-əʎt-s-əʎe
I.ACC heavy sack-LOC lift-CAUS-PST-3SG>SG
'(S)he loaded me with a heavy sack.' [I.Moldanova p.c.]

(192) a. Vasja-jen iki-ja mašina want-ʎta-s,
Vasya-POSS.2SG.[NOM] man-DAT car.[ACC] see-CAUS-PST.[3SG]
imi-ja pa pøsəntijəʎ-ti mašina want-ʎta-s.
woman-DAT ADD wash-NPST.NFIN machine.[ACC] see-CAUS-PST.[3SG]

b. Vasja-jen ik-en mašina-jən
Vasya-POSS.2SG.[NOM] man-POSS.2SG.[ACC] car-LOC
want-ʎta-s-ʎe, im-en pøsəntijəʎ-ti
see-CAUS-PST-3SG>SG woman-POSS.2SG.[ACC] wash-NPST.NFIN
mašina-jən want-ʎta-s-ʎe.
machine-LOC see-CAUS-PST-3SG>SG
'Vasya showed the man a car and the woman a washing machine.'

Causative derivation in Kazym Khanty is unproductive, i.e. only a closed group of verbs has a causative counterpart.[30] Since the derivation is unproductive, it could be claimed that all causative verbs have been reinterpreted and are now simple lexical ditransitives. If so, they behave exactly the same as the low applicative derivation in the previous section. However, if lexicalized causative verbs still have the (underlying) causative syntax, a question arises as to whether the proposed analysis of indirective alignment can be extended to causative clauses.

The analysis in Section 3.2.3 links the structural dative case to the presence of HighApplP-projection in the clause. Causative clauses might be seen as a potential problem for this analysis. The current section argues that indirective alignment in causative clauses in Kazym Khanty can be captured in the same manner as in applicative clauses.

3.3.1 Typology of causatives

The long research tradition of causative derivations has established significant cross- and intralinguistic differences between them. These differences concern, in particular, the presence of agentivity properties of the causee, the mono- or bi-eventive semantics of the causative derivation, as well as what can be the base of the causative derivation (cf. Harley 2013, 2017; Alexiadou, Anagnostopoulou, & Schäfer 2015; Akkuş 2021, 2022, also Nie 2020 for the overview and discussion).

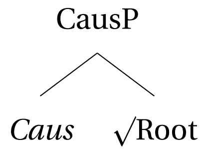
In her seminal book, Pylkkänen (2008) distinguishes between three main types of causative clauses that are present cross-linguistically. Syntactic and semantic differences between these types are motivated by selectional properties of different types of CausP-projection. In root-selection causatives 193, the Caus^{0} in merged directly with the verbal root. Therefore, root-selecting causatives are only possible with intransitive verbs, and the causee lacks agentive semantics.

In verb- or VP-selecting causatives 194, the causative projection merges above the VP but below Voice. This type can also be derived from transitive verbal roots. However, the causee lacks agentive semantics.

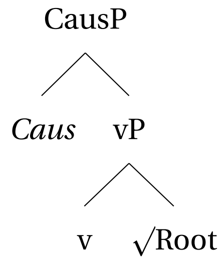
For phase-selecting causatives 195, Pylkkänen (2008) argues that CausP can be merged above VoiceP. In this scenario, the causee is introduced not in SpecCausP but in the specifier of the embedded VoiceP. For this reason, the causee is simultaneously the agent of the (embedded) caused event.

[30] Kazym Khanty has causative derivation from nouns, adjectives, and adverbials, but they are irrelevant for the present purposes. Only causative verbs derived from transitive verbs have both direct and indirect objects, which are necessary for indirective-secundative alternation.

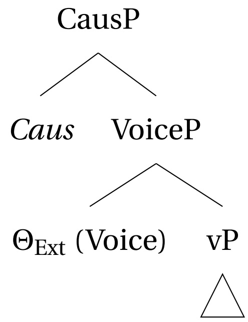
(193) Root-selecting



(194) Verb-selecting

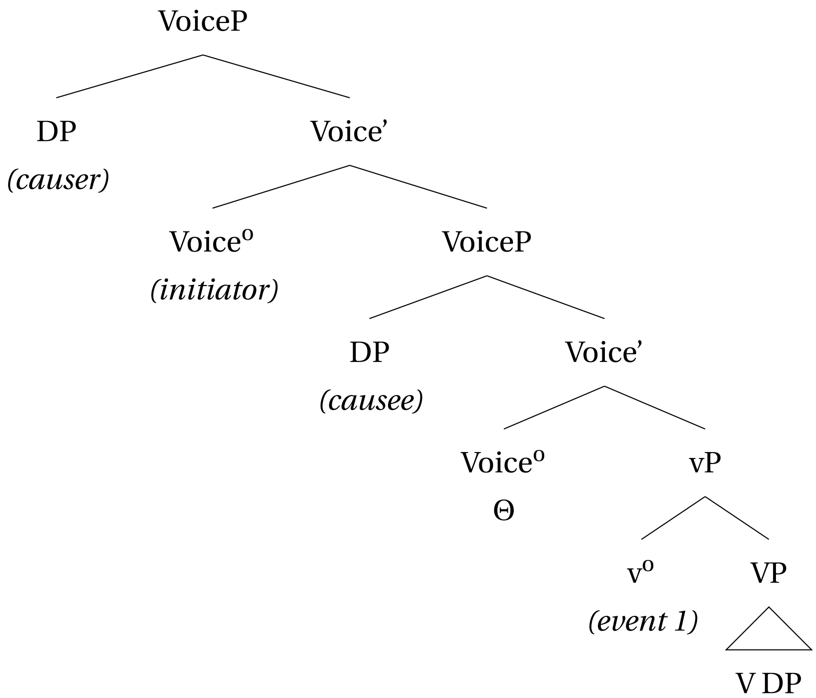


(195) Phase-selecting



An additional distinction can be made between mono- and bi-eventive causative derivations.[31] In mono-eventive causatives, the causation and the caused action form one event, whereas in bi-eventive causatives, they remain two separate events. The most straightforward way to account for the bi-eventivity of a causative derivation is to postulate the presence of two *v*P-projections (cf. Nie 2020; Akkuş 2022). Bi-agentivity can also be captured if the causer and the causee are introduced in two distinct VoiceP-projections. Thus, phase-selecting causatives are divided into two subtypes in 196. Mono-eventive causatives have just two VoicePs, one above the other, and bi-eventive causatives have a second vP.

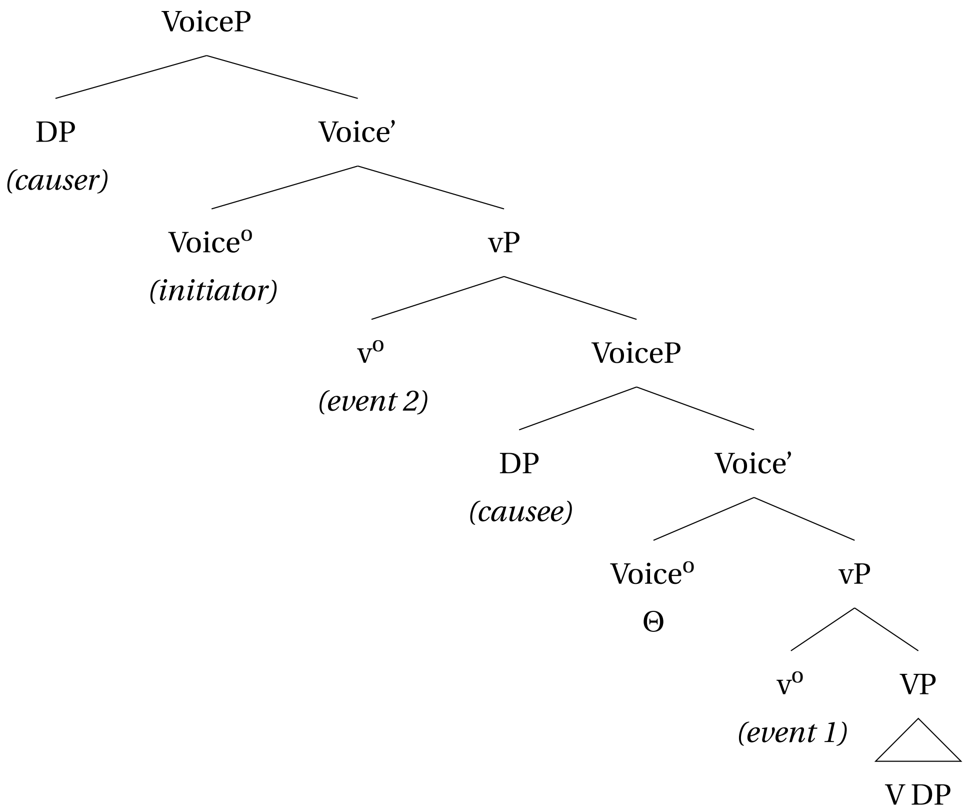
(196) a. mono-eventive bi-agentive causatives (phase-selecting, type 1)



(adopted from Akkuş 2022: p.322, ex.67)

[31] Contra to Pylkkänen's (2008) model, Nie (2020) shows that mono-/bi-agentivity and mono-/bi-eventivity do not necessarily correlate in causative derivations and that all four logically possible combinations of these parameters are attested cross-linguistically. She proposes an analysis alternative to Pylkkänen (2008) and argues that all types of causatives are restructuring predicates with an embedded VoiceP. The embedded VoiceP can have different flavors, which derives the variation between mono-/bi-agentivity. I leave this alternative approach to causatives outside the scope of the current discussion.

b. bi-eventive bi-agentive causatives (phase-selecting, type 2)



In the remainder of this section, I discuss causative derivation in Kazym Khanty and indirective-secundative alternation. The Kazym Khanty causatives are shown to have all the properties of a VP-selecting derivation. Furthermore, it is shown that the analysis of the indirective alignment presented in the previous section, where dative is assigned by a HighAppl°, can be easily extended to causative clauses.

3.3.2 Kazym Khanty causatives are VP-selecting

Causative constructions in Kazym Khanty have the properties of verb-selecting derivation in Pylkkänen's (2008) classification. First, transitive verbs can be causativized, which suggests causative in Kazym Khanty is not a root-selecting derivation.

(197) ʎawərt χir mănɛm aʎm-əʎt-s-əʎe
heavy sack-[ACC] I.DAT lift-CAUS-PST-3SG>SG
'(S)he loaded me with a/the heavy sack.' [Moldanova 2018]

Second, the causee in phase-selecting causatives is introduced in SpecVoiceP and is argued to have the agent θ-role. However, the Kazym Khanty causees do not pass the agentivity test. As 198 shows, an agent-oriented adverbial can only refer to the causer and not to the causee. In phase-selecting causative derivation, such adverbials can be merged to the embedded VoiceP as well as to the matrix VoiceP and refer either to the causer or to the causee.

(198) agent-oriented adverb (test from Pylkkänen 2008)

Putər-əλ **amət-man** mănɛma χθλ-m-əλt-s-əλλe
talk-POSS.3SG be_happy-CVB I.DAT hear-MOM-CAUS-PST-3SG>SG

'She happily made me to hear this talk.'

NOT: 'She did so that I happily heard this talk.'

[I.Moldanova p.c.]

Furthermore, 199 shows that two manner adverbs cannot co-occur in a causative clause in Kazym Khanty. Therefore, it can be concluded that only one vP is present in the structure and that causative derivation is mono-eventive.

(199) two manner adverbs

* **Sora** mănɛma **jăma** in ar-əλ χθλ-m-əλt-s-əλλe
fast I.DAT well this song-POSS.3SG hear-MOM-CAUS-PST-3SG>SG

Intend.: 'She quickly made me [to listen to this song well].'

[I.Moldanova p.c.]

The interpretation of causative clauses also provides evidence for their mono-eventivity. As reported by a native speaker consultant, 200 can only mean 'I was made to hear a song because a person sang it to me', i.e. the causation event and the caused event took place simultaneously. An interpretation that the causer sends a recording (e.g., in the morning) and the causee listens to it later (e.g., in the evening) is not available.

(200) In ar-əλ mănɛma χθλ-m-əλt-s-əλλe.
one song-POSS.3SG.[ACC] I.DAT hear-MOM-CAUS-PST-3SG>SG
'She sang this song to me.'

Not: 'She made me to hear this song (by sending it no me in a messenger).' (I. Moldanova, p.c.)

As the tests above suggest, the causative dervation in Kazym Khanty has a single VoiceP (mono-agentivity) and a single vP (mono-eventivity). Therefore, I assume that Kazym Khanty causatives are of Pylkkänen's verb-selecting type.

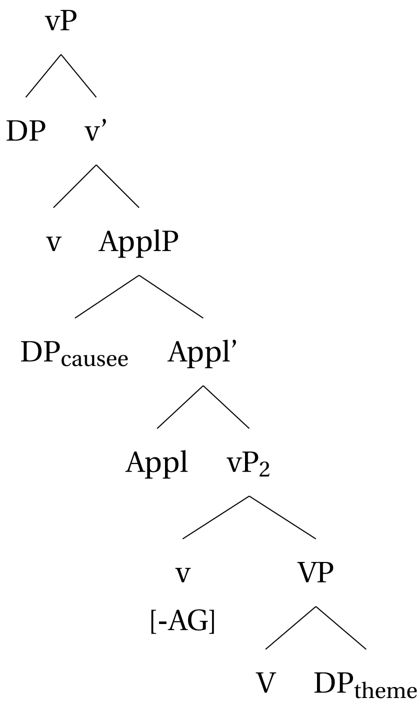
3.3.3 Indirective alignment in Khanty causatives

There are two main options for capturing the DAT-marking on the causee in Kazym Khanty. As shown in the previous subsection, causative derivation in Kazym Khanty is both mono-agentive and mono-eventive, i.e. VP-selecting in Pylkkänen (2008)'s system.

The analysis of indirective alignment, proposed in Section 3.2, links the dative case to HighAppl-projection. I argue that this analysis can be easily extended to causative clauses. Several studies have observed that in many languages, the causee is in complementary distribution with high applied arguments, which suggests a connection between applicative and causative derivations.

One group of researchers has proposed that causee can be introduced in ApplP (e.g. Zubizarreta 1985; Ippolito 2000; Nash 2017, and a somewhat similar proposal by Legate 2014). Georgian is an example of such a language (Nash 2017). Causees in Georgian fail the agentivity test and cannot co-occur with applied arguments. However, benefactives are allowed in clauses, where the causee is existentially closed/not introduced in the structure. Nash (2017) proposes that the causee is merged in SpecApplP, as shown in 201. This applicative projection is located between two vP-projections, where only the higher one introduces an agentive external argument.

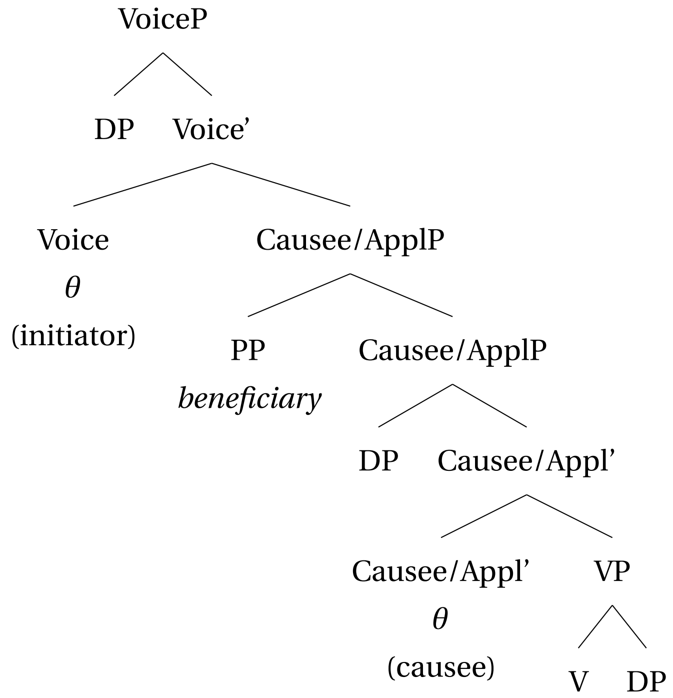
(201) (Nash 2017: p.18, ex. (53))



This analysis cannot be directly extended to Kazym Khanty caustives. Crucially, the structure in 201 has two vPs, which predicts that the clause is mono-agentive and bi-eventive. Even though such causatives are present cross-linguistically (see e.g. the typology in Nie 2020), this is not the case in Kazym Khanty, as argued in Subsection 3.3.2.

A slightly different approach is developed by Akkuş (2021, 2022). He distinguishes among three types of causative constructions in Sason Arabic. In the first one, the causee is introduced in CauseeP, which allows active-passive alternation but fails agentivity tests. Akkuş proposes that the causee-introducing projection CauseeP is bundled with HighApplP, which captures the fact that beneficiaries (i.e., high applied arguments) are only allowed as PP-adjuncts in this type of causative construction. The proposed structure is illustrated in 202.

(202) (Akkuş 2022: p.329, ex.86)

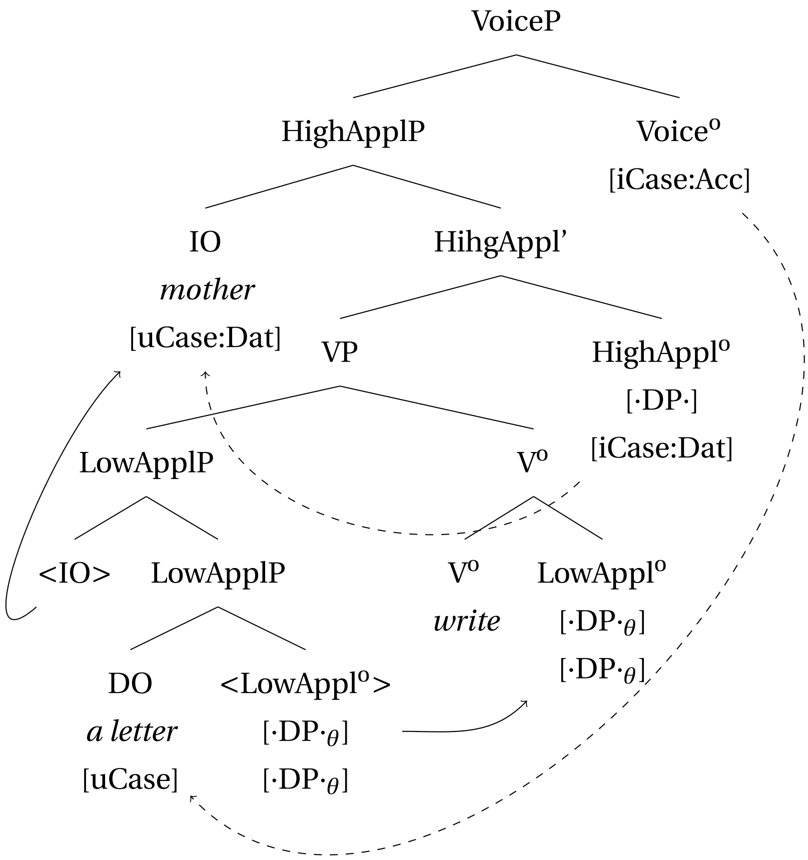


The approaches by Nash (2017) and Akkuş (2021, 2022) share a close connection between applicative and causative derivations. In particular, applicative projection is present in a causative clause without adding a proper applied argument. Moreover, in both proposals, ApplP is responsible for introducing the causee in the structure. Further supporting evidence for this approach can be found in Kinyarwanda (Bantu), where causees and instruments are in complementary distribution. Moreover, the same derivational suffix is used in causative and intrumetal applicative clauses (Jerro 2016, 2017).

I suggest that indirective-secundative alternation in causative clauses in Kazym Khanty can and should be modelled uniformly with this alternation in low applicative clauses.[32] In Section 3.2.3, I suggest that a non-thematic flavor of HighAppl^{0} is responsible for DAT assignment to the indirect object and, thus, for indirective alignment in a low applicative clause. The derivation is repeated in 203. The a-thematic flavor of HighAppl^{0} is merged with the projection that has two [·DP·$_{\theta}$]-features. In a low applicative clause, it will be the VP. A [·DP·]-feature on HighAppl^{0} attracts IO to SpecHighApplP and assigns it the dative case.

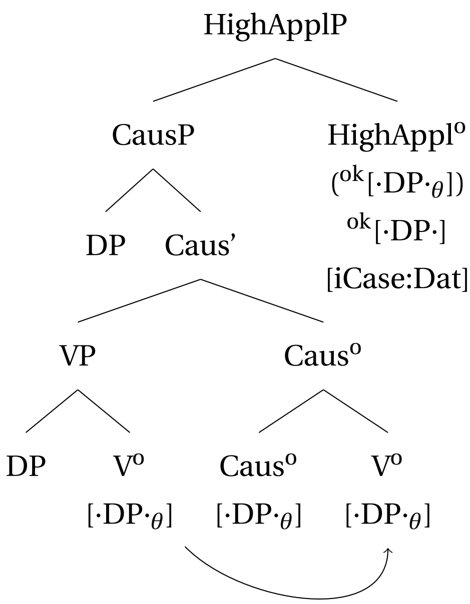
[32]Preliminary, this analysis can be extended to other languages with indirective-secundative alternation in causative clauses without major modifications, in particular to Eastern Khanty, Eastern and Northern Mansi, Tundra Nenets, and Inuit languages. Unfortunately, causative derivations in all these languages are severely under-researched. A detailed typological discussion of indirective-secundative alternation in causative clauses is therefore left for further research.

(203) (simplified from 180)



A similar derivation can be assumed for causative clauses. As discussed in Subsection 3.3.2, causative derivation in Kazym Khanty is verb-selecting (Pylkkänen 2008), i.e. CausP is merged above the VP. I also suggest that the head movement of V at least up to AspP in Kazym Khanty (cf. arguments in Belkind 2023). Hence, V^o moves to Caus^o, and the complex head has two [·DP·]-features. This allows the merger of the a-thematic flavor of HighApplP. The abstract derivation is presented in 204.

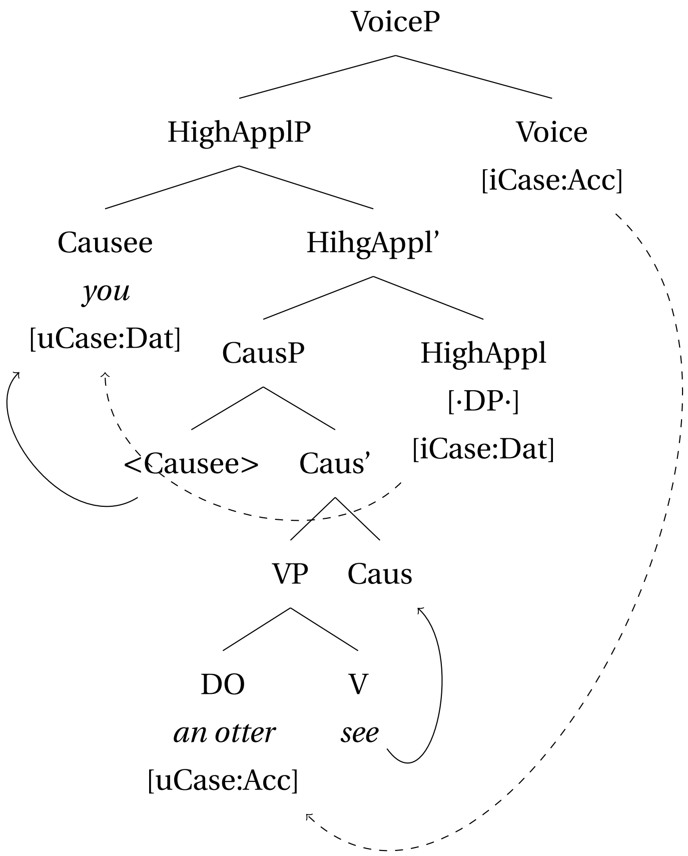
(204) (repeated from 188)



The rest of the derivation is identical to that of low applicative clauses. As shown in 205, HighAppl⁰ attracts the closest DP (the causee) and assigns dative to it. Subsequently, VoiceP is merged and Voice⁰ probes for the highest caseless DP. Since the causee has already received a case, Voice⁰ agrees with the theme and assigns accusative to it.

(205) a. Ma năŋena wɵntər want-λta-λ-əm
I.NOM you.DAT otter.[ACC] see-CAUS-NPST-1SG
'I'll show your an otter.'

b.



The derivation in 205 differs from the analyses proposed by Nash (2017) and Akkuş (2021, 2022) in that HighApplP is distinct from CausP and does not introduce the causee argument. As in low applicative clauses, its only function is to license the highest argument in its c-command domain and allow Voice to reach the lower one. Note that the bundling analysis by Akkuş (2021, 2022) can derive the indirective alignment in Kazym Khanty equally well. Importantly, however, the bundling analysis predicts that applicative projection should invariably be present in causative clauses. If the higher applicative head is a source of the dative, as argued in Section 3.2.3, then the causee should behave similarly to the high applied arguments, i.e. it should be marked with the inherent dative case. Hence, the bundling analysis predicts that only indirective alignment is available in causative clauses. However, causative clauses in Kazym Khanty alternate between indirective and secundative alignments. The novel analysis in 205 separates the case assigning projection from the argument introducing projection and can account for alignment alternation. The derivation of secundative alignment is discussed in detail in Chapter 4.

Chapter 4

Secundative alignment

In secundative alignment, the recipient/causee (IO) is marked with ACC and triggers object agreement on the verb. As 206 illustrates, plural IO, rather than singular DO, controls number agreement on the verb.

(206) Ma ńawrɛm-**ət** i šokolatka-jən mă-s-**λam/*-ɛm** pa
I.NOM child-PL.[ACC] one choclate_bar-LOC give-PST-1SG>NSG/*-1SG>SG ADD
ort-ti lup-s-əm
share-NPST.NFIN say-PST-1SG
'I gave children a chocolate bar and told to share.'

In Khanty, as in some other languages, secundative alignment alternates with the indirective alignment, which was discussed at length in the previous section. There are languages, where secundative alignment is the only available option in ditransitive clauses, e.g. Nez Perce 207 (see Rude 1986; Deal 2006, 2010, 2013 et seq. for a detailed discussion).

(207) Weet *pro*$_{agent}$ 'e-tkuytuu'-∅-ye Angel-ne$_{goal}$ poxpok'ala$_{theme}$?
Y/N pro 3OBJ-throw-P-REM.PST Angel-OBJ ball.NOM
'Did you throw Angel the ball?' (Deal 2013: p.396, ex.9)

There is a crucial difference between the secundative patterns in Nez Perce and Khanty. In Nez Perce, the IO is marked with the object case, and the DO is unmarked. In contrast, the secundative DO in Khanty is marked with an oblique case (LOC/INSTR). However, the difference cannot be reduced solely to the oblique vs. unmarked opposition. As discussed by Deal (e.g. Deal 2010), Nez Perce has differential object marking (DOM) in monotransitive clauses, i.e. the single object in a monotransitive clause can be marked with the object case (OBJ), as in 208a, or zero-marked, as in 208b.[33]

(208) a. pit'íin-im páa-'yax̂-na picpíc-ne.
girl-ERG 3/3-find-PERF cat-OBJ
'The girl found the cat.' (Deal 2010: p.75, ex.2b)

[33] Deal (2006; 2010) analyzes 208b as a case of semantic antipassivization. Namely, zero-marked objects are too small to agree. Therefore, they cannot receive case (cf. approaches to DOM, discussed in Chapter 6, Section 6.2). However, from the translation in 207, it seems that the theme in the applicative clauses can have definite interpretation, i.e. can be full DP. If this is true, Deal (2006)'s analysis must be revised accordingly.

b. pit'íin hi-'yáax̂-na picpíc.
girl.NOM 3SUBJ-find-PERF cat.NOM
'The girl found the cat.' (Deal 2010: p.75, ex.3b)

On the other hand, Khanty does not have DOM. The direct object in monotransitive clauses is universally marked with a ∅-accusative marker on nouns and overtly on pronouns. As shown in 209, a direct object in monotransitive clauses cannot be marked with the locative case if the verb does not assign locative as a lexical case.

(209) a. λʉw pɵšas λeśit-s.
he.NOM fence[.ACC] fix-PST.[3SG]
'He fixed a fence.'

b. *λʉw pɵšas-ən λeśit-s.
he.NOM fence-LOC fix-PST.[3SG]
Intend.: 'He fixed a fence.'

This chapter focuses primarily on the properties of the secundative theme (DO) in Khanty. The oblique marking of the theme is shown to be a default case marking used on the theme when it is not licensed via case assignment by a syntactic mechanism.

As mentioned in Chapter 2, Section 2.4, secundative IOs are usually secondary topics. However, as shown in the same chapter, the topicality of the IO is not sufficient to explain the distribution of indirective and secundative alignments. First, not all topical IOs can be promoted, e.g. secundative alignment is strongly ungrammatical with unaffected goals (directions) and high applied arguments, irrespective of their information-structural status (see the corresponding data and discussion in Chapter 3).

Second, secundative alignment also poses certain restrictions on oblique themes. As discussed in Section 4.2, the secundative themes in Kazym Khanty systematically lack the DP-layer. Therefore, the DP-layer modification on the theme makes secundative alignment unavailable.

This chapter proceeds as follows: the first section presents an analysis of the secundative alignment. The second section discusses structural restrictions on secundative themes and suggests a way to account for them. Finally, in the third section, alternative analyses are discussed and ruled out.

4.1 Secundative alignment derivation

An analysis of secundative alignment in Kazym Khanty should account for the promotion of the recipient to a direct object, i.e. derive ACC-marking and cross-reference by ϕ-agreement on the verb. It should also account for the properties of the secundative themes: oblique case marking and size restriction. As shown in Section 4.2, secundative themes are structurally reduced and lack a DP-projection. In an optimal analysis, these two properties of secundative alignment should be treated together as two sides of one phenomenon. The approach presented in this chapter can account for both the promotion of the IO and the size restriction on the theme. The mechanism deriving the oblique marking of the theme is discussed in detail in the next chapter.

In a framework where each case is assigned by a certain functional head under Agree, it follows trivially that an ACC-marked IO in secundative alignment receives its case from the same functional head as an ACC-marked DO in indirective alignment or in a monotransitive clause. Voice^{o} is an obvious candidate for the assignment of accusative (cf., e.g., Chomsky (2001); Sigurðsson (2017) a.o.). As discussed in Chapter 3, case assignment under Agree obeys Relativized Minimality (Rizzi 1990), i.e. no other potential case assigner should intervene between a case assigner and a noun receiving the respective case. Recall that a HighAppl^{o} is present in the indirective alignment. It agrees with IO and removes it from the list of potential goals for Voice^{o}. Hence, Voice^{o} can skip the IO and reach the DO.

I suggest that HighApplP is missing in secundative alignment. Furthermore, this is the only syntactic difference between the two alignments. The derivation of a secundative clause is illustrated in 210. The LowAppl-projection introduces the two objects but is unable to check their Case features. Similarly, V^{o} only introduces the verb in the derivation but does not have a Case-probe. Therefore, when a Case probe on Voice^{o} is merged, neither the IO nor the DO has a checked Case feature. Since IO c-commands DO in the base-position, it is closer to the probe. Hence, Voice^{o} assigns ACC to IO and can further agree with it in ϕ-features (number).[34] In the subsequent derivation, the external argument is merged, and T^{o} assigns NOM to it.

In the remainder of this chapter, I turn to the issue of the oblique theme. The analysis of the oblique theme I am going to argue for is already previewed in the derivation in 210. As can be seen, there are three arguments and only two heads that can assign case. This creates a configuration where there is no available head to assign a case to the theme. The absence of case assignment is crucial part of the current analysis. As discussed below, this allows a derivation of the DP-restriction and the simultaneous capture of oblique case marking. Crucially, it is argued that oblique case marking is the default case that can be spelled out on an argument if no external head has assigned the case to it.

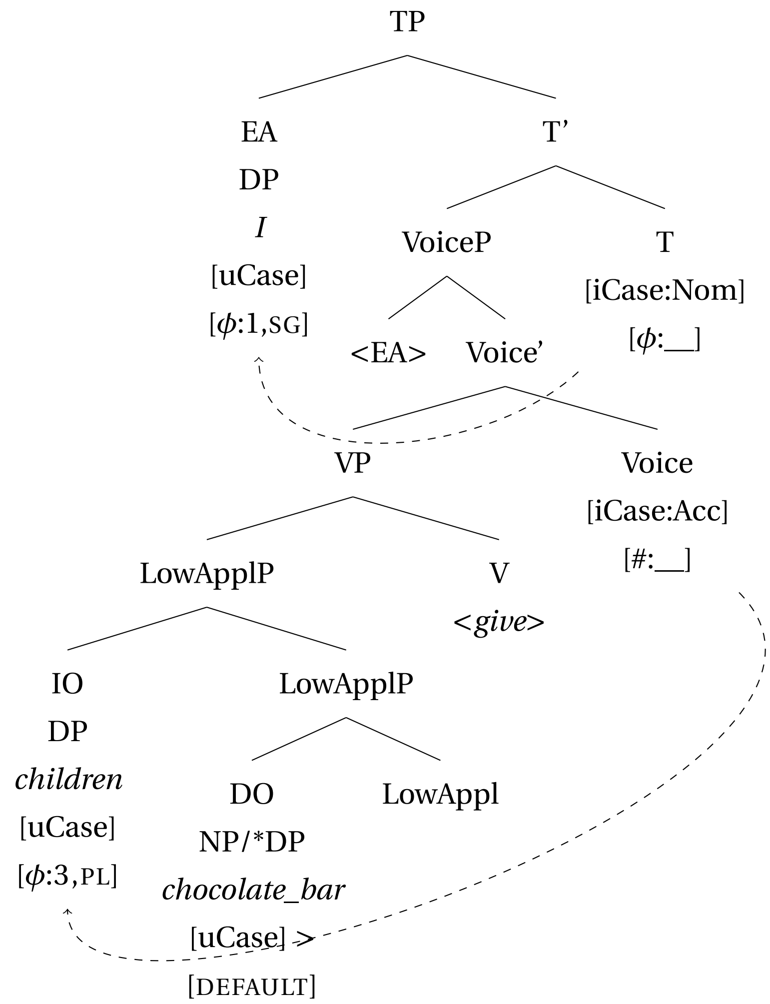
The rest of this chapter deals solely with secundative themes. In the next section, I discuss the structure of Khanty nominals and provide evidence that secundative themes are indeed structurally reduced compared to other arguments. Furthermore, I argue that this size restriction can and should be analyzed as a consequence of the absence of case-licensing. This leads to a non-obvious assumption that the oblique morphological case

[34] For now, I abstract away from the interaction of object agreement with topicality. Object agreement in Khanty is traditionally assumed to be sensitive to topicality (e.g. Nikolaeva 1999, 2001; Dalrymple & Nikolaeva 2011; É. Kiss 2021 etc.). Moreover, agreement with the IO in secundative alignment is obligatory, which leads researchers to argue that the topicality of the IO triggers the secundative alignment (most notably Dalrymple & Nikolaeva 2011). However, secundative alignment in Kazym Khanty is exceptionally possible even when the IO is a contrastive focus (see discussion in Chapter 2, Section 2.4). Object agreement in these cases remains obligatory. Note also that high aspect and actionality affect the presence of object agreement in a non-obvious way (e.g. Kozlov 2022, additionally supported by my own fieldwork data). I leave a detailed discussion of the link between (high and low) aspect, topicality, and object agreement for further research.

on the theme can surface in the absence of syntactic case assignment. A way to account for this, as well as the theoretical consequences of this assumption, are discussed in detail in the next chapter. In the last section of the current chapter, alternative analyses of the secundative themes are discussed and ruled out.

(210) a. Ma ńawrɛm-ət i šokolatka-jən mă-s-ʎam
I.NON child-PL.[ACC] one chocolate_bar-LOC give-PST-1SG>NSG
'I gave children a chocolate bar.'

b.



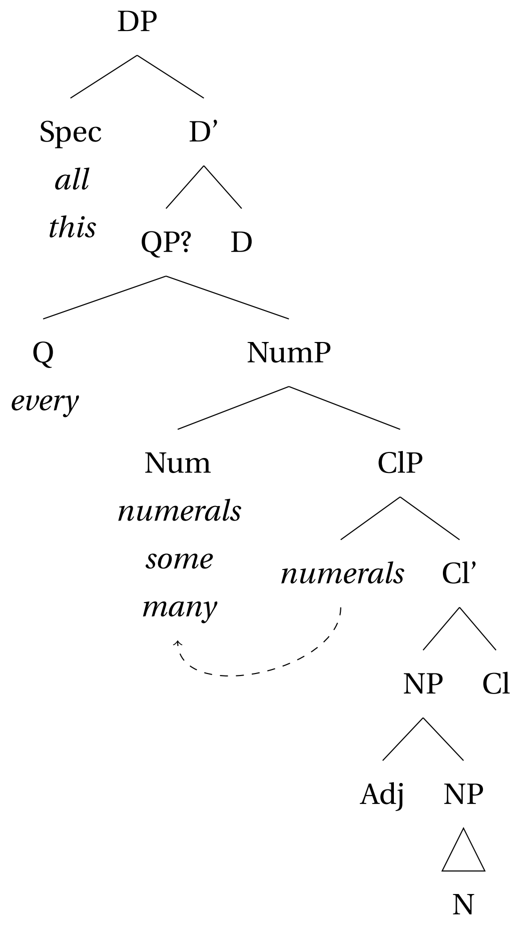
4.2 Size of secundative theme

Since the introduction of the DP-hypothesis by Abney (1987), the question of whether the DP-projection is universal across languages remains open. The structure of nominals in articleless languages is a matter of ongoing debate. Bošković (2005; 2008 et seq.), Bošković & Gajewski (2011) and Bošković & Şener (2014) argue that nouns in articleless languages systematically lack the DP-layer. On the other hand, many other studies have provided arguments in favor of a DP projection in various articleless languages (e.g. Pereltsvaig 2006, 2021; Kagan & Pereltsvaig 2011; Rudnev 2024 for Russian, Yuan 2013; Langr 2014b,a; Manlove 2015 for Inuit, Lyutikova & Pereltsvaig 2015; Lyutikova 2017 for Tatar, Erschler 2019 for Ossetic, Salzmann 2018 for general discussion). In the next two subsections, I argue that Kazym Khanty also has a DP-projection and that themes in secundative clauses are restricted being smaller than DP.

4.2.1 Structure of the Kazym Khanty nominals

The structure of DP in (Kazym) Khanty has not yet been systematically studied. However, the available data suggest that Khanty has a structure similar to that proposed by Dékány (2011, 2021) for the Hungarian DP (cf. Burov et al. 2024). I assume that Khanty nominals can project the structure in 211.

(211) Structure of DP in Kazym Khanty[35]
(adapted from Dékány 2011, 2021 for Hungarian and Burov et al. 2024)



The linear order of modifiers in Kazym Khanty is presented in 212 (after Pleshak 2020). Modifiers that are separated by & can appear in the reversed order

(212) DEM/UQ & Possessor > NUM & attributivizer & ADJ > bare nominal modifier > N

Pleshak (2018, 2020) shows that the order of an adjective and a numeral is reversible in some, but not all cases (213 vs. 214). I assume that the always available order NUM > ADJ corresponds to the order of the respective projections. The reverse order ADJ > NUM is derived via movement of the adjective (see Belkind 2023 for some observations on

[35] PossP is placed above NumP based on data from attributivizers, which can be modified with numerals but not with possessors (Čeremisinova 2019).

movement inside the DP). It is beyond the scope of the current discussion to determine what factors regulate this movement and why the order of the modifiers is frozen in 214. However, the upper part of the structure in 211 plays a crucial role for an analysis of the theme in secundative alignment.

(213) [OK]wet jaŋ χuraməŋ / [OK]χuraməŋ wet jaŋ wʉλ-et ma muχt-ɛm-ə
five ten beautiful beautiful five ten reindeer-PL.[NOM] I by-POSS.1SG-?
măn-s-ət
go-PST-3PL
'Fifty beautiful reindeers ran by me.' (Pleshak 2018: p.2, ex.7)

(214) [OK]χɵλəm wɵn / *wɵn χɵλəm nʉw-ən ma išn-ɛm lăp
three big big three branch-LOC I window-POSS.1SG.[NOM] PREV
wʉ-s-i
take-PST-PASS.[3SG]
'Three big branches cover my window.' (Pleshak 2018: p.3, ex.8)

The quantifier *kašəŋ* 'every' differs from low quantifiers, such as *(i)tɵχ* 'some' and *ar* 'many', in that it can co-occur with numerals. Hence, it should be base-generated in a projection separate from NumP. The order of the elements in 215 suggests that *kašəŋ* 'every' merges above NumP. Analogous to Dékány (2021)'s analysis of Hungarian, I assume that *kašəŋ* is merged in QP, above NumP and below DP.

(215) kašəŋ kăt ewi pɛlt-s-əλλən akan-λ-ən
every two girl[.NOM] change-PST-3DU>NSG doll-PL-POSS.2NSG.[ACC]
'Each two girls gave each other their dolls.' (E.Kozlova p.c.)

Importantly, the universal quantifier *χuλ* 'all' cannot co-occur with demonstratives, as 216 illustrates. This is straightforwardly accounted for if both elements compete for the same position. It has been suggested for other languages that demonstratives merge in DemP, and universal quantifiers merge in QP. However, neither DemP nor QP is able to introduce both demonstratives and quantifiers. On the other hand, they can both be generated in the DP-layer (cf. Dékány (2021) for Hungarian[36]). Therefore, I assume that a DP-projection exists in Kazym Khanty, and that both demonstratives and universal quantifiers occupy SpecDP.

(216) a. [#]Tăm χuλ χot-ət
this all house-PL
Intend.: 'All these houses'
Actual meaning: 'These are all houses.'

b. *χuλ tăm χot-ət
all this house-PL

In contrast to the universal quantifier, other quantifiers can freely co-occur with demonstratives 217, as is expected if they are generated below the DP-layer.

[36]However, note that she places another type of demonstrative in a separate DemP projection, similar to Manlove (2015)'s analysis of Inuit

(217) tăm măttə χop-əλ wʉs-əŋ
this some boat-POSS.3SG.[NOM] hole-ATTR
'One of these boats is leaky' (Pleshak 2018: p.2, ex.4)

Another aspect of the DP architecture that is important for the discussion of secundative alignment concerns the position of possessors and possessive markers. Bárány & Nikolaeva (2021) argue that possessors in Tundra Nenets (Samoyedic, Uralic) can occupy several positions inside the DP. Similar to Khanty, possessors in Tundra Nenets can optionally control agreement on the possessee. Furthermore, possessors can occupy two different positions inside the matrix DP. Bárány & Nikolaeva (2021) propose the derivation in 218. First, a possessor DP is merged in the specifier of PossP, where it is case-marked. This is the final position of low possessors. Crucially, PossP is located just below DP since even non-agreeing possessors precede quantifiers, adjectives, and other nominal modifiers (Nikolaeva 2014: p.171-173). High possessors (PIPs, prominent internal possessors) are moved higher: first, to ArgP, where they trigger possessive agreement on Arg^{o}, and then to a position adjoined to DP. This is supported by the fact that agreeing possessors are able to c-command out of the matrix DP.

(218) Position of possessors in the Tundra Nenets DP
[$_{DP}$ PIP-GEN [$_{DP}$ D [$_{AgrP}$ ~~PIP~~ [$_{Agr'}$ Agr [$_{PossP}$ LOW POSS-GEN [$_{nP}$ n NP]]]]]]

Analogous to Tundra Nenets, Kazym Khanty also seem to have two possible positions for possessor DPs. As shown by Masliukov (2024), a possessor DP can both precede and follow a demonstrative after linearization. Importantly, only a pre-demonstrative position obligatorily correlates with the presence of the possessive suffix on the possessee, as 219 illustrates. Masliukov (2024) argues that topical possessors are attracted to outer SpecDP and are obligatory coreferenced by a possessive suffix on the head noun, while non-topical ones can stay in situ and the possessive suffix is optional.[37]

(219) a. ma [tăm Maša-jen jermas-(əλ)] kăt pʉš
I.NOM this Masha-POSS.2SG dress-POSS.3SG.[ACC] two time
pɵ-s-ɛm.
wash-PST-1SG>SG
'I washed this dress of Masha's two times.'

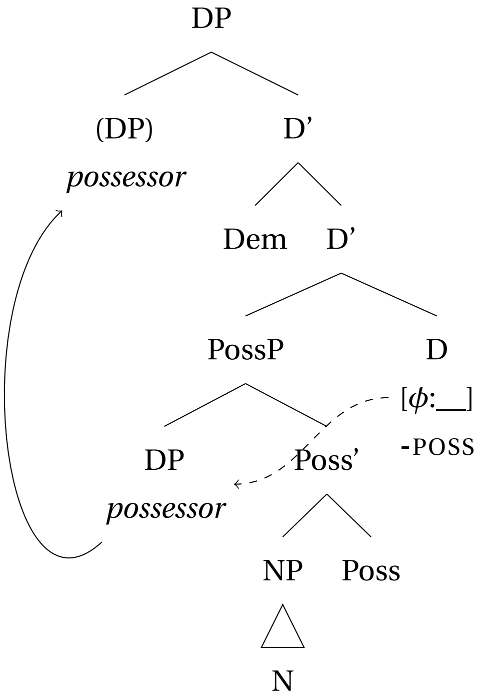
b. ma [Maša-jen tăm jermas-*(əλ)] kăt pʉš
I.NOM Masha-POSS.2SG this dress-POSS.3SG.[ACC] two time
pɵ-s-ɛm.
wash-PST-1SG>SG
'I washed this dress of Masha's two times.' (Masliukov 2024: p.1, ex.1)

In Hungarian, a possessive marker on the possessee noun is obligatory. Therefore, Dékány (2021) suggests that it spells out Poss^{o}, which attracts the possessor DP from NP. In Khanty,

[37] I abstract away from differences between alianable and inalianable possessors and remain agnostic, if possessors are base-generated in SpecPossP or moved there from inside the NP (e.g. Alexiadou 2003; Georgi 2014 etc.).

however, the possessive suffix is optional, i.e. the PossP projection can be present even in the absence of a possessive suffix.[38] Hence, the possessive suffix is spelled out in another projection. In contrast to Bárány & Nikolaeva (2021) and Dékány (2021), I assume that possessive morphology in Kazym Khanty is spelled out on D^o that agrees with the possessor in ϕ-features and can optionally attract the possessor to SpecDP, as illustrated in 220. Since the DP in Khanty is head-final, the possessive marker is spelled out as a suffix on the head noun.

(220)



The analysis of the Khanty possessors in 220 is minimally different from the analysis for Tundra Nenets in 218. The main difference is the presence of an additional AgrP projection in Bárány & Nikolaeva (2021)'s analysis. Crucially, I argue that the possessive agreement probe is located on D, at least in Khanty. There are two main arguments in favor of this approach. First, from a syntactic perspective, ArgP is redundant. In a minimalist approach that does not have AgrS- and ArgO-projections in the clausal spine, there is no reason to postulate another ArgP-projection inside the DP. Second, possessive markers in Kazym Khanty often have definiteness-related and article-like functions (Mikhailov 2023). These types of possessive morphology should be spelled out on D^o for semantic reasons (e.g. Abney 1987). Additionally, locating possessive agreement on D^o allows for a simpler generalization that accounts for the modification restriction on secundative themes.

4.2.2 Secundative theme is smaller than DP

Multiple studies argue that structurally reduced nominals co-exist with full DPs in various languages, including articleless languages (e.g. Pereltsvaig 2006 or Heck & Richards 2010). Richards (2014) advocates the idea that the presence of a DP-layer in a nominal structure

[38] Possessive suffix is obligatory with prononimal possessors and optional with nominal possessors (Mikhailov 2023).

is tied to the [+Person]-feature. [+Person] is closely connected to the animacy and definiteness prominence scales in 221. The lower a nominal is on the scales, the more likely it is to lack the [+Person]-feature and, therefore, a DP-layer.

(221) [Richards 2014: p.141, ex. (4-5)]

a. Person/animacy scale
[+Person] (=DP) | [-Person](=NP)
1/2-person pron. > animate (3-person, pron./noun) > inanimate (3-person pron./noun)
← *(likelihood/obligatoriness of) animacy*

b. Person/definiteness scale
[+Person] (=DP) | [-Person](=NP)
1/2-person (pron.) > 3-person (pron.) > definite > specific > nonspecific
← *(likelihood/obligatoriness of) definiteness*

This idea has been adopted in certain approaches to DOM in Romance (e.g. Irimia 2022) and beyond (Lyutikova & Pereltsvaig 2015).

In addition, Lyskawa & Ranero (2022) argue that only nominals in a complement position can be merged as NPs, while all nominals merged in a specifier position are obligatorily full DPs. They discuss an agreement pattern in Santiago Tz'utujil where ϕ-agreement is optional with inanimate goals only if they are base-generated in a complement position, and obligatory with goals in specifier positions. This pattern is captured if (i) only goals in complement positions can be merged as NPs, and (ii) full DPs must be licensed via ϕ-agreement, while NPs does not need licensing. Therefore, restriction on structurally-reduced nominals is a selectional difference between syntactic positions and a licensing-requirement on DPs.

Secundative themes are not separately discussed in the literature on Kazym Khanty and other Ob-Ugric languages. For other languages with the indirective-secundative alternation, it is argued that secundative themes tend to have generic or indefinite interpretation (e.g. Fortescue 1984 for Kalaallisut, Nikolaeva 2014 for Tundra Nenets). This subsection presents novel fieldwork data and discusses restriction on the modification of a theme argument in secundative alignment in Kazym Khanty. It is argued that the restrictions on the secundative theme in Khanty are structural rather than semantic.

As briefly mentioned in Chapter 2, Kazym Khanty shows certain restrictions on theme modification in secundative alignment. In particular, a LOC-marked theme in secundative alignment cannot be modified with possessive markers, demonstratives, or universal quantifiers. This is illustrated in 222-224 for each of these modifiers.

1. No possessive markers on DO

(222) *Vasja-jen λɵχs-əλ χot-**ɛm**-ən wanλta-s-λe.
Vasya-POSS.2SG friend-POSS.3SG house-POSS.1SG-LOC show-PST-3SG>SG
Intend.: 'Vasya showed my house to his friend.'

2. No demonstratives on DO

(223) a. *Nurum **tăm** kińška-(jət)-n tɛλ pun-s-ɛm
shelf this book-(PL)-LOC entierly put-PST-1SG>SG
Intend.: 'I've filled the shelf with this book(s)' [REI]

b. *Ma Vasja-jen **śi** kińśka-ən mă-s-ɛm.
1SG Vasya-POSS.2SG-[ACC] DEM book-LOC give-PST-1SG>SG
Intend.: 'I gave that book to Vasya.'

3. No universal quantifiers on DO (*χuλ/χuλijewa* 'all')

(224) *Toxtər-en mөšəŋ ut-λ **χuλ** purtөŋ-ən mă-s-λe.
doctor ill something-POSS.3SG all medicine-LOC give-PST-3SG>SG
intend.: 'Doctor gave the patient all medicine.'

Note that these are the only three types of modification that are strictly ungrammatical on secundative themes. For instance, all quantifiers except the universal *χuλ* are grammatical: e.g., *kašəŋ 'every', itөχ 'some' (225), taλaŋ* 'whole' (226), *ar* 'many' (227).

(225) okToχtər-en mөšəŋ ut-λ **kašəŋ / itөχ** purtөŋ-ən
doctor ill something-POSS.3SG every some medicine-LOC
mă-s-λe.
give-PST-3SG>SG
'Doctor gave the patient every/some medicine.'

(226) Maša-jen ńawrɛm **tăλaŋ** šokoladka-jən
Masha-POSS.2SG.[NOM] child[.ACC] whole chocolate_bar-LOC
mă-s-λe
give-PST-3SG>SG
'Masha gave the child a whole bar of chocolate.'

(227) Vasja-jen kampanija **ar** rөpatnək-ən tө-s-λe
Vasya-POSS.2SG.[NOM] company[.ACC] many worker-LOC bring-PST-3SG>SG
'Vasya brought many workers to the company.'

Lower modifiers, i.e. numerals in 228, adjectives in 229, and overt number marking in 230, are also grammatical on secundative themes.

(228) Maša-jen ik-eλ χөλəm puχ-ən
Masha-POSS.3SG.[NOM] man-POSS.3SG.[ACC] three son-LOC
tө-s-λe
bring-PST-3SG>SG
'Masha gave her husband three sons.'

(229) λiw aškola-jew nuwi ow-ən λөt-s-əλ
they.PL.NOM school-POSS.1PL.[ACC] white door-LOC buy-PST-3PL>SG
'They bought white door(s) for your school.'

(230) okToχtər-en mөšəŋ ut-λ purtөŋ-ət-ən mă-s-λe.
doctor ill something-POSS.3SG medicine-PL-LOC give-PST-3SG>SG
'Doctor gave the patient (multiple) medicines.'

This restriction on theme modification is highly rigid, and native speakers immediately rule out the stimuli in 222-224 as completely ungrammatical. Note that all three modifiers – the universal quantifier, demonstratives, and possessive markers – are merged at the DP-level, as shown in 211 and 220. There are two possible ways to analyze this restriction. First, it can be explained semantically, i.e. as a requirement for secundative themes to be indefinite. Second, there can be a structural restriction on DPs in this position. As DP is a functional projection associated with definiteness (Abney 1987), it is not easy to separate these two options. Indeed, definite themes are ungrammatical in secundative alignment. This holds for both strong and weak definiteness (Schwarz 2013), even though only strong definiteness is marked overtly by a possessive suffix. The ungrammaticality of overtly marked (strong definite) themes in secundative alignment is illustrated in 231.

(231) In kińška-jen χuti? – *Ma Vasja-jen
this book-POSS.2SG.[NOM] how I.NOM Vasya-POSS.2SG.[ACC]
kińška-jen-ən mă-s-ɛm.
book-POSS.2SG.LOC give-PST-1SG>SG
Intend.: 'What about the book? – I gave it to Vasya.'

Weak definite themes, i.e. situationally or generally unique items, are not marked overtly by a possessive marker or any other kind of article. Nevertheless, semantic definiteness is sufficient to trigger the ungrammaticality of secundative alignment. The museum in 232 is generally unique because there is only one museum in the village of Kazym. Equally, the glasses in 233 are situationally unique, since the addressee most probably has only one pair of glasses. In both scenarios, secundative alignment is ungrammatical.

(232) *Olja-jen kašəŋ χujat Kasəm musej-ən
Olya-POSS.2SG.[NOM] every someone.[ACC] Kazym museum-LOC
wanλtə-s-λe.
show-PST-3SG>SG
'Olya showed everyone the museum in Kazym.'

(233) *Iśńi χop sɛm karti-jən ăλ pun-a.
window boat.[ACC] eye iron-LOC PROH put-IMP
'Don't put (your) glasses on a window.'

These data might be seen as evidence that the restriction on the theme in secundative alignment is semantic in nature, i.e. that the secundative theme must be indefinite. However, there are two pieces of evidence that the restriction on secundative themes is structural and not semantic. First, as discussed by Mikhailov (2023) and illustrated in 234, a proper possessive marker (i.e., reflecting ϕ-features of a possessor) does not entail definiteness.

(234) muλsər an-en mij-a
some cup-POSS.2SG.[ACC] give-IMP.[SG]
'Give me one of your cups.' (Mikhailov 2023: p.20, ex.7)

However, themes with a proper possessive marker are equally ungrammatical in secundative alignment. As shown in 235, even an indefinite theme bearing a proper possessive marker causes ungrammaticality.

(235) *Vasja-jen aŋk-eʎ muʎsər an-əʎ-ən
Vasya-POSS.2SG.[NOM] mother-POSS.3SG.[ACC] some cup-POSS.3SG-LOC
mă-s-ʎe
give-PST-3SG>SG
'Vasya gave his mother one of his cups.' (repeated from 92)

Second, semantic definiteness inside the theme does not block secundative alignment as long as the head noun lacks the DP-layer. In 236, the theme is modified by a superlative adjective that presupposes definiteness. Note that the superlative adjective is definite, but not necessarily the noun (as argued, e.g., by Coppock & Beaver 2015 for Swedish). As an adjective, it is merged at the NP-level. Hence, modification with a superlative adjective does not require the presence of a DP-layer in the structure of the nominal argument, and secundative alignment is grammatical.

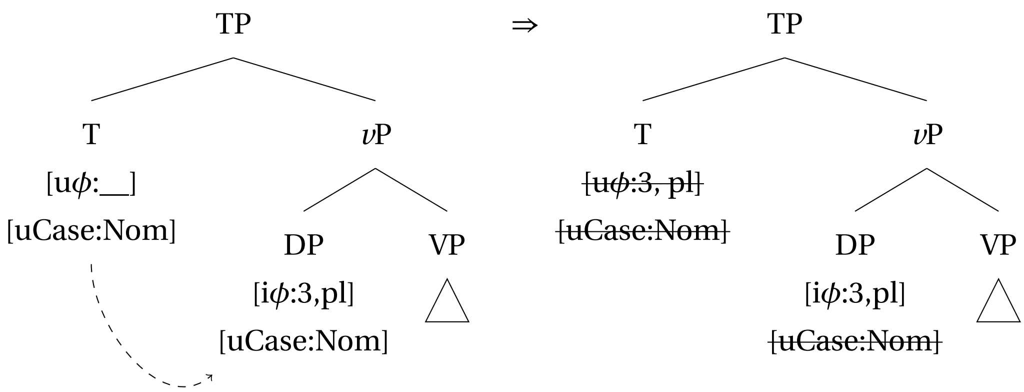
(236) Maša-jen ʎɵχs-əʎ mɛt χʉw kina-jən wanʎta-s-ʎe.
Masha-POSS.2SG friend-POSS.3SG-[ACC] most long film-LOC show-PST-3SG>SG
'Masha showed her friend the longest film.'

Other types of prominence (e.g., specificity, animacy, and topicality) are also unable to capture the restrictions above, as has been discussed in detail in Chapter 2, Section 2.4. Therefore, it can be safely concluded that the restriction on secundative theme is syntactic in nature. In particular, the theme in secundative alignment must lack a DP layer, though projections up to QP and PossP do not cause ungrammaticality. In the next section, I propose a formal analysis of secundative alignment and explain the DP-restriction through the absence of case-licensing.

4.2.3 Nominal licensing and DP-restriction

Numerous studies following Chomsky's (2000; 2001) original proposal have argued that nominals in argument positions require additional licensing by some type of Agree operation. Chomsky originally proposed that arguments are licensed by ϕ-agreement. An unvalued and unintrepretable ϕ-probe on a functional head agrees with the closest nominal in its c-command domain and copies its ϕ-features. At the same time, an uninterpretable Case feature on the goal is checked and deleted during the same Agree operation. The mechanism is illustrated in 237, where T^o agrees with the external argument and assigns the nominative case to it.

(237)



The presence of an uninterpretable feature ensures that a nominal is an available goal for agreement (Activity Condition). Importantly, every uninterpretable feature should be deleted under Agree. If the uninterpretable Case feature on a noun is not checked and deleted, it remains active and crashes the derivation at LF. This can be reformulated as a requirement for every noun to be licensed. Hence, nominal licensing equals ϕ-agreement accompanied by the checking of a [uCase]-feature. This approach has been adopted straightforwardly, for example, by Kalin (2018) and Nie (2019, 2020).

Later, important modifications have been proposed for the original Chomskian analysis. Two of these modifications are particularly important for the current discussion. First, Preminger (2014) argues that Agree can fail, i.e. an unvalued (and uninterpretable) feature does not necessarily cause the derivation to crash if it has not been valued and deleted under Agree. Furthermore, Preminger concludes that the notion of uninterpretability is superfluous, since its only function in Chomsky's (2000; 2001) original proposal was to ensure that a successful Agree is obligatory.

Another important observation concerns the idea that ϕ-agreement and case assignment are two sides of a single operation. This view presupposes that case assignment and ϕ-agreement are always in a one-to-one relation, i.e. that a head X^o can only agree with a nominal it assigns case to. However, several studies have shown that ϕ-agreement should be separated from the case assignment. First, it has long been observed that some languages with ergative-absolutive case alignment show nominative-accusative alignment in ϕ-agreement (e.g. Bobaljik 1993; Woolford 2006a; Coon 2017). As illustrated in 238, the verb in Nepali agrees with the external argument in both transitive and intranstive clauses. Importantly, Nepali is an ergative language where the external argument is marked with the ergative case in transitive clauses and the absolutive case in intransitive clauses. For the original Chomskyan system in 237, it would mean that the same T^o-head that agrees with the subject is able to assign two different cases.

(238) Nepali (after Coon 2017: p.363, ex.7)

a. **maile** yas pasal-mā patrikā kin-**ē**
2SG.ERG DEM store-LOC newspaper.ABS buy-1SG
'I bought the newspaper in this store.'

b. **ma** thag-ī-**ē**
1SG.ABS cheat-PASS-1SG
'I was cheated.'

A similar dissociation has been discussed for object agreement by Bárány (2021b, 2023). For example, in Amharic, only the recipient agrees with the verb, irrespective of whether it is marked with the accusative (239a) or dative case (239b).

(239) Amharic (Baker 2012: 261, 258, 261)

a. Ləmma [$_R$ Aster-textbarin] [$_T$ hɨs'an-u-n] asaj-at.
Lemma.M Aster.F-ACC baby-DEF-ACC show.3.M.S-3.F.O
'Lemma showed Aster the baby.'

b. Ləmma [$_R$ l-Almaz] [$_T$ tarik-u-n] nəggər-at.
Lemma.M DAT-Almaz.F story.M-DEF-ACC tell.3.M.S-3.F.O
'Lemma told Almaz the story.'

Scenarios where case and agreement do not align together pose a serious problem for the Chomskyan model in 237 where ϕ-agreement and case assignment are two sides of a single Agree operation and are predicted to be in a one-to-one relation. Hence, ϕ-agreement and case assignment are two independent operations, and the original Chomskyan model of nominal licensing cannot be maintained. A natural question arises: which of the two operations – case assignment or ϕ-agreement – should be analyzed as a nominal licensing operation?

In many languages, ϕ-agreement cross-references only nominative-marked subjects, i.e. external arguments in active clauses and internal arguments in passive clauses. No other argument can be cross-referenced on the verb. This is the case in, e.g., German, Russian, French, and many other languages. Similarly, ϕ-agreement in Khanty cross-references only subjects and *some* ACC-marked objects. Other ACC-marked objects, as well as DAT-marked indirect objects (including recipients, experiencers, and causees), also need to be licensed but are never cross-referenced by ϕ-agreement. Consequently, the licensing function should be attributed to another type of Agree operation. Indeed, a large part of the literature on differential argument marking suggests that nominal licensing happens via assignment of syntactic case (e.g. Danon 2006; López 2012; Ormazabal & Romero 2013; Lyutikova & Pereltsvaig 2015; Levin 2019; Kalin 2018; Irimia 2022; Driemel 2023 etc.).[39] In the majority of DOM languages, structurally reduced objects or objects without a prominence feature are unmarked for case or marked with a nominative and not

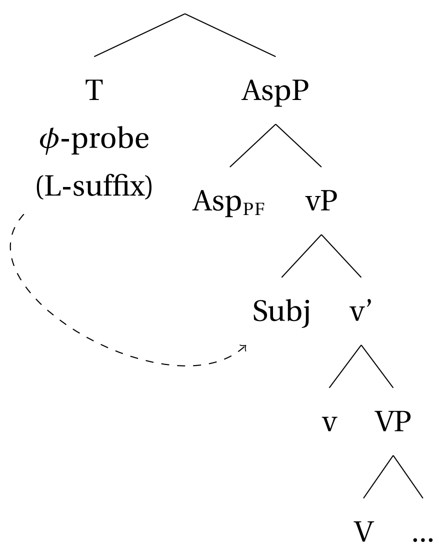
[39]An alternative research tradition dissociates syntactic nominal licensing and case. Morphological case is treated as a postsyntactic phenomenon, independent of DP-licensing (e.g. McFadden 2004 and Bobaljik 2008). But see e.g. Preminger (2014); Levin (2015) for counterarguments and in favor of C/case and ϕ-agreement being present in Syntax.

accusative case exponent. Many studies on DOM assume that such unmarked objects do not require syntactic licensing and are never assigned a case in syntax. A similar analysis has been suggested for a "reversed" DOM pattern in Fijian (van Urk 2020). In this language, an independent case-licensing operation is not available. Therefore, more prominent arguments are licensed via morphological Merger and are obligatorily verb-adjacent. Less prominent, structurally reduced arguments do not need case-licensing, are not morphologically merged with the verb, and can occupy various syntactic positions. Furthermore, abstract case assignment has been argued to be related to nominal licensing, even in morphologically caseless languages. Sheehan & van der Wal (2018) and van der Wal (2022) argue that even in Bantu languages, which do not have any morphological case, nominals are licensed by the assignment of an abstract Case. This is even more justified for languages that mark abstract Case overtly in morphology.

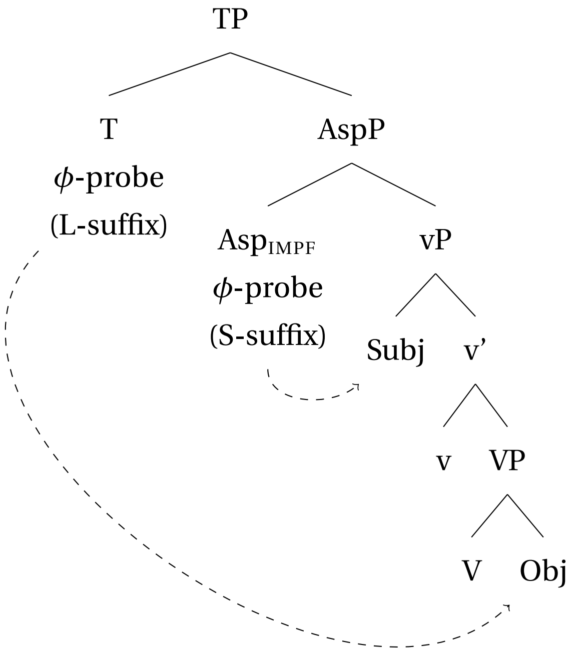
The size restriction is crucial for the analysis of secundative alignment. As discussed in Section 4.2.2, a full DP is ungrammatical as a secundative theme. A similar size restriction has been observed for several other phenomena, most notably for differential object marking and pseudo-noun incorporation (see a detailed discussion in Chapter 6). In both cases, multiple analyses suggest that the size restrictions are motivated by the absence of case-licensing. To name a few examples, Lyutikova & Pereltsvaig (2015), Kalin (2018), and Irimia (2022) argue for Tatar, Amharic and Romance respectively that NPs can survive derivation unlicensed, while DPs must be obligatorily licensed under an Agree operation.

In Senaya (Neo-Aramaic), only non-specific objects are grammatical in perfective clauses, whereas imperfective clauses do not show this restriction. Kalin & van Urk (2015) and Kalin (2018) suggest that there is only one ϕ-probe on T^{o} in perfective clauses 240a. Therefore, only one argument controls ϕ-agreement and is licensed with the abstract Case. In imperfective clauses, on the other hand, there are two ϕ-probes – on T^{o} and on Asp^{o} – and both the subject and object can be case-licensed under agreement 240b.

(240) a. (Kalin 2018: p.141, ex.37)

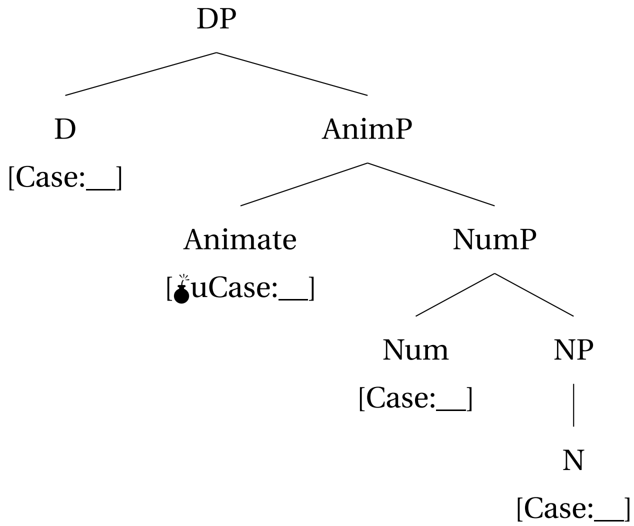


b. (Kalin 2018: p.141, ex.39)



Kalin (2018) argues that only non-specific nominals in Senaya can survive the derivation unlicensed, and specific nominals must be licensed via syntactic case assignment. Following the proposals by Pesetsky & Torrego (2007) and Danon (2011), Kalin (2018) assumes that an unvalued Case feature is shared between all projections in the DP, as shown in 241.[40] Hence, all nominals can be potentially case-licensed, i.e. they have at least one unvalued Case feature. However, only a subset of nominals necessarily needs to be licensed. She attributes this to the presence of an unvalued and *uninterpretable* [uCase] feature.

(241) (Kalin 2018: p.138, ex.35)



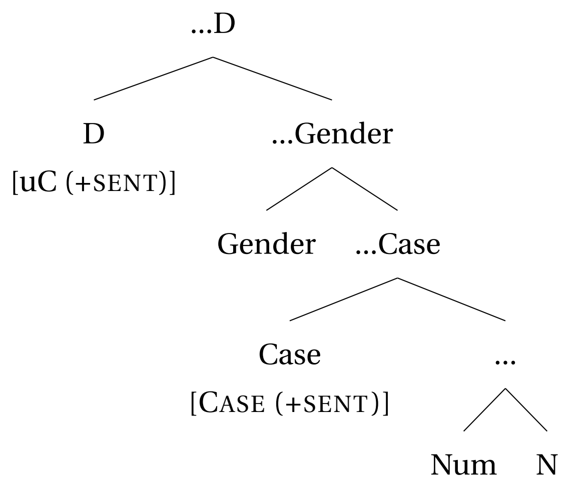
In Senaya, the unvalued Case feature is interpretable on all projections except AnimP, which is related to specificity. Non-specific nouns do not have AnimP-projection, i.e.

[40] The theoretical notion of feature sharing has been introduced to minimalist syntax by Frampton & Gutmann (2006). Danon (2011) employs this theory to avoid a problem with case assignment being counter-cyclic.

they only have interpretable unvalued Case features and lack the uninterpretable [uCase]-feature. Hence, such nominals can survive the derivation without syntactic case-licensing. On the other hand, specific DPs have an animacy projection. The unvalued uninterpretable Case feature causes the derivation to crash unless it is valued and deleted under agreement.

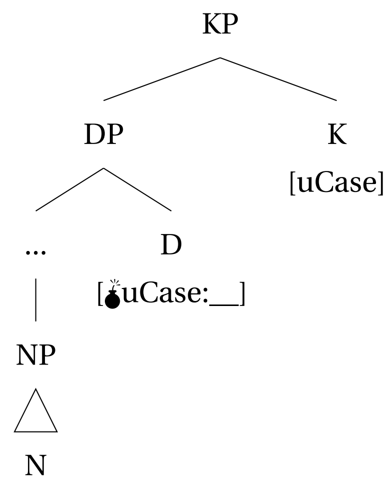
Irimia (2022) proposes a similar analysis for differential object marking in Spanish and Romanian. She argues that there are two projections bearing a Case feature inside the DP: D^{o} with an uninterpretable [uC(ase)]-feature and a Case-projection with unvalued but interpretable Case feature, as 242 shows. The uninterpretable unvalued [uC(ase)]-feature on the D-head must be valued during the derivation to avoid derivation crash. The lower Case feature can but does not have to be valued. If Case is valued and the noun is high on the prominence scale (i.e., [+sent]), the noun is differentially case-marked. If the noun is not marked for case, the Case feature is either absent or remains unvalued.

(242) (Irimia 2022: p.4, ex.10)



Therefore, DP-restriction on objects arises when syntactic case-licensing is not taking place. Unlicensed nominals are structurally reduced and lack an externally assigned case value. The absence of syntactic licensing, i.e. agreement for case, provides a motivation for the DP-restriction on secundative themes. Following the analyses just discussed, I propose that there are two Case features inside the DP in Khanty: one on the KP-projection and one on the D^{o}, as shown in 243. For technical reasons discussed in the next chapter, I argue that both features are uninterpretable. Note that an argument must be licensed under Agree with an external case assigner only when two Case features (on K^{o} and on D^{o}) are simultaneously present in the structure. Following Kalin (2018), it is indicated with a 💣-sign on D^{o}. The nature of this "derivational time-bomb" is discussed in detail in Chapter 5. Importantly, the Case feature on the K^{o} can but does not need to be externally licensed. This correctly predicts that all objects in monotransitive clauses in Khanty are accusative-marked, irrespective of the presence or absence of the DP-projection.

(243)



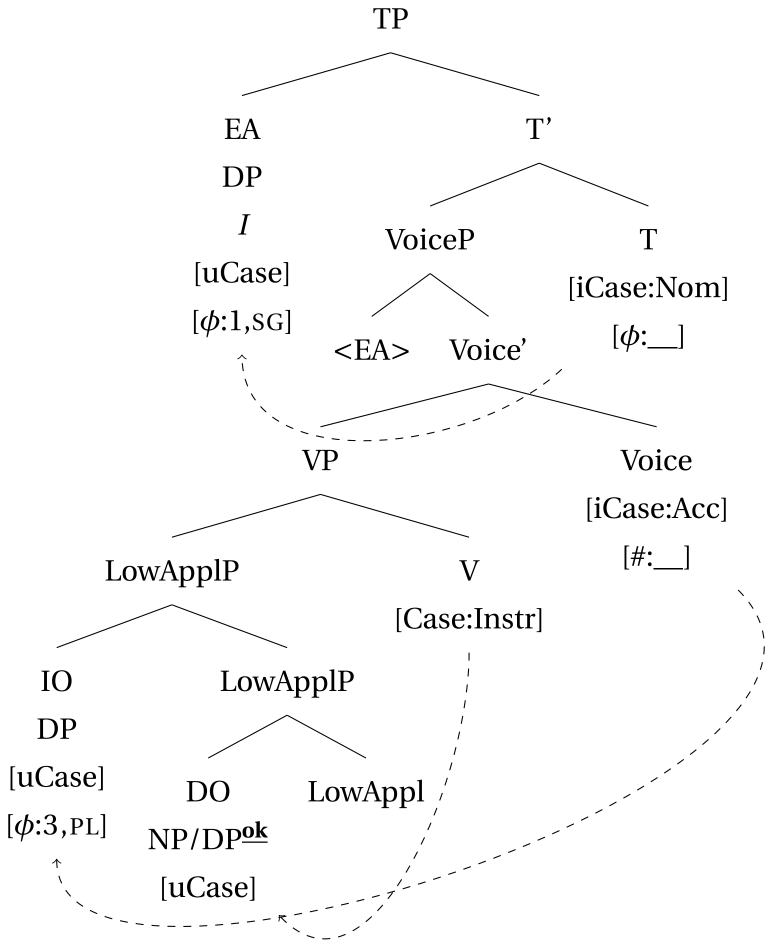
Note that secundative themes have one very important difference from unmarked objects in DOM. Namely, secundative themes bear an oblique case marking, while structurally reduced objects are less marked than more prominent objects in DOM systems. In Chapter 5, I argue that oblique marking on secundative themes should be analyzed as default marking that surfaces when no external head assigns a specific case value to the [uCase]-feature on K^o. Furthermore, I argue that all case values are present on K^o. Therefore, the most marked case in the language can be used as the default, if it spells out an unchecked Case feature on K^o.

4.3 Alternative analyses of oblique themes

There are two possible alternative ways to analyze the oblique marking on secundative themes. In this subsection, I briefly discuss both of them and show that they cannot account for all the relevant properties of the oblique themes.

First alternative suggests that the oblique marking on the theme is an inherent/lexical case. Namely, a last-resort probe on V^o or $LowAppl^o$ is activated in secundative alignment and assigns an inherent/lexical case to the theme. A possible derivation is illustrated in 244. The mechanism of additional licensing probes that are activated as a last resort to save the derivation from a crash has been proposed e.g. for Tagalog by Nie (2020).

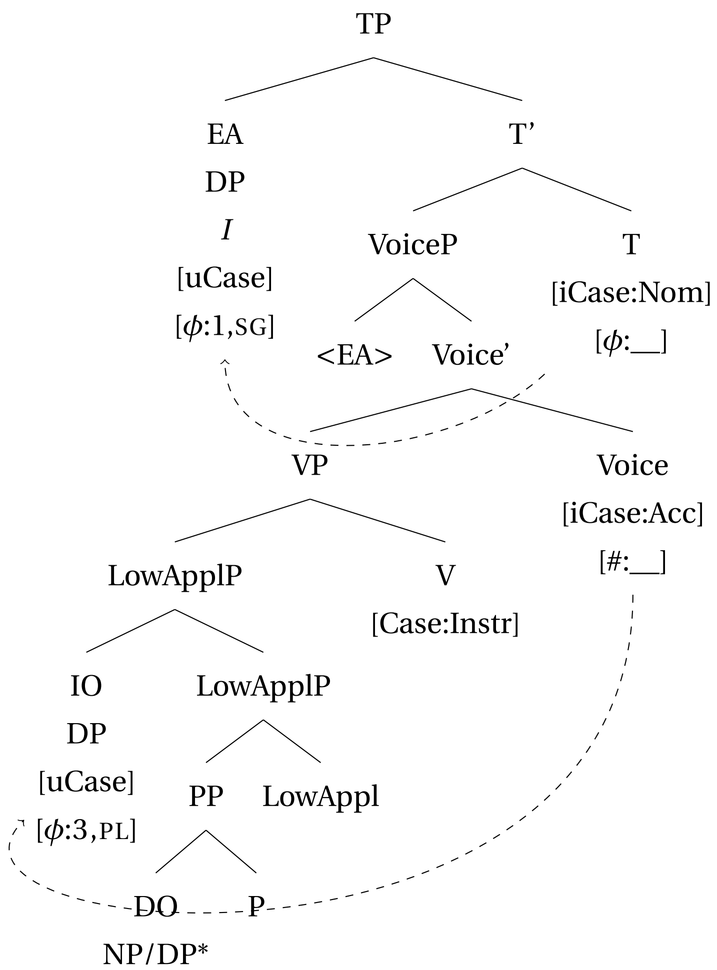
(244)



The derivation in 244 explains the oblique marking on the secundative theme in a straightforward manner. However, it is cannot account for the DP-restriction. As the theme is licensed by a last-resort probe, nothing should prevent it from being a full DP. Inherently case-marked nominals do not show a DP-restriction, neither in Kazym Khanty nor crosslinguistically.Therefore, the last-resort analysis of secundative alignment can only account for half of the data: it explains the oblique marking but fails to explain the DP-restriction.

The second alternative is based on the suggestion that all inherent cases are (underlyingly) PPs (e.g. den Dikken 1995; McFadden 2004; Rezac 2008; Alexiadou et al. 2014; Anagnostopoulou & Sevdali 2020). Under this analysis, the secundative theme is introduced as a PP, and the oblique case marking is either a cliticized postposition or a case assigned by a silent P-head, as in 245. This analysis can account both for the oblique case marking and for the DP-restriction. Adpositions with selectional restrictions on DPs are indeed attested crosslinguistically, e.g. the comitative preposition in Ossetic (Erschler 2019).

(245)



However, this analysis faces two problems. First, LOC-marking is not accompanied by DP-restriction in any other environment. Locative and instrumental adjuncts, as well as demoted agents in passives, are marked with locative, but can be full DPs. 246 illustrates this for adjuncts, and 247 for passive agents. Therefore, it must be concluded that the locative case itself has no selectional restrictions on DPs.

(246) Lexical Case

a. Ar joχ **tăm** woš-ən wəλ-λ-ət.
many people[.NOM] this town-LOC be-NPST-3PL
'There live many people in this village/town.'

b. Kartλʉŋk **ma** săŋkəp-**ɛm**-ən śiw sɛŋk-e.
nail[.ACC] I.NOM hammer-POSS.1SG-LOC PREV_there hit-IMP
'Hit (the/a) nail with my hammer.'

(247) Passive agent

Maw-λ-am Maša-**jen**-ən ńawrɛm-ɛm-a
candy-PL-POSS.1SG.[NOM] Masha-POSS.2SG-LOC child-POSS.1SG-DAT
mă-s-i-jət.
give-PST-PASS-3PL

'My candy was given by Masha to my kid.' (Colley & Privoznov (2020))

A silent postposition that assigns LOC and has the selectional DP-restriction can be a way out. However, such a postposition does not have clear semantics and is used in a single, very restricted context. Furthermore, this postposition would be unique in Khanty because it would be the only postposition with the selectional DP-restriction and the only postposition that assigns case to its complement. All overt postpositions in Khanty allow DPs as complements and never mark them with a morphologically visible case, as shown in 248.

(248) Aj was-ət ank-eʎ jʉpij-n mănʎ-ʎ-ət
small duck-PL.[NOM] mother-POSS.3PL after-LOC swim-NPST-3PL
'Ducklings swim after their mother.' (Loginopulo 2019: p.3, ex. 1)

The bunch of unique features, as well as the fact that this postposition is silent, should make it hard for children to learn it. Therefore, much more variation with respect to DP-restriction is expected in native speakers' judgments. However, DP-restriction is highly rigid. This makes the PP-based analysis improbable.

The second problem arises from the fact that the oblique marking on the theme can be overwritten. This is predicted to be impossible under both the PP-analysis and the inherent case analysis. Consider the following corpus example in 249. The clause is in secundative alignment, which is evident from the ACC-marking on the IO. Crucially, the theme is also NOM-/ACC-marked, and topicalized, i.e. clause-initial.

(249) śit aj wɛr, ma năŋ-ti *(t)?* wɛr-ʎ-ɛm
this small business I you.ACC do-NPST.1SG>SG
'This small business I do for you.' (Kaškin & Muraviev 2014)

Topicalization is only marginally present in Khanty. Nevertheless, I assume that it has mixed A/A' properties (see, e.g., van Urk 2015; Colley & Privoznov 2020 for complex A/A'-probes). The topicalizing head in Khanty can assign ACC/NOM case to the nominal it attracts.[41]

[41] Unfortunately, topicalization in Kazym Khanty is marginal, and other A/A'-properties cannot be properly tested. Furthermore, topicalization coexists with focus fronting, which complicates the picture. However, these two operations are clearly distinct. Focus fronting shows the properties of A'-movement. In (i), the recipient is focus-fronted, but the dative marking is preserved. More importantly, focus fronting does not create new binding configuration. The possessive marker on the subject *up-eʎ* 'his/her sister' cannot be bound by the fronted recipient.

(i) CONTEXT: To whom did his sister show new toys?
Kašəŋ ńawrɛm-a$_i$ up-eʎ$_{k/*i}$ jiʎəp juntut-ət want-ʎta-s
every child-DAT sister-POSS.3SG.[NOM] new toy-PL.[ACC] show-CAUS-PST.[3SG]
'His sister showed new toys TO EVERY CHILD.'

Therefore, I conclude that focus fronting is A'-movement and differs from topicalization, that is A-movement. This is not unique to Khanty. Georgi & Amaechi (2020, 2023) show that in Igbo, topicalization is also distinct from focus fronting. Focus fronting is an instance of A'-movement. The Igbo topicalization is the base-generation of a topic DP at the left periphery of the clause, from where it controls a resumptive pronoun in the canonical argument position. The base-generation analysis cannot be adopted for Khanty topicalization, where the secundative theme cannot be realized as a silent *pro* (see the discussion of 251). Therefore, I analyze Khanty focus fronting as A'-movement and Khanty topicalization as A-movement.

Note that when a secundative theme stays in situ, only a LOC-marked form is allowed, as demonstrated in 250.

(250) a. *ma năŋ-ti śit aj wɛr nuχ wɛr-λ-ɛm
I you.ACC this small business[.ACC] PREV do-NPST.1SG>SG
'This business is small. I'll do it for you.'

b. ^{ok}ma năŋ-ti ($^{?}$śit) aj wɛr-ən nuχ wɛr-λ-ɛm
I you.ACC this small business-LOC PREV do-NPST.1SG>SG
'I'll do such small business for you.'

Also note that 249 cannot be analyzed as a base-generated topic, binding a *pro* in the matrix clause (cf. such analysis for Igbo by Georgi & Amaechi 2020, 2023). If there were a *pro* in the base position of DO, it could be coreferent with the R-expression in a preceding finite clause. As 251 demonstrates, this is not borne out. If *śit wɛr* is put into a finite clause, the following clause with a *pro* as a secundative theme is ungrammatical. Hence, the secundative theme cannot be a silent *pro*.

(251) śit wɛr aj *copula*. *Ma năŋ-ti *pro* nuχ wɛr-λ-ɛm
this business small is I you.ACC PREV do-NPST.1SG>SG
'This business is small. I'll do it for you.'

Therefore, I conclude that topicalization in 249 is movement rather than base-generation and is able to overwrite the oblique case on the secundative theme. This provides a strong argument against any alternative analysis that treats locative on the secundative theme as a lexical or inherent case.

Chapter 5

Marked default and inherent Case overspecification

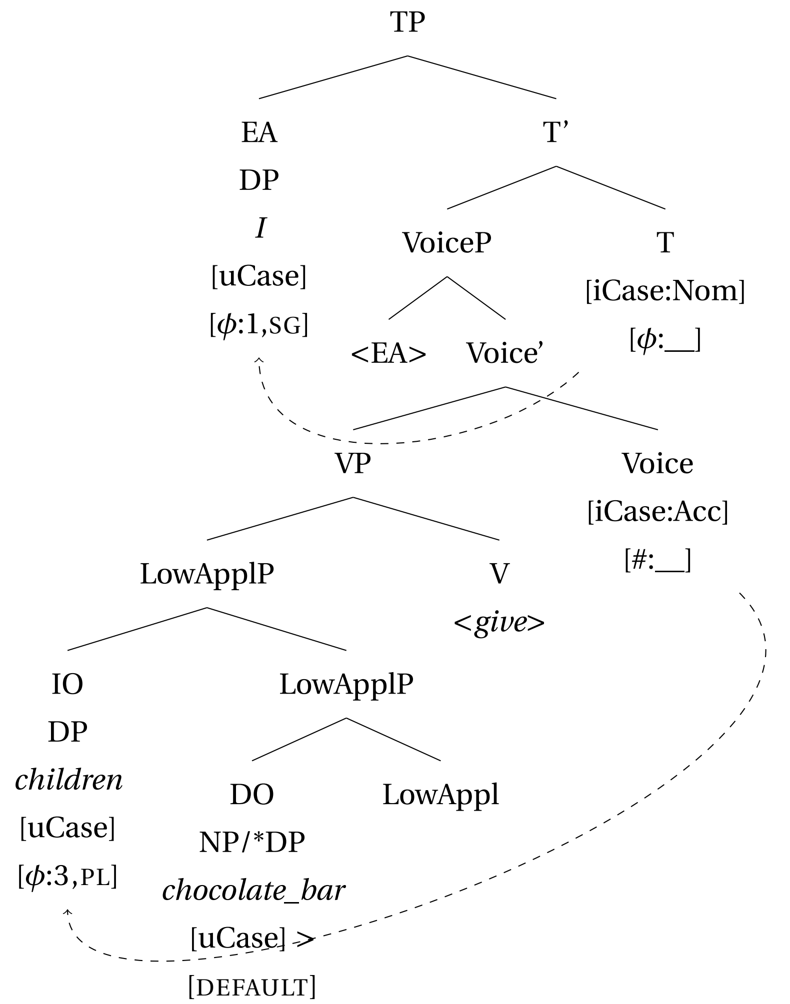
As argued in Chapter 4, size restriction on the direct object (theme) in secundative alignment can be motivated by the absence of syntactic licensing. Full DPs must be assigned case in Syntax to be licensed. Structurally reduced nominals, on the other hand, can survive derivation unlicensed. The proposed analysis of secundative alignment is repeated in 252: T^{o} agrees with the external argument (agent) and assigns case to it, while $Voice^{o}$ agrees and assigns case to the indirect object. Crucially, there is no case-assigning head that agrees with the theme. The theme remains unlicensed and does not receive a case value from an external head. Hence, secundative theme must be marked with a default case. In current approaches to default case marking, unlicensed nominals are marked with the least marked case (nominative or absolutive). However, the secundative theme bears an oblique morphological case. Therefore, an oblique case can be used as default.

In this chapter, I show that the oblique case on the secundative theme is always the most marked case in the case hierarchy. Furthermore, I argue that maximal marking can indeed be used as a default case value. I propose a new model of case assignment that motivates why a maximally marked case is used as a default case marker and review the consequences of this assumption for the theory of Case and the theory of Agree. The proposal consists of three main parts: (i) hierarchical organization of case features, (ii) inherent case overspecification and distribution of case features inside the DP, and (iii) case checking as a subtype of Agree. In particular, I argue that all case values are inherently present on nominals in argument positions from the beginning of the derivation. A subset of the case values is checked under Agree with a case-assigning projection and can be spelled out at PF. If Agree does not take place, the derivation either crashes or all unchecked case values survive to the Spell Out.

(252) (repeated from 210)

a. Ma ńawrɛm-ət i šokolatka-jən mă-s-λam
I.NON child-PL.[ACC] one chocolate_bar-LOC give-PST-1SG>NSG
'I gave children a chocolate bar.'

b.



This Chapter proceeds as follows. In the first section, I discuss existing approaches to the decomposition of case features. In the second section, I introduce the inherent case overspecification hypothesis and present a modification to the theory of Agree that allows to account both for case assignment and for ϕ-agreement. This section also suggests a motivation for the size restriction on unlicensed arguments. Section 3 discusses the distribution of marked and unmarked defaults in the language. Section 4 provides supporting evidence from Eastern Khanty that the oblique default is not bound to a specific case value but is always the most marked case available in a given language. Finally, Section 5 shows that Dependent Case Theory runs into problems when trying to account for indirective-secundative alignment alternation.

5.1 Case feature hierarchy

Some approaches treat case features as simplex values (e.g. Chomsky 2000, 2001 or Pesetsky & Torrego 2007). There is, however, a long tradition of extensive research, based on structuralist analysis by Roman Jakobson (Jakobson 1962; Lotz 1967), that decomposes case into smaller features. Within this tradition, there are two main groups of approaches. The first group decomposes case values into combinations of binary features. In the most widely adopted analysis of German(ic), the case system is reduced to two binary features, [±oblique] and [±governed] or [±hr] and [±lr] (Bierwisch 1967 and Wunderlich (1997)) 253.

(253) Structural cases (Wunderlich 1997: p. 48,ex. 41)

Dative	[+hr, +lr]
Accusative	[+hr]
Ergative	[+lr]
Nominative/Absolutive	[ ]

This approach has been adopted by Stiebels (1999); Morimoto (2002); Müller (2002); Keine & Müller (2015) both within and beyond Germanic. For richer case systems, different sets of features have been used, with further feature(s) added to the system. Müller (2004) decomposes the Russian case system into three binary features: [±subj(ect)], [±gov(erned)], and [±obl(ique)], as in 254.

(254) Russian case system (Müller 2004: p.201, ex.2)

Nominative	[+subj,-gov,-obl]
Accusative	[–subj,+gov,–obl]
Dative	[–subj,+gov,+obl]
Genitive	[+subj,+gov,+obl]
Instrumental	[+subj,-gov,+obl]
Locative	[–subj,-gov,+obl]

Kiparsky (2001) adds an underspecified [±SC] to account for non-structural cases in Finnish. Note also that the feature decomposition of structural cases in Finnish, proposed by Kiparsky (2001), differs from both the German case feature specifications in 253 and the Russian case feature specifications in 254. Kiparsky's approach is shown in 255.

(255) Structural cases in Finnish (Kiparsky 2001: p.14, ex.14)

Nominative	[ ]
Partitive	[-hr]
Accusative	[-hr,-lr]
Genitive	[+hr]

Yet another way to decompose case values into more primitive features has been suggested by Calabrese (2008) and Glushan (2010). They develop a system of four semantically motivated binary features [±peripheral], [±source], [±location] and [±motion] to account for Russian case system.

(256) (Calabrese 2008: p.171, ex. 6.19)

	Nom	Acc	Gen	Dat	Loc	Abl	Instr
Peripheral	–	–	+	+	+	+	+
Source	–	–	+	–	–	+	–
Location	–	–	–	–	+	+	+
Motion	–	+	–	+	–	+	+

Another influential line of research has developed a radically different approach to decomposing case features. These analyses argue that all cases should be decomposed into privative features and organized into a hierarchy in which less marked cases are contained within more marked ones (the Case Contiguity Hypothesis). Crucially, the case hierarchy is expected to be universal. Harðarson (2016); Smith, Moskal, Xu, Kang, & Bobaljik (2019); Zompì (2019) argue for a weak version of the Case Contiguity Hypothesis, where individual cases are joint in three groups – unmarked vs. dependent vs. all oblique cases – and only these groups are organized into a hierarchy, as in 257.

(257) unmarked > dependent > oblique

On the other hand, McFadden (2004); Caha (2009, 2013, 2024); Bárány (2017, 2018); Irimia (2023) argue for a strong version of the Case Contiguity Hypothesis, where each case feature is decomposed into a set of abstract privative features (A,B,C etc.).[42] Case inventory is organized into a universal hierarchy, where every more marked case includes all the features of the less marked cases, but not vice versa 258.

(258) The case sequence for nominative language (adopted from Caha 2009: p.21,23)

NOM	>	ACC	>	GEN	>	DAT	>	INSTR	>	COMIT...
[A]		[A,B]		[A,B,C]		[A,B,C,D]		[A,B,C,D,E]		[A,B,C,D,E,F]

Caha (2009, 2013) also argues that there are several instances of locative case that are

[42] McFadden (2004) uses non-abstract feature values in (i), but otherwise his analysis is almost identical to that of Caha (2009) in 258. Note that despite a seemingly binary notation ([+F]), he uses privative rather than binary features, since negative values are not present in the system.

(i)

Nom	[+case]
Acc	[+case, +inferior]
Dat	[+case, +inferior, +oblique]
Gen	[+case, +inferior, +oblique, +genitive]

Another difference from the hierarchy in 258 is the order of dative and genitive. I do not address this issue here because it is orthogonal to the main claim of this chapter. Nevertheless, the order [genitive > dative] makes correct predictions for (Kazym) Khanty. Khanty does not have a separate genitive morpheme. Nominals in positions where structural Genitive is expected, e.g. possessors, are zero-marked, which makes genitive syncretic with nominative and accusative. In all variants of the case hierarchies, only the cases next to each other can have a syncretic exponent. Therefore, the hierarchy in 258 captures the absence of the morphological genitive in Khanty straightforwardly, while McFadden's hierarchy does not.

located in different positions in the hierarchy, as shown in 259, and languages can have different instances of locative in their inventory.

(259) NOM > ACC > LOC_1 > GEN > LOC_2 > DAT > LOC_3 > INSTR > COMIT ...
(Caha 2009: p.10, 130,292)

This hierarchy explains why some cases are more marked than others. Although the hierarchy in 258-259 is argued to be universal[43], not all languages realize every case with a separate exponent. Crucially, the hierarchy in 258 allows syncretic case markers, but only for cases in neighboring cells. It is also possible that a separate exponent for a certain case is absent in a particular language and a PP is used instead. Importantly, however, if a certain case is absent from the language and is replaced by a PP, then all other cases to its right should be realized as PPs as well.

For the analysis below, I follow the strong version of the Case Contiguity Hypothesis and assume that specific case values correspond to sets of privative features, where more marked cases contain all the features of the less marked cases plus some additional features.

For ease of demonstration, I use a reduced case hierarchy in 260 for Kazym Khanty. Recall that nominative and accusative are syncretic on nouns but have two separate exponents for pronouns (see Chapter 2, Section 2.2.1), which motivates the presence of both cells. There is no separate genitive exponent, so it is ommitted from the hierarchy in 260. Intrumental and locative cases are syncretic in Kazym Khanty. Hence, I assume that the Khanty locative correspond to LOC_3 in Caha's hierarchy. Finally, comitative is realized as an overt postposition, i.e. is absent from the case inventory in Kazym Khanty and from the hierarchy in 260.

(260) Case hierarchy in Kazym Khanty

NOM	>	ACC	>	DAT	>	LOC/INSTR
[A]		[A,B]		[A,B,C]		[A,B,C,D,(E)]

Crucially for the current discussion and the proposed analysis, secundative theme bears the most marked case in the hierarchy in 260. In Kazym Khanty, it is locative. As will be discussed below, other languages indeed use different morphological cases to mark secundative themes. I argue that this case-marking is not random, and that the secundative theme should bear the most marked case available in the language.

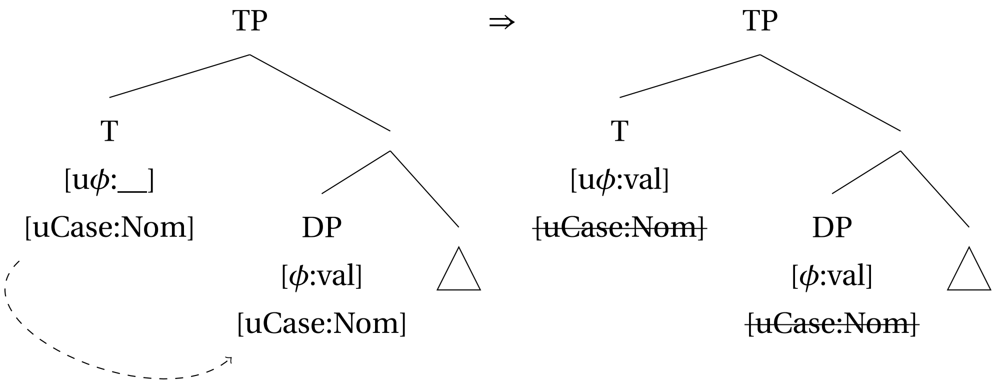
[43] I leave ergative languages outside of the discussion for now and return to them in Chapter 6.

5.2 Case assignment

5.2.1 Current theories of case assignment

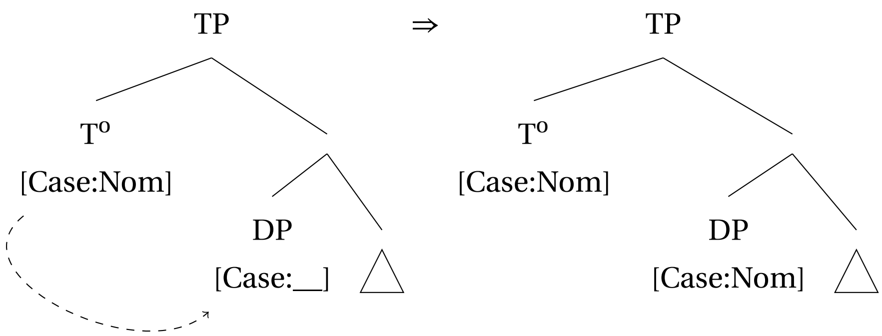
In the current minimalist approach to language, there exist three main theories of case assignment.[44] The first approach was developed by Chomsky (2000, 2001) and is illustrated in 261. In this model, both the argument and the case-assigning head have identical valued, but uninterpretable abstract Case features. Agree triggers mutual deletion of both features. If not deleted, the uninterpretable features crash the derivation at LF. Note that a Case feature with a simple value is inherently present on each argument. Case features in this approach are not decomposed into more primitive features, and the derivation can only converge if and only if the inherent feature on the argument coincides with the feature on the case-checking head.

(261)



More recent research adopts a valuation approach to Case. In contrast to the previous approach, the Case feature on a noun is inherently unvalued. During the derivation, it enters into Agree with a valued Case feature and copies the case value from it (e.g. Assmann et al. 2015; Bárány 2017; Bárány & Sheehan 2021 among others). The derivation is illustrated in 262. As argued by Preminger (2014), the (un)interpretability of a feature is irrelevant for syntactic derivation, i.e. uninterpretable features do not cause crash at LF. The only relevant requirement for Agree to succeed is that one of the agreeing features is valued and the other is unvalued.

(262)

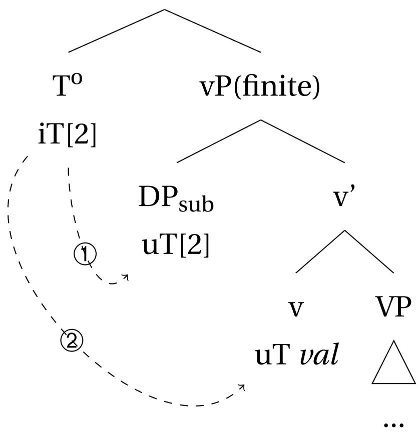


[44] I set aside a discussion of approaches that argue that case is a completely postsyntactic phenomenon (e.g. Bobaljik 2008; McFadden 2004). It seems to be a consensus in the more recent research that C/case is, at least partially, present in Syntax.

However, Bošković (2006b) explicitly argues against the valuation approach to case assignment and provides arguments in favor of the checking approach to Case.

Pesetsky & Torrego (2007) propose a modification of Chomsky's model, suggesting that the Case feature spells out the values of a verbal projection on the noun, e.g. T=NOM. Hence, the Case feature is interpretable on the verb and uninterpretable on the noun. This model of case assignment is illustrated in 263. First, the interpretable unvalued iT-feature on T agrees with the uninterpretable unvalued uT-feature on the subject DP. In the second step, the shared feature agrees with the uninterpretable but valued feature on v. In the third step, the T-value is copied to both T^{o} and the subject DP. The valued T-feature on the DP is spelled out as nominative.

(263) T^{o} and nominative case in a finite clause (Pesetsky & Torrego 2007: 278, ex. 17)



The general idea of this analysis is adopted by Clem & Deal (2024). They suggest that ϕ-features are copied from arguments to a verbal projection are subsequently copied back to an argument. These "secondary" ϕ-features are spelled out as case markers.

Finally, an alternative approach to case is formulated in Dependent Case Theory (Marantz 1991, 2000; Baker & Vinokurova 2010; Baker 2015, also Levin 2015). Dependent Case Theory radically differs from approaches that derive case marking from any kind of Agree operation. In contrast to Agree-based approaches, Dependent Case Theory derives case marking from the c-command relation between two arguments. Two rules for the dependent case in languages with nominative-accusative alignment are given in 264. Note that agreement with a verbal projection does not play a role for assignment of a dependent case.

(264) Dependent Case (Baker & Vinokurova 2010)

a. If there are two distinct argumental NPs in the same VP-phase such that NP_1 c-commands NP_2, then value the case feature of NP_1 as dative unless NP_2 has already been marked for case.

b. If there are two distinct argumental NPs in the same phase such that NP_1 c-commands NP_2, then value the case feature of NP_2 as accusative unless NP_1 has already been marked for case.

Levin & Preminger (2015) and Preminger (2020) extend the Dependent Case mechanism to lexical and oblique cases, arguing that these are not assigned via Agree with a verbal root/functional projection or P-head, but simply because a nominal is in a c-command relation with a head of a certain kind.

Secundative themes are a challenge to all current theories of Case. The DP-restriction suggests that secundative themes are unlicensed, i.e. they are not assigned case in syntax (via Agree or by Dependent Case rules). However, they have an oblique case morphology, which is unexpected in the direct object position unless this is a lexical case. In the previous Chapter, I argued that oblique marking on the secundative theme cannot be analyzed as a lexical case. Case-valuation theories predict that unlicensed nominals completely lack case values (see e.g. Legate (2008) and discussion in Section 5.3). Hence, unlicensed themes cannot be oblique-marked. Case-checking theories (Chomsky 2000, 2001; Bošković 2006b) assume that nominals must have a simplex case value, which is determined by their theta-role and requires checking. Therefore, unlicensed themes must either be ungrammatical or have a canonical "object" case, i.e. accusative. In the remainder of this chapter, I propose a modified theory of Case and Agree that accounts for the oblique marking of the secundative theme while retaining the generalization that DP-restriction arises due to the lack of syntactic (Case-)licensing. The challenges that secundative alignment poses for the Dependent Case Theory are discussed separately in Section 5.5.

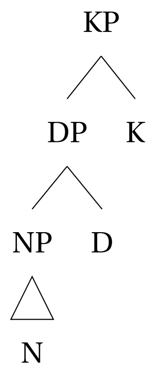
5.2.2 KP and inherent case overspecification

In this Subsection, I present the central proposal of this thesis and argue that arguments are inherently overspecified for case values. This assumption enables a straightforward derivation of the marked default case.

I assume that all nominals in argument positions have a KP projection with an inherent Case feature (Levin 2015). The structure of an argument DP/KP in Kazym Khanty is illustrated in 265. The position of the KP projection above the DP is supported by morphological evidence. The possessive marker in D^o is closer to the root than the case marker. Therefore, according to the Mirror Principle (Baker 1985), a KP is merged above a DP.

(265) Structure of an argument DP in Khanty (to be revised)

a. amp-əλ-a
dog-POSS.3SG-DAT

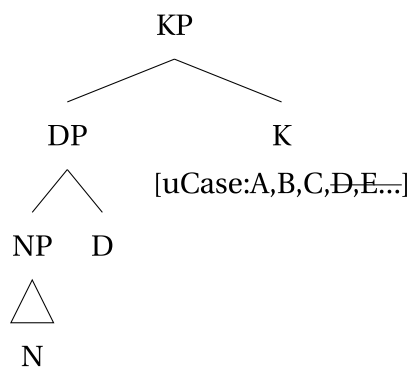
b. 

An analysis of case assignment in Kazym Khanty must account for two facts: (i) that the most marked case in the hierarchy is used as the default case-marking, and (ii) that DP-licensing is related to case assignment.

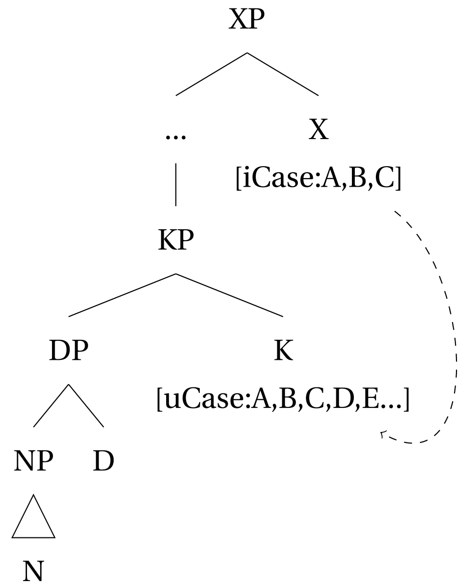
The central claim of this study is that K^o has an inherent uninterpretable case feature, which is specified for the full set of case values, as illustrated in 266b. The case assignment process is illustrated in 266c. The uninterpretable feature on K^o agrees with a valued and interpretable Case feature on a functional head X^o. Under Agree, a subset of "matching" case values on K^o is checked and can be further spelled out.

(266) Structure of an argument DP in Khanty (to be revised)

a. amp-əλ-a
dog-POSS.3SG-DAT

b. 

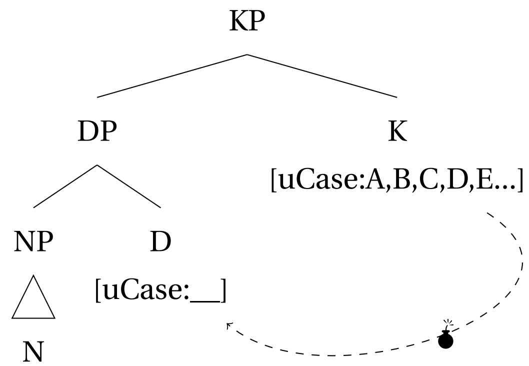
c.



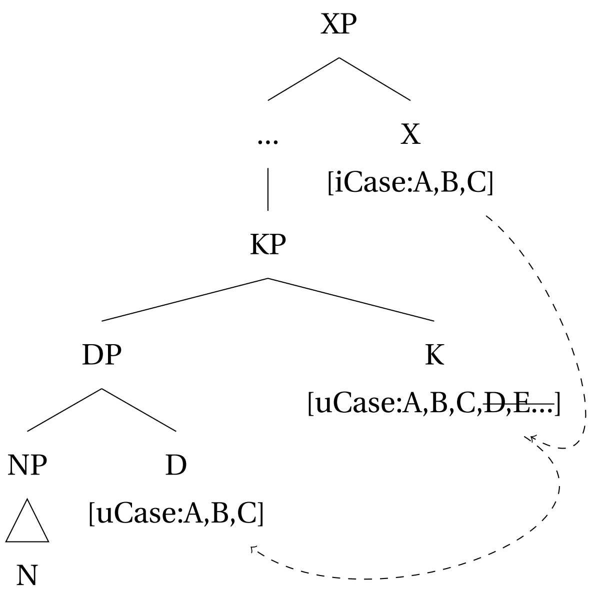
This inherent overspecification of the Case feature on K^o is the core of the current proposal. This derives the oblique marking on unlicensed nominals without assuming language-specific default case values. Although it may seem a radical theoretical innovation, overspecification has been recently proposed for various linguistic features, e.g. number (Harbour 2014), case resolution in Icelandic (Snorrason, Wood, & Sigurðsson 2024; Sigurðsson & Wood 2024), phonological features (Stojković & Simonović 2024), and for pre-syntactic morphology in general (Müller 2024, 2025).

The requirement for DPs to be licensed via case-assignment is modelled as follows. First, I assume that D^o has an uninterpretable unvalued Case feature and attempts to agree with the uninterpretable overspecified Case feature on K^o (contra, e.g., Norris 2014, 2017, who argues that all nominal concord is postsyntactic). This creates a "derivational time-bomb" (cf. Kalin 2018), which must be further checked towards an interpretable Case feature on X^o, as shown in 267. If Agree with X^o fails or X^o is not present in the structure, the "time-bomb" causes the derivation to crash.

(267) a. – requires further checking



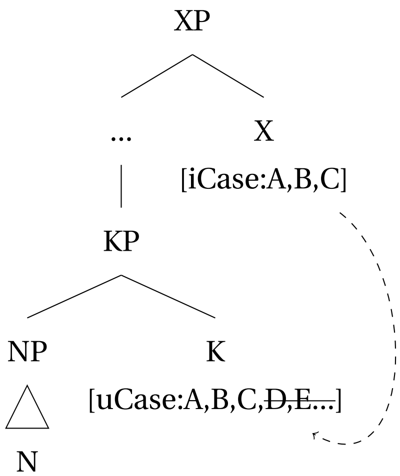
b.



– ok

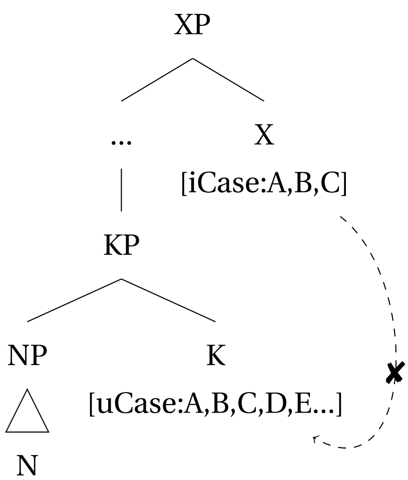
Structurally reduced arguments, on the other hand, do not have a DP-prjection. Therefore, a “time-bomb” in 267a is not created. If a case-assigning head X^{o} is present and successfully agrees with K^{o}, as in 268a, the Case feature on K^{o} is checked and its case value matches the value on X^{o}. If Agree fails, as in 268b, the Case feature on K^{o} remains unchecked, and the whole set of case values can be spelled out.

(268) a.



– ok

b.



– ok

I argue that the derivation in 268b illustrate what happens in secundative alignment to the theme argument: no X° agrees with the theme KP, but the derivation can proceed, since the DP projection is absent and a time-bomb is not created.

In the next subsection, I discuss the consequences of this proposal for the theory of Agree and suggest some adjustments to the definition of Agree. In particular, the proposed definition allows for Agree between two valued features, and the "derivational time-bombs" are formally defined.

5.2.3 Agree-Link and Case checking

The definition of Agree has changed over the years of minimalist research, as modifications to the original Chomskian approach have been suggested. In most studies, Agree has been discussed in context of agreement for ϕ-features. Chomsky (2000, 2001) argued that an uninterpretable and unvalued probe enters Agree with an interpretable and valued goal. A potential goal must have another uninterpretable feature to be accessible for Agree (Activity condition). Under Agree, values are copied from the goal to the unvalued probe, and all uninterpretable features are deleted. In this model, uninterpretable features cause the derivation to crash if not deleted, i.e. Agree cannot fail.

Preminger (2014) argues at length that Agree can fail without causing the whole derivation to crash. The (un)interpretability of features is redundant in his model, specifically because its only function is to ensure that the probe must agree. If Agree can fail, the only relevant characteristic of features is valuedness (cf. also Béjar 2003). In this approach, the definition of a probe and a goal is straightforward. All unvalued features are probes because they are defective and therefore actively want to agree. Valued features are syntactically inert and can only be agreed with, i.e. they are goals. This proposal initiated a long discussion about the direction of Agree, specifically wrt. Case assignment, wh-agreement, and negative concord, where the higher feature is valued and the lower one is unvalued. Zeijlstra (2012), Wurmbrand (2012a,b) and Bjorkman & Zeijlstra (2019) argued for Agree being directed upward, while other scholars, e.g. Preminger & Polinsky (2015); Rudnev (2021), argue for a strictly downward Agree, first of all for ϕ-agreement. Georgi (2014), Himmelreich (2017), Murphy & Puškar (2018), and Bárány & van der Wal (2022) use a mixed model in which Agree is predominantly downward but may be directed upward in certain scenarios.

Deal (2015b, 2022) proposes a completely different model of Agree, where neither interpretability nor valuedness are relevant factors. In her model, a probe is defined by the presence of interaction and satisfaction features. The interaction feature triggers the copying of feature values to the head and satisfaction feature signals when probing must stop. In this model, problematic cases, such as negative concord, can be derived with downward Agree.

A group of studies in the valuation approach argues that Agree must be divided into two operations: Agree-Link, which creates a pointer from a functional head to a nearby DP, and Agree-Copy, which transfers features from the goal to the probe and deletes the pointer (Arregi & Nevins 2012; Bhatt & Walkow 2013; Marušič et al. 2015; Himmelreich 2017; Atlamaz 2019; Baker & Camargo Souza 2020; Lyskawa 2021). Agree-Link is a syntactic operation, while Agree-Copy takes place postsyntactically.

Crucially, current approaches assume that Agree results in valuation, i.e. copy of the feature values. Feature checking under Agree is only allowed in the initial Chomskian model, but even there, checking is parasitic on ϕ-agreement, i.e. on a valuation process. After Preminger (2014), it has been assumed that only the valuedness of a feature is relevant for syntax. Therefore, the only possible scenario is Agree between an unvalued probe and a valued goal. Importantly, the overspecification hypothesis introduced in Section 5.2.2 presupposes that a case is assigned under Agree between two valued features. This poses a challenge to the current approaches to Agree, which explicitly disallow such configurations. To resolve this problem, I propose a modified definition of Agree that accounts for (i) Agree between an unvalued probe and a valued goal (ϕ-agreement) and (ii) Agree between a valued probe and a valued goal (case assignment).

I assume that language has only three types of features: valued interpretable (e.g., ϕ-features on nouns), valued uninterpretable (e.g., Case feature on K^o) and unvalued uninterpretable (e.g., ϕ-features on T^o). Unvalued interpretable features are not likely to exist in a language for conceptual reasons.[45] Note that I follow Zeijlstra (2014, 2019) and assume that interpretability is a formal characteristics of a feature and only indirectly corresponds to semantic meaning (see more on that below). The feature inventory is summarized in 269. All uninterpretable features are considered to be defective. As in existing approaches, the defectiveness of syntactic features is a factor that allows Agree to complete.

(269) Feature inventory:

a. Valued features can be both interpretable and uninterpretable;
b. All unvalued features are uninterpretable;
c. Uninterpretable features (both valued and unvalued) are defective;

In the proposed model, Agree is defined as in 270. It is based on the approach that divides Agree into two operations. On the syntactic side, Agree-Link creates a link between two features, at least one of which must be defective. The requirement for at least one feature to be defective prevents overgeneration, e.g. scenarios where two valued interpretable features agree with an unknown result. However, note that the definition in 270 allows Agree between two valued features if at least one of them is uninterpretable. As Agree-Copy in the original model by Arregi & Nevins (2012), the postsyntactic Agree-Match ensures that the value of a defective feature matches the non-defective one and subsequently deletes the pointer. Matching of the two features can be achieved in two ways: (i) either value is copied to the unvalued feature, or (ii) redundant feature values on the uninterpretable feature are impoverished. Crucially, Agree-Match is bi-directional and does not care whether

[45] Note that such features have been proposed by Pesetsky & Torrego (2007), but their existence was not independently motivated.

a defective feature is a probe or a goal. I assume that interpretable features are inert, i.e. the value of an interpretable feature can never be changed, both on a probe and on a goal.

(270) a. AGREE-LINK:
Feature A is linked to feature B with a pointer, iff (i) A is a probe on the head of a functional projection, and (ii) B is the closest available goal in the maximal projection of A, and (iii) either A or B is defective (lacks value and/or interpretation);

b. AGREE-MATCH:
The value of the defective feature is modified to match the value of the interpretable feature. The pointer is deleted.

Note that under the present definition, Agree always happens downwards or in a Spec-Head-configuration[46], but the transfer of the value can happen from a probe to a goal, as well as from a goal to a probe. In this model, the probe is not defined as an unvalued feature (cf. Preminger 2014). Rather, any feature on the head of the highest functional projection acts as a probe. This means that any type of feature from 269 can be a probe. For instance, an interpretable valued feature can act as a probe, scan through its c-command domain, and enter into Agree-Link as soon as it finds a suitable defective goal. Similarly, a defective as well as a non-defective feature can be the goal of Agree. Therefore, probe is not an independently defined notion, as it was in the previous approaches to Agree (cf. Heck & Himmelreich 2017 for the notion of valued probes). The single function of probehood is to determine the domain of Agree and to restrict possible Agree-configurations.[47]

Crucially, semantic interpretation of a feature is not relevant for PF operations and Agree-Match is not expected to have access to the information about which feature is interpretable and which is not.[48] This is indeed so under the classical understanding of feature interpretability (Chomsky 2000, 2001). However, Zeijlstra (2014, 2019) argues that syntactic (un)interpretability is fully independent form semantic content and is a purely syntactic characteristic of features. His definition of interpretability is given in 271. Under this definition, "interpretable formal features are purely formal features that have the capacity to check off uninterpretable features, but that lack any semantic interpretation. <...> A more proper way to refer to them would be using "independent" and "dependent" formal or categorial features." (Zeijlstra 2019: p.37f.)

(271) The element carrying [iF] must be taken to carry the semantics of F as part of its lexical semantics; this means that it is not the feature [iF] itself that is being interpreted at LF.
(Zeijlstra 2014: p.120, ex.25)

[46] Compare e.g. a proposal by Clem (2022) that maximal projections can be probes

[47] I leave cases of multiple Agree outside of the current discussion. Preliminarily, they can be accounted for in a manner similar to what has been proposed in the existing approaches (e.g., Béjar & Rezac 2009; Deal 2022).

[48] I am grateful to Gereon Müller for pointing this out.

Under this view, (un)interpretability (or (in)dependence) of a feature can still be accessibly at the PF. Hence, independent (aka. interpretable) features are seen by Agree-Match as a standard of comparison, while dependent (aka. uninterpretable) features can be freely modified.

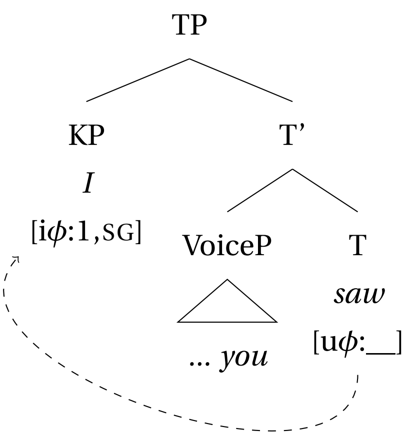
Note also that Zeijlstra's (2014) proposal about independence of synatactic interpretability from semantics is especially beneficial for an analysis of Case. Case features are assumed by almost all researchers to lack proper semantic meaning. Pesetsky & Torrego (2007) is the only study suggesting that Case is semantically meaningful (NOM = T, ACC = Voice/v, etc.). This view is rarely followed, as it is conceptually hard to tie verbal semantic categories to nominal domain (though see Wiltschko 2003). Under Zeijlstra's (2014) definition in 271 Case can be tied to the semantics of verbal functional projections without being completely identified with it.

The definition in 270 can account for different types of Agree relations. The three main types are discussed below. First, this definition allows to keep the analysis of ϕ-agreement as the valuation of an unvalued feature on a verbal projection, i.e. similar to all current analyses (e.g., Chomsky (2000, 2001); Béjar & Rezac (2009); Preminger (2014); Arregi & Nevins (2012); Bárány (2017) etc.). I illustrate the derivation for a simple case of subject-verb agreement in 272. The unvalued ϕ-probe on T^{o} agrees with the highest available goal in the maximal projection of TP, i.e. with a pronoun in SpecTP. The goal has a valued interpretable ϕ-feature. In syntax, Agree-Link creates a pointer from T^{o} to the pronoun in SpecTP. At PF, Agree-Match copies feature values from the goal to the probe and deletes the pointer.

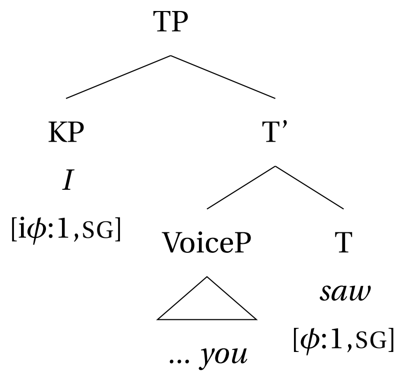
(272) Agree between unvalued uninterpretable F_A and a valued interpretable F_B: ϕ-agreement

a. ma naŋ-e:n wa:n-s-əm
I.NOM you-ACC see-PST-1SG
'I saw you.' (Nikolaeva 1999: ex.176)

b. Agree-Link



c. Agree-Match

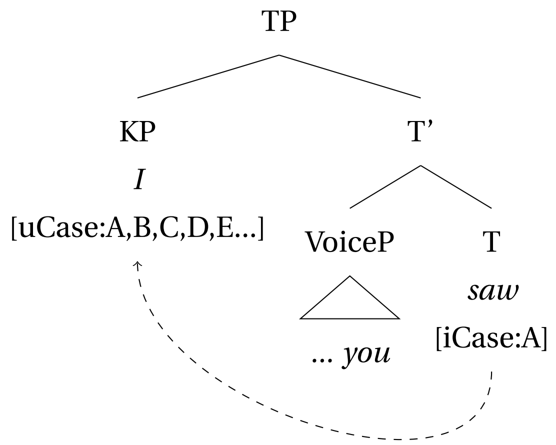


As argued in this chapter, both the case-assigner and the argument DP have valued Case features. An advantage of the definition in 270 is that it captures agreement between two valued features in the same way as Agree for valuation, i.e. it can account both for case assignment and ϕ-agreement. As 273 illustrates, a valued Case feature on T^o can enter into an Agree-Link relation with a valued Case feature on the KP of the subject, because the latter is uninterpretable, i.e. defective, and therefore a legitimate goal for the interpretable probe-feature on T^o. When the pointer is created, Agree-Match can be applied at PF. It seeks to unify the value on the two agreeing features. Because the interpretable feature on T^o cannot be changed, the only option is to impoverish the "unmatching" feature values on K^o. Hence, case assignment is checking of a subset of inherent feature values and impoverishment of the unchecked values.

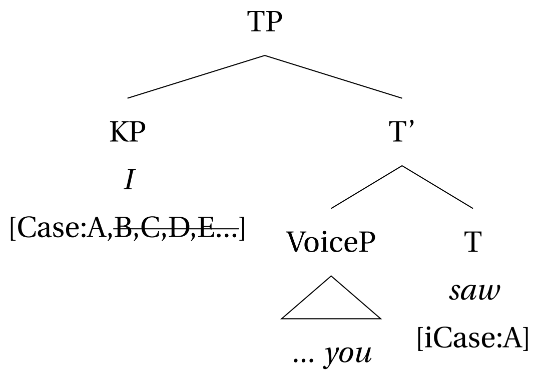
(273) Agree between valued interpretable F_A and valued uninterpretable F_B: Case assignment

a. ma naŋ-e:n wa:n-s-əm
I.NOM you-ACC see-PST-1SG
'I saw you.' (Nikolaeva 1999: ex.176)

b. Agree-Link

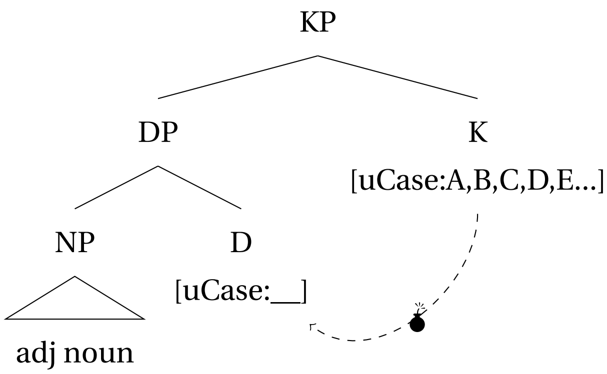


c. Agree-Match



Finally, I assume that Agree-Link between an unvalued uninterpretable feature and a valued uninterpretable feature is also possible. However, this results in a "derivational time-bomb" (Kalin 2018), which must be defused by a subsequent Agree-Link to an interpretable feature. In 274, an Agree between two uninterpretable Case features on K^o and D^o is illustrated. The current approach allows to define the nature of the "derivational time-bomb" as the inability of the system to define the direction of the Agree-Match operation. Because neither of the features is of the inert interpretable kind, Agree-Match cannot decide whether it should copy case values to D^o or delete them on K^o. The simultaneous availability of two conflicting operations causes the derivation to crash. Crucially, the configuration in 274 allows for an additional instance of Agree with an interpretable Case feature, as in 273. If this happens, the derivation is stabilized, and Agree-Match can adjust the [uCase]-features on K^o and D^o to the [iCase]-feature of a case-assigning head.

(274) Agree between two uninterpretable features: Case concord inside KP (for a head final language)



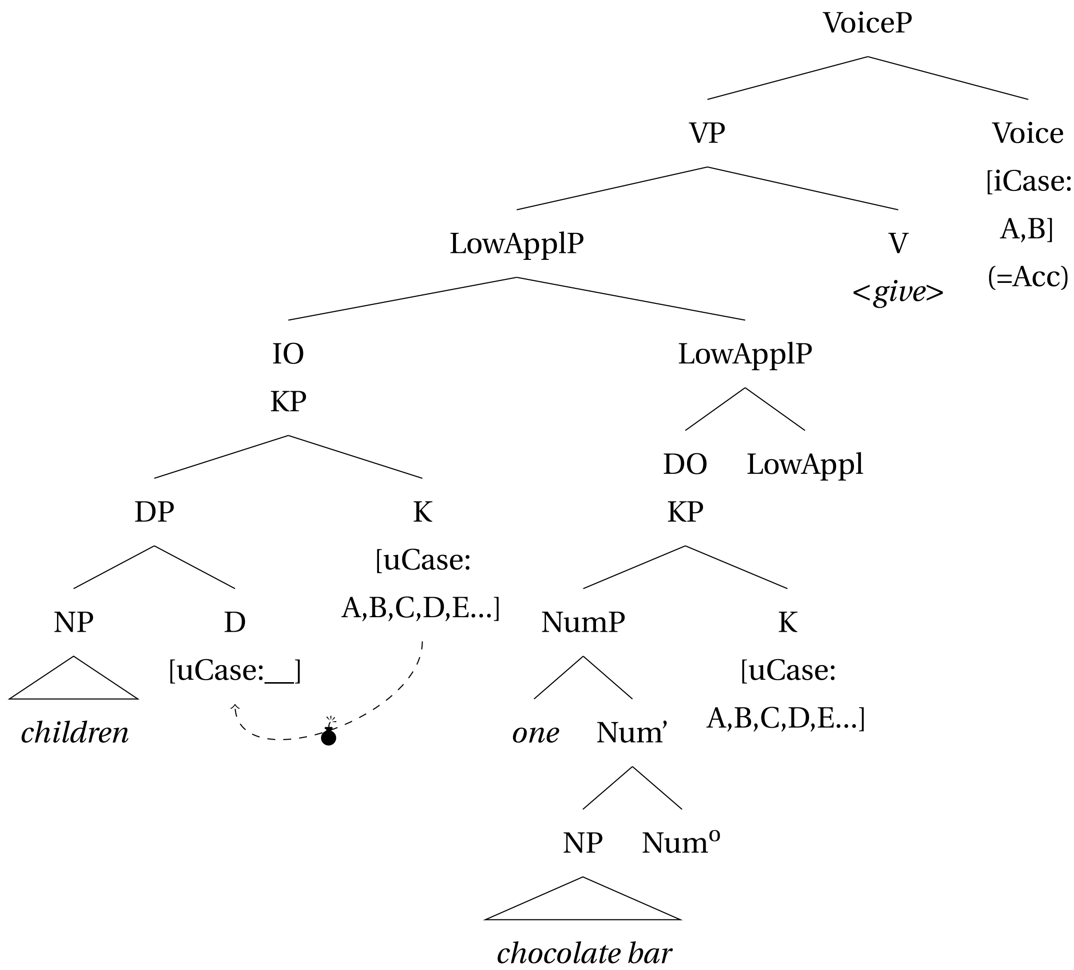
The presence of an unvalued Case feature on functional projections inside the DP is a matter of crosslinguistic variation. Some languages have the [uCase:__]-feature on every functional projection inside DP, while others might have it only on D^o or on some other projection. Whether the DP-restriction is a universal phenomenon is a matter for further research. If the [uCase] is present only on K^o and no other nominal projection has it, this predicts that DPs, as well as structurally reduced nominals, can receive the marked default case. On the other hand, some languages might have an obligatory case-concord between K^o and N^o. In this scenario, all nominals in argument positions must be assigned case.

5.2.4 Secundative alignment in Kazym Khanty: unifying oblique default and DP-restriction

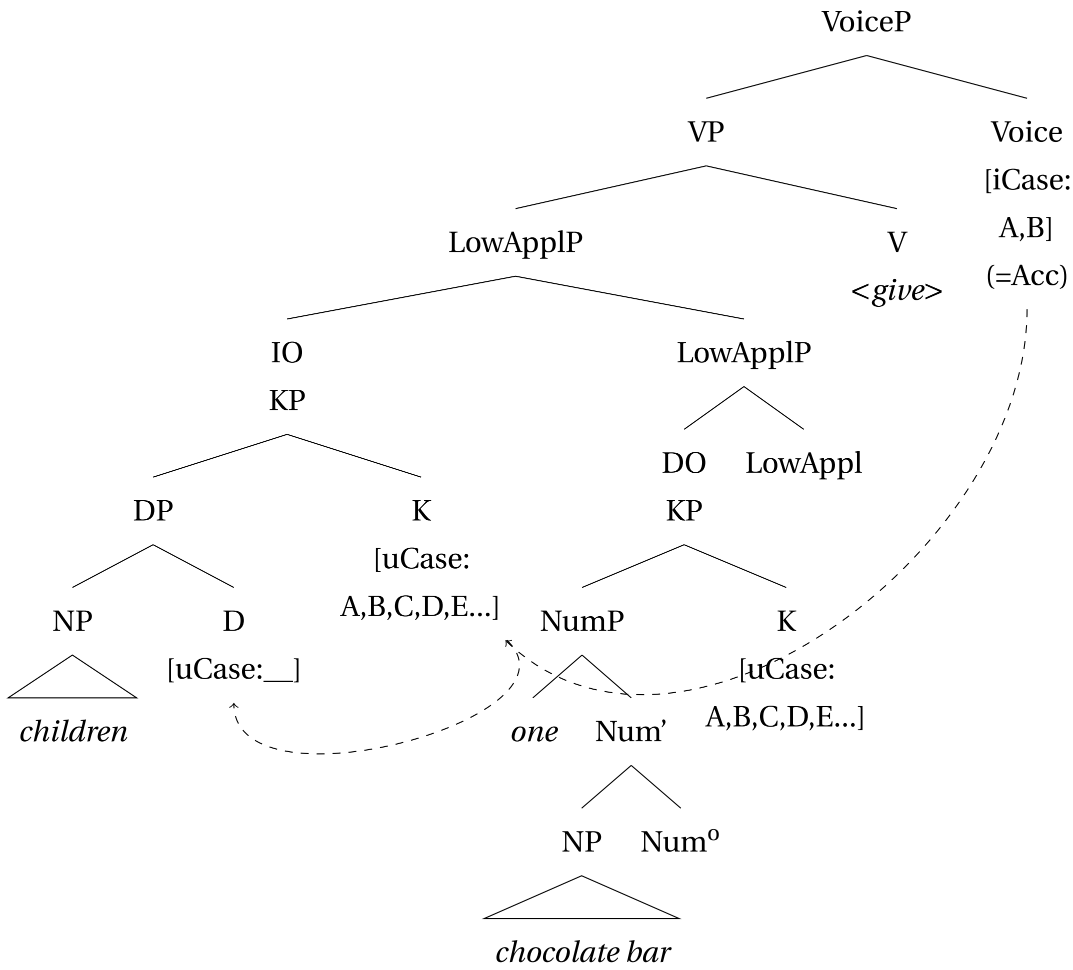
The hypothesis of inherent case overspecification and case checking allows to account for oblique case marking on the unlicensed secundative theme. The derivation in 275 shows how Agree for Case features proceeds in the lower phase in a clause with secundative alignment. In the first step in 275b, Agree-Link creates a pointer between Case features inside every argument KP. Since the IO has a DP-projection, Agree-Link takes place in this KP and creates a "derivational time-bomb" for IO. The DO does not have a DP-projection, and other functional projections do not have an unvalued [uCase]-feature. Therefore, Agree-Link fails within the DO. The next step of the derivation is shown in 275c: an interpretable Case feature on Voice^o probes and enters into Agree-Link with IO. A pointer is created between Voice^o and KP$_{IO}$. The Case feature on DO is not agreed with. The final step of the derivation in 275d takes place in the postsyntax. Agree-Match impoverishes superfluous features on KP$_{IO}$ and copies case values from Voice^o to DP$_{IO}$. Agree-Match cannot take place in the second KP$_{DO}$. Hence, all case values are preserved on the K$^o{}_{DO}$ and remain active for the Vocabulary Insertion.

(275) a. Ma ńawrɛm-ət i šokolatka-jən mă-s-λam
I.NON child-PL.[ACC] one chocolate_bar-LOC give-PST-1SG>NSG
'I gave children a chocolate bar.'

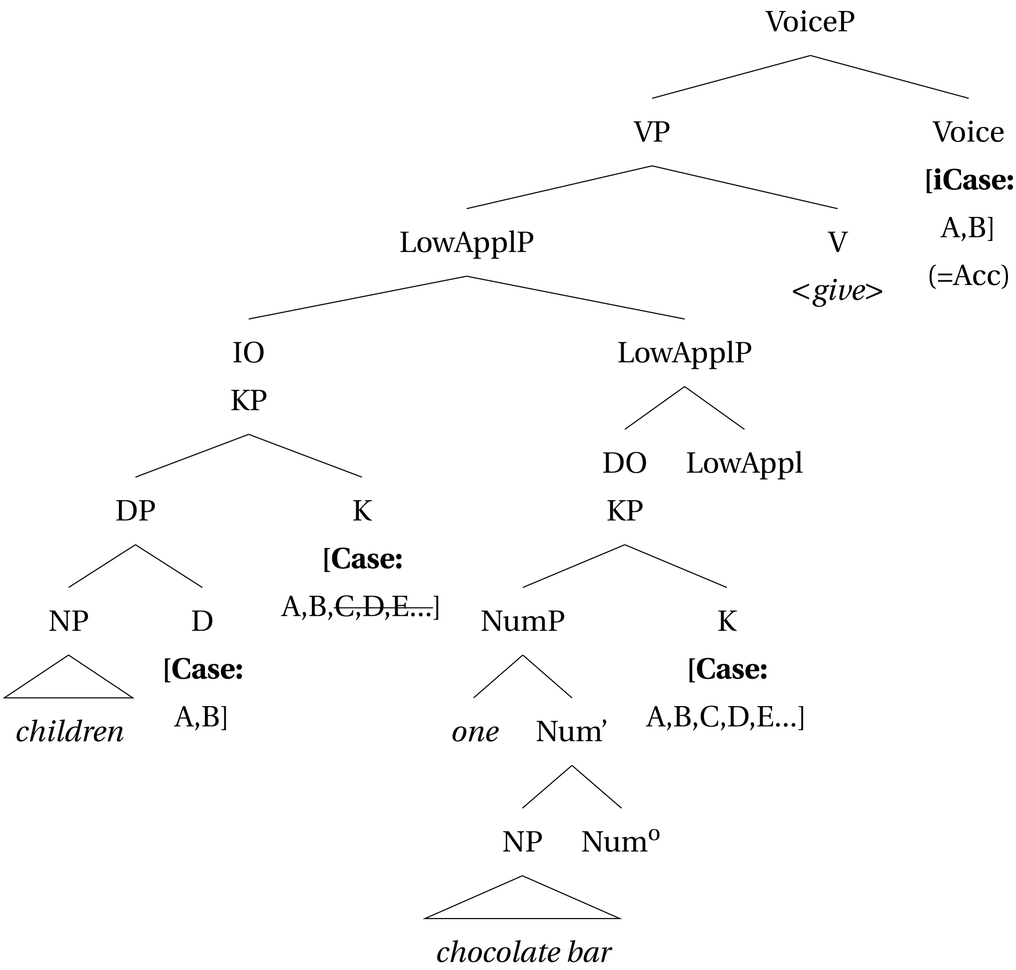
b. Agree-Link for case inside KP



c. Agree-Link for case with an interpretable Case feature on an external head



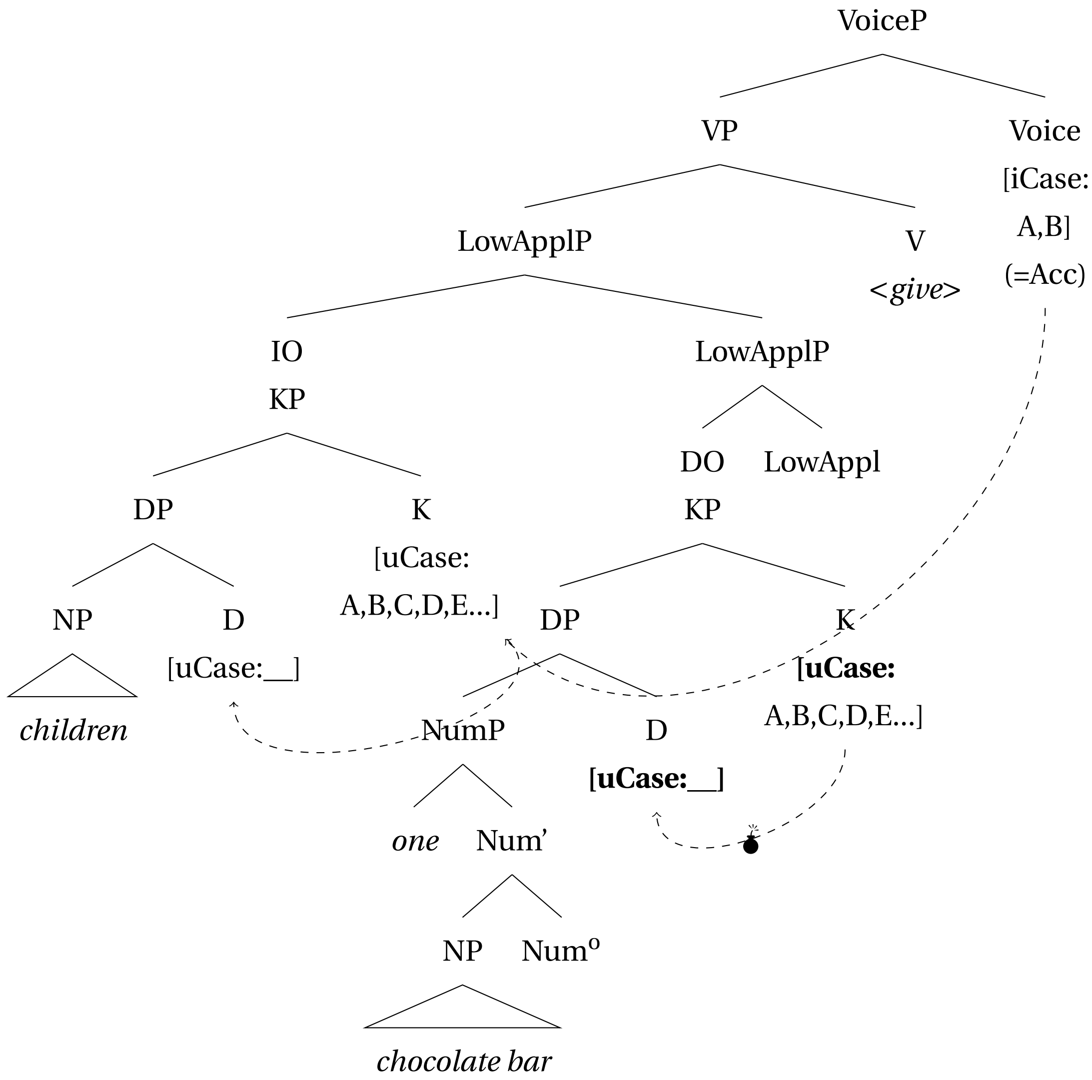
d. Agree-Match



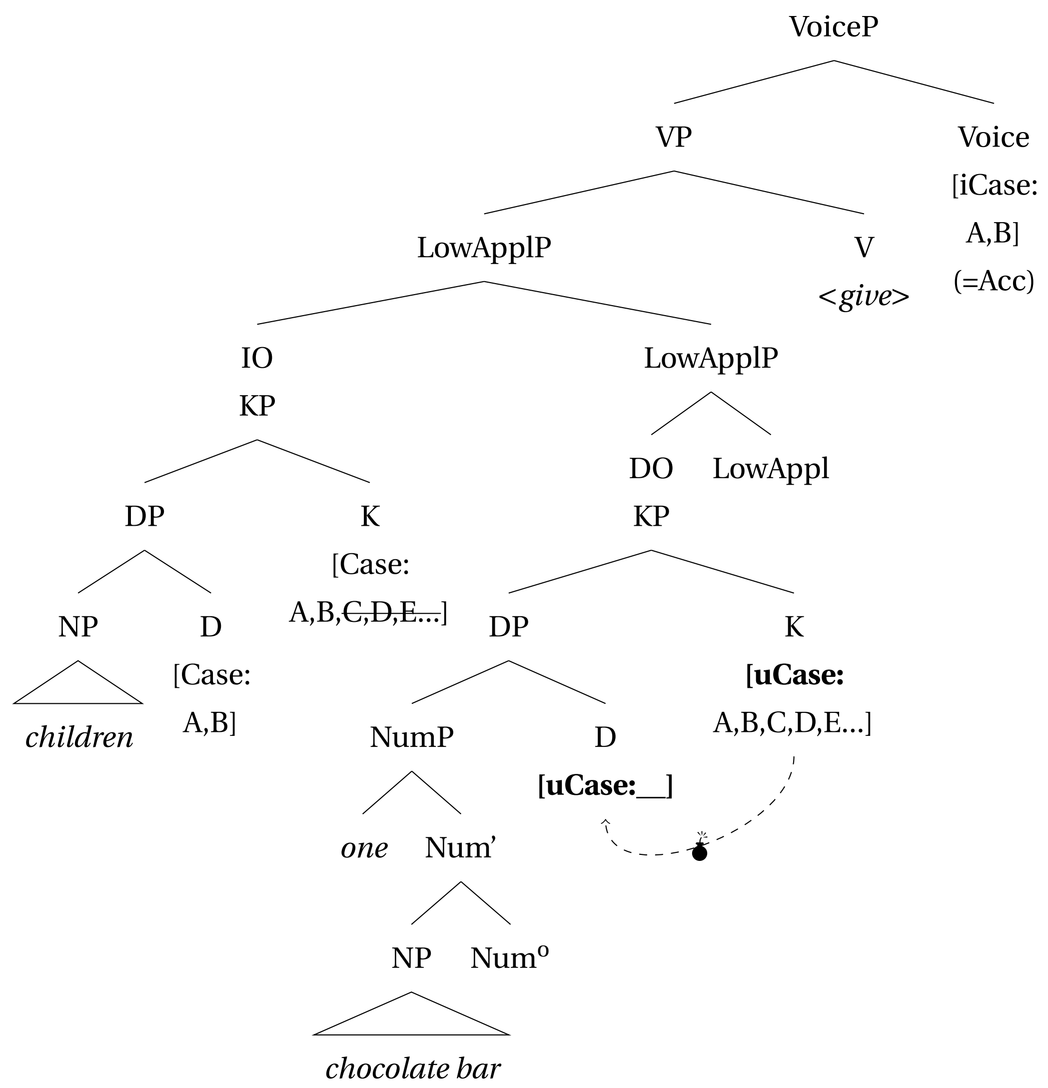
Importantly, this analysis straightforwardly captures the DP-restriction for the unlicensed secundative theme. If the theme in 275 had a DP-layer, both arguments would have a "time-bomb". As 276a illustrates, Agree-Link with Voice^o defuses only one of the two "time-bombs". Therefore, Agree-Match in 276b crashes the derivation when it attempts to resolve the feature conflict between KP and DP on the theme argument.

(276) Secundative derivation with a DP-theme

a. Agree-Link



b. Agree-Match



Note also that the derivation in 276 cannot be saved if IO is structurally reduced. Even when lacking a DP-layer, an IO is an argument and has a KP projection with an uninterpretable [uCase]-feature. Every KP is an available goal for Agree, even when lacking the internal "time-bomb". Therefore, the [iCase]-feature on Voice° would still agree with IO, even in absence of a DP-projection, and the derivation would crash as in 276.[49]

5.3 Unmarked and marked default case

The notion of default case was first introduced by Fanselow (1986, 1991) and has been systematically used in linguistic research after Schütze's (2001) seminal paper.[50] Schütze identifies a number of contexts where there is no obvious case-assigner, and notices that nominals have the same morphological case marking throughout all these contexts. Therefore, this case value is treated as default, used as repair in absence of a syntactically

[49] Lyskawa & Ranero (2022) propose that only complements, i.e. only internal arguments (themes), can be both full DPs and structurally reduced NPs. External arguments, as well as applied arguments, are merged as specifiers and are obligatory DPs. If this proposal holds in Khanty, it ensures that only 275 and 276 are possible scenarios.

[50] The abstract notion of default was introduced in phonology by Chomsky & Halle (1968).

assigned case value, as defined in 277.

(277) "The default case forms of a language are those that are used to spell out nominal expressions (e.g., DPs) that are not associated with any case feature assigned or otherwise determined by syntactic mechanisms." (Schütze 2001: p.206)

In Schütze's original approach, different languages use different cases as default marking, e.g. hanging topics are nominative in German 278a and accusative in English 278b. Despite a crosslinguistic tendency to use nominative/absolutive as the default marking, any case can be randomly assigned the default value.

(278) a. Der/*Den Hans, an den erinnere ich mich nicht.
the.NOM/*ACC of him.ACC remember I myself not
'Hans, I don't remember him.' (Schütze 2001: p.223, ex.25a)

b. Me/*I, I like beans. (Schütze 2001: p.210, ex.4a)

Following this line, Pesetsky (2013) argues that genitive is the default case in Russian, inherently present on all nominals. THe default genitive surfaces, in particular, under genitive of quantification, when a QP blocks case assignment to N^o by an external head (but see e.g. Bošković (2006a) for an alternative approach, where genitive is assigned by Q^o).

On the other hand, many studies argue that not every case value can be randomly chosen as the default. Rather, the default case is always the least marked case or even the complete absence of a morphological case marker (Legate 2008; McFadden & Sundaresan 2011; Weisser 2020; Caha 2024 etc.). This follows straightforwardly in a valuation approach, where a Case feature on a noun is inherently unvalued and receives a value during the derivation. As no valuation takes place in default contexts, the default case corresponds to the absence of syntactic case value. Thus, only nominative and absolutive can be analyzed as the proper default cases. As discussed by Weisser (2020) and Caha (2024), "accusative" forms are used as the default only on pronouns and only in languages with a poor morphological case inventory. They suggest that languages like English and Italian have strong forms of pronouns that are syncretic between nominative and accusative. These strong forms are used in Schütze (2001)'s contexts, which creates the illusion that accusative marking is the default form.

Crucially, in all existing approaches to the default case, there is no size restriction on nominals which lack a syntactically assigned case. Indeed, hanging topics are the most uncontroversial example default case. however, as illustrated in 278a for German, nominals in this position can have articles, i.e. are full DPs. The analysis developed in this thesis for secundative alignment differs from the existing approaches to the default case in two respects. First, default marking is tied to size restriction on the non-case-licensed nominal. Second, the most marked case in the case hierarchy is used as the default, contra to predictions of the existing approaches to the default case marking. I argue that this is the correct analysis of oblique secundative themes and that the maximally marked case can be treated as the default marking. Importantly, however, I do not intend to dismiss the traditional approach to the default case completely. The current hypothesis is that the maximally marked default is restricted to argument positions, while the unmarked default case is found in adjoined positions, e.g. under left dislocation and by hanging topics (Schütze 2001; McFadden 2004; Alexiadou 2006; Caha 2024 etc.), and possibly on some adverbials (e.g. Bošković 2006b).

The co-existence of two default cases – the most marked one and the unmarked one – can be easily accounted for if KP-projection is only present in argument positions. Indeed, if KP is absent on adjuncts, then only a DP with an unvalued Case feature is present in the structure. Therefore, no inherent Case features are present on adjuncts, and in the absence of case assignment by an external head, the default case value is the absence of case-marking or the least marked case exponent.

The approach with two default cases allows for the solution of an overgeneration problem, which was noticed by Fanselow (1986, 1991). Fanselow (1991: p.113) points out that the definition in 277 predicts a default nominative on external arguments in nominalizations. Nominalizations in German include structure up to VoiceP. Hence, an external argument can be merged in SpecVoiceP. However, there is no head X^{o} that can assign a case to it. Under the definition in 277, external arguments in nominalizations should be marked with the default nominative, as they are not assigned case by a syntactic mechanism. However, 279 is ungrammatical, i.e. the default nominative is not freely available, as suggested by the definition in 277.

(279) *die [Eroberung Galliens] Caesar
the.F.NOM conquest Gaul.GEN Caesar.NOM
Intend.: 'The conquest of the Gaul by Ceasar' (Fanselow 1991: p.113, ex. 19a)

The notion of a marked default case can easily capture the ungrammaticality of 279. If only KPs can be merged in argument positions, arguments are always inherently overspecified for case, and the unmarked default case can never be found in an argument position. This explains the ungrammaticality of 279. In contrast, adjuncts do not have a KP-projection and inherent case values. Hence, if they do not receive a case value from an external source, the unmarked default, i.e. nominative or absolutive, is inserted at PF. This correctly predics that hanging topics and other adjuncts are indeed marked with the default nominative.

The current proposal is further supported by the fact that two default cases behave differently with respect to the presence or absence of the size restriction on unlicensed nominals. Unlicensed nominals in adjunct positions have no size restrictions and can be full DPs, as in 278a. Indeed, if there is no KP, Agree-Link with an overspecified [uCase]-feature does not happen, and no "time-bomb" is created, even when a DP-layer is present in the structure. In contrast, unlicensed arguments must be structurally reduced, as discussed in detail in Section 5.2.4.

5.4 Surgut Khanty: Oblique marking of the theme

Surgut (Eastern) Khanty has a much richer case inventory than the Northern Khanty dialects. There are nine morphological cases in the noun paradigm, as shown in Table 5.1 (Csepregi 1998, Sosa 2017, and Schön & Gugán 2022).[51] Pronouns additionally differentiate between nominative vs. accusative and lative vs. dative.

NOM/ACC	∅
Lative/Dative	-a
Locative	-nə
Ablative	-i
Approximative	-nam
Translative	-ɣə
Instrumental-Comitative	-nat
Abessive	-λəɣ
Instructive-final (=instrumental object)	-at/-(t)ə

Table 5.1: Case inventory for nouns in Eastern Khanty (Sosa 2017: p.37, after Csepregi 1998)

Like all Ob-Ugric languages, Surgut Khanty has indirective-secundative alignment alternation. In contrast to Kazym Khanty, locative or instrumental/comitative case is not used to mark the secundative theme. Instead, it is marked with instructive-final case, as 280b illustrates.

(280) (Sosa 2017: p.118)

a. λüw mantem kat quλə-ɣən məj.
(s)he.NOM I.DAT two fish-DU.[ACC] give.PST.3SG
'She gave two fish to me.'

b. λüw mant kat quλə-ɣən-at məj.
(s)he.NOM I.ACC two fish-DU-INSFIN give.PST.3SG
'She provided me with 2 fish.'

Apart from secundative alignment, instructive-final case has several other usages: (i) it is used as a lexical case, assigned to the complements of some verbs 281, and (ii) it marks direct objects with partitive semantics in monotransitive clauses 282b.

(281) jəmnam mantemat λăɣλəqsə-λ.
in.vain 1SG.INSFIN wait-NPST.[3SG]
'She waits for me in vain.' (Sosa 2017: p.43, ex.3.11.c)

[51] There exist several research traditions among studies of Eastern Khanty dialects, which single out different number of morphological cases. For instance, Solovar et al. (2016) list 8 cases for Surgut Khanty and (partially overlapping) 8 cases for Vakh Khanty. For an unknown reason, they leave out the instructive-final in the Surgut dialect. Filchenko (2007: p.84-85) provides a unified list of 11 cases for both Eastern Khanty varieties. He has two additional cases compared to Table 5.1: allative (*-pa*) and translative (*-ka*). A detailed discussion of the Eastern Khanty case system is beyond the scope of this study, since these differences do not play a role in the current discussion.

(282) a. ma čaj λăŋq-λ-əm.
1.SG tea.[ACC] want-NPST.1SG
'I like tea (e.g. not coffee).' (Sosa 2017: p.44, ex.3.12b)

b. ma čaj-at λăŋq-λ-əm.
1.SG tea-INSFIN want-NPST.1SG
'I want to have some tea.' (Sosa 2017: p.44, ex.3.12c)

Note that secundative alignment in Surgut Khanty seems to show the same restriction on the size of the oblique theme: Sosa (2017) reports that themes with possessive suffixes are ungrammatical in secundative alignment. However, this is not a (selectional) property of the instructive-final case. As 281 shows, personal pronouns can be marked with instructive-final case in other contexts.

Crucially, secundative themes again bear the most marked case in the hierarchy. I assume the case hierarchy in 283 for Surgut Khanty. The hierarchy is illustrated with the declension paradigm for a personal pronoun (1SG). This provides a morphological motivation for some parts of the hierarchy. In this hierarchy, instructive-final is the most marked case.[52] Note that locative or instrumental/comitative case is not used to mark a theme argument in secundative alignment, as might have been expected if secundative theme were assigned case by a functional projection.

(283) Case hierarchy in Surgut (Eastern) Khanty[53]

NOM >	LOC >	(B)ACC >	DAT >	LAT >		
ma	manə	mant	mantem	mantema		
ABL /	COMIT /	APPROX /	TRANS /	ABESS	>	INSFIN
mantemi	mantemnat	mantemnam	mantemɣə	mantemλəɣ		mantemat

Based on the language-specific case hierarchies for Kazym Khanty (Section 5.1) and Surgut Khanty (in 283), I argue that the secundative theme bears the most marked case available in every particular language. In Kazym Khanty, the most marked case is locative. Surgut Khanty has a richer case inventory, and the most marked case is instructive-final.

5.5 Complications for the Dependent Case theory

The current analysis of indirective-secundative alternation is developed under the assumption that nominals receive case under Agree with a functional head that is able to assign a specific case value. There exists, however, an important alternative approach to

[52] Caha (2007) argues that partitive case (in Russian) is located between genitive and dative. However, it is not certain whether instructive-final in Eastern Khanty should be treated as a true partitive case.

[53] Somewhat unorthodox ordering – LOC before ACC – is motivated morphologically. Since all oblique cases, except LOC, are based on the ACC form *man*-**t**, ACC should be adjacent to DAT and other oblique cases. Irimia (2023: p.38, ex. 73) argues for an enriched case hierarchy with two entries for accusative: the unmarked SACC at the left edge and a marked BACC that can follow some inherent cases. To avoid the conflict between this hierarchy and the original hierarchy in Caha (2007, 2009), I assume that ACC on pronouns in Surgut Khanty corresponds to BACC.

case assignment – Dependent Case Theory (Baker & Vinokurova 2010; Baker 2015). This approach to Case is adopted in many studies, (e.g. McFadden & Sundaresan 2011; Levin & Preminger 2015; Yuan 2018, 2021, 2022; Anagnostopoulou & Sevdali 2020 etc.). Therefore, it is important to discuss why the analysis of indirective-secundative alternation proposed in this thesis is not based on the Dependent Case Theory.

The Dependent Case Theory can easily predict indirective alignment (DAT-ACC); in fact, it was designed to derive this alignment. The dependent case rules in 284 state that the higher of the two arguments in the lower phase (VP=VoiceP in the notation used in this thesis) receives dative case.

(284) Dependent Case (Baker & Vinokurova 2010)

a. If there are two distinct argumental NPs in the same VP-phase such that NP_1 c-commands NP_2, then value the case feature of NP_1 as dative unless NP_2 has already been marked for case.

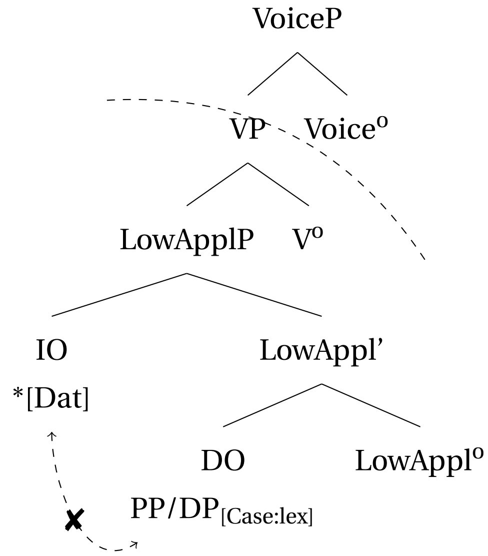
b. If there are two distinct argumental NPs in the same phase such that NP_1 c-commands NP_2, then value the case feature of NP_2 as accusative unless NP_1 has already been marked for case.

Secundative alignment is much more problematic for a derivation with dependent case rules. In fact, it cannot be derived without serious modifications that can potentially compromise the original proposal.

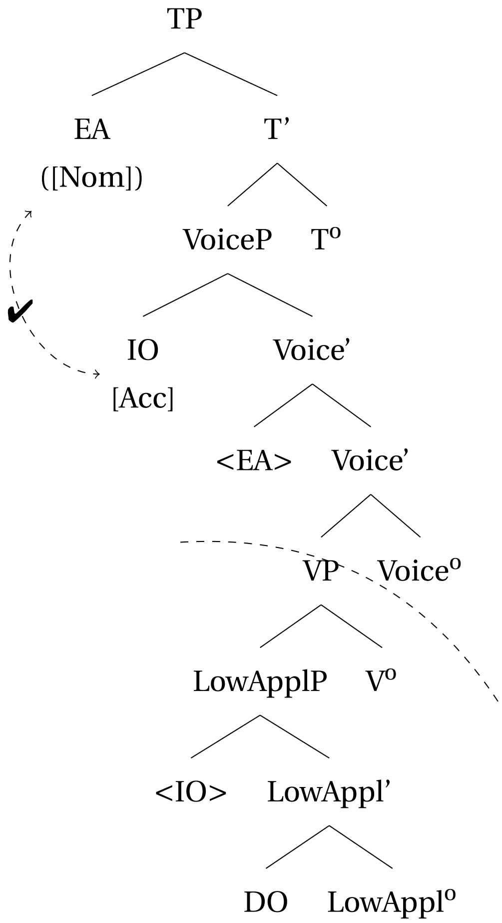
The dependent rules in 264 can predict ACC-marking on the IO in two scenarios. First, if the theme is already marked for case, i.e. if oblique marking of the theme is a lexical case or a PP. As shown in 285a, the rule for dependent dative does not apply in this configuration, and the IO remains caseless, i.e. available for accusative assignment in the higher phase. Second, if the promoted IO raises out of the VoiceP, as in 285b, it escapes the dative-assigning configuration in 284a and enters an accusative-assigning configuration in 284b.

(285) IO promotion accusative under Dependent Case rule

a.



b.

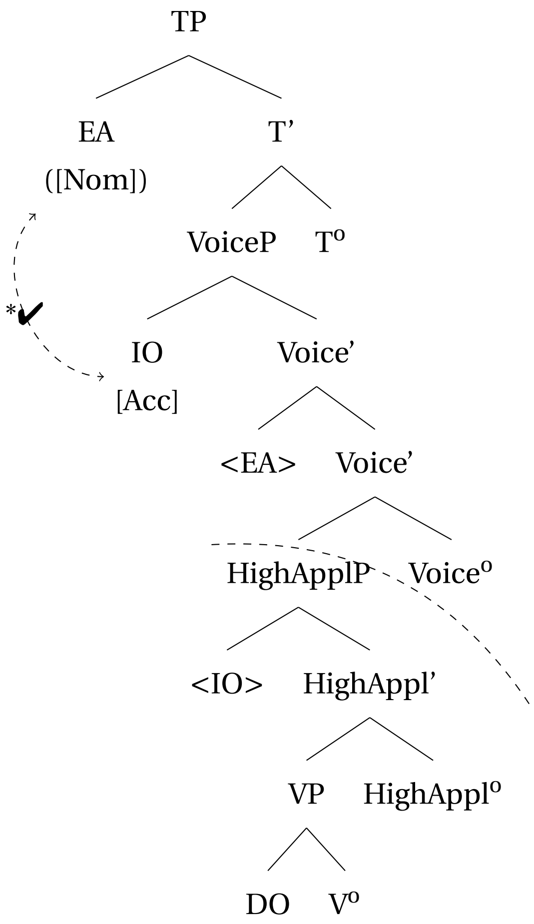


However, the oblique marking of the theme is highly problematic for the Dependent Case Theory. As discussed in Chapter 4, Section 4.3, oblique themes are neither PPs nor marked with a lexical case, which rules out the derivation in 285a. The configuration in 285b predicts that DO, as a single argument in the lower phase, must remain caseless, i.e. be marked with nominative. However, secundative themes are marked with an oblique case. Hence, an additional rule should be introduced that regulates the assignment of oblique. This rule must state that an NP, located inside a VP-phase and c-commanded by an ACC-marked DP in the next phase is assigned the most marked oblique case.

This modification has severe conceptual downsides. The additional rule for the dependent oblique case differs from the two original rules in that it makes an explicit reference to the size of an argument. The two original rules do not distinguish between DPs and NPs. In addition, the dependent oblique would be assigned across the phase boundary, which is again very different from the other two dependent case rules.

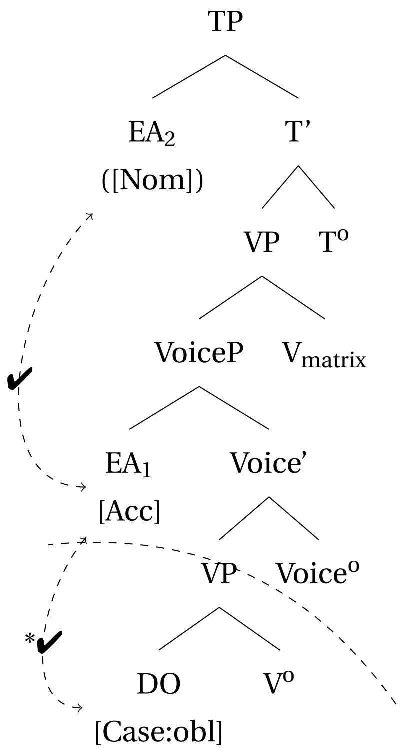
Furthermore, this new rule overgenerates by making two incorrect predictions. First, the additional rule does not capture the asymmetry between high and low applied arguments (see discussion in Chapter 3) and predicts that high applied arguments must also alternate between dative and accusative. The derivation in 286 is minimally different from the low-applicative scenario and is falsely predicted by the theory.

(286) Falsely predicted secundative alignment in HighApplP



The second incorrect prediction states that the oblique dependent case can emerge in Voice-restructuring contexts (e.g. Wurmbrand 2014). When the embedded subject is marked with accusative, it creates an environment for the direct object to be assigned the dependent oblique if it has not yet been marked with dependent accusative. This scenario is illustrated in the abstract derivation in 287, but seems to be unattested crosslinguistically.

(287) Falsely predicted secundative alignment in Voice-restructuring



At this stage of research, I do not see how Dependent Case Theory can be modified to account for the oblique marking of the secundative theme and avoid the problems mentioned above. It can be concluded that secundative alignment provides a counterargument to this approach to case assignment. Further problems for Dependent Case Theory are discussed by e.g. Bárány & Sheehan (2021) and Šereikaitė (2020).

Chapter 6

Typological parallels

Indirective-secundative alignment alternation is not unique to Kazym Khanty. It is present in all Ob-Ugric languages and their dialects: in the Obdorsk dialect of Northern Khanty, in Eastern (Surgut) Kahnty, and in Northern and Eastern Mansi (Nikolaeva 1999, 2001; Bíró & Sipőcz 2017; Virtanen 2012; Sosa 2017; Virtanen & Sosa 2018, etc.). Beyond Ob-Ugric, the indirective-secundative alternation is found in some other Uralic languages, e.g. Tundra Nenets (Nikolaeva 2014), and in a broader typological perspective, e.g. in Inuit languages (Fortescue 1984; Johns 1984; Baker 1988). An exhaustive typological review of indirective-secundative alternation and its properties across different languages is beyond the scope of the present study. However, in this chapter, I look beyond Kazym Khanty and test whether the proposed analysis holds for languages with different grammatical properties, e.g. in languages with ergative alignment. Another goal of this chapter is to distinguish indirective-secundative alternation from other linguistic phenomena, associated with oblique marking or demotion of the internal argument (theme).

This chapter addresses three major topics that could be seen as potential complications for the case overspecification hypothesis formulated in Chapter 5. The first two sections are organized as case studies of two individual languages.

The first section discusses Kalaallisut (aka. West Greenlandic, Inuit). As a classical example of an ergative language, Kalaallisut is an important parallel to Kazym Khanty. Both languages make use of indirective-secundative alternation. Therefore, Kalaallisut allows to test the analysis of the indirective-secundative alternation that is proposed in this thesis.

Furthermore, Kalaallisut data provide evidence that antipassive and secundative alignment are two distinct phenomena, despite some superficial similarities, e.g. the oblique case marking of the object.

The second section concerns with Eastern Mansi, an Ob-Ugric language that has both indirective-secundative alternation and differential object marking. Differential object marking (DOM) is often analyzed as the presence vs. absence of a Case feature on two types of direct objects. Unmarked direct objects are often structurally reduced, similar to the theme in secundative alignment. One influential approach to the DOM analyzes unmarked objects as lacking syntactic case assignment. This again parallels the analysis advocated in this thesis for the secundative themes. Importantly, however, unmarked objects in DOM languages often lack case marking completely, while the case on secundative themes is overspecified. Eastern Mansi provides a necessary ground for discussion of the existing analyses of DOM and how DOM can be reconciled with inherent case overspecification hypothesis and secundative alignment.

The third section contains a brief discussion of "applicative constructions" in the sense of Baker (1988), where case alternation on the applied argument and the theme is accompanied by the presence or absence of a morphological reflex on the verb. There are three crucial differences from indirective-secundative alternation in Ob-Ugric and Inuit: (i) the presence of an additional morphological marker on the verb in clauses with "promoted" applied arguments, (ii) the absence of the DP-restriction on the instrumental themes, and (iii) the fact that this construction is restricted to the closed lexical group of verbs. These differences allow to argue that "applicative constructions" in German, Russian and Hungarian are distinct from indirective-secundative alternation in Khanty and beyond.

6.1 Kalaallisut (Inuit)

Kalaallisut (aka. West Greenlandic, Inuit) is an important parallel to indirective-secundative alternation in Khanty. First, it is an unrelated language,[54] which suggests that indirective-secundative alternation should not be modelled in a language-specific way. Second, Kalaallisut is an ergative language, which requires a successful analysis of indirective-secundative alternation to be independent of the nominative or ergative alignment of the language. As will be shown below, the approach to the indirective-secundative alternation proposed in the previous sections is indeed independent from the overall alignment of transitive clauses, i.e. it is able to capture both Khanty and Kalaallisut data with minimal adjustments. Third, Kalaallisut has an antipassive construction. Many properties of secundative themes have also been observed on antipassive objects, including, importantly, oblique case marking (e.g. Polinsky 2017 for an overview). However, this sections shows that the antipassive in Kalaallisut cannot be unified with secundative alignment and requires a different analysis (e.g., the one proposed by Yuan 2018).

Kalaallisut (West Greenlandic Inuit) is a polysynthetic language with free word order (Fortescue 1984). The pragmatically neutral order is SOV. Case alignment in Kalaallisut is ergative-absolutive. Overall, there are eight morphological cases, as listed in Table 6.1. All cases are marked with suffixes on the head noun and freestanding postnominal modifiers, as shown in 288, where both the head noun 'student' and the postnominal universal quantifier are marked with the modalis case. Freestanding postnominal modifiers also undergo number concord with the head noun, as 289 illustrates.

[54]The Eskimo-Uralic hypothesis about a genetic relation between Uralic and Eskimo (aka. Inuit) languages (e.g. Fortescue 1998, 2016; Janhunen 2023) lacks sufficient evidence. Furthermore, even if this hypothesis is correct, differences between the syntax of Inuit and Uralic languages are too significant to argue that indirective-secundative alignment alternation can be a feature inherited from a common ancestor.

	SG	PL
absolutive	(-q/t/k/∅)	-(i)t
ergative	-(u)p	-(i)t
locative	-mi	-ni
allative (= dative)	-mut	-nut
ablative	-mit	-nit
modalis (= comitative/instrumental)	-mik	-nik
prosecutive (= vialis)	-kkut	-tigut
equative (= similaris)	-tut	-tut

Table 6.1: Case inventory of Kalaallisut(after Fortescue 1984: p.206; case labels adjusted)

(288) [atuartu-**nik** tama-**nik**] uqaluqatigi-nnig-sima-vu-q
[student-PL.MOD all-PL.MOD] talk.with-AP-PERF-INTR.IND-3SG.S
'He has talked with all the students.' (Manlove 2015: p.334, ex.27, glosses adjusted)

(289) qimmi-t qaqurtu-t marluk taakku
dog-PL white-PL two.PL those.PL
'those two white dogs' (Fortescue 1984: p.118, glosses adjusted)

Finite verbs can have multiple suffixes, spelling out modality, aspect, and tense, as well as ϕ-agreement in person and number. An example of a finite verbal form is provided in 290. The agreement exponents cross-reference the subject (ERG or ABS) and the direct object (ABS) if the clause is transitive (Yuan 2021, 2022).

(290) Schema of Inuit verb complex (Yuan 2022: p.514, ex.2)

a. puiur-sinnaa-sima-ssa-vaa
forget-can-PERF-FUT-INTER.3SG.S>3SG.O
'Who could ever forget it (the great plain)?' (Fortescue 1984: p.194)

b. [[[[[Verb] Modal] Asp] Tense] Mood.Agr]

As argued by Compton (2016) and Yuan (2021), the ϕ-probe in Inuit is located in the CP(MoodP)-domain rather than on Voice^{o} or T^{o}. Furthermore, Yuan (2021, 2022) argues that absolutive-marked arguments obligatorily raise from their base-generation positions to CP. Evidence for this assumption come from the fact that absolutive DPs obligatorily take scope above negation, as shown in 291.

(291) a. Atuagaq ataasiq tikis-sima-nngi-laq.
book.ABS one.ABS come-PERF-NEG-3SG.S
'There is one (particular) book that hasn't arrived.' (ABS subj.)

Available reading: ∃ > NEG; *NEG > ∃

b. Suli Juuna-p atuagaq ataasiq tigu-sima-nngi-laa.
still Juuna-ERG book.ABS one.ABS get-PERF-NEG-3SG.S>3SG.O
'There is one (particular) book that Juuna hasn't received yet.' (ABS obj.)

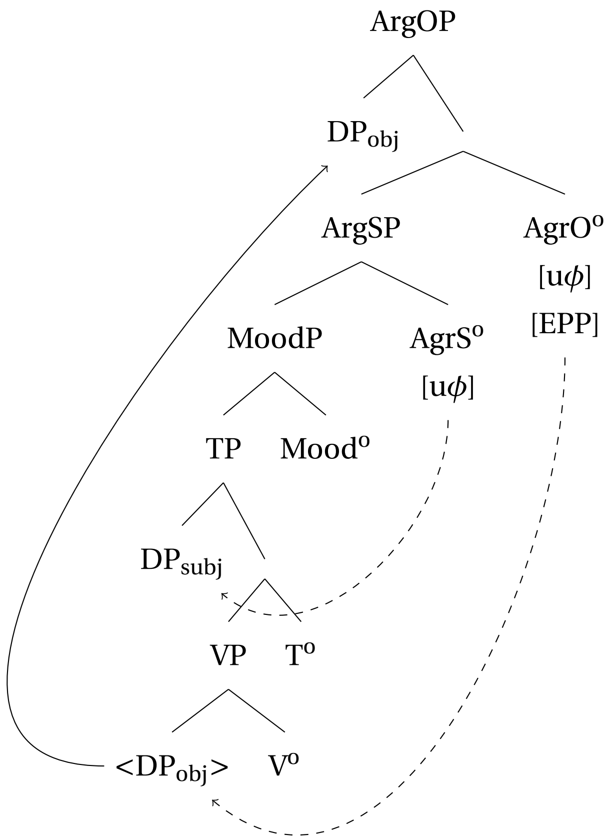
Available reading: ∃ > NEG; *NEG > ∃

(Yuan 2021: p.160, ex.8)

Yuan (2022) proposes the following analysis of agreement and case assignment in

Inuit, particularly in Kalaallisut. As illustrated in 292, MoodP has two functional extended projections, ArgS, controlling subject agreement, and ArgO, controlling agreement with an ABS-marked object. If object agreement is successful, DP_{obj} is attracted to SpecArgOP by the EPP feature.

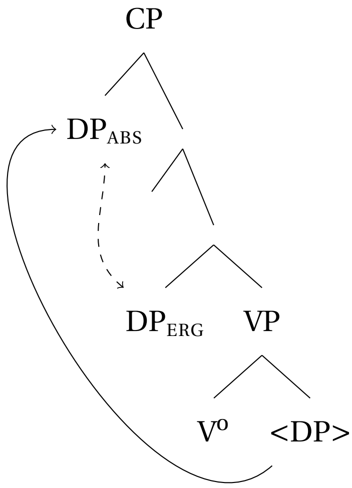
(292)



(Yuan 2022: p.516, ex.4, combined with p.523, ex. 15)

For case assignment, Yuan (2022) employs a Dependent Case mechanism, where a c-commanded argument (transitive subject) receives the ergative case and the higher argument (transitive object) receives the absolutive case 293.

(293) (Yuan 2018: p.103, ex. 2a)



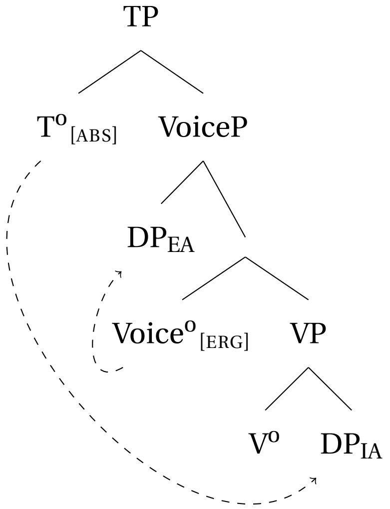
As illustrated in 293, Yuan (2018) analyzes ergative as a dependent case, assigned in the higher phase to a c-commanded DP (similar to accusative in Baker & Vinokurova 2010). She discusses and rules out alternative analyses of ergative in favor of the Dependent Case approach. All three main alternatives are formulated under a general approach to Case, where it is assigned under Agree by a functional or lexical head. In the first analysis, ERG is treated as an inherent case, assigned by Voice^{o} (v^{o}) (see Woolford 2006b; Aldridge 2008; Legate 2008, 2012; Mahajan 2012, etc.). In the second approach, ERG is a structural case assigned by T^{o} (Bobaljik 1993; Pittman 2005). Finally, in the third approach, the external argument is treated as a possessor in a nominalized transitive clause. Ergative is therefore genitive, assigned to the possessor of a nominalization (Johns 1987, 1992; Bok-Bennema 1991; Yuan 2013). For a detailed discussion of each of these approaches, the reader is referred to Yuan (2018).[55]

The analysis of Kazym Khanty indirective-secundative alternation in Chapters 3 and 4 is formulated in terms of a Case-under-Agree approach. Therefore, I concentrate now on the theories ruled out by Yuan (2018) and argue that at least one of them can be modified to account for the Inuit data.

In studies that treat ERG as an inherent case, it is assumed that the transitive Voice^{o} assigns case to its specifier. As 294 shows, it allows T^{o} to probe across SpecVoiceP and assign absolutive to the internal argument.

[55]A general overview of the existing approaches to ergativity can be found in Deal (2015a).

(294) Inherent case assignment (after Yuan 2018: p.105, ex.3)



However, this analysis can only account for a part of the Inuit data. Yuan (2018) provides examples where ergative case is present in applicativized unaccusative clauses. The single argument of the unaccusative clause in 295a is marked with ABS, as expected. When the clause is applicativized, an additional applied argument is merged in SpecHighApplP. As shown in 295b, this additional argument is marked with ERG, similar to the external argument in a transitive clause. However, the analysis in 294 presupposes that only a transitive flavor of Voice° can assign ERG. Since the applied argument is not merged in SpecVoiceP, an intransitive flavor of Voice° must be merged in 295b, and the presence of the ergative case value is unexpected.

(295) Inuktitut: ERG on unaccusative subject (Yuan 2018: p.106, ex.5)

a. Baseline:

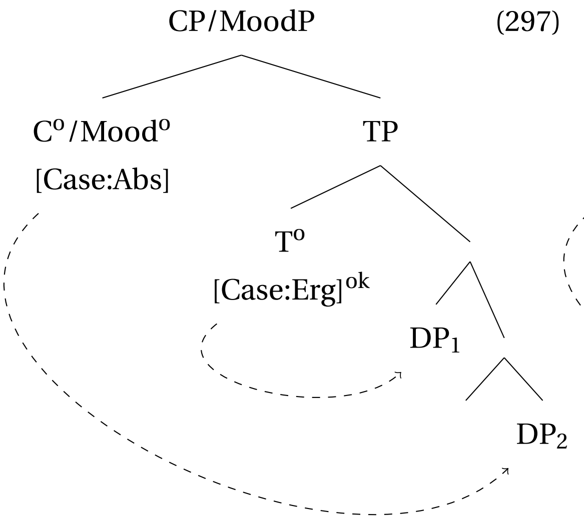
Jiisusi tuqu-lauq-tuq
Jesus.ABS die-PST-3SG.S
'Jesus died.'

b. Applicative of unaccusative:

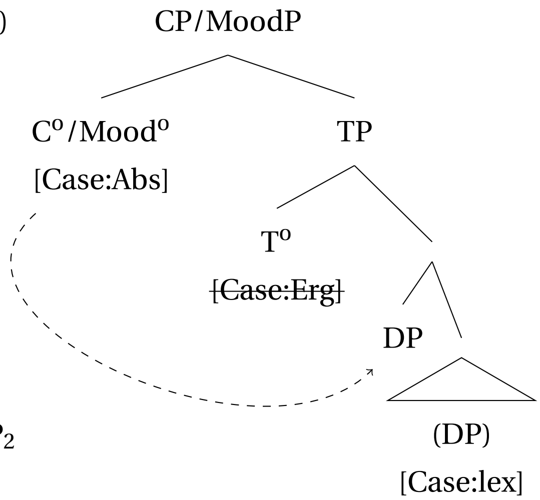
Jiisusi-up tuqu-jjutigi-lauq-taatigut
Jesus-ERG die-APPL-TR-PST-3SG.S>1PL.O
'Jesus died for us.'

The analysis by Bobaljik (1993) and Pittman (2005) can avoid this problem while still maintaining the baseline that case is assigned under Agree. They argue that ergative is a structural case assigned by T°. I additionally suggest that the presence of the egrative-assigning probe on T° is sensitive to the number of caseless arguments in its c-command domain. If there is only one caseless DP (the internal or external argument), the ergative probe is absent, as shown in 297. In clauses with two caseless DP-arguments in the maximal projection of VoiceP, as in 296, the ergative Case probe is added and assigns case to the highest DP. For present purposes, I assume that the optional ergative probe is located on T° and the obligatory absolutive probe is located on C/Mod°.

(296)



(297)



A similar proposal has been made for for ϕ-agreement by Kalin (2018) and Nie (2020), who argue that a last resort ϕ-probe can be activated (added) in the derivation for licensing purposes.[56]

In this model, the ergative probe is present on T° only if there are two caseless DPs in VoiceP. It accounts for both canonical transitive clauses with agent DP in SpecVoiceP and applicativized unaccusative clauses, as in 295b. The correctness and details of this analysis require further research. In particular, the exact location of the ergative assigner in the clausal structure requires further investigation. Importantly, however, case assignment in Kalaallisut can be accounted for in a Case-under-Agree model.

Another aspect of Kalaallisut grammar important for the subseuqnt discussion is the antipassive construction. Antipassive is a theme-demotion operation available in transitive clauses. An example of a antipassive construction is provided in 298. The direct object is marked with the oblique modalis case, and the whole clause is detransitivized, i.e. the external argument is marked with absolutive, as in intransitive clauses.

(298) Suli Juuna atuakka-mik ataatsi-mik tigu-si-sima-nngi-laq.
still Juuna.ABS book-MOD one-MOD get-AP-PERF-NEG-3SG.S
'Juuna hasn't received (even) one book yet.' (MOD obj.)

Available reading: NEG > ∃; *∃ > NEG

(Yuan 2021: p.160, ex.8)

As 298 shows, the oblique object in an antipassive clause obliqatorily takes a narrow scope with respect to negation. Yuan (2021) takes this as evidence that the antipassive object stays in situ. She argues that an additional aP-projection (for antipassive) is merged inside the VoiceP-domain, as illustrated in 299. The a°-head is spelled out with antipassive morphology. It merges on top of an argument-introducing projection (VP) and licenses

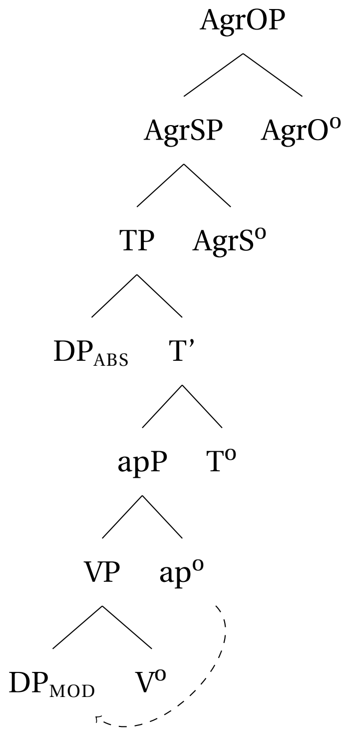
[56]Compare also Deal (2010) who treats ergative as a product of two instances of ϕ-agreement on the verb. If the verb agrees twice, the two bundles of ϕ-features are realized as the ergative marker on the second agreement goal.

the closest nominal by assignment of MOD/INSTR (Yuan 2018). Since it is already licensed for case, the antipassive object cannot agree with CP/MoodP and is forced to stay in situ.[57]

(299) Derivation of antipassive construction (Yuan 2018: p.217, ex.71)

a. Taiviti surak-si-juq igalaar-mik
Taiviti.ABS break-AP-3SG.S window-MOD
'David broke the window.'

b.



Subsection 6.1.1 discusses indirective-secundative alternation in Kalaallisut. The analysis developed for Kazym Khanty in Chapter 3 extends to Kalaallisut data straightforwardly, with some minor modifications. Kalaallisut provides supporting morphological data for HighApplº being the source of dative/allative case.

Subsection 6.1.2 provides supporting evidence for the analysis of secundative alignment in Chapter 4. Inuit is argued to be similar to Kazym Khanty in that the secundative theme is not case-licensed in syntax and bears the oblique default case. First, Inuit is shown to have the DP-restriction on secundative themes, similar to Khanty. Second, the differences between oblique objects in antipassive clauses and secundative themes are discussed. Despite their identical case morphology (instrumental/modalis), the source of case marking is different. Yuan (2018) argues that antipassive objects are licensed by the functional projection aP (antipassive) in the clausal spine. The antipassive object in Inuit can be a full DP, which is consistent with the analysis, which includes syntactic case-licensing. In contrast, secundative themes are structurally reduced and marked with a default case.

Finally, Section 6.1.3 discusses another environment where the marked default case is

[57] In subsequent work, Yuan (2022) suggests a different analysis and argues that MOD can be modeled with the Dependent Case mechanism. Therefore, Modalis is the unmarked case in the *v*P-internal domain.

found. Stranded modifiers under noun incorporation are marked with instrucmental/modalis, similar to secundative themes. I argue that this is another context where the marked default case is used.

6.1.1 Indirective alignment and High Applicative Phrase

Kalaallisut, as other Inuit languages, exhibits an alternation between indirective and secundative alignment, as illustrated in 300. In 300a, the low applied argument (recipient, IO) is marked with the allative/dative case, and the theme is marked with absolutive. In secundative alignment in 300b, the IO is marked with absolutive, while the theme has oblique modalis marking.

(300) a. anguti-p aqerluusaq meeqqa-mut tuni-**up**-paa
man-ERG pencil.[ABS] child-ALL give-**uti**-3SG.S>3SG.O
'The man gave the pencil to the child.'

b. anguti-p aqerluusa-mik meeraqq tuni-paa
man-ERG pencil-MOD child.[ABS] give-3SG.S>3SG.O
'The man gave the pencil to the child.'

(Johns 1984: p. 167, ex. 9, glosses adapted)

Note that 300b is a proper secundative alignment and cannot be analyzed as an antipassive construction. A proper antipassive clause is always marked with the suffix *-si-*, as in 301. In 300b, however, *-si-* is absent. Hence, 300b is a proper secundative alignment clause.

(301) qimmit marluk arna-nik pingasu-nik kii-**si**-pput
dog.PL.ABS two.ABS woman-PL.MOD three-MOD bite-AP-3PL.S
'Two dogs bit three women.' (Yuan 2018: p.45, ex.45)

Crucially, the indirective-secundative alternation in 300 provides additional evidence that allative/dative is assigned by a separate functional projection in the clausal spine. There appears an additional morphological marker (*-uti-*) in indirective alignment, in 300a. The projection spelled out by *-uti-* is distinct from the argument-introducing one (LowApplP), since the marker is absent in secundative alignment 300b, even though the recipient, and therefore LowApplP, is present.[58]

High applicative clauses, such as a clause with a benefactor argument in 302, provide crucial morphological evidence for HighAppl^{o} as the source of the allative/dative case. As discussed by Johns (1984), indirective-secundative alternation in Kalaallisut is present with low applicatives, but not with high applicatives. Recall that high applicatives in Khanty also do not alternate between the two alignments either. In contrast to Khanty, where high applicative clauses are obligatorily indirective, high applicative clauses in Kalaallisut are obligatorily secundative, as in 302.

[58] The indirective-secundative alternation is present an all Inuit languages. However, Kalaallisut differs from the rest of the Inuit varieties, since it marks the high applicative projection with the overt morphological marker *-uti-*. Other Inuit varieties use a silent allomorph in both indirective alignment and high applicative clauses (Johns 1984). Therefore, Kalaallisut provides crucial morphological evidence in favor of the current analysis of indirective-secundative alternation.

(302) anguti-p meeraq aderluusa-mik pini-**up**-paa.
man-ERG child.[ABS] pencil-MOD hunt-**uti**-3SG.S>3SG.O
'The man is looking for a pencil for the child.' (Johns 1984: p.168, ex. 11b, glosses adapted)

Note that *-uti-* is present in obligatory secundative high applicative clauses and in indirective alignment in low applicative clauses.[59] Unless the morphological marker *-uti-* spells out HighAppl°, its presence in high applicative clauses is hard to motivate.[60] If *-uti-* indeed spells out HighAppl°, its presence in the indirective alignment provides further support for (i) the presence of an additional functional projection, assigning allative/dative to the low applied argument, and (ii) for the analysis where this head is HighAppl°. Hence, indirective-secundative alternation in Kalaallisut is derived in the same manner as the similar alternation in Kazym Khanty (cf. Chapters 3 and 4).

In contrast to Khanty, Kalaallisut is an ergative language. Crucially, indirective-secundative alternation is derived in the same manner in languages with nominative and ergative transitive alignment. As demonstrated in the derivations below, the assignment of ALL/DAT by HighAppl° is completely independent of the assignment of other syntactic cases. In 303a, HighAppl° assigns a case to the recipient in SpecLowApplP and T° assigns a case to the external argument. I abstract away from the potential movement of the ergative argument outside of VoiceP and assume that it is assigned case in situ. As a result, both higher arguments are case-licensed, and C°/Mood° can reach to the lowest argument, attract the theme to SpecCP/MoodP, and assign ABS to it. In 303b, T° assigns case to the external argument, but HighApplP is absent. Therefore, when C°/Mood° probes, the applied object is the closest argument with an unchecked Case. C°/Mood° successfully agrees with the applied object, attracts it to SpecCP/MoodP, and assigns ABS to it. The Case feature on the theme stays unchecked and spells out the default case value, similarly to Khanty.

[59] Despite Johns (1984)'s generalization, indirective-secundative alternation does not always correspond to the presence vs. absence of the *-uti-* morpheme. E.g.:

(i) [Fortescue 1984: p. 88, glosses adapted]

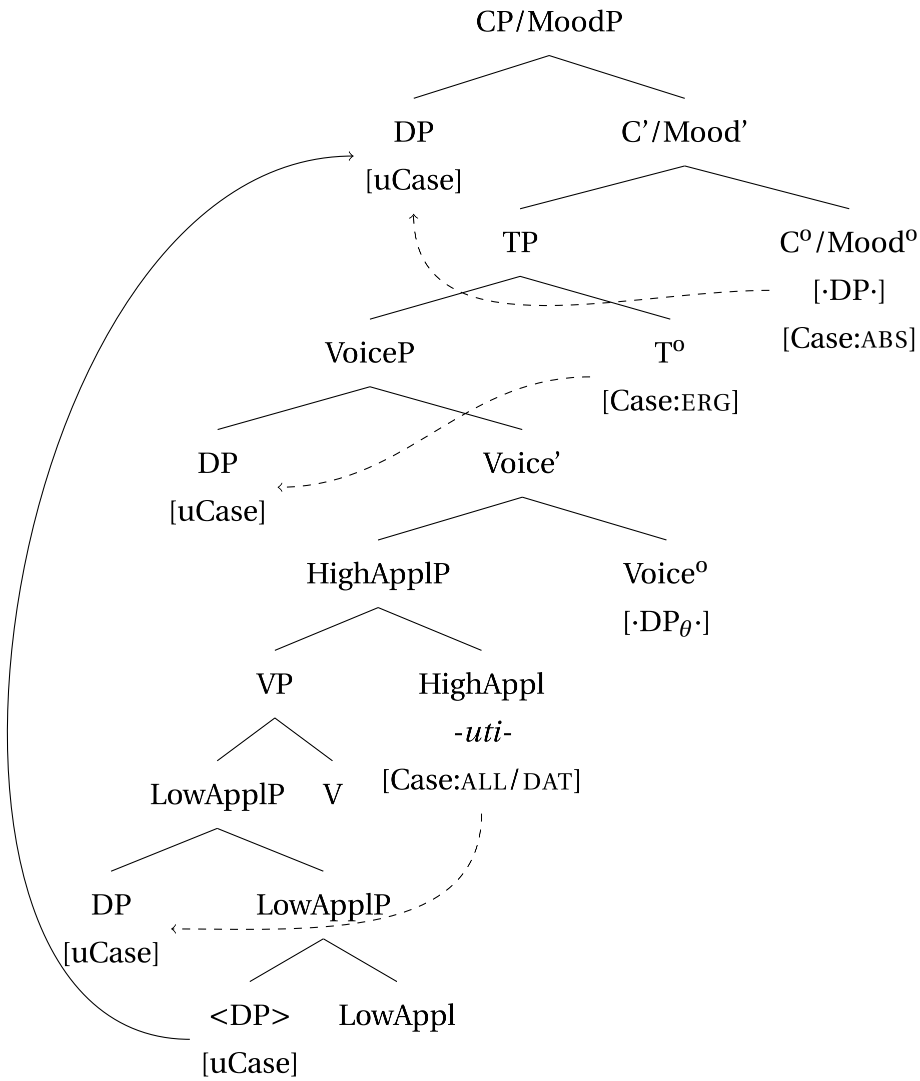
a. niviarsia-mut uqar-put
girl-ALL say-IND.3PL.S
'The said (smth.) to the girl.'

b. niviarsiaq uqarvig-aat
girl.[ABS] tell.to-IND.3PL.S>3SG.O
'They told (it to) the girl.'

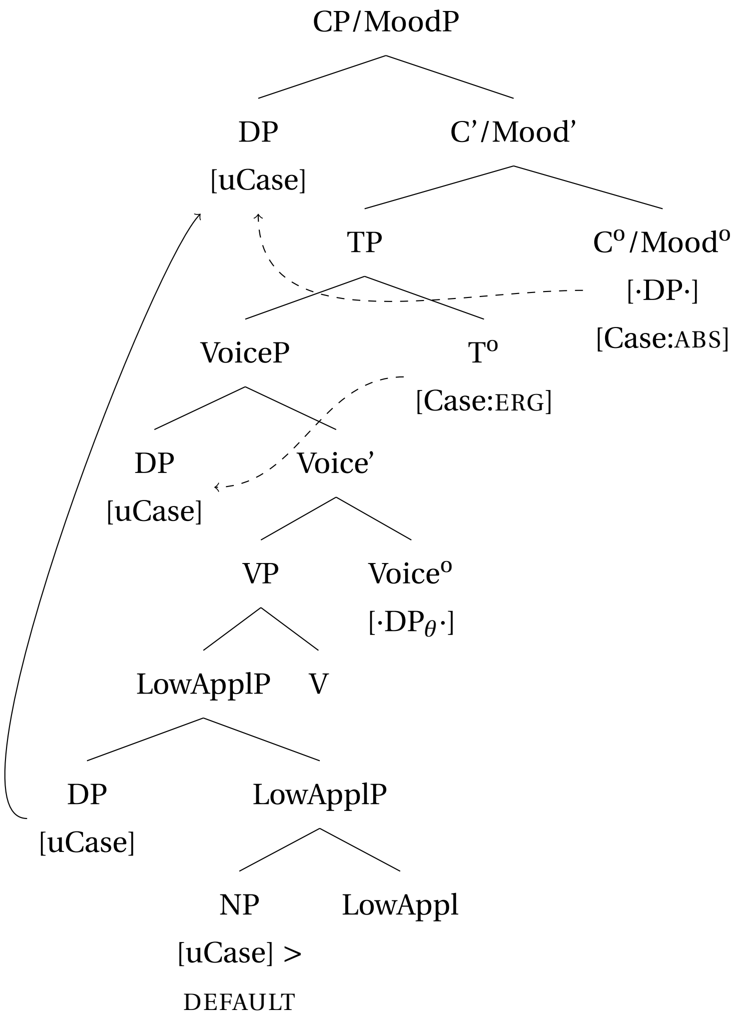
Nevertheless, the verb is not identical in these two sentences. Sadock (2003) glosses *-vig(E)-* as "do.at" or locative applicative. Fortescue (1984: p. 92) treats *-(v)vigi* as a 'transitivizer' and provides a list of derived verbs with locative applicative semantics and ABS-marked objects. If *-vig(E)-* in (ib) is another allomorph of HighAppl and 'girl' is introduced in its Spec, there are no counterarguments to Johns's (1984) generalization.

[60] Compare also an analysis by Baker 1988, who suggests high applied arguments are merged as PPs and *-uti-* is an incorporated P.

(303) a. indirective alignment



b. secundative alignment

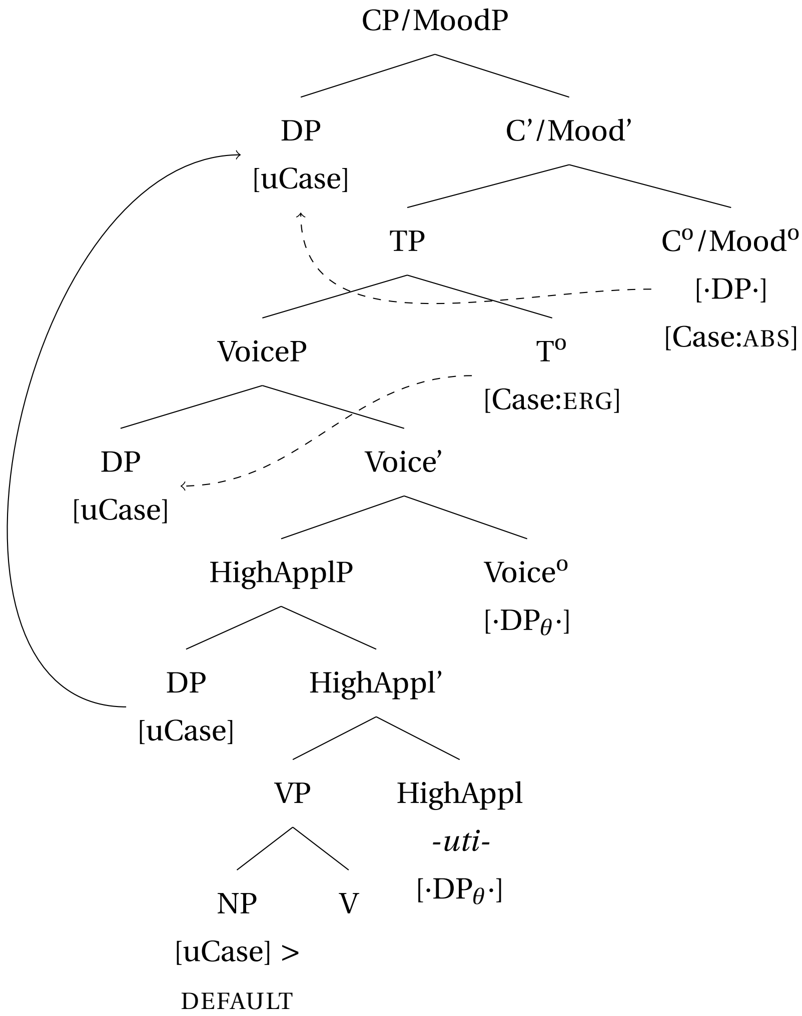


There is one important difference between Kazym Khanty and Kalaallisut (and Inuit, in general). In Khanty, high applied arguments are marked with DAT (=ALL), whereas in Kalallisut, they are obligatorily marked with ABS. This difference can be easily accounted for. As discussed in Chapter 3, HighAppl° can have different flavors – thematic and a-thematic. In Khanty, both flavors have a Case probe and differ only by the DP-selecting/-attracting feature. While in Khanty [·DP$_\theta$·] and [Case:Dat] can co-occur on the head, they are in complementary distribution in Kalaallisut. The thematic flavor of HighAppl° in 304a has only the argument-selecting [·DP$_\theta$·]-feature and lacks the Case probe [Case:ALL/DAT]. Therefore, there are three caseless arguments in the clause and only two case-checking heads. T° checks case on the external argument, and the C°/Mood° agrees with the next highest caseless argument, i.e. the applied object in SpecHighApplP. This correctly predicts that whenever HigAppl° introduces a DP, the lower argument is not case-licensed.

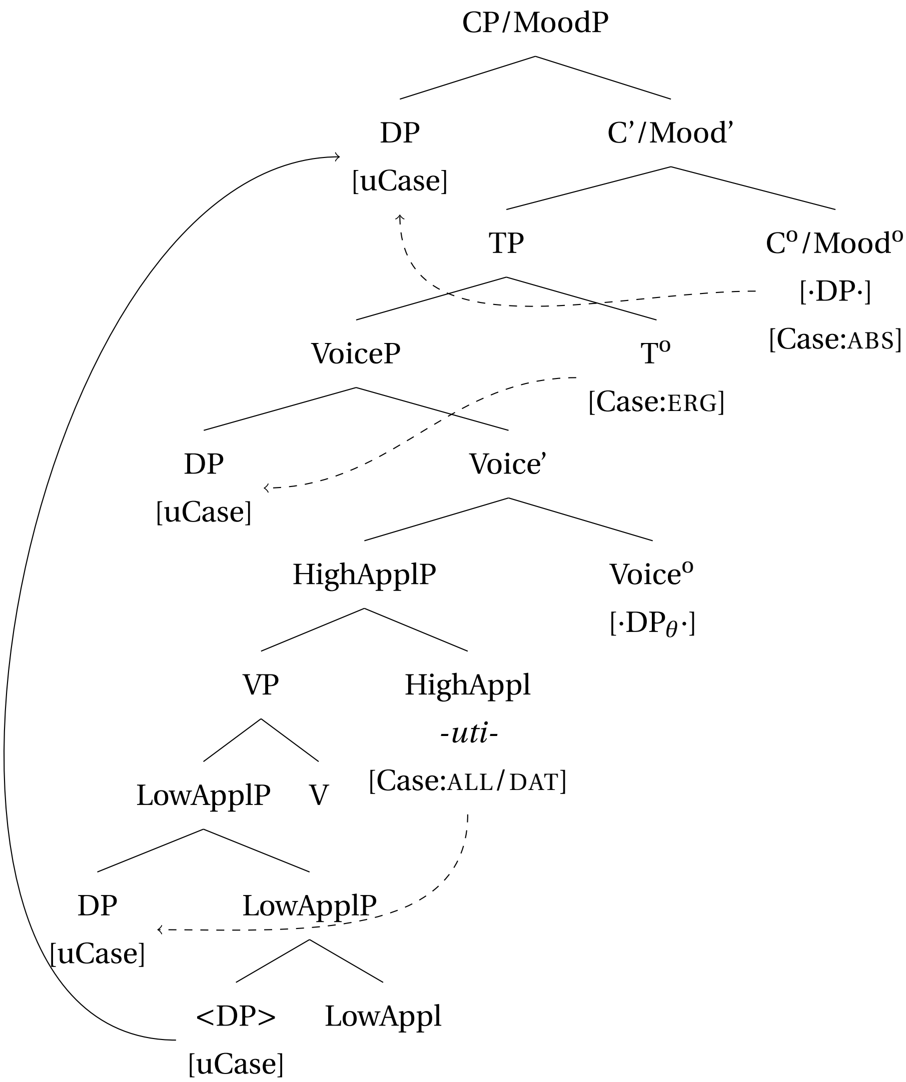
The a-thematic flavor of HighAppl° has only the Case probe [Case:ALL/DAT] and lacks the [·DP·]-feature. It checks case on the closest argument in its c-command domain, i.e. on the IO (recipient), as illustrated in 304b. After T° has agreed with the external argument, the probe on C°/Mood° can reach the theme DP.

(304) Two flavors of HighAppl° in Kalaallisut

a. HighAppl° [·DP$_\theta$·], *[Case:ALL/DAT]



b. HighAppl^{o} [Case:ALL/DAT], *[·DP$_{(\theta)}$·]



c. HighAppl^{o} [·DP$_{(\theta)}$·]+[Case:ALL/DAT] – unattested in Inuit (but present in Uralic languages)

The third possible flavor of HighAppl^{o}, where both [·DP$_{(\theta)}$·]- and [Case:ALL/DAT]-features are present, is not available in Inuit languages. In contrast, it is present in Khanty and other Uralic languages, as discussed in detail in Chapter 3.

Crucially, Kalaallisut provides a useful parallel to Khanty, showing that indirective-secundative alternation is independent of the overall ergative or nominative alignment of the language. Furthermore, Kalaallisut adds a morphological piece of evidence for high applicative projection as a source of the dative/allative case and indirective alignment.

6.1.2 Antipassive vs. secundative alignment: MOD/INSTR as default and the DP-restriction

An important part of the current analysis of secundative alignment is the idea of case overspecification and the marked default case. As discussed in Chapters 4-5, the oblique marking of the secundative theme is tightly connected to the size restriction on the theme argument. I have argued that the existence of the DP-restriction supports the analysis where the secundative theme lacks syntactic (case-)licensing. Consequently, the notion of the marked default and inherent case overspecification of arguments has been suggested. Alternative analyses, where oblique marking on the theme is due to a last-resort Case probe/lexical case marking or PP-insertion, cannot capture the DP-restriction properly. Kalaallisut also provides supporting evidence for treating LOC/INSTR (= MOD) on themes as the default case.

The crucial assumption for the analysis is the idea that syntactic (and consequently, morphological) cases are organized in a hierarchical structure. I adopt the strong version of the Case Contiguity Hypothesis (cf. Chapter 5, Section 5.1). Case hierarchy for Khanty is repeated in 305 (following universal hierarchy by Caha 2009, 2024; Bárány 2017, 2021a). Of the four cases, instrumental/locative is the most marked and contains all abstract case features. As expected under the case overspecification hypothesis, when a nominal is not case-checked, all (unchecked) case values can be exceptionally spelled out, and the most marked case exponent (locative) is inserted.

(305) The case sequence for nominative language (adopted from Caha 2009: p.23)

NOM	>	ACC	>	DAT	>	LOC/ INSTR
[A]		[A,B]		[A,B,C]		[A,B,C,(D etc.)]

No studies have been conducted on case hierarchy in Inuit languages. For the current purposes, I adopt the hierarchy in 306 proposed by Irimia (2021) for Gujarati, another ergative-absolutive language.[61]

(306) Case sequence for an ergative language (adopted from Irimia 2021: 41)

ABS	>	ERG	>	LOC	>	ALL	>	ABL	>	MOD
						(DAT)				(INSTR)
[A]		[A,B]		[A,B,C]		[A,B,C,D]		[A,B,C,D,E]		[A,B,C,D,E,F]

In both Khanty and Kalaallisut, the most marked case in the respective hierarchy surfaces on secundative themes. In the remainder of this subsection, I show that the secundative themes in Kalaallisut obey the DP-restriction, similar to Khanty. Additionally, I provide arguments that MOD on the secundative theme is not assigned by an antipassive apP-projection (see analysis by Yuan 2018). Section 6.1.3 discusses another context in which MOD-marking surfaces in the absence of case-licensing.

[61] On a descriptive level, Kalaallisut has eight cases. I remain agnostic as to whether prosecutive and equative are proper cases or cliticized postpositions. The postposition analysis seems plausible because the phonological forms of the exponents for prosecutive and equative differ from those for other oblique cases. The exponents for locative, allative, ablative, and modalis all have a similar structure: -*m*V(C) in the singular and -*n*V(C) in the plural. The exponents for prosecutive (-*kkut*/-*tigut*) and equtive (-*tut*/-*tut*) look radically different. However, note that if these two forms are proper syntactic cases, they should precede modalis in the hierarchy.

I. DP-restriction

The main argument for treating locative/instrumental/modalis as a default case marking, which surfaces in absence of syntactic case checking comes from the fact that secundative themes in Khanty obligatorily lack DP-layer. Kalaallisut is an articleless language, and the DP-projection is not visible on the surface, but a number of studies have argued in favor of its presence on Inuit nominals (Langr 2014a,b; Manlove 2015; Yuan 2013).[62] Unfortunately, secundative alignment has been discussed very little in the literature on Inuit, and there are no reliable data about the size of secundative themes. Nevertheless, indirect data support the conclusion that DP-restriction is not unique to Khanty and holds in Inuit as well.

First, Fortescue (1984) reports that themes in secundative alignment usually have a non-specific or generic interpretation. "...<A>n instrumental (*modalis – AB.*) case object will be less definite than an absolutive case one (the latter, as also an allative case indirect object, will usually have a specific - if not actually definite - sense)" (Fortescue 1984: p.89). To illustrate this statement, he provides the example in 307 as a typical context for secundative alignment. Here, the secundative theme is indeed non-specific and generic.

(307) Niisi aningaasa-nik tuni-vaa
Niisi.ABS money-PL.MOD give-IND.3SG.S>3SG.O
'He gave money to Niisi.' (Fortescue 1984: p.89, glosses adapted)

This semantic restriction can be seen as indirect evidence that secundative themes lack a DP-projection. Since Abney's (1987) original proposal, DP has been analyzed as a projection associated with definiteness (see also e.g. Coppock & Beaver 2014, 2015 and Lyutikova & Pereltsvaig 2015; Lyutikova 2017). Therefore, definite nominals obligatorily include a DP-projection, while structurally reduced nominals can only have an indefinite or non-specific interpretation.

Another piece of evidence for DP-restriction being active on secundative themes in Inuit is provided by external possessor constructions. The external possessor argument is ABS-marked and the theme (possessee) is obligatorily incorporated into the verb. Crucially, secundative alignment is never used in clauses with external possessors. I argue that the reason behind obligatory incorporation is DP-restriction on secundative themes. The theme in clauses with external possessors has a definite interpretation and contains a projection for the internal possessor, i.e. must be a full DP. If the DP-restriction on secundative themes were not present in Kalaallisut, nothing would prevent secundative alignment in clauses with external possessors. However, if the theme obligatory has DP-projection, is causes ungrammaticality in secundative alignment and require a repair operation, e.g. noun incorporation, as in external possessor constructions. Therefore, I conclude that DP-restriction on secundative themes is active in Kalaallisut, as well as in Khanty.

[62] Compton & Pittman (2010) suggest that only freestanding arguments have a DP-projection in Inuit, while incorporated nouns are obligatorily NPs. However, Yuan (2013) provides counter-evidence and shows that incorporated nominals can be full DPs.

As illustrated in 308, an absolutive-marked argument (IO) in ditransitive clauses with an incorporated theme has two possible interpretations. In the first interpretation, 'Kaali' is the owner of the bike, but it is not necessarily the case that the bike was taken from him. In the second interpretation, 'Kaali' is a malefactive argument who has lost a bike in the described event. However, he is not necessarily the owner of the bike, as indicated by the availability of the indefinite interpretation 'a bike'. The first interpretation corresponds to the external possessor construction.

(308) Piita-p Kaali cykili-irut-p-a-a
Peter-ERG Kaali.ABS bike-take.by.force-IND.-[+TR]-3SG.S>3SG.O
i. 'Peter took Kaali's bike by force.'
ii. 'Peter took a/the bike from Kaali by force.'
(Van Geenhoven 2002: p.761, ex.2)

Note that the interpretation in (i) is only available if the theme is incorporated. When *cykili* 'bike' is a freestanding nominal, only the malefactive interpretation in (ii) remains available. The subsequent discussion concerns only the external possessor interpretation in (i).

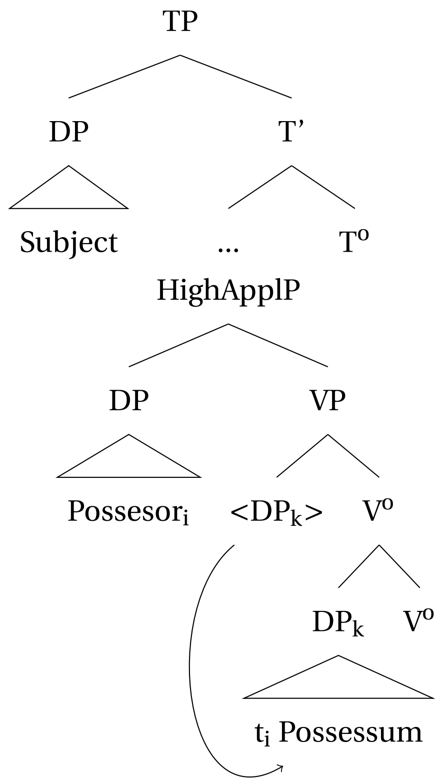
There are two possible analyses of external possessor construction. First, the possessor can be base-generated within the possessum DP. When the possessum undergoes incorporation, the possessor is stranded and raised to a higher position (e.g. SpecVP or SpecμP), where it receives the absolutive case, as in 309. The raising analysis has been proposed for a number of languages, e.g. Nez Perce and Hebrew, even in the absence of possessum incorporation (Deal 2013, 2017). On the other hand, Van Geenhoven (2002) argues that external possessors in Kalaallisut are base-generated in a projection outside of DP (= Spec(High)ApplP in the derivation suggested in this thesis) and bind a variable on the incorporated possessee (see also Baker 1988, Lee-Schoenfeld 2006 for German and Deal 2017 for general overview).[63]

[63] Deal (2017) differentiates between four possible types of external possessors, listed in (i). The exact type of external possessor construction in Kalaallisut does affect my argument. In all four variants, the possessum is always a DP containing either a trace of the possessor or a bound variable/*pro*.

(i) A Typology of External Possession (Deal 2017: p.18, ex.51)

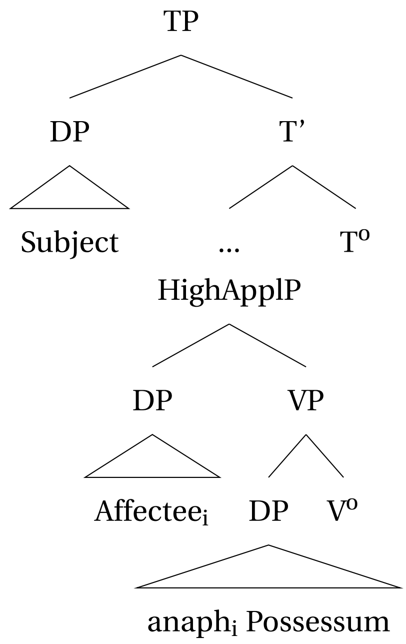
		Does the DP move?	
		Yes	No
Does the possessor receive an additional θ-role?	Yes	Hybrid analysis: movement to θ-position	Binding analysis
	No	Possessor raising analysis	Possessor government analysis

(309)



(adapted from Van Geenhoven 2002: p.763, ex.7)

(310)



(adapted from Deal 2017: p.12, ex.30)

Note that in the both analyses above, the incorporated possessum is not a bare NP and contains a bigger amount of structure. This is a necessary assumption because the incorporated noun contains either a trace of the raised possessor or an anaphor bound by the external possessor, i.e. is at least as big as PossP. Furthermore, Sadock (1980: p.310) mentions that "external possessors of incorporated nouns have always been allowed when the possessive suffix of the noun survives after incorporation", as e.g. in 311.[64]

(311) Kalaallit nunaa-li-ar-poq.
Greenlander.PL country-3PL-go-INDIC.3SG
'He went to the Greendlanders' country.' (Sadock 1980: p.314, ex. 60; modified)

Crucially, Langr (2014b) and Manlove (2015) argue that possessive suffix spells out an agreement probe on D^{o}. Hence, its presence in 311 provides evidence that the incorporated possessum is a full DP.[65]

[64] If the incorporated nominal does not bear a possessive suffix, the possessor tends to have a generic interpretation (Sadock 1980; Johns 2009). As this piece of data is not relevant for the current discussion, I leave a formal analysis for further research. Preliminary, it can be noted that Sadock 1980's examples allow to analyze such "generic possessors" as nominal modifiers, cf. (i).

(i) Tutt-up neqi-tor-punga.
caribou-REL meat-eat-INDIC.1SG
'I ate reindeer meat.'
(Sadock 1980: p.309, ex.33, glosses after Johns 2009)

[65] Yuan (2013); Johns (2009) show that different Inuit varieties allows for incorporation of full DPs.

The current analysis of high applicative constructions in Inuit, therefore, correctly predicts that noun incorporation is obligatory in external possessor constructions. Indeed, if a possessor occupies SpecHighApplP (either via raising or base-generation), it intervenes between C^{o}/Mood^{o} and the possessum DP. Since case-licensing is derived via Agree, a Case probe on C^{o}/Mood^{o} agrees with the possessor DP and cannot reach the possessum DP due to the Minimal Link Condition (Rizzi 1990). As suggested above, HighAppl^{o} in Inuit does not have a [iCase]-feature, if a DP is merged in its specifier. As a result, the possessum DP remains unlicensed, and secundative alignment is expected. However, only structurally reduced nominals without the DP-projection can survive the derivation unlicensed. A way to save a derivation with a DP-sized theme is noun-incorporation, since incorporated nouns do not require external licensing via case-checking. Hence, the possessum DP is incorporated, no longer requires case-checking, and the derivation does not crash.

II. Objects in antipassive clauses

Finally, the difference between antipassive objects and secundative themes must be addressed. Note that the theme in antipassive constructions is also marked with MOD/INSTR. Quite often, antipassive construction is accompanied by a non-specific or generic interpretation of themes (Polinsky 2017). Since Yuan (2018) argues that an additional functional apP-projection is merged in antipassive contexts and assigns MOD/INSTR to the theme, it seems a plausible hypothesis that the same (apP-)projection is merged in secundative alignment. I argue that antipassive and secundative alignment are two completely different derivations in Inuit. First, antipassive clauses have a special morphological marker *-si-*, which in Yuan (2018)'s analysis spells out a separate projection in the clausal spine (cf. 312). Importantly, the antipassive marker is absent in the secundative clause in 313.

(312) qimmit marluk arna-nik pingasu-nik kii-**si**-pput
dog.PL.ABS two.ABS woman-PL.MOD three-MOD bite-AP-3PL.S
'Two dogs bit three women.' (Yuan 2018: p.45, ex.45, repeated from 301)

(313) anguti-p aqerluusa-mik meeraqq tuni-paa
man-ERG pencil-MOD child.[ABS] give-3SG.S>3SG.O
'The man gave the pencil to the child.'
(Johns 1984: p. 167, ex. 9b, repeated from 300b)

Second, a demoted argument in antipassive clauses can be specific and definite, e.g. it can be a proper name. In 314, antipassive serves for a demotion of the recipient argument 'Carol'. As a proper name, 'Carol' is a DP and thus requires syntactic licensing. In Yuan's (2018) analysis, it is licensed by the antipassive projection and, therefore, nothing prevents it from being a DP.

(314) Jesusi-mik taku-si-vuq
Jesus-MOD see-AP-3SG.S
'He saw Jesus.' (Yuan 2018: p.246, ex.3a)

Hence, it can be safely concluded that the apP-projection is absent in secundative alignment, but present in antipassive clauses. Therefore, the secundative theme truly lacks case-licensing and the MOD marker is used as a marked default.

6.1.3 Marked default case under noun incorporation

Noun incorporation constructions provide evidence that the oblique case (MOD, INSTR, or LOC) is indeed a default case marker that is used when no case value has been assigned in syntax.

In a clause with noun incorporation in 315, the ergative probe is inactive/absent, and the external argument is marked with ABS. Because the absolutive probe agrees with the external argument, there is no probe left to assign case to the object position. The object is incorporated into the verb and does not require case-licensing to survive derivation.

(315) Suulut biili-liur-sima-v-u-q
Søren.ABS car-make-PERF-IND-DETR-3SG.S
'Søren made a car.' (Van Geenhoven 2002: p.765, ex.9a, glosses adapted)

Note that different types of modifiers can be stranded in Kalaallisut when the direct object is incorporated, e.g. adjectives (316a), numerals (316b), relative clauses (316c), or wh-words (316d).

(316) (Van Geenhoven 2002: p.766, ex.(i-iv), glosses adapted)

a. suluut qisuk-mik timmisartu-liur-p-u-q
Søren.ABS wooden-SG.MOD airplane-make-IND-DETR-3SG.S
'Søren made a wooden airplane.'

b. Suulut marluk-nik timmisartu-liur-p-u-q
Søren.ABS two-PL.MOD airplane-make-IND-DETR-3SG.S
'Søren made two airplanes.'

c. Arne qatannguti-qar-p-u-q Canada-mi
Arne.ABS sister-have-IND-DETR-3SG.S Canada-LOC
najuga-lik-mik
dwelling.place-have.REL.[-TR]-SG.MOD
'Arne has a sister who lives in Canada.'

d. qassi-nik qimmi-qar-p-i-t?
how many-PL.MOD dog-have-INTER-DETR-2SG.S
'How many dogs do you have?'

The stranded modifiers in the examples above are marked with MOD, similar to the secundative themes. As in secundative alignment, there is no obvious probe that would be able to assign case to them.[66] Under the analysis advocated in this thesis, all argument

[66] Compton & Pittman (2010) analyze free-standing adjectives in noun incorporation as adverbial modifiers

positions (i.e., θ-positions) contain a KP with all case values inherently present on K^o. If a nominal has not been licensed/checked by an external head, all inherent case values are spelled out, giving rise to the marked default morphology. As expected under the inherent case overspecification hypothesis, stranded modifiers show default case overspecification: when there is no available case assigner, e.g. in noun-incorporation clauses, stranded adjectives are marked with an oblique case.

6.1.4 Intermediate summary

The data from Kalaallisut are an important parallel to the Kazym Khanty data. First, Kalaallisut provides morphological evidence that HighApplP is the source of the dative/allative case in ditransitive clauses. Second, DP-restriction is shown to be a defining feature of secundative alignment cross-linguistically. This additionally allows to distinguish secundative alignment from the antipassive. Finally, stranded modifiers in noun-incorporation clauses support the suggestion that the most marked oblique case is used as the default marker in argument positions.

6.2 Eastern Mansi: DOM and case-licensing

Differential Object Marking (DOM) is another important parallel to the theme in secundative alignment. The restrictions on unmarked objects in DOM languages are very similar to those on themes in secundative alignment. In particular, it has been argued for many languages that only marked objects in DOM languages have a DP-layer, while unmarked objects are structurally reduced. Furthermore, as discussed below, many studies have linked the absence of case marking to the absence of the DP-layer. The approach to secundative themes proposed in this study is based on the opposite assumption. It argues that structurally reduced nominals can survive without being assigned a case. However, since all arguments are inherently overspecified for case values, an oblique morphological marking will surface on arguments in the absence of syntactic case-checking. Therefore, DOM and its coexistence with secundative alignment require further discussion.

Crosslinguistically, DOM is argued to be dependent on prominence features such as definiteness, specificity, or animacy (e.g. Morimoto 2002; Aissen 2003; De Swart & de Hoop 2007; Sinnemäki 2014; Levin 2019 etc.). Other features have also been argued to trigger

adjoined to the VP. However, they do not provide any supporting evidence for this analysis. Moreover, the adverbial analysis cannot straightforwardly explain why stranded modifiers obligatorily agree with the incorporated noun in number (i).

(i) a. angisuu-nik qamuti-qar-p-u-q
big-PL.MOD sled.PL-have-IND-DETR-3SG.S
'He has big sleds.'
b. *angisuu-mik qamuti-qar-p-u-q
big-SG.MOD sled.PL-have-IND-DETR-3SG.S
(Van Geenhoven 2002: p.767, ex.10-11, glosses adapted)

If modifiers are in the argument position, number coreference is explained straightforwardly.

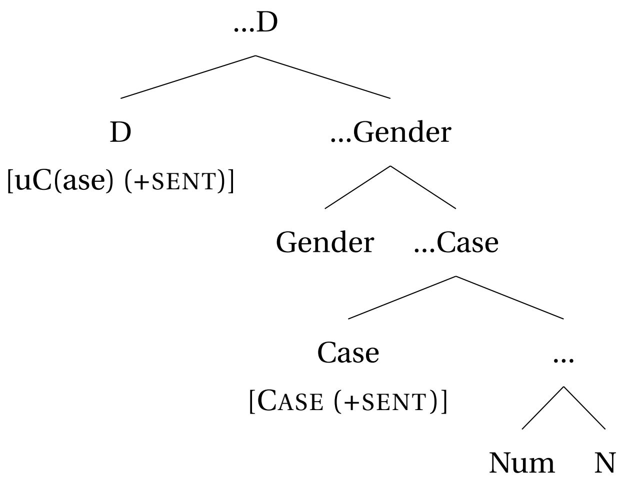
DOM, such as affectedness (Næss 2004), topicality (Dalrymple & Nikolaeva 2011), and even focus and number (Woolford 1999). Overall, there are four main groups of approaches to DOM in the linguistic research tradition.

In the first approach, DOM is dependent on the syntactic position of the object in the same local domain as a second caseless DP (i.e., the external argument) and consequently on the movement of the object out of the vP-phase. Baker & Vinokurova (2010, et seq.) argue that ex situ objects in Sakha are marked with overt accusative, while in situ objects are unmarked. Similar proposals have been made for a number of other languages, e.g. Hindi (Bhatt & Anagnostopoulou 1996), Spanish (López 2012), and Amharic (Baker 2012, 2015). Smith (2020) and É. Kiss (2021) suggest the same analysis for (Northern) Khanty. Objects that trigger object agreement are located in a higher structural position compared to objects that do not trigger verbal agreement. However, not all languages follow the lines of this analysis. In particular, Kalin & Weisser (2019) discuss that some languages allow DOM inside a coordination, which is incompatible with a movement-based analysis. Similarly, DOM in Hungarian does not depend on the syntactic position of the object (Bárány 2017).

The second approach to DOM suggests that the difference in nominal marking depends from the presence or absence of syntactic licensing, as well as from the structural properties of the objects. More specifically, DPs and NPs in object position are marked differently because they are not equally licensed. DP-objects must be licensed via ϕ-agreement or through case assignment. Therefore, the DP-sized objects obligatorily control agreement and/or bear a case marker. On the other hand, NP-objects lack the Case feature and can remain unlicensed. Unlicensed and structurally reduced direct objects are unmarked for case and/or do not control ϕ-agreement on the verb. This kind of analysis is advocated by Aissen (2003), Heck & Richards (2010), Richards (2014), Danon (2006) for Hebrew, Ormazabal & Romero (2013) for Spanish, Lyutikova & Pereltsvaig (2015) for Tatar, Kalin (2018) for Senaya (Neo-Aramaic), Levin (2019) for Palauan (Austronesian), and Driemel (2023).

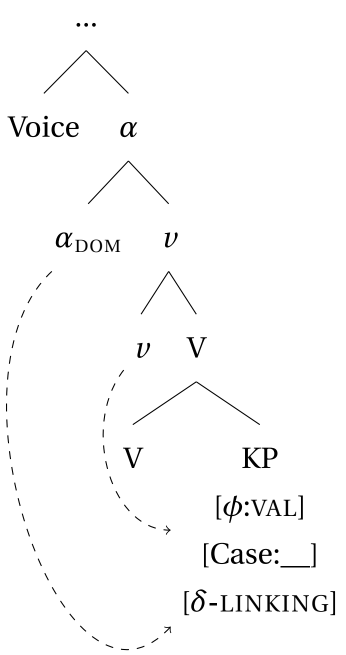
The third approach derives differential object marking by the interaction of case features with inherent prominence features on the nominal. Bárány (2017) argues that DOM (in case marking or verbal agreement) is triggered by the interaction of agreement for Case or ϕ-features with inherent prominence features on the object, i.e. definiteness, specificity, and animacy. The difference between languages with and without DOM comes from the fact that only some languages have prominence features that are formally represented in the syntax (Bárány 2017: p.272). If prominence features are active in syntax, they can interact with ϕ-features and affect case morphology at the PF. In languages without DOM, prominence features are semantic, i.e. inactive in the syntax, and cannot have effect at the PF. Irimia (2022) makes a similar proposal, arguing that the presence of a discourse-related prominence feature triggers a differential marker on the object. In Romance languages, prominence features are obligatorily present on the DP projection. Hence, all DP-sized nominals are DOM-marked. Crucially for her proposal, some bare nominals are DOM-marked as well, despite the absence of the DP-layer. For these nouns, she suggests that prominence features can be exceptionally bundled with a Case-projection below DP. The full structure is given in 317.

(317) (Irimia 2022: p.4, ex.10)



Furthermore, Irimia (2023) explicitly argues that both marked and unmarked objects in DOM languages are syntactically accusative, even when the DOM marker is syncretic with oblique marking. Irimia (2023) uses enriched case hierarchies with two accusative/dative cases. As shown in 318, the "oblique" accusative contains additional δ-linking features (prominence- or discourse-related), which require additional licensing beyond Case proper.

(318)



[uCase] ↔ /ACC/

[uCase & δ] ↔ /OBL/

(Irimia 2023: p.23, ex.38)

A similar route is taken in Nikolaeva (1999, 2001) and Dalrymple & Nikolaeva (2011) for Northern Khanty data.

Finally, there is a group of approaches that analyze DOM as a morphological phenomenon (e.g. Glushan 2010; Keine & Müller 2015). Keine & Müller argue that the same case value is assigned in syntax to both marked and unmarked objects. A differential marking is triggered postsyntactically, when a subset of case values is impoverished on the unmarked/less marked objects. Among other supporting evidence, Keine & Müller (2015) discuss a pattern of non-binary DOM opposition in several languages. One such language is Trumai (isolate, Brazil). This language has a tripartite opposition in differential object case marking that depends on individuation and prominence. The three exponents marking direct objects are illustrated in 319. Strikingly, none of the available object markers is phonologically zero. Keine & Müller conclude that DOM cannot be analyzed as the presence vs. absence of case values. Rather, differential marking should be derived at PF by impoverishment of a subset of features inherited from syntax and the existence of several exponents with different feature specifications.

(319) a. ha hu'tsa chï_in kasoro-**tl**
I see Foc/Tens dog-DAT
'I saw the dog (I know it).'
(individuated, identifiable, prominent)

b. ha hu'tsa chï_in kasoro yi-**ki**
I see Foc/Tens dog YI-DAT
'I saw a dog/the dog (I do not know it well).'
(individuated, but not identifiable/prominent, or non individuated indentifiable)

c. ha hu'tsa chï_in kasoro-**s**
I see Foc/Tens dog-DAT
'I saw dogs.'
(non individuated, non identifiable, not prominent)
(Keine & Müller 2015: p.30, ex.48)

Similarly, Legate (2008) argues that zero case marking is not uniform crosslinguistically and can result either from the choice of exponence in the morphology or from the absence of abstract case values.

The most widely adopted analyses of DOM rely on the assumption that the absence of morphological case marking indicates the absence of syntactic case licensing and of a case value (e.g. Aissen 2003; Baker & Vinokurova 2010; Lyutikova & Pereltsvaig 2015; Lyutikova 2017; Driemel 2023 etc.). Importantly, these analyses are incompatible with the hypothesis of inherent case overspecification introduced in Chapter 5. If arguments are inherently overspecified for case values, then the absence of syntactic case-licensing should lead to the spell-out of the maximally marked case on structurally reduced objects, similar to oblique themes in secundative alignment.

The main goal of this section is to argue that a DOM pattern cannot be used as a counterargument to the overspecification hypothesis. Before proceeding to the main discussion, a brief comment should be made about the relation of DOM to indirective-secundative alignment alternation.

6.2.1 Secundative alignment and DOM

As Table 6.2 demonstrates, DOM and the indirective-secundative alternation are two independent phenomena, and the presence or absence of one does not entail the presence or absence of the other. Some languages have DOM but do not have alignment alternation. Spanish, Sakha, Tatar, Hungarian, Nez Perce, and many other languages belong to this type (Aissen 1999, 2003; Morimoto 2002; Deal 2006, 2010; Baker & Vinokurova 2010; Baker 2012, 2015; Lyutikova & Pereltsvaig 2015; Bárány 2017, 2018; Irimia 2022, 2023, etc.). The second group of languages has alignment alternation and does not have DOM. Some examples are varieties of Khanty, Northern Mansi, Tundra Nenets, and Inuit languages. Yet another group of languages has neither DOM nor alignment alternation, e.g. German, Latin, and Greek. Finally and most importantly, at least one language – Eastern Mansi – has both DOM and indirective-secundative alignment alternation.

		Ind/Sec alignment alternation?	
		yes	no
DOM?	yes	Eastern Mansi (Virtanen 2012, 2013, 2014, 2015)	Sakha (Baker & Vinokurova 2010) Tatar (Lyutikova & Pereltsvaig 2015; Lyutikova 2017) Nez Perce (Rude 1986; Deal 2010) Romance (e.g. Irimia 2022) etc.
	no	Khanty (Nikolaeva 1999; Filchenko 2007) Tundra Nenets (Nikolaeva 2014) Northern Mansi (Bíró & Sipőcz 2017; Rombandeeva 2017) Kalaallisut (Fortescue 1984; Johns 1984)	e.g. German, Latin, Greek etc.

Table 6.2: Correlation of DOM and indirective-secundative alignment alternation

A language that uses both DOM and indirective-secundative alignment alternation is an ideal starting point for the discussion of the relation between DOM and indirective-secundative alignment alternation. This allows for a comparison of the two phenomena without addressing the individual differences between languages. In the remainder of this section, I discuss Eastern Mansi, the only language I know with both DOM and indirective-secundative alternation. Without the ambition to propose a universal analysis for DOM in all languages, I argue that the Eastern Mansi DOM cannot be used as counter-evidence for the case overspecification hypothesis. In particular, I propose that unmarked direct objects in Eastern Mansi should not be analyzed as lacking the Case feature or case values and that DOM-effects in Eastern Mansi are purely morphological in nature.[67]

6.2.2 Eastern Mansi

Eastern Mansi is an Ob-Ugric, Uralic language, closely related to (Kazym) Khanty. There are no significant grammatical differences between the two languages which are relevant for the current discussion. Eastern Mansi is also an SOV head-final language with nominative-accusative alignment in case and agreement (Forsberg 2007). Similar to Khanty, it has obligatory subject and optional object agreement, as illustrated in 320.

(320) a. suj-nø köäl-ååm, suj-sot oos-**øm**
moor-LAT step-1SG moor-lord be-1SG
'I step on the moor, I am the lord of the moor' (Virtanen 2012: p.122, ex. 1)

b. sos-mø uus köält-øs-**tø**
moose-ACC again frighten-PST-3SG>SG
'He frightened the moose again' (Virtanen 2012: p.122, ex. 2)

As the rest of Ob-Ugric languages, Eastern Mansi has indirective-secundative alternation in ditransitive clauses; see 321.

(321) a. indirective alignment

om kurøm lyõx äk° näg-nöän tåt-s-øm
I.NOM three message only you-LAT bring-PST-1SG
'I brought three messages just for you' (Virtanen 2012: p.126, ex. 8)

b. secundative alignment

am nää-n tat-øs-lø nee-l
I.NOM you-POSS.2SG>SG.[ACC] bring-PST-1SG>SG woman-INSTR
'I brought you a wife' (Virtanen 2012: p.125, ex. 5)

Eastern Mansi is the only language in the Ob-Ugric group that has a separate marker for accusative case. Moreover, not all direct objects in active clauses are marked with accusative, which makes Eastern Mansi a DOM language. Virtanen (2012) argues that Eastern Mansi DOM is dependent on topicality, similar to differential object agreement in Khanty (Nikolaeva 1999, 2001; Dalrymple & Nikolaeva 2011). Moreover, topical direct objects are marked by agreement on the verb, while case marking has "a complementing function" in the Eastern Mansi DOM.

The existence of Eastern Mansi is important for the analysis proposed in Chapter 4, not only because it allows for the complete separation of DOM from indirective-secundative al-

[67]Note that I do not argue that DOM is a morphological phenomenon in all DOM-lanagues. Whether structurally reduced objects in some languages indeed lack the Case feature and the KP-projection altogether and why this may be the case remains a matter for further research.

ternation. The Eastern Mansi system also provides means to test the case overspecification hypothesis in Chapter 5 and whether it can be conciliated with the existence of unmarked objects. In the rest of this section, I argue that DOM in Eastern Mansi does not provide a counterargument to the ihnerent case overspecification analysis of oblique secundative themes.

In Eastern Mansi, both accusative marking and object agreement are optional, showing DOM effects (Virtanen 2013, 2014). In the majority of contexts, the two markers conspire: the direct object *luj-ootr-äg-***mø** 'two princes of the underworld' in 322 is marked with the overt accusative marker and triggers object agreement on the verb, while in 323 the object *nee* 'woman' is morphologically zero-marked and does not control agreement on the verb.

(322) luj-ootr-äg-**mø** wot-**öän**!
down-prince-DU-ACC call-IMP.2SG>DU
'Call the two princes of the underworld here!' (Virtanen 2013: p.311, ex.10)

(323) nee öät uusyøntöäl-i.
woman NEG see-3SG
'He does not see any woman.' (Virtanen 2013: p.309, ex.4)

"Differential accusative" in Eastern Mansi is spelled out as "absolute accusative" form on nouns without possessive agreement in Table 6.3, as opposed to the zero "nominative" marking. If there is a possessive agreement on the noun, a portmanteau form spells out the accusative case and possessive features simultaneously. Note that the portmanteau paradigm in Table 6.4 is defective, and separate accusative forms are available only in two cells of the whole paradigm. In all empty cells only unmarked possessive forms are available and accusative is syncretic with nominative.[68]

SG	DU	PL
-mø/ -m / -øm	-iimø / -ägmø	-tmø

Table 6.3: "Absolutive accusative" morphology (Eastern Mansi)

Possessor \ Possessee	Sg obj	Du obj	Pl obj
3Sg	-ääm, -øtääm	-iimø	-
3Du/Pl	-	-	-
2	-	-	-
1	-	-	-

Table 6.4: Possessive accusative portmanteau morphology (Eastern Mansi)

Virtanen (2013, 2014, 2015) argues at length that these DOM effects depend on the topicality of the direct object rather than its definiteness or animacy.

Crucially, the two types of DOM (verbal agreement and case marking) do not always conspire. As Table 6.5 demonstrates, of all clauses with object agreement, only 54% have

[68] For a full paradigm of the Eastern Mansi possessive forms, see Forsberg (2007).

over ACC marking on the direct object. In clauses with subject agreement, the object is expected to be unmarked for DOM, and indeed, the majority of the objects are NOM- or ø-marked. Nevertheless, in 7% of all cases, ACC marking can surface without object agreement on the verb.

	Obj-agreement	Subj-agreement
Acc	54%	**7**%
Nom	**46**%	93%

Table 6.5: Interaction of two types of DOM in Eastern Mansi (after Virtanen 2014)

I will leave aside the conspiring scenarios and concentrate on cells with conflicting parameters.

A. **Nom + Obj-agree**:

In 46% of all transitive clauses examined by Virtanen (2014), the direct object lacks differential case marking (ACC) in the presence of object agreement. However, all examples mentioned by Virtanen (2014) where an unmarked object controls agreement on the verb are of the same type. These objects have a possessive marker with a value where no DOM-marked form is present in the paradigm (see Table 6.4). The example in 324 illustrates this for the object with POSS.2SG-marker which lacks a separate DOM-allomorph. Hence, these are not cases of a proper mismatch in the two types of DOM.

(324) øt mõõ-nø toont loåw-s-øløm, öätøm-ään tøxt-øx^{o}.
this earth-LAT PART order-PST-1SG>SG human-PL.POSS.2SG feed-INF
'I ordered (you) to this place to feed your people.' (Virtanen 2014: p.405, ex.10)

B. **Acc + Sub-agree**:

More interesting are the rare cases where DOM is marked on the noun, but object agreement is absent from the verb. Virtanen (2014) provides a single example of this sort, repeated in 325.

(325) nääk-øng pum jesøx-t-ään-ääm kurøm kotøl worii-nø
joint-ADJ grass rub-FREQ-GER-ACC.POSS.3SG>SG three day distance-LAT
koontl-ii.
listen-PRS.3SG
'The humming (rubbing) of the wind through the joint grass can be heard three days off by foot.' (Virtanen 2014: p.410, ex.27)

It is difficult to determine whether this single example reflects the general pattern of the disjoint DOM in Eastern Mansi. However, if it does, the pattern is similar to that of Kazym Khanty, where a high aspect, for instance habitual, bleeds object agreement on the verb in contexts where it is otherwise obligatory.

In sum, Eastern Mansi has two types of DOM-marking: case marking on the noun and object agreement on the verb. The first can be obscured by impoverishment/homophony in the morphological paradigm, but otherwise seems to be the main way to reflect DOM. The second type (object agreement) can be bled by external factors, but otherwise coincides with case marking. As discussed at the beginning of this section, DOM is a potential counterargument to the case overspecification hypothesis. The two leading theories of DOM reduce it to an opposition between the presence and absence of case values on the noun. In the next subsection, I demonstrate that the two main analyses of DOM cannot capture the Eastern Mansi data. Therefore, an alternative analysis has to be adopted and DOM can no longer be used as an argument against the case overspecification hypothesis.

6.2.3 Testing existing analyses of DOM

As discussed above, there are two main approaches to DOM in the literature, both of which are incompatible with the case overspecification hypothesis. Importantly, Eastern Mansi is an ideal candidate for testing the hypothesis because, apart from the indirective-secundative alignment alternation, it also has differential object marking. I will test the two theories of DOM in turn and show that they cannot be maintained in Eastern Mansi.

The first approach to DOM suggests that it is dependent on the syntactic position of the object. In particular, more marked objects are raised ex situ to a higher projection (cf. Smith 2020 for Surgut Khanty), where they can receive a more marked case (e.g., via the Dependent Case mechanism). This prediction is not borne out in Eastern Mansi. As 326 shows, a ø-marked object can be scrambled above an adjunct adverbial.

(326) om kurøm lyõx äk^{o} näg-nöän tåt-s-øm
1SG three message only 2SG-LAT bring-PST-1SG
'I brought three messages just for you.' [Virtanen 2012, ex. 8]

Unfortunately, Eastern Mansi is an extinct language, and the corpus data not to determine the landing site of the object. Nevertheless, if Virtanen (2013, 2014)'s conclusion is correct, DOM is triggered not by word order but merely by the topicality of the object.

The second approach to DOM ties the contrast in case marking to the size of the object (e.g., Kalin 2018; Levin 2019 etc.). Specifically, it is argued that structurally reduced objects (NPs) lack the [uCase]-feature and cannot receive a case value altogether. In contrast, DP-sized objects have the [uCase]-feature and can be assigned a case.

Most unmarked objects in Eastern Mansi are indefinites, which can point towards an analysis based on NP-DP opposition. However, Eastern Mansi DOM cannot be reduced to NP–DP variation. As 327 illustrates, the direct object can be unmarked even when bearing a possessive suffix. I assume that possessive markers in Eastern Mansi are hosted by D^{o}, similar to Kazym Khanty (see Chapter 4, Section 4.2.1 for discussion). Hence, 327 provides evidence that unmarked objects in Eastern Mansi can be of DP-size and that the Eastern Mansi DOM is not dependent on the size of the object.

(327) oltøn-wity-øng, suurøny-wity-øng jälpøng toågl-äät
silver-water-ADJ gold-water-ADJ sacred cloth-POSS.3SG>SG

nok-posyg-øs
up-wrap-PST.3SG
'He wrapped his silver and gold sacred cloth (an item of clothing) around his shoulders.' (Virtanen 2013: p.314, ex.19)

Both of these analyses are incompatible with the case overspecification hypothesis proposed for secundative alignment. However, neither can capture the Eastern Mansi data, a language that has both indirective-secundative alternation and the DOM. Therefore, the existence of DOM in Eastern Mansi does not provide an argument against the case overspecification hypothesis. In the next subsection, I propose a morphological analysis of the Eastern Mansi DOM. Under this analysis, both marked and unmarked objects are assigned the same case in syntax, and the presence or absence of morphological case marking depends on the specification of case exponents present in the language.

6.2.4 Analysis of Eastern Mansi DOM

As Virtanen (2013, 2014) argues, the DOM in Eastern Mansi is triggered solely by the topicality of the object, rather than by its definiteness and specificity. Crucial for the current proposal, topicality is not a defining feature of the DP-projection. Therefore, it cannot be argued that non-topical nouns have a reduced structure and lack the DP-layer. Therefore, both marked and unmarked nouns in Eastern Mansi can have both a DP-layer and a Case projection (KP). Moreover, in the absence of intervention effects, nothing can prevent case assignment to the object, whether it is marked or unmarked for DOM. Nevertheless, DOM in Eastern Mansi has to be modeled.

In a nutshell, I propose that DOM effects in Eastern Mansi emerge at the PF due to the featural specification of morphological exponents. Voice^{0} always assigns a structural case (accusative) to the direct object in its c-command domain. However, the overt realization of the case value is dependent on the presence or absence of the [+top]-feature on the noun. Feature specifications for the case exponents are shown in 328. If the [+top]-feature is absent, a default, phonologically zero exponent is used. Apart from the default, there are several exponents that are specified for the [+top]-feature. According to the Subset Principle (Halle 1997), they can only be used if [+top] is present in the syntax. If the nominal is [-top], only the default ∅-exponent can be used.

(328) a. /ø/ ↔ [A (=NOM)]

b. /-mø/, /-m/, /-øm/ ↔ [A,B(=ACC), SG, +top]

c. /-iimø/, /-ägmø/ ↔ [A,B(=ACC), DU, +top]

d. /-tmø/ ↔ [A,B(=ACC), PL, +top]

In the possessive paradigm, only two accusative exponents are specified for [ACC] and

[+top], as shown in 329a and 329b. The exponent underspecified for case can be used in all other cells in the possessive paradigm (e.g., 329c).

(329) a. /-ääm/, /-øtääm/ ↔ [A,B(=ACC), POSS3SG>SG, +top]
vs.
/-ø/, /-e/, /-tø/, /-äät/ ↔ [(A), POSS3SG>SG]

b. /-iimø/ ↔ [A,B(=ACC), POSS3SG>DU, +top]
vs.
/-äö/, /-ii/, /-ää/ ↔ [(A), POSS3SG>DU]

c. -øn ↔ [(A), POSS2SG>SG]

Importantly, there is no independent accusative exponent for the possessive paradigm, even when the topicality-feature is present. As illustrated in 329c, a case- and topicality-neutral, underspecified exponent is the only option available for the 2nd person possessive marker.

There is a potential downside to the morphological approach to DOM. Under this approach, it seems to be completely random that differential marking is only attested for accusative case, but not throughout the entire case system. Nevertheless, I argue that the differential object marking in Eastern Mansi is too inconsistent to be considered a syntactic phenomenon. Recall that most cells in the possessive paradigm do not show differential object marking. Hence, the differential accusative in Eastern Mansi is not a systematic feature and can hardly be explained in syntactic terms. Therefore, I suggest that it is not tied to syntactic nominal licensing, and is rather a morphological phenomenon triggered by bundling for information structural features and case features.[69]

6.2.5 Intermediate summary

Under a closer investigation, DOM in Eastern Mansi is most adequately analyzed as a morphological phenomenon and not as the absence of syntactic case values on unmarked objects. Consequently, Eastern Mansi does not provide supporting evidence for existence of the unvalued Case feature on nouns in syntax. Therefore, the analysis of oblique themes in secundative alignment as inherently overspecified for case values can be maintained.

6.3 "Applicative construction" (Baker 1988)

A number of languages, including German, Russian and Hungarian, have an alternation, that are somewhat similar to the indirective-secundative alternation in Khanty. In particular, the "themes" in (b)-examples in 330-332 are marked with the oblique instrumental case,

[69] See also Arkadiev & Testelets (2019) for a language that shows differential argument marking in all argument positions and for all structural cases (nominative, accusative, and dative).

and the recipient is promoted to accusative. It is also worth noting that the (b)-examples were originally defined as applicative constructions by Baker (1988). Pylkkänen (2008) redefined the notion of applicative as a functional projection able to introduce an applied argument in the clause, and her definition is used in the majority of modern linguistic studies, including this thesis.

(330) German

a. Klaus schenk-te seiner Mutter diese Blumen.
Klaus.NOM present-PST.3SG his.DAT mother.DAT this.PL.ACC flowers.ACC

b. Klaus **be**schenkte seine Mutter mit diesen Blumen.
Klaus.NOM present-PST.3SG his.ACC mother.ACC with this.PL.DAT flowers.PL.DAT
'Klaus gave his mother flowers.'

(331) Russian

a. Drug **po**dari-l Lid-e eti zvet-y.
friend.SG.NOM present-PST.M Lida-SG.DAT this.PL.ACC flower-PL.ACC
'Her friend presented flowers to Lida.'

b. Drug **o**dari-l Lid-u etimi zvet-ami.
friend.SG.NOM present-PST.M Lida-SG.ACC this.PL.INSTR flower-PL.INSTR
'Her friend presented Lida with flowers.'

(332) Hungarian

a. ajándékoz vki-nek vmi-t
present.PRES.3SG someone-DAT something-ACC

b. **meg**ajándékoz vki-t vmi-vel
meg.present.PRES.3SG someone-ACC something-INSTR
'(S)he gives a gift'. (Bíró & Sipőcz 2017)

However, there are three crucial differences between these alternations and indirective-secundative alternation in Khanty. First, in all these languages, the verb is not identical in the two contexts: a prefix is added in contexts that correspond to secundative alignment. If an additional morphology signals the presence of an additional syntactic projection, the data in 330-332 show exactly the reversed pattern from the one predicted by my analysis. In my analysis of Khanty, an additional projection is present in indirective alignment, where the verbs in 330-332 do not have a prefix.

Secondly, there is no size restriction on oblique-marked arguments in (b)-examples. This is particularly straightforward in the German example 330b where the PP-theme is modified by a demonstrative. Similarly, in Russian example 331b the oblique-marked theme/instrument is modified by a demonstrative. Moreover, 333 shows that even a personal pronoun is possible in this position.

(333) ona <...> snjala doroguščij starinnyj persten' i... **o**dari-la
she took_off expensive.M.ACC old.M.ACC ring.M.ACC and present-PST.F

im ošelomlennogo toreadora.
it.SG.INSTR stunned.ACC bullfighter.ACC
'... she <...> took off an expensive old ring and... presented the stunned bullfighter with it.' (National corpus of Russian Language)

It can be safely concluded the oblique argument in German and Russian is a fully licensed DP. As discussed in Chapter 4, the DP-restriction in secundative alignment in Khanty is very robust. The discussion of Inuit data in Section 6.1.2 allows to assume that DP-restriction is an important property of secundative alignment.

Finally, this alternation in all three languages is lexically restricted to a small number of verbs, while the indirective-secundative alternation in Khanty is lexically unrestricted. Therefore, it can be concluded that the alternation in German, Russian, and Hungarian is a different phenomenon from indirective-secundative alignment alternation. An analysis of the pattern in 330-332 is beyond the scope of the present study. A preliminary analysis is that prefixes spell out a functional projection that assigns instrumental to the theme.

6.4 Summary

This chapter tests whether the analysis developed for Kazym Khanty extends to two other languages with indirective-secundative alternation: Kalaallisut (Inuit, Eskimo-Aleut) and Eastern Mansi (Ob-Ugric, Uralic). Kalaallisut data provide additional morphological support for the analysis of indirective-secundative alternation developed in this dissertation. HighApplP is morphologically marked on Kalaallisut verbs. Hence, it can be shown that this projection is present in clauses with high applied arguments, as well as in low applicative clauses with indirective alignment. Crucially, high applicative morphology is absent from clauses with secundative alignment. This confirms the conclusion that indirective-secundative alternation is dependent on the presence or absence of HighApplP in the structure. Furthermore, the Kalaallisut data provide additional evidence for the existence of the marked default case. The marked default case is found not only on the secundative themes, but also on stranded modifiers under noun incorporation.

In this chapter, I also argue that secundative alignment requires an analysis distinct from oblique objects in antipassive clauses and unmarked objects in DOM-systems. A systematic comparison of these phenomena in Inuit and Eastern Mansi provides evidence of their different natures. Although antipassive objects are marked with the same oblique case, they do not have a DP-restriction and are argued to be assigned case in syntax, which allows to distinguish them from secundative themes.

Eastern Mansi, on the other hand, allows secundative alignment to be distinguished from differential object marking. This is particularly important in light of multiple analyses of DOM that treat unmarked objects as missing case values (Ormazabal & Romero 2010; López 2012; Bárány 2017, 2018 a.o.). If this is universally true, it would create a problem for the hypothesis about the inherent case overspecification in argument positions. Eastern Mansi has both DOM and secundative alignment. I show that DOM in Eastern Mansi should not be analyzed as an opposition between nominals with and without syntactic case-licensing (cf., e.g., Keine & Müller 2015; Irimia 2023). Rather, Eastern Mansi has two allomorphs for accusative case. This solves the potential problem for the approach to secundative alignment developed in this dissertation and allows the inherent case overspecification hypothesis to be maintained.

Finally, the "applicative constructions" in the sense of Baker (1988) are reviewed and shown to be a distinct phenomenon from indirective-secundative alternation.

Chapter 7

Conclusion

This dissertation addresses three aspects of the theory of Case. First, it discusses the mode of case assignment and provides arguments in favor of the checking approach and against the valuation approach. Second, it models how case assignment is connected to the abstract licensing of nominal arguments and size restrictions on unlicensed arguments. Third, it investigates default case marking, specifically the default case in argument positions and motivates the existence of a marked default case.

The empirical basis of the discussion is indirective-secundative alignment alternation in low applicative and causative clauses. This study is primarily based on the data from the Kazym dialect of Northern Khanty. The proposed analysis is supported by data from Eastern Khanty, Eastern Mansi (Uralic), and Kalaallisut (Inuit).

The main theoretical proposal of this study comprises three core parts. First, it provide evidence for existence of two types of default case marking in the language. It is argued that the default case in argument positions is different from the default case in adjunct positions, e.g. hanging topics (cf. Schütze 2001). In adjunct positions, the unmarked default case is found, which is spelled out by the morphology of the least marked case in the language. This follows trivially if nominals in adjunct positions lack inherent case values. In contrast, nominals in argument positions can have oblique morphology in the absence of syntactic Case licensing. Therefore, a second type of default case has to be postulated: a marked default case. The marked default case can be found on theme arguments in secundative alignment and on stranded modifiers of an incorporated noun. There is another important difference between the two types of the default case. The unmarked default can be found on both NPs and DPs, whereas the marked default is accompanied by a DP-restriction, i.e. it can be found only on structurally reduced nominal arguments. The analysis suggested in this dissertation accounts for this asymmetry by suggesting that arguments have a KP-projection with an inherent uninterpretable [uCase]-feature, which is overspecified for case values. If a noun is not assigned a case value under Agree with an interpretable [iCase]-feature, all inherent case values are spelled out, which gives rise to oblique morphology being used as the default case marking.

The second part of the proposal discusses the mechanism of case assignment. As it is argued that arguments have an inherent case value, case assignment must be modeled as checking rather than valuation. In contrast to the original Chomskian model (Chomsky 2000, 2001), the current proposal models case assignment as Agree between two valued features with different sets of values. This is accounted for by the theory of two-step Agree (Arregi & Nevins (2012)). A modification to the original definition of the two-step Agree has been proposed, so that it can account for checking as well as valuation. In particular, it is suggested that two features can enter into Agree-Link if at least one of the features is uninterpretable and/or unvalued. Additionally, Agree-Copy is replaced by a more general Agree-Match that modifies the value of the uninterpretable [uF], so that it matches the values of the interpretable [iF].

The third part of the proposal addresses the nature of "derivational time-bombs" (Kalin 2018) and DP-restriction on unlicensed arguments. The novel definition of Agree provides a motivation for the DP-restriction on secundative themes. I propose that D^o in Khanty has an unvalued [uCase]-feature, which enters into the Agree-Link relation with a valued [uCase]-feature on K^o. Crucially, both features are uninterpretable, and Agree-Match cannot prefer impoverishment over copying of the case values. Such derivation crashes at PF unless a subsequent Agree-Link with an interpretable case feature takes place. If this happens, Agree-Match can modify both uninterpretable features on K^o and D^o, and the case is successfully assigned.

This study also makes several smaller claims to the case theory. First, dative case in indirective alignment is argued to be a structural case assigned by HighAppl^o. An applied argument, base-generated in SpecHighApplP, is correctly predicted to be obligatory marked with an (inherent) dative. In low applicative and causative clauses, an HighApplP is merged optionally, which gives rise to indirective-secundative alignment alternation. If HighApplP is present, the indirect object receives dative case, and the clause has indirective alignment. This analysis is supported by the raising of the dative-marked indirect object to a position outside the VP, as well as by the inavailability of secundative alignment for clauses with high applied arguments. I additionally distinguish among the three flavors of HighAppl^o. The thematic flavor of HighAppl^o externally merges an applied argument. In Kazym Khanty, it also assigns dative case to this argument. In Kalaallisut, Case-probe is absent in the thematic flavor of HighAppl^o, so high applied arguments are never marked with dative. The a-thematic flavor of HighAppl^o does not externally merge a new argument in the structure but is able to assign dative case. This flavor can be merged in low applicative and causative clauses and assigns dative case to the highest argument in its c-command domain.

Secondly, themes in secundative alignment share some properties with demoted objects in antipassive clauses and unmarked objects under differential object marking. I argue that secundative alignment is distinct from both antipassive and differential object marking. Antipassive objects in Inuit are marked with the same case as secundative themes. Therefore, it is a plausible hypothesis that antipassive objects also bear a marked default case. However, this hypothesis cannot be maintained. Antipassive objects can be full DPs and are case-licensed by functional antipassive (aP) projection (cf. Yuan 2018). Unmarked objects in languages with differential object marking are often argued to obey the same type of DP-restriction as secundative themes. However, in contrast to secundative themes, unmarked objects in DOM are unmarked morphologically. The absence of case morphology is often analyzed as the absence of any abstract Case value, i.e. the absence of a syntactically assigned case. However, I show for Eastern Mansi that unmarked direct objects receive an abstract case value, that is “weak” and is not realized morphologically.

This study leaves several questions for further research. First, the proposed model of case assignment assumes that case features are uninterpretable on arguments and interpretable on case-assigning functional heads. However, I remain agnostic regarding the actual interpretation of the case features. The semantics of case is a question that is rarely addressed in linguistic studies (though see e.g. Kagan 2007) and requires further research.

Second, the current study discusses scenarios in which marked default is found on the internal argument (secundative theme and modifiers of incorporated direct object). It remains an open question whether the marked default case can be found in other argument positions. This is closely connected to the question of whether non-internal arguments can be structurally reduced. Lyskawa & Ranero (2022) argue that only complements can be NPs, while specifiers are always full DPs. If this is universally true, the marked default case can only be found on the internal argument. Nevertheless, the current state of research on this matter does not allow to exclude the existence of structurally reduced specifiers and a wider distribution of the marked default case. On the other hand, current analysis leaves open the possibility that DP-restriction is not a necessary condition of the marked default case. In a language exists that has a [uCase]-feature only on K^o, but not on any other projection, the marked default should not be accompanied by a DP-restriction.

Another open issue is whether inherent overspecification is unique to the [Case]-feature. It can be relatively safely assumed that ϕ-agreement does not show signs of inherent overspecification on probes. In particular, default agreement is always the least marked exponent (Preminger 2014). If Case is unique in this respect, the question arises about the reason behind its special status. If Case is not unique, the question is what other features have the same property and whether they form a natural class with the Case feature. Each of these questions goes beyond the scope of this study and requires extensive further research.

Appendix A

Case system in dialects of Northern Khanty

As mentioned in Chapter 2, the Northern Khanty dialects have very different case systems. Additionally, the mismatch between the nominal and pronominal case systems is different in all of them. This is surprising, since each of the Northern Khanty dialects has only three to four morphological cases. The Kazym Khanty system is described in detail in Chapter 2. The case system in Eastern Khanty dialects is discussed in Chapter 5. This Appendix provides an overview of the case morphology in two other dialects of Northern Khanty.

Shuryshkary dialect of Northern Khanty

the case exponents for nominal declension in the Shuryshkary dialect are listed in Table 1. The morphological case system for nouns is very similar to that of Kazym Khanty. The only difference is that Shuryshkary Khanty has two exponents for Dative *-a* and *-i*, while Kazym Khanty has only the first one. I could not find any report on the distribution of these two exponents, e.g. Solovar et al. (2016) list them both as dative.

NOM/ACC	-∅
DAT	-a/-i
(LOC)-INSTR	-ən

Table 1: Case exponents (nominal declension; Shuryshkary Khanty) (Solovar et al. 2016: p. 104)

Personal pronouns have a reduced case system with only two morphological forms. Pronominal declension is shown in Table 2. In the Shuryshkary dialect, nominative and accusative are syncretic in both nominal and pronominal declension. I remain agnostic about whether this is a purely morphological syncretism or whether the nominative-accusative distinction is absent in the syntax of this dialect as well. As for the two dative allomorphs, Solovar et al. (2016) do not comment on their distribution, but maybe, it depends on the information structure (cf. Koškareva 2002).

	NOM/ACC	DAT
1SG	ma	manɛm/manema
2SG	ŋăn	ŋănen/ŋănena
3SG	λʉw	λʉweλ/λʉweλa
1DU	min	minemən/minməna
2DU	nin	ninan/ninana
3DU	λin	λinan/λinana
1PL	mʉŋ	mʉŋew/mʉŋewa
2PL	niŋ	niŋan/niŋana
3PL	λiw/λij	λiweλ/λiweλa
		λijeλ/λijeλa

Table 2: Case exponents (pronominal declension; Shuryshkary Khanty) (Solovar et al. 2016: p.109)

To the best of my knowledge, there is no grammar of the Shuryshkary dialect of Northern Khanty. Therefore, it remains unclear whether grammatical functions of the cases differ in the Kazym and Shuryshkary dialects.

Obdorsk dialect of Northern Khanty

The Obdorsk dialect also has three case exponents in nominal declension, which are listed in Table 3. Crucially, this case system differs from both the Kazym and the Shuryshkary dialects. Note that the Obdorsk dialect has no dative case, but the translative exponent (terminology after Nikolaeva 1999) resembles one of the dative exponents in the Shuryshkary dialect.

NOM/ACC	∅
LOC	-na
TRANS	-ji/-Ci

Table 3: Case exponents (nominal declension; Obdorsk Khanty)

The Obdorsk pronominal case system in Table 4 is very different from the nominal case system in this dialect, as well as from the pronominal case system in the other dialects. On one hand, personal pronouns in Obdorsk Khanty do not have a translative form. However, in contrast to the Kazym and Shuryshkary dialects, there is a locative form of pronouns. On the other hand, personal pronouns in Obdorsk Khanty have a syncretic dative-accusative form used to mark both recipients and direct objects. Recall from Table 3 that nouns do not have a special morphological marker for either of these cases.

	NOM	DAT-ACC	LOC
1SG	ma	mănem	mănemna
2SG	ŋăn	ŋănen	ŋănenna
3SG	λʉw	λʉweλ	λʉweλna
1DU	min	minemən	minemənna
2DU	nin	ninan	ninanna
3DU	λin	λinan	λinanna
1PL	mʉŋ	mʉŋew	mʉŋewna
2PL	niŋ	niŋan	niŋanna
3PL	λuwət	λuweλ	λuweλna

Table 4: Case exponents (pronominal declension; Obdorsk Khanty) (Solovar et al. 2016: p.110-111)

The unmarked nominative is found on subjects, possessors, and complements of postpositions. In this respect, Obdorsk Khanty is similar to other Northern Khanty dialects. Adjuncts denoting a period of time are also marked with nominative, as 334 illustrates.

(334) ka:t tal xo:p we:r-s-əlli
two year.[NOM] boat.[ACC] do-PST-3SG>SG
'He was making a boat for two years.' ((Nikolaeva 1999: p.13, ex. 20))

Nominal direct object are zero-marked, as in 335. Accusative in nominal declension is syncretic with nominative, similar to the other Northern Khanty dialects.

(335) o:xsar so:wər we:l-əs
fox.[NOM] hare.[ACC] kill-PST.[3SG]
'The fox killed the hare...' ((Nikolaeva 1999: p.50, ex. 142a))

Personal pronouns have an "object" form, which is syncretic between dative and accusative. The "object" form marks direct objects, as in 336, and recipients, as in 337-338.

(336) luw jir-əs ma:-ne:m anta naŋ-e:n
(s)he.NOM tie_down-PST.[3SG] I-ACC/DAT NEG you-ACC/DAT
'He tied down me, not you.' ((Nikolaeva 1999: p.74, ex. 195a))

(337) mola naŋ kutpe:-l śi iwe:nŋa li-l-e:n
why you.NOM middle_part-POSS.3SG.[ACC] FOC I-ACC/DAT always
feed-NPST-2SG>SG
'Why do you always feed me a middle part [of the fish]?' ((Nikolaeva 1999: p.54))

(338) to:rəm ma:n-e:m mosa śi part-əs
God.[NOM] I-ACC/DAT something.[ACC] EMPH order-PST.[3SG]
'God has ordered me something.'

There is no dative exponent in the nominal declension. Moreover, the dative marker is completely absent from the paradigm rather than being syncretic with some other case marker. Nominal recipients are merged as PPs, as in 339.

(339) ma a:n Petra e:lti ma-s-əm
I.NOM cup.[ACC] Peter to give-PST-1SG
'I gave a/the cup to Peter.' ((Dalrymple & Nikolaeva 2011: p.180, ex.8))

Nominal experiencers, which are also expected to bear dative, are merged as PPs, similar to recipients. Note that the same postposition is used to introduce a recipient in 339 and an experiencer in 340).

(340) xu:j-əm a:mp-ət e:lti ka:wrəm u:l-m-al
sleep-NFIN.PST dog-PL.[NOM] to hot be-NFIN.PST-3SG
'Sleeping dogs were hot.' ((Nikolaeva 1999: p.43, ex. 115c))

Nikolaeva (1999) has several example clauses with pronominal experiencers in her grammar. In 341a, the experiencer bears a locative case, while in others (341b), the dative-accusative form is used.

(341) a. ma:n-e:mna iśki
I-LOC cold
'I am cold.' ((Nikolaeva 1999: p.43, ex. 115b))

b. naŋ-e:n tam muw-na u:l-ti a:təm
you-ACC that land-LOC live-INF bad
'It's bad for you to live in this land.' ((Nikolaeva 1999: p.43, ex. 117))

Translative marks nominals in predicative use, as in 342a. It also marks a "result or final point of movement or other verbal action" (Nikolaeva 1999: p. 14). An example is provided in 342b. The translative exponent is similar to one of the dative exponents in the Shuryshkary dialect.

(342) a. ma luw-e:l ma jik-e:m-mi lu:ŋət-l-e:m
I he-ACC I son-POSS.1SG-TRANS count-NPST-1SG>SG
'I consider him my son.' (Nikolaeva 1999: ex.23a)

b. saŋxəŋ-ŋi le:l-s-əŋən
top-TRANS load-PST-3DU
'They loaded (it) till the top.' (Nikolaeva 1999: ex.23b)

Locative is used to mark both location and direction, as in 343, as well as some other adjunct types, e.g. instrument, measure, point in time. It can also be used as a lexical case on the objects of some verbs, e.g. 'be proud' in 344.

(343) (Solovar et al. 2016: p.106)

a. Mʉn was-na ol-l-ew
we town-LOC live-NPST-1PL
'We live in the town.'

b. Am-em was-na măn-əs
Mother-1SG town-LOC go-PST.[3SG]
'My mother went to the town.'

(344) jik-əl-naj ara:ś-l
son-POSS.3SG-LOC proud-NPST.[3SG]
'He is proud of his son.' (Nikolaeva 1999: ex.22)

Locative also marks demoted arguments, e.g. the passive agent in 345a and the theme in a secundative construction in 345b. Exeptionally, agents in active constructions can be locative marked as well, e.g. in 346.[70]

(345) a. wo:rona:j-na śikənśa ittam wo:lək-ə-l xo:ll-əpt-əl-a
crow-LOC so that wolf-POSS.3SG cry-CAUS-NPST-PASS.[3SG]
'So the crow made this wolf cry.' (Nikolaeva 1999: ex.75b)

b. pe:tra:j-e:n xo:p-na mo:jl-ə-s-a
Peter-POSS.2SG boat-LOC give-EP-PST-PASS.[3SG]
'He was given a boat by Peter.' (Nikolaeva 1999: ex.75a)

(346) so:rńi-na pos-ij-əl
gold-LOC flow-IMPF-NPST.[3SG]
'Gold is floating down.' (Nikolaeva 1999: ex.118a)

Note that locative forms of personal pronouns are used only as temporal or locative adjuncts. Personal pronouns are illicit as passive subjects in Obdorsk Khanty, similar to the Kazym dialect, despite the presence of the exponent. Therefore, the restriction on pronouns as passive agents is likely not morphological in nature. Rather, it is explained by the fact that passive agents must be (i) focal, (ii) new information, and (iii) less animate than the promoted internal argument (cf. Muravyev 2023). Pronouns do not satisfy these requirements and hence cannot be used as passive agents.

[70]Agent demotion in active clauses is very common in Eastern Khanty (e.g. Filchenko 2007). But it is rather an exception in Northern Khanty dialects. The speakers on Kazym Khanty do not allow 346 at all. For Obdorsk Khanty, Nikolaeva (1999) provides this example, but reports that such constructons are rare.

Appendix B

Quantifier scope in double object constructions

The relative scope of two quantifiers in argument positions and, more precisely, the possibility of inverse scope has been used as a test for the base-generation order of two arguments (Frey 1993; Aoun & Li 1993; Bruening 2001; Harley & Jung 2015; Collins 2020 among others). Kazym Khanty data on the relative scope of two quantified objects does not align with other tests in Chapter 3 and seems to be independent of the surface word order.[71]

In secundative alignment, the universal quantifier on IO_{ACC} obligatorily scopes over the numeral on DO_{LOC}, irrespective of linear word order. In 347, each child has received exactly one math problem, different for every child. In 348, the mother has read three books to every child, but those books are different in each case. Recall from Chapter 3 that the scrambling of the LOC marked themes does not affect variable binding and is considered an instance of A'-movement. However, the frozen scope in 347-348 cannot be explained with A/A'-opposition, since A'-movement should be able to affect quantifier scope.

(347) a. Uśitel [kašəŋ ńawrɛm] [i primer-ən]
Teacher.[NOM] every child.[ACC] one maths.example-LOC
mă-s-λe
give-PST-3SG>SG

b. Uśitel [i primer-ən] [kašəŋ ńawrɛm]
Teacher.[NOM] one maths.example-LOC every child.[ACC]
mă-s-λe
give-PST-3SG>SG
'The teacher gave every child one mathematical example.'

*(every > Num; *Num > every)*

(348) a. Aŋk-eλ [kašəŋ ńawrɛm-əλ] [χɵλəm kinśka-jən]
mother-POSS.3SG.[NOM] every child-POSS.3SG.[ACC] three book-LOC
λʉŋət-s-əλλe
read-PST-3SG>SG

[71] Clauses with causative verbs behave similarly to lexical ditransitives and applicativized monotransitives. Therefore, I will not provide examples of all three types of clauses in the remainder of this Appendix.

b. Aŋk-eλ [χθλəm kinśka-jən] [kašəŋ ńawrɛm-əλ]
mother-POSS.3SG.[NOM] three book-LOC every child-POSS.3SG.[ACC]
λʉŋət-s-əλλe
read-PST-3SG>SG
'Mother read every child three books.'
*(every > Num; *Num > every)*

The same behavior is attested in indirective alignment with non-agreeing direct objects. The non-agreeing direct object obligatorily scopes below the DAT-marked IO, irrespective of linear word order. The books in 349 can be different for each child. There is a considerable degree of inter-speaker variation in this regard. Some native speakers allow both direct and inverse scope in secundative alignment and in indirective alignment without object agreement. Determining the exact factors regulating this variability is beyond the scope of the present study. However, note that both secundative themes and ACC-marked non-agreeing direct objects tend to be focal and stay in situ (Smith 2020; É. Kiss 2021) or undergo A'-movement (see Chapter 3). Importantly, the behavior of LOC-marked and ACC-marked non-agreeing direct objects is identical. In both cases, it is independent of the surface word order. Since word order does affect Q-scope, I assume that non-agreeing direct objects (both LOC and ACC marked) obligatorily reconstruct for Q-scope and are interpreted in their base position.

(349) a. Aŋk-eλ [kašəŋ ńawrɛm-a] [χθλəm kinśka]
mother-POSS.3SG.[NOM] every child-DAT three book.[ACC]
λʉŋət-əs
read-PST.[3SG]

b. Aŋk-eλ [χθλəm kinśka] [kašəŋ ńawrɛm-a]
mother-POSS.3SG.[NOM] three book.[ACC] every child-DAT
λʉŋət-əs
read-PST.[3SG]
'Mother read every child three books.'
*(every > Num; *Num > every)*

Interestingly, clauses with agreeing direct objects also have a frozen scope, but here, the DO obligatorily scopes over the IO. For instance, the three books in 350 must be the same for every child. Again, the overt order of the two objects does not influence the relative scope of the two quantifiers.

(350) a. Aŋk-eλ [kašəŋ ńawrɛm-a] [χθλəm kinśka]
mother-POSS.3SG.[NOM] every child-DAT three book.[ACC]
λʉŋət-s-əλλe
read-PST-3SG>SG

b. Aŋk-eλ [χɵλəm kinśka] [kašəŋ ńawrɛm-a]
mother-POSS.3SG.[NOM] three book.[ACC] every child-DAT
λʉŋət-s-əλλe
read-PST-3SG>SG
'Mother read every child three books.'

*(*every > Num; Num > every)*

It has long been known that A- and A'- movement behave differently with respect to reconstruction for quantifier scope. Boeckx (2001) suggests that A-moved quantifiers do not reconstruct because arguments are interpreted in their case-position. He follows the assumption that A-moved nominals receive case in their ex situ position. Therefore, only the higher copy is relevant for the interpretation. Hence, A-scrambled (ex-situ, agreeing) DOs must have a wide scope over the IO. However, this approach is unable to account for the data above because the agreeing quantified DOs have a wide scope, even if they remain in situ, i.e. lower than an IO. Furthermore, both agreeing and non-agreeing DOs obligatorily reconstruct for Condition C (see Chapter 3). This, in turn, suggests that direct objects always receive case in their base position below IO (Bhatt & Keine 2019).

To sum up, the Kazym Khanty data suggest that the relative scope of the two quantified objects does not depend on the overt word order, but rather on the presence of object agreement, i.e. on the definiteness/topicality of the direct object. If the DO is topical/definite (i.e., triggers object agreement), it takes an obligatory wide scope over the IO, irrespective of the overt word order. In contrast, if the DO is focal/indefinite, it takes a narrow scope below the IO or (for some speakers) allows optional variation with a wide scope interpretation. Since the variation between wide/narrow scope does not correlate with linear word order, it cannot be analyzed in terms of reconstruction into the base-generation position. Hence, the availability of the inverse scope cannot be considered a reliable test for examining the base-generation order of the two objects in applicative and causative clauses. Note that, according to Frey (1993: p.176-178), numerals and universal quantifiers, as well as the quantifier 'every', do not exclude a referential reading in German. Rather, they indicate the difference between definite and indefinite DPs. Therefore, they cannot be used to reliably test quantifier scope and the base-generation order. A deeper analysis of the relation between the frozen quantifier scope and object agreement, i.e. (in)definiteness of the direct object, in Kazym Khanty is a matter for future research.

References

Abney, Steven P. 1987. The English noun phrase in its sentential aspect. Doctoral dissertation, Massachusetts Institute of Technology.

Aissen, Judith. 1999. Markedness and subject choice in Optimality Theory. *Natural Language & Linguistic Theory* 17:673–711.

Aissen, Judith. 2003. Differential object marking: Iconicity vs. economy. *Natural language & linguistic theory* 21:435–483.

Akkuş, Faruk. 2021. (Implicit) argument introduction, Voice and causatives. Doctoral dissertation, University of Pennsylvania.

Akkuş, Faruk. 2022. On Causee in Sason Arabic. *Syntax* 25:299–334.

Aldridge, Edith. 2008. Generative approaches to ergativity. *Language and Linguistics Compass* 2:966–995.

Alexiadou, Artemis. 2003. Some notes on the structure of alienable and inalienable possessors. *From NP to DP* 2:167–188.

Alexiadou, Artemis. 2006. Left dislocation (including CLLD). In *The Blackwell companion to syntax*, ed. Henk van Riemsdijk Martin Everaert, 668–699. Wiley Online Library.

Alexiadou, Artemis, Elena Anagnostopoulou, & Florian Schäfer. 2015. *External arguments in transitivity alternations: A layering approach.* Oxford: Oxford University Press.

Alexiadou, Artemis, Elena Anagnostopoulou, & Christina Sevdali. 2014. Opaque and transparent datives, and how they behave in passives. *The Journal of Comparative Germanic Linguistics* 17:1–34.

Anagnostopoulou, Elena. 2003. *The syntax of ditransitives: Evidence from clitics*, volume 54. Walter de Gruyter.

Anagnostopoulou, Elena, & Christina Sevdali. 2020. Two modes of Dative and Genitive case assignment: Evidence from two stages of Greek. *Natural Language & Linguistic Theory* 38:987–1051.

Aoun, Joseph, & Yen-Hui Audrey Li. 1989. Scope and constituency. *Linguistic inquiry* 20:141–172.

Aoun, Joseph, & Yen-hui Audrey Li. 1993. *Syntax of scope*, volume 21. MIT Press.

Aristova, Maria V. 2023. Linguistic situation and language shift in the Kazym Khanty-speaking community. Master's thesis, Higher Schhol of Econimics, Moscow.

Arkadiev, Peter M., & Yakov G. Testelets. 2019. Differential nominal marking in circassian. *Studies in Language* 43:715–751.

Arregi, Karlos, & Andrew Nevins. 2012. *Morphotactics: Basque auxiliaries and the structure* *of spellout*, volume 86. Springer Science & Business Media.

Assmann, Anke, Doreen Georgi, Fabian Heck, Gereon Müller, & Philipp Weisser. 2015. Ergatives move too early: On an instance of opacity in syntax. *Syntax* 18:343–387.

Asztalos, Erika, Lena Borise, Katalin Gugan, Nikolett Mus, Andreas Schmidt, & Balázs Surányi. 2022. From rigid to flexible verb-finality: A prosodically motivated information-structural account of word order change. ICHL 25.

Atlamaz, Ümit. 2019. Agreement, case, and nominal licensing. Doctoral dissertation, Rutgers University.

Bailyn, John Frederick. 2001. On scrambling: A reply to Bošković and Takahashi. *Linguistic inquiry* 32:635–658.

Baker, Mark. 1985. The mirror principle and morphosyntactic explanation. *Linguistic Inquiry* 16.

Baker, Mark. 1988. *Incorporation: A theory of grammatical function changing*.

Baker, Mark. 2012. On the relationship of object agreement and accusative case: Evidence from Amharic. *Linguistic Inquiry* 43:255–274.

Baker, Mark. 2015. *Case: Its principles and its parameters*. 146. Cambridge University Press.

Baker, Mark, & Livia Camargo Souza. 2020. Agree without agreement: Switch-reference and reflexive voice in two Panoan languages. *Natural Language & Linguistic Theory* 38:1053–1114.

Baker, Mark, & Nadya Vinokurova. 2010. Two modalities of case assignment: Case in Sakha. *Natural Language & Linguistic Theory* 28:593–642.

Balandin, Aleksey N. 1955. O jazykakh i dialectakh khanty. [About languages and dialects of Khanty people.]. *V pomošč učitelju škol Krainego Severa* 5:75–92.

Baltin, Mark, Rose-Marie Déchaine, & Martina Wiltschko. 2015. The irreducible syntax of variable binding. URL: ling.auf.net/lingbuzz/002425.

Baltin, Mark R. 1995. Floating quantifiers, PRO, and predication. *Linguistic inquiry* 199–248.

Bárány, András. 2017. *Person, case, and agreement*. Oxford: Oxford University Press.

Bárány, András. 2018. DOM and dative case. *Glossa: a journal of general linguistics* 3:1–40.

Bárány, András. 2021a. Partially ordered case hierarchies. *Glossa: a journal of general linguistics* 6.

Bárány, András. 2021b. A typological gap in ditransitive constructions. In *Proceedings of the 38th West Coast Conference on Formal Linguistics*, ed. Rachel Soo, Una Y. Chow, & Sander Nederveen, 43–53. Somerville, MA: Cascadilla Proceedings Project. URL http://www.lingref.com/cpp/wccfl/38/paper3549.pdf.

Bárány, András. 2023. Interaction, satisfaction, and (a)symmetric object agreement. In *Proceedings of the 53th annual meeting of the North East Linguistic Society (NELS53)*, ed. Suet-Ying Lam & Satoru Ozaki, 39–51. GLSA.

Bárány, András, & Irina Nikolaeva. 2021. On adjoined possessors. *Linguistic Inquiry* 52:181–194.

Bárány, András, & Michelle Sheehan. 2021. Challenges for dependent case. *On the place of* *case in grammar* .

Bárány, András, & Jenneke van der Wal. 2022. We don't agree (only) upward. *Linguistic Inquiry* 53:501–521.

Barss, Andrew, & Howard Lasnik. 1986. A note on anaphora and double objects. *Linguistic inquiry* 17:347–354.

Béjar, Susana. 2003. *Phi-syntax: A theory of agreement*. University of Toronto Toronto, ON.

Béjar, Susana, & Milan Rezac. 2009. Cyclic agree. *Linguistic inquiry* 40:35–73.

Belkind, Aleksandra. 2023. Prominence as an anchor for a clitic: Prosody-sensitive placement of the conditional subordinator ki in Kazym Khanty. *Glossa: a journal of general linguistics* 8.

Benmamoun, Elabbas. 1999. The syntax of quantifiers and quantifier float. *Linguistic inquiry* 30:621–642.

Bhatt, Rajesh, & Elena Anagnostopoulou. 1996. Object shift and specificity: Evidence from ko-phrases in Hindi. *Papers from the main session of CLS* 32:11–22.

Bhatt, Rajesh, & Stefan Keine. 2019. Secondary strong crossover in Hindi and the typology of movement. In *49th Annual Meeting of the North East Linguistic Society*, 125–135.

Bhatt, Rajesh, & Martin Walkow. 2013. Locating agreement in grammar: an argument from agreement in conjunctions. *Natural Language & Linguistic Theory* 31:951—-1013.

Bierwisch, Manfred. 1967. Syntactic features in morphology: General problems of so-called pronominal inflection in German.

Bikina, Daria, Denis Rakhman, Vsevolod Potseluev, Aleksey Starchenko, & Svetlana Toldova. 2022. Non-finite constructions in Khanty: Their unity and diversity. *Folia Linguistica* 56:625–665.

Bíró, Bernadett, & Katalin Sipőcz. 2017. The Mansi ditransitive constructions. *Finno-Ugric Languages and Linguistics* 6.

Biskup, Petr. 2006. Scrambling in Czech: Syntax, semantics, and information structure. *Proceedings of NWLC 21* 1–15.

Bjorkman, Bronwyn, & Hedde Zeijlstra. 2019. Checking up on (ϕ-)Agree. *Linguistic Inquiry* 50:527–569.

Bobaljik, Jonathan. 1993. On ergativity and ergative unergatives. *MIT Working papers in Linguistics* 19:334–385.

Bobaljik, Jonathan. 2003. Floating quantifiers: Handle with care. *The second Glot international state-of-the-article book* 107–148.

Bobaljik, Jonathan. 2008. Where's phi? Agreement as a post-syntactic operation. In *Phi-theory: Phi features across modules and interfaces*, ed. Daniel Harbour, David Adger, & Susana Béjar, 295–328. Oxford: Oxford University Press.

Boeckx, Cedric. 2001. Scope reconstruction and a-movement. *Natural Language & Linguistic Theory* 19:503–548.

Bok-Bennema, Reineke. 1991. *Case and agreement in Inuit*. 38. Walter de Gruyter.

Borik, Olga, & Berit Gehrke. 2015. An introduction to the syntax and semantics of pseudo-incorporation. In *The syntax and semantics of pseudo-incorporation*, 1–43. Brill.

Borise, Lena, & Tamás Halm. 2024. The curious case of a two-headed phrase: the symmetrical syntax of co-compounds. presentation on Syntax of Uralic Lnaguages 5 (SOUL5).

Bošković, Željko. 2004. Be careful where you float your quantifiers. *Natural language & linguistic theory* 22:681–742.

Bošković, Željko. 2005. On the locality of left branch extraction and the structure of NP. *Studia linguistica* 59:1–45.

Bošković, Željko. 2006a. Case and agreement with genitive of quantification in Russian. *Agreement System* 99–120.

Bošković, Željko. 2006b. Case checking versus case assignment and the case of adverbial NPs. *Linguistic inquiry* 37:522–533.

Bošković, Željko. 2008. What will you have, DP or NP? In *Proceedings of the 37th annual meeting of the North East Linguistic Society (NELS37)*, 101–114.

Bošković, Željko, & Jon Gajewski. 2011. Semantic correlates of the NP/DP parameter. In *Proceedings of the 39th annual meeting of the North East Linguistic Society (NELS39)*, 121–134.

Bošković, Željko, & Serkan Şener. 2014. The Turkish NP. In *Crosslinguistic studies on noun phrase structure and reference*, 102–140. Brill.

Bowers, John S. 1981. *The theory of grammatical relations*. Cornell University Press.

Brisson, Christine M. 1998. *Distributivity, maximality, and floating quantifiers*. Rutgers The State University of New Jersey, School of Graduate Studies.

Bruening, Benjamin. 2001. QR obeys superiority: Frozen scope and ACD. *Linguistic inquiry* 32:233–273.

Bruening, Benjamin. 2010a. Ditransitive asymmetries and a theory of idiom formation. *Linguistic inquiry* 41:519–562.

Bruening, Benjamin. 2010b. Double object constructions disguised as prepositional datives. *Linguistic inquiry* 41:287–305.

Burov, Daniil, Anna Kulikova, Denis Pisarenko, Alexey Starčenko, Varvara Tyutyunnikova, & Maria Čeremisinova. 2024. Mal mala menshe: malyje imennyje gruppy v kazymskom dialekte khantyjskogo jazyka [small and even smaller: small nominals in Kazym Khanty]. Paper presentated at the conference Tipologija morphosintaksicheskikh parametrov.

Caha, Pavel. 2007. Case movement in PPs. *Nordlyd* 34.

Caha, Pavel. 2009. The nanosyntax of Case. Doctoral dissertation, Universitetet i Tromsø.

Caha, Pavel. 2013. Explaining the structure of case paradigms by the mechanisms of nanosyntax: The classical armenian nominal declension. *Natural Language & Linguistic Theory* 31:1015–1066.

Caha, Pavel. 2024. A default theory of default case. *Glossa: a journal of general linguistics* 9(1).

Calabrese, Andrea. 2008. On absolute and contextual syncretism: remarks on the structure of case paradigms and on how to derive them. In *Inflectional identity*, ed. Asaf Bachrach & Andrew Nevins, volume 18 of *Oxford Studies in Theoretical Linguistics*, 156–205. Oxford: Oxford University Press.

Choi, Hye-Won. 1998. Optimizing structure in context: The case of German scrambling. In *Proceedings of the Twentysixth Western Conference on Linguistics*, ed. Vida Samiian, volume 9, 56–71.

Choi, Hye-Won. 1999. *Optimizing structure in context: Scrambling and information structure*. Stanford: CSLI Publications.

Chomsky, Noam. 1981. *Lectures on government and binding: The Pisa lectures*. Foris Publications.

Chomsky, Noam. 2000. Minimalist Inquiries: The Framework. In *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik*, ed. Roger Martin, David Michaels, Juan Uriagereka, & Samuel Jay Keyser, 89–155. Cambridge, Mass.: MIT Press.

Chomsky, Noam. 2001. Derivation by Phase. In *Ken Hale: A Life in Language*, ed. Michael Kenstowicz, 1–52. Cambridge, Mass.: MIT Press.

Chomsky, Noam, & Morris Halle. 1968. *The sound pattern of English*. New York: Harper & Row.

Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford: Oxford University Press.

Clem, Emily. 2022. Cyclic expansion in Agree: Maximal projections as probes. *Linguistic Inquiry* 54(1):39–78.

Clem, Emily, & Amy Rose Deal. 2024. Dependent case by Agree: Ergative in Shawi. *Linguistic Inquiry* 1–75.

Colley, Justin, & Dmitry Privoznov. 2018. Passive in Khanty. Fieldwork handout.

Colley, Justin, & Dmitry Privoznov. 2020. On the topic of subjects: composite probes in Khanty. In *Proceedings of the 50th annual meeting of the North East Linguistic Society (NELS50)*. GLSA Publications.

Collins, Chris. 2020. A smuggling approach to the dative alternation. In *Smuggling in syntax*, ed. Adriana Belletti & Chris Collins, 96–107. Oxford: Oxford University Press.

Compton, Richard. 2016. Mutually conditioned mood and object agreement in Inuit. In *Proceedings of the 46th annual meeting of the North East Linguistic Society (NELS46)*, 241–250.

Compton, Richard, & Christine Pittman. 2010. Word-formation by phase in Inuit. *Lingua* 120:2167–2192.

Coon, Jessica. 2013. *Aspects of split ergativity*. Oxford: Oxford University Press.

Coon, Jessica. 2017. Two types of ergative agreement: Implications for case. In *A pesky set: Papers for David Pesetsky*, volume 80, 361–370.

Coon, Jessica, & Omer Preminger. 2017. Split ergativity is not about ergativity. In *The Oxford handbook of ergativity*, ed. Jessica Coon, Diane Massam, & Lisa Travis, 226–252. Oxford: Oxford University Press.

Coppock, Elizabeth, & David Beaver. 2014. A superlative argument for a minimal theory of definiteness. In *Semantics and Linguistic Theory*, 177–196.

Coppock, Elizabeth, & David Beaver. 2015. Definiteness and determinacy. *Linguistics and philosophy* 38:377–435.

Csepregi, Márta. 1998. Szurguti osztják chrestomathia. *Studia uralo-altaica* 1–183.

Csirmaz, Aniko. 2004. Perfective and imperfective aspect in Hungarian:(Invisible) differences. In *Proceedings of CONSOLE*, volume 12.

Däbritz, Chris Lasse. 2020. *Topik, Fokus und Informationsstatus: Modellierung am Material nordwestsibirischer Sprachen*, volume 17. Walter de Gruyter.

Dalrymple, Mary, & Irina Nikolaeva. 2005. Nonsubject agreement and discourse roles. *Oxford Working Papers in Linguistics, Philology and Phonetics* 71–91.

Dalrymple, Mary, & Irina Nikolaeva. 2011. *Objects and information structure*. 131. Cambridge University Press.

Danon, Gabi. 2006. Caseless nominals and the projection of DP. *Natural Language & Linguistic Theory* 24:977–1008.

Danon, Gabi. 2011. Agreement and DP-internal feature distribution. *Syntax* 14:297–317.

Dayal, Veneeta. 2011. Hindi pseudo-incorporation. *Natural Language & Linguistic Theory* 29:123–167.

De Swart, Peter, & Helen de Hoop. 2007. Semantic aspects of differential object marking. In *Proceedings of Sinn und Bedeutung*, volume 11, 598–611.

Deal, Amy. 2006. Case-marking and object interpretation in Nez Perce. In *Proceedings of the Annual Meeting of the Chicago Linguistic Society*, volume 42, 17–28. Chicago Linguistic Society.

Deal, Amy Rose. 2010. Ergative case and the transitive subject: A view from Nez Perce. *Natural Language & Linguistic Theory* 28:73–120.

Deal, Amy Rose. 2013. Possessor raising. *Linguistic Inquiry* 44:391–432.

Deal, Amy Rose. 2015a. Ergativity. In *Syntax–theory and analysis. an international handbook*, ed. Artemis Alexiadou & Tibor Kiss, volume 1, chapter 20, 654–708. Berlin: Mouton de Gruyter.

Deal, Amy Rose. 2015b. Interaction and satisfaction in φ-agreement. In *Proceedings of the 45th annual meeting of the North East Linguistic Society (NELS45)*, 179–192.

Deal, Amy Rose. 2017. External possession and possessor raising. In *The Wiley-Blackwell companion to syntax*, ed. Henk van Riemsdijk Martin Everaert, 1509—-1540. Wiley-Blackwell, 2nd edn. edition.

Deal, Amy Rose. 2022. Interaction, satisfaction, and the PCC. *Linguistic inquiry* 1–56.

Dékány, Éva. 2011. A profile of the Hungarian DP: The interaction of lexicalization, agreement and linearization with the functional sequence. Doctoral dissertation, University of Tromsø, Tromsø.

Dékány, Éva. 2021. *The Hungarian nominal functional sequence*. Springer.

Déprez, Viviane Marie. 1989. On the typology of syntactic positions and the nature of chains: move [alpha] to the specifier of functional projections. Doctoral dissertation, Massachusetts Institute of Technology.

Diesing, Molly. 1992. *Indefinites*, volume 20. MIT press Cambridge, MA.

den Dikken, Marcel. 1995. *Particles: On the syntax of verb-particle, triadic, and causative constructions*. Oxford: Oxford University Press.

den Dikken, Marcel. 2023. High and low applicatives of unaccusatives: Dependent case and the phase. *Linguistic Inquiry* 54:479–503.

Doetjes, Jenny. 1992. Rightward floating quantifiers float to the left. *The Linguistic Review* 9:313–332.

Driemel, Imke. 2020. Pseudo-noun incorporation across languages. Doctoral dissertation, Universität Leipzig.

Driemel, Imke. 2023. *Pseudo-Noun Incorporation and Differential Object Marking*, volume 82 of *Oxford Studies in Theoretical Linguistics*. Oxford: Oxford University Press.

Dryer, Matthew S. 1986. Primary objects, secondary objects, and antidative. *Language* 62:808–845.

É. Kiss, Katalin. 2003. Argument scrambling, operator movement, and topic movement in Hungarian. In *Word order and scrambling*, ed. Simin Karimi, 22–43. Blackwell Publishers.

É. Kiss, Katalin. 2019. Fused grammatical and discourse functions in Ob-Ugric: Case, agreement, passive. *Proceedings of the Workshop Clause Typing and the Syntaxto Discourse Relation in Head Final Languages* 163–174.

É. Kiss, Katalin. 2021. What determines the varying relation of case and agreement? Evidence from the Ugric languages. *Acta Linguistica Academica* 67:397–428.

Eberhard, David M., Gary F. Simons, & Charles D. Fennig (eds.). 2024. Ethnologue: Languages of the World. Twenty-seventh edition. Online version: http://www.ethnologue.com.

Erschler, David. 2019. A new argument for existence of the DP in languages without articles. *Journal of Linguistics* 55:879–887.

Fanselow, Gisbert. 1986. On the sentential nature of prenominal adjectives in German. *Folia Linguistica* 20(3-4):341–380.

Fanselow, Gisbert. 1991. *Minimale syntax*, volume 32 of *GAGL: Groninger Arbeiten zur germanistischen Linguistik*.

Fedotova, Idaliya V. 2022. Dialektnoje chlenenie khantyjskogo jazyka po dannym bazisnoj leksiki. [Dialectal classification of Khnaty in the light of basic vocabulary]. *Ural-Altaic studies* 47:117–166.

Filchenko, Andrey Y. 2007. A grammar of Eastern Khanty. Doctoral dissertation, Rice University.

Filchenko, Andrey Y. 2012. Continuity of information structuring strategies in Eastern Khanty. In *Argument structure and grammatical relations. a crosslinguistic typology*, ed. P.Suihkonen, B.Comrie, & V.Solovyev, 115–131. Amsterdam: John Benjamins.

Folli, Raffaella, & Heidi Harley. 2007. Causation, obligation, and argument structure: On the nature of little v. *Linguistic inquiry* 38:197–238.

Forsberg, Ulla-Maija. 2007. *Eastern Mansi (Konda) Grammar*. Société Finno-Ougrienne.

Forsberg, Ulla-Maija. 2024. Khanty dialects. In *The languages and linguistics of Northern Asia*, volume 1. Language families, chapter 5, 253–279. De Gruyter Mouton.

Fortescue, Michael D. 1984. *West Greenlandic*. Croom Helm London.

Fortescue, Michael D. 1998. *Language relations across Bering Strait: Reappraising the* *archaeological and linguistic evidence*. Open linguistics series. Cassell.

Fortescue, Michael D. 2016. How the accusative became the relative: A Samoyedic key to the Eskimo-Uralic relationship? *Journal of Historical Linguistics* 6:72–92.

Frampton, John, & Sam Gutmann. 2006. How sentences grow in the mind. In *Agreement systems*, ed. Cedric Boeckx, volume 92 of *Linguistik Aktuell/Linguistics Today*, 121–157. Amsterdam: John Benjamins.

Frey, Werner. 1993. *Syntaktische Bedingungen für die semantische Interpretation: über Bindung, implizite Argumente und Skopus*. De Gruyter.

Georgala, Effi. 2011. Why German is not an exception to the universal <IO, DO> base order of double object constructions. In *Proceedings of the 28th West Coast Conference on Formal Linguistics (WCCFL 28)*, 96–105.

Georgala, Efthymia. 2012. Applicatives in their structural and thematic function: A minimalist account of multitransitivity. Doctoral dissertation, Cornell University.

Georgi, Doreen. 2014. Opaque interactions of Merge and Agree: On the nature and order of elementary operations. Doctoral dissertation, Leipzig University.

Georgi, Doreen, & Mary Amaechi. 2020. Resumption and islandhood in Igbo. In *Proceedings of the 50th annual meeting of the North East Linguistic Society (NELS 50)*, 261–74. GLSA Amherst.

Georgi, Doreen, & Mary Amaechi. 2023. Resumption in Igbo: Two types of resumptives, complex phi-mismatches, and dynamic deletion domains. *Natural Language & Linguistic Theory* 41:961–1028.

Giusti, Giuliana. 1990. Floating quantifiers, scrambling, and configurationality. *Linguistic Inquiry* 21:633–641.

Glushan, Zhanna. 2010. Deriving case syncretism in differential object marking systems.

Gong, Zhiyu Mia. 2023. Case in wholesale late merger: Evidence from Mongolian scrambling. *Linguistic Inquiry* 1–43.

Hale, Ken, & Samuel Jay Keyser. 1993. On argument structure and the lexical expression of syntactic relations. In *The view from building 20: Essays in linguistics in honor of Sylvain Bromberger*, ed. Ken Hale & Samuel Jay Keyser, 53–110. MIT press.

Halle, Morris. 1997. Distributed morphology: Impoverishment and fission. In *MIT Working Papers in Linguistics 30: Papers at the Interface*, ed. B. Bruening, Y. Kang, & M. McGinnis, 425–449. Cambridge, Massachustets: MIT.

Hallman, Peter. 2015. Syntactic neutralization in double object constructions. *Linguistic inquiry* 46:389–424.

Hallman, Peter. 2024. Argument structure hierarchies and alternations in causative and double object constructions. *Glossa: a journal of general linguistics* 9(1).

Harðarson, Gísli Rúnar. 2016. A case for a Weak Case Contiguity hypothesis — a reply to Caha. *Natural language & linguistic theory* 34:1329–1343.

Harbour, Daniel. 2014. Paucity, abundance, and the theory of number. *Language* 90(1):185–229.

Harley, Heidi. 1997. If you have, you can give. In *Proceedings of the 15th West Coast*

Conference on Formal Linguistics, 193–207. Citeseer.

Harley, Heidi. 2002. Possession and the double object construction. *Linguistic variation yearbook* 2:31–70.

Harley, Heidi. 2013. External arguments and the Mirror Principle: On the distinctness of Voice and v. *Lingua* 125:34–57.

Harley, Heidi. 2017. The "bundling" hypothesis and the disparate functions of little v. In *The verbal domain*, ed. Roberta D'Alessandro, Irene Franco, & Ángel J. Gallego, volume 3, 3–28. Oxford: Oxford University Press.

Harley, Heidi, & Hyun Kyoung Jung. 2015. In support of the P-have analysis of the double object construction. *Linguistic inquiry* 46:703–730.

Harley, Heidi Britton. 1995. Subjects, events, and licensing. Doctoral dissertation, Massachusetts Institute of Technology.

Heck, Fabian, & Anke Himmelreich. 2017. Opaque intervention. *Linguistic Inquiry* 48:47–97.

Heck, Fabian, & Gereon Müller. 2007. Extremely local optimization. In *Proceedings of WECOL*, ed. E. Brainbridge & B. Agbayani, volume 26, 170–183. Fresno: California State University.

Heck, Fabian, & Marc Richards. 2010. A probe-goal approach to agreement and non-incorporation restrictions in Southern Tiwa. *Natural Language & Linguistic Theory* 28:681–721.

Heim, Irene, & Angelika Kratzer. 1998. *Semantics in generative grammar*. Oxford: Blackwell.

Henry, Alison. 2012. Phase edges, quantifier float and the nature of (micro-)variation. *Iberia: An International Journal of Theoretical Linguistics, 4 (2), 23-39.* .

Himmelreich, Anke. 2017. Case matching effects in free relatives and parasitic gaps. Doctoral dissertation, Leipzig University.

Hinterhölzl, Roland. 2012. Some notes on scrambling and object shift. In *Discourse and grammar: A Festschrift in honour of Valeria Molnar*, 305–321.

Holmberg, Anders, Michelle Sheehan, & Jenneke van der Wal. 2019. Movement from the double object construction is not fully symmetrical. *Linguistic Inquiry* 50:677–722.

Ippolito, Michela. 2000. Remarks on the argument structure of Romance causatives. Ms.

Irimia, Monica Alexandrina. 2021. *ABA, DOM and other Accusatives. Paper presented at GLOW 43, Humboldt-Universität zu Berlin. URL https://osf.io/snuyf/.

Irimia, Monica Alexandrina. 2022. DOM and nominal structure: Some notes on DOM with bare nouns. *Languages* 7:175.

Irimia, Monica Alexandrina. 2023. Oblique DOM in enriched case hierarchies. *Glossa: a journal of general linguistics* 8(1).

Jakobson, Roman. 1962. Beitrag zur allgemeinen Kasuslehre. Gesamtbedeutungen der russischen Kasus. In *Selected writings*, volume 2, 23–71. The Hague and Paris: Mouton.

Janhunen, Juha A. 2023. Linguistic Connections across the Bering Strait: Facts and Fantasy. *International Journal of Eurasian Linguistics* 5:143–151.

Jerro, Kyle. 2017. The causative–instrumental syncretism. *Journal of Linguistics* 53:751–788.

Jerro, Kyle Joseph. 2016. The syntax and semantics of applicative morphology in Bantu. Doctoral dissertation, University of Texas at Austin.

Johns, Alana. 1984. Dative Movement in Eskimo. In *Papers from the Parasession on Lexical Semantics*, ed. V. Mishra D. Testen & J.Drogo. Chicago Linguistic Society.

Johns, Alana. 1987. Transitivity and grammatical relations in Inuktitut. Doctoral dissertation, University of Ottawa (Canada).

Johns, Alana. 1992. Deriving ergativity. *Linguistic Inquiry* 23:57–87.

Johns, Alana. 2009. Additional facts about noun incorporation (in inuktitut). *Lingua* 119:185–198.

Kagan, Olga. 2007. On the semantics of structural case. Doctoral dissertation, Hebrew University Jerusalem.

Kagan, Olga, & Asya Pereltsvaig. 2011. Syntax and semantics of bare NPs: Objects of intensive reflexive verbs in Russian. *Empirical issues in syntax and semantics* 8:221–238.

Kaksin, A.D. 2010. *Kazymskij dialekt khantyjskogo jazyka. [Kazym dialect of Khanty]*. Khanty-Mansijsk, 2 edition.

Kalin, Laura. 2018. Licensing and differential object marking: The view from Neo-Aramaic. *Syntax* 21:112–159.

Kalin, Laura, & Coppe van Urk. 2015. Aspect splits without ergativity: Agreement asymmetries in Neo-Aramaic. *Natural language & Linguistic theory* 33:659–702.

Kalin, Laura, & Philipp Weisser. 2019. Asymmetric DOM in coordination: A problem for movement-based approaches. *Linguistic Inquiry* 50:662–676.

Kardos, Eva, & Imola-Ágnes Farkas. 2021. The syntax of inner aspect in Hungarian. *Journal of Linguistics* 58(4):807–845.

Karjalainen, Kustaa Fredrik. 1902. Zur ostjakischen und wogulischen dialektkunde. statistisches. *Anzeiger der Finnisch-Ugrischen Forschungen* 2:77–78.

Kasjanova, Polina. 2019. čja eto mukha? [whose fly is it?]. Fieldwork handout.

Kayne, Richard S. 1998. Overt vs. covert movements. *Syntax* 1:128–191.

Kaškin, Egor, & Nikita Muraviev. 2014. Western Khanty corpus. https://osf.io/uraqx/files/.

Keine, Stefan. 2018. Case vs. positions in the locality of A-movement. *Glossa: a journal of general linguistics* 3(1):1–34.

Keine, Stefan, & Gereon Müller. 2015. Differential argument encoding by impoverishment. In *Scales and hierarchies: A cross-disciplinary perspective*, ed. Ina Bornkessel-Schlesewsky, Andrej Malchukov, & Marc D. Richards, 75–130. Berlin: Mouton de Gruyter.

Kiparsky, Paul. 2001. Structural case in Finnish. *Lingua* 111:315–376.

Kiss, Katalin É. 2023. The (non-) finiteness of subordination correlates with basic word order: Evidence from Uralic. *Acta Linguistica Academica* 70:171–194.

Klumpp, Gerson. 2012. Differential object marking and information structure: On the function of two different pronominal accusatives in Komi and Khanty dialects. *Eesti ja soome-ugri keeleteaduse ajakiri. Journal of Estonian and Finno-Ugric Linguistics* 3:343–372.

Koopman, Hilda, & Dominique Sportiche. 1991. The position of subjects. *Lingua* 85:211–

258.

Kozlov, Alexej A. 2022. Aspect and object agreement in Kazym Khanty. Paper presented at SOUL 4, online.

Koškareva, Natalja Borisovna. 2001. The case system of nous and pronouns in Khanty. In *Variierende Markierung von Nominalgruppen in Sprachen unterschiedlichen Typs*, volume 4 of *Studia Slavica Oldenburgensia*, 235—-254.

Koškareva, Natalja Borisovna. 2002. Kommunikativnaja paradigma khantyjskogo predloženija. [A communicative paradigm of Khanty sentence.]. *Jazyki korennykh narodov Sibiri* 12:29–44.

Koškareva, Natalja Borisovna. 2013. Antualnyje voprosy khantyjskoj grafiki. [Current problems in imprvement of the graphics and orthography of Khanty.]. *Vestnik ugrovedenija* 14(3):47–78.

Koškareva, Natalja Borisovna. 2016. Khantyjskj jazyk. [Khanty language.]. In *Jazyk i obshchestvo: Entsiklopedija*, 535–539.

Kulonen, Ulla-Maija. 1989. The passive in Ob-Ugrian. Doctoral dissertation, Universisty of Helsinki.

Langr, Kathleen. 2014a. Evidence for a DP projection in Inuit. Ms., University of Washington.

Langr, Kathleen. 2014b. Possessive structures as evidence for DP in West Greenlandic. In *In Proceedings of the 40th Annual Meeting of the Berkeley Linguistics Society*, ed. S. Bakst, H. Leung, A. Lutzross, J. Manker, Z. O'Hagan, O. Pusey, N. Rolle, & K. Sardinha, 273–288.

Larson, Richard K. 1988. On the double object construction. *Linguistic inquiry* 19:335–391.

Larson, Richard K. 2014. *On shell structure*. Routledge.

Lee-Schoenfeld, Vera. 2006. German possessor datives: Raised and affected. *The Journal of Comparative Germanic Linguistics* 9:101–142.

Legate, Julie Anne. 2008. Morphological and abstract case. *Linguistic inquiry* 39:55–101.

Legate, Julie Anne. 2012. Types of ergativity. *Lingua* 122:181–191.

Legate, Julie Anne. 2014. *Voice and v: Lessons from Acehnese*, volume 69. MIT Press.

Levin, Theodore. 2019. On the nature of differential object marking: Insights from Palauan. *Natural Language & Linguistic Theory* 37:167–213.

Levin, Theodore, & Omer Preminger. 2015. Case in Sakha: Are two modalities really necessary? *Natural Language & Linguistic Theory* 33:231–250.

Levin, Theodore Frank. 2015. Licensing without case. Doctoral dissertation, Massachusetts Institute of Technology.

Loginopulo, Daria. 2019. Prostranstvennyje poslelogi. [spatial postpositions.]. Fieldwork handout.

Lohninger, Magdalena, Iva Kovač, & Susanne Wurmbrand. 2022. From prolepsis to hyperraising. *Philosophies* 7(2):32–72.

Lohninger, Magdalena, & Ka-Fai Yip. 2023. (In)dependence of features on composite probes. In *Proceedings of the 53th annual meeting of the North East Linguistic Society (NELS53)*, 155–165.

López, Luis. 2001. Head of a projection. *Linguistic Inquiry* 32(3):521–532.

López, Luis. 2012. *Indefinite objects: Scrambling, choice functions, and differential marking*, volume 63. MIT press.

Lotz, John. 1967. Jakobson's Case Theory and the Russian Prepositions. In *To honor Roman Jakobson*, volume 1, 1207–1212.

Lyskawa, Paulina. 2021. Coordination without grammar-internal feature resolution. Doctoral dissertation, University of Maryland.

Lyskawa, Paulina, & Rodrigo Ranero. 2022. Optional agreement as successful/failed Agree: Evidence from Santiago Tz'utujil (Mayan). *Linguistic Variation* 22:209–267.

Lyutikova, Ekaterina A. 2017. Agreement, case and licensing: Evidence from Tatar. *Ural-Altaic studies* 25–45.

Lyutikova, Ekaterina A., & Asya Pereltsvaig. 2015. The Tatar DP. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 60:289–325.

Mahajan, Anoop. 2012. Ergatives, antipassives and the overt light v in Hindi. *Lingua* 122:204–214.

Mahajan, Anoop Kumar. 1990. The A/A-bar distinction and movement theory. Doctoral dissertation, Massachusetts Institute of Technology.

Malchukov, Andrej, Martin Haspelmath, & Bernard Comrie. 2010. Ditransitive constructions: a typological overview. In *Studies in ditransitive constructions: A comparative handbook*, ed. Andrej Malchukov, Martin Haspelmath, & Bernard Comrie, 1–64. Walter de Gruyter Berlin/New York.

Manlove, Kathleen. 2015. Evidence for a DP-projection in West Greenlandic Inuit. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 60:327–360.

Marantz, Alec. 1991. Case and licensing. In *Proceedings of ESCOL 91*.

Marantz, Alec. 2000. Case and licensing. In *Arguments and case: Explaining Burzio's generalization*, ed. Eric J. Reuland, 11–30.

Marušič, Franc, Andrew Ira Nevins, & William Badecker. 2015. The grammars of conjunction agreement in Slovenian. *Syntax* 18:39–77.

Masliukov, Vsevolod. 2024. Possessive agreement and external possessors in Kazym Khanty. Paper presented at the 5th Conference on syntax of Uralic languages (SOUl 5), Universität Hamburg. URL https://www.slm.uni-hamburg.de/ifuu/forschung/tagungen/soul5/abstracts/masliukov-kazym-khanty-agr-poss-soul5.pdf.

Massam, Diane. 2009. Noun incorporation: Essentials and extensions. *Language and linguistics compass* 3:1076–1096.

McCloskey, James. 2000. Quantifier float and wh-movement in an Irish English. *Linguistic inquiry* 31:57–84.

McFadden, Thomas. 2004. The position of morphological* case in the derivation: A study on the syntax-morphology interface. Doctoral dissertation, University of Pennsylvania.

McFadden, Thomas, & Sandhya Sundaresan. 2011. Nominative case is independent of finiteness and agreement. Paper presented at BCGL 5: Case at the interfaces. URL http://ling.auf.net/lingbuzz/001350.

McGinnis, Martha. 1999. A-scrambling exists. In *Proceedings of the 23rd Annual Penn Linguistics Colloquium*, 283–97. Citeseer.

Merchant, Jason. 1996. Object scrambling and quantifier float in German. In *Proceedings of the 26th annual meeting of the North East Linguistic Society (NELS26)*, 14.

Michelioudakis, Dimitris. 2012. Dative arguments and abstract case in greek. Doctoral dissertation, University of Cambridge.

Mikhailov, Stepan Kirillovich. 2023. Severnokhantyjskaja possessivnost' i tipologija determinatsii. [Northern Khanty possessives and determiner typology]. *Rodnoj jazyk* 2:6–51.

Moldanova, Irina Maksimovna. 2018. Semanticheskije tipy kauzativnykh glagolov khantyjskogo jazyka (na materiale kazymskogo dialekta). [Semantic types of causative verbs in the Kazym dialect of the Khanty language (in the Khazym dialect)]. *Rodnoj jazyk* 2:81–105.

Morimoto, Yukiko. 2002. Prominence mismatches and differential object marking in bantu. In *Proceedings of the LFG02 Conference*, 292–314. CSLI Stanford, CA.

Müller, Gereon. 1995. *A-bar Syntax: A study in movement types*, volume 42 of *Studies in generative grammar*. Walter de Gruyter.

Müller, Gereon. 1999. Optimality, markedness, and word order in German. *Linguistics* 37:777–818.

Müller, Gereon. 2002. Remarks on nominal inflection in German. In *More than words: A Festschrift for Dieter Wunderlich*, ed. Ingrid Kaufmann & Barbara Stiebels, 113–145.

Müller, Gereon. 2004. On decomposing inflection class features: Syncretism in Russian noun inflection. In *Explorations in nominal inflection*, ed. Gereon Müller, Lutz Gunkel, & Gisela Zifonun, 189–227. Berlin: Mouton de Gruyter.

Müller, Gereon. 2009. Ergativity, accusativity, and the order of Merge and Agree. In *Explorations of phase theory: Features and arguments*, ed. Kleanthes K. Grohmann, 269–308. Mouton de Gruyter Berlin/New York.

Müller, Gereon. 2024. Pre-syntactic impoverishment: Premature exponence in Harmonic Serialism. *Word Structure* 17:134–184.

Müller, Gereon. 2025. *German syntax: A structure removal approach.* Cambridge University Press.

Muravyev, Nikita. 2021. Simultaneity and epistemic access in Kazym Khanty temporal subordination. *Linguistica Uralica* 57:39–55.

Muravyev, Nikita. 2023. Passive in Kazym Khanty and the interaction of givenness, topicality and animacy. *Linguistica Uralica* 59:49–66.

Muravyev, Nikita. 2024. Subjective and objective conjugation in Northern Khanty: A lexically conditioned choice? Presented at the talk series of the Uralic Information center.

Murphy, Andrew, & Zorica Puškar. 2018. Closest conjunct agreement is an illusion. *Natural Language & Linguistic Theory* 36:1207–1261.

Næss, Åshild. 2004. What markedness marks: The markedness problem with direct objects. *Lingua* 114:1186–1212.

Nash, Léa. 2017. Causees are not agents. Unpublished paper. Presented at Linguistic Perspectives on Causation. URL https://causalityconference.files.wordpress.com/2017/06/lea-nash-handout.pdf.

Neeleman, Ad, & Hans Van de Koot. 2008. Dutch scrambling and the nature of discourse templates. *The Journal of Comparative Germanic Linguistics* 11:137–189.

Newman, Elise Sophia Bershad. 2021. The (in) distinction between wh-movement and c-selection. Doctoral dissertation, Massachusetts Institute of Technology.

Nie, Yining. 2019. Raising applicatives and possessors in Tagalog. *Glossa: a journal of general linguistics* 4(1):139.

Nie, Yining. 2020. Licensing arguments. Doctoral dissertation, New York University.

Nikolaeva, Irina. 1999. *Ostyak*. Lincom Europa.

Nikolaeva, Irina. 2001. Secondary topic as a relation in information structure. *Linguistics* 39(1):1–50.

Nikolaeva, Irina. 2014. *A grammar of Tundra Nenets*. De Gruyter Mouton.

Norris, Mark. 2014. A theory of nominal concord. Doctoral dissertation, UC Santa Cruz.

Norris, Mark. 2017. Description and analyses of nominal concord (Pt. I-II). *Language and Linguistics Compass* 11.

Ormazabal, Javier, & Juan Romero. 2010. The derivation of dative alternations. In *Argument structure and syntactic relations*, 203–232. John Benjamins.

Ormazabal, Javier, & Juan Romero. 2013. Differential object marking, case and agreement. *Borealis – An international journal of Hispanic linguistics* 2(2):221–239.

Ott, Dennis. 2015. Symmetric merge and local instability: Evidence from split topics. *Syntax* 18(2):157–200.

Overfelt, Jason. 2022. Asymmetrical symmetry in Tigrinya Object Marking. In *Angles of Object Agreement*, ed. Andrew Nevins, Anita Peti-Stantić, Mark de Vos, & Jana Willer-Gold. Oxford: Oxford University Press.

Pereltsvaig, Asya. 2006. Small nominals. *Natural Language & Linguistic Theory* 24:433–500.

Pereltsvaig, Asya. 2021. Noun phrases, big and small. In *The size of things I: Structure building*, ed. Sabine Laszakovits & Zheng Shen, Open Generative Syntax, 323–337. Berlin: Language Science Press.

Pesetsky, David. 2013. *Russian case morphology and the syntactic categories*, volume 66. MIT Press.

Pesetsky, David, & Esther Torrego. 2007. The syntax of valuation and the interpretability of features. *Phrasal and clausal architecture* 262–294.

Pisarenko, Denis. 2020. Markirovanije množestvennogo chisla v kazymskom dialekte khantyjskogo jazyka. [Plural marking in the Kazym dialect of Khanty.]. In *Proceedings of Malyje jazyki v bolshoj lingvistike*, 181–185.

Pisarenko, Denis. 2022. Množestvennoe čislo: revised. [Plural: revised.]. Fieldwork handout.

Pittman, Christine. 2005. Non-canonical switch-reference in Inuktitut. In *the 2005 Annual Meeting of the Canadian Linguistics Association*, ed. Claire Gurski.

Pleshak, Polina. 2018. Struktura imennoj gruppy v khantyjskom jazyke. [Structure of the NP in Khanty.]. Fieldwork handout.

Pleshak, Polina. 2020. To, čto vy khoteli znat' o strukture khantyjskoj IG, no bojalis' sprosit'. [What you wanted to know about the structure of the Khanty NP, but was too afraid to ask.]. Fieldwork handout.

Polinsky, Maria. 2017. Antipassive. In *Oxford handbook of egrativity*, ed. Diane Massam Jessica Coon & Lisa deMena Travis. Oxford: Oxford University Press.

Preminger, Omer. 2014. *Agreement and its failures*, volume 68. MIT press.

Preminger, Omer. 2020. Taxonomies of case and ontologies of case. In *On the Place of Case in Grammar*, ed. Christina Sevdali, Dionysios Mertyris, & Elena Anagnostopoulou, Oxford Studies in Theoretical Linguistics, 73–92. Oxford: Oxford University Press.

Preminger, Omer, & Maria Polinsky. 2015. Agreement and semantic concord: A spurious unification. *Ms., University of Maryland* .

Pylkkänen, Liina. 2008. *Introducing arguments*, volume 49. MIT press.

Rezac, Milan. 2008. Phi-agree and theta-related case. In *Phi-theory: Phi features across interfaces and modules*, ed. David Adger Daniel Harbour & Susana Béjar, 83–129. Oxford: Oxford University Press.

Richards, Marc. 2008. Two kinds of variation in a minimalist system. *Varieties of competition* 133–162.

Richards, Marc. 2014. Defective agree, case alternations, and the prominence of person. In *Scales and hierarchies: A cross-disciplinary perspective*, ed. Ina Bornkessel-Schlesewsky, Andrej Malchukov, & Marc D. Richards, volume 277 of *Trends in Linguistics. Studies and Monographs*, 173–196. Berlin, München, Boston: Walter de Gruyter.

Rizzi, Luigi. 1990. *Relativized minimality*. MIT Press.

Roberts, Ian G. 2010. *Agreement and head movement: Clitics, incorporation, and defective goals*, volume 59. MIT Press.

Rombandeeva, Evdokia Ivanovna. 2017. *Sintaksis sovremennogo mansijskogo jazyka. [Syntax of modern Mansi]*. OOO "Format".

Rude, Noel. 1986. Topicality, transitivity, and the direct object in Nez Perce. *International Journal of American Linguistics* 52:124–153.

Rudnev, Pavel. 2021. Against upwards Agree. *The Linguistic Review* 38:65–99.

Rudnev, Pavel. 2024. Categorial selection and functional structure in the noun phrase: revisiting Russian small nominals. *Russian Linguistics* 48:1.

Ryžowa, Darja A. 2021. Kodirovanije argumentov dvukhmestnykh predikatov v kazymskom dialekte khantyjskogo jazyka. [Argument encoding in ditransitive predicates in Kazym dialect of Khanty]. In *Proceedings of malyje jazyki v bol'šoj lingvistike*, ed. Ksenia Semenova, volume 3, 143–154.

Sadock, Jerrold M. 1980. Noun incorporation in greenlandic: A case of syntactic word formation. *Language* 56:300–319.

Sadock, Jerrold M. 2003. *A grammar of Kalaallisut (West Greenlandic Inuttut)*. LINCOM Europa.

Saito, Mamoru. 1985. Some asymmetries in Japanese and their theoretical implications. Doctoral dissertation, NA Cambridge.

Salzmann, Martin. 2018. Revisiting the NP vs. DP debate. *Unpublished Ms., University of Leipzig* .

Schoenmakers, Gert-Jan, Marjolein Poortvliet, & Jeannette Schaeffer. 2022. Topicality and anaphoricity in Dutch scrambling. *Natural language & linguistic theory* 40:541–571.

Schön, Zsófia, & Katalin Gugán. 2022. Eastern Khanty. In *The Oxford guide to the Uralic languages*, ed. Marianne Bakró-Nagy, Johanna Laakso, & Elena Skribnik, 608–635. Oxford: Oxford University Press.

Schütze, Carson T. 2001. On the nature of default case. *Syntax* 4:205–238.

Schwarz, Florian. 2013. Two kinds of definites cross-linguistically. *Language and linguistics compass* 7:534–559.

Šereikaitė, Milena. 2020. Voice and case phenomena in Lithuanian morphosyntax. Doctoral dissertation, University of Pennsylvania.

Sheehan, Michelle, & Jenneke van der Wal. 2018. Nominal licensing in caseless languages. *Journal of Linguistics* 54:527–589.

Shlonsky, Ur. 1991. Quantifiers as functional heads: A study of quantifier float in Hebrew. *Lingua* 84:159–180.

Sigurðsson, Einar Freyr. 2017. Deriving case, agreement and voice phenomena in syntax. Doctoral dissertation, University of Pennsylvania.

Sigurðsson, Einar Freyr, & Jim Wood. 2024. Resolution by case syncretism in Icelandic passives. Paper presented at Workshop On Morphology at Princeton (WOMP 2024).

Sinnemäki, Kaius. 2014. A typological perspective on Differential Object Marking. *Linguistics* 52:281–313.

Sipőcz, Katalin. 2015. Ditransitivity in the Ob-Ugric languages. In *Congressus Duodecimus Internationalis Fenno-Ugristarum*, ed. Harri Mantila et al., 133–157.

Sipos, Mària. 2022. North Khanty. In *The Oxford guide to the Uralic languages*, ed. Marianne Bakró-Nagy, Johanna Laakso, & Elena Skribnik, 582–607. Oxford: Oxford University Press.

Skribnik, Elena. 2001. Pragmatic structuring in Northern Mansi. In *Congressus nonus internationalis fenno-ugristarum. pars iv. dissertationes sectionum: Linguistica iii*, 222–239. Tartu: Tartu University.

Smith, Peter W. 2020. Object agreement and grammatical functions: A re-evaluation. In *Agree to Agree: Agreement in the Minimalist Programme*, ed. Peter W. Smith, Johannes Mursell, & Katharina Hartmann, volume 6 of *Open Generative Syntax*, 17–147. Berlin: Language Science Press.

Smith, Peter W., Beata Moskal, Ting Xu, Jungmin Kang, & Jonathan David Bobaljik. 2019. Case and number suppletion in pronouns. *Natural Language & Linguistic Theory* 37:1029–1101.

Snorrason, Oddur, Jim Wood, & Einar Freyr Sigurðsson. 2024. Resolution by case syncretism in Icelandic passives. In *Proceedings of the 54th Annual Meeting of the North East Linguistic Society (NELS 54).*

Solovar, Valentina N. 2016. Semantičeskaja klassifikayija glagolov po sposobu osuščestvlenija dviženija (na materiale kazymskogo dialekta khantyjskogo jazyka). [Semantic classification of verbs according to the mean of movement (on the material of Kazym dialect of the Khanty language)]. *Vestnik ugrovedenija* 27(4):58–67.

Solovar, Valentina N., Galina L. Nakhracheva, & Anastasija A. Shijanova. 2016. *Dialekty khantyjskogo jazyka. [Dialects of Khanty.]*. OOO "Print-2".

Solovieva, Anita. 2019. χuλ vs. χuλijewa. psevdopartitivnyje konstruktsii. [χuλ vs. χuλijewa. pseudopartitive constructions.]. Fieldwork handout.

Solovieva, Anita. 2020. Nominal ellipsis in Kazym Khanty: UQ $\chi u\lambda$(ijewa). Paper presented at Malyje jazyki v bolshoj lingvistike.

Sosa, Sachiko. 2017. Functions of morphosyntactic alternations, and information flow in Surgut Khanty discourse. Doctoral dissertation, University of Helsinki.

Sportiche, Dominique. 1988. A theory of floating quantifiers and its corollaries for constituent structure. *Linguistic inquiry* 19:425–449.

Sportiche, Dominique. 2006. Reconstruction, binding, and scope. In *The Blackwell companion to syntax*, ed. Henk van Riemsdijk Martin Everaert, volume 4, 35–93. Wiley Online Library.

Steinitz, Wolfgang. 1937. Khantyjskij (ostjatskij) jazyk. [Khaty (Ostyak) lnaguage.]. In *Jazyki i pismennist' narodov Severa*, 193–227.

Stiebels, Barbara. 1999. Noun-verb symmetries in Nahuatl nominalizations. *Natural language & linguistic theory* 17:783–834.

Stojković, Jelena, & Marko Simonović. 2024. Allomorphy reveals the trace of a (different) theme vowel. Paper presented at Traces in Phonology and Morphology. Workshop in honour of Jochen Trommer's 50th birthday.

Takahashi, Shoichi, & Sarah Hulsey. 2009. Wholesale late merger: Beyond the A/Ā distinction. *Linguistic Inquiry* 40:387–426.

Tereškin, Nikolay Ivanovič. 1974. Sistema sklonenija v dialectakh obsko-ugorskikh jazykov. [Declension system in the dialects of O-Ugric languages]. In *Sklonenije a paleoaziatskikh i samogijskikh jazykakh*, ed. Petr Jakovlevič Skorik, Georgij Alekseevich Menovshchikov, Nikolay Ivanovič Tereshchenko, & Aleksandr Pavlovich Volodin, 67–77. Leningrad: Nauka.

Tereškin, Nikolay Ivanovič. 1981. *Slovar' vostočnokhantzjskikh dialektov. [Eatern Khanty dialectal dictionary]*. Nauka.

Tiutiunnikova, Varvara. 2024. Bare nouns in Kazym Khanty feature differentiated scope. Unpublished manuscript.

Tiutiunnikova, Varvara, Stepan Mikhailov, & Fedor Golosov. 2024. No iota type-shift in Kazym Khanty. Presented at Sinn und Bedeutung 29.

Torrego, Esther. 1996. On quantifier float in control clauses. *Linguistic inquiry* 27(1):111–126.

van Urk, Coppe. 2015. A uniform syntax for phrasal movement: A case study of Dinka Bor. Doctoral dissertation, Massachusetts Institute of Technology.

van Urk, Coppe. 2020. Object licensing in Fijian and the role of adjacency. *Natural Language*

& Linguistic Theory 38:313–364.

Van Geenhoven, Veerle. 2002. Raised possessors and noun incorporation in West Greenlandic. *Natural Language & Linguistic Theory* 20:759–821.

Van Riemsdijk, H. C., & E. S. Williams. 1986. *Introduction to the theory of grammar*. Cambridge: MIT Press.

Virtanen, Susanna. 2012. Variation in three-participant constructions in Eastern Mansi. *Linguistica Uralica* 48:120–130.

Virtanen, Susanna. 2013. Contextual function of noun marking in the direct object marking system in Eastern Mansi. *Uralisztikai Tanulmányok* 2012(20).

Virtanen, Susanna. 2014. Pragmatic direct object marking in Eastern Mansi. *Linguistics* 52:391–413.

Virtanen, Susanna. 2015. Transitivity in Eastern Mansi. Doctoral dissertation, University of Helsinki.

Virtanen, Susanna, & Sachiko Sosa. 2018. Remarks on areal linguistics in the information structure of the Ob-Ugrian languages. *Folia Uralica Debreceniensia* 25:233–260.

Volkova, Anna, & Svetlana Toldova. 2023. Reflexivity in Kazym Khanty. In *Reflexive constructions in the world's languages*, ed. Nicoletta Puddu Katarzyna Janic & Martin Haspelmath, chapter 10, 259–291. Language Science Press.

van der Wal, Jenneke. 2022. *A featural typology of Bantu agreement*. Oxford: Oxford University Press.

Weisser, Philipp. 2020. On the symmetry of case in conjunction. *Syntax* 23:42–77.

Wiltschko, Martina. 2003. On the interpretability of tense on D and its consequences for case theory. *Lingua* 113:659–696.

Woolford, Ellen. 1999. Ergative agreement systems. In *Proceedings of the Workshop on Morphology. University of Maryland Working Papers in Linguistics*, volume 10, 157–191.

Woolford, Ellen. 2006a. Case-agreement mismatches. In *Agreement systems*, ed. Cedric Boeckx, volume 92 of *Linguistik Aktuell/Linguistics Today*, 317–339. Amsterdam: John Benjamins.

Woolford, Ellen. 2006b. Lexical case, inherent case, and argument structure. *Linguistic inquiry* 37:111–130.

Wunderlich, Dieter. 1997. Cause and the structure of verbs. *Linguistic inquiry* 28(1):27–68.

Wurmbrand, Susi. 2012a. Parasitic participles in germanic: Evidence for the theory of verb clusters. *Taal en tongval* 64:129–156.

Wurmbrand, Susi. 2012b. The syntax of valuation in auxiliary-participle constructions. In *Coyote working papers: Proceedings of the 29th West Coast Conference on Formal Linguistics*, ed. Jaehoon Choi, E. Alan Hogue, Jeffrey Punske, Deniz Tat, Jessamyn Schertz, & Alex Trueman, 154–162. University of Arizona Linguistics Circle (Tucson, Arizona).

Wurmbrand, Susi. 2014. Restructuring across the world. In *Complex visibles out there. Proceedings of the Olomouc Linguistics Colloquium*, 275–294.

Yuan, Michelle. 2013. On the phasal status of DP in Inuktitut. *McGill Working Papers in Linguistics* 23.

Yuan, Michelle. 2018. Dimensions of ergativity in Inuit: Theory and microvariation. Doctoral dissertation, Massachusetts Institute of Technology.

Yuan, Michelle. 2021. Diagnosing object agreement vs. clitic doubling: An Inuit case study. *Linguistic Inquiry* 52:153–179.

Yuan, Michelle. 2022. Ergativity and object movement across Inuit. *Language* 98:510–551.

Zeijlstra, Hedde. 2012. There is only one way to agree. *The linguistic review* 29:491–539.

Zeijlstra, Hedde. 2014. On the uninterpretability of interpretable features. In *Minimalism and beyond: Radicalizing the interfaces*, 109–128. John Benjamins Publishing Company.

Zeijlstra, Hedde. 2019. Labeling, selection, and feature checking. In *Agree to agree: Agreement in the minimalist programme*, ed. Peter W. Smith, Johannes Mursell, & Katharina Hartmann, 31–70. Language Science Press.

Zompì, Stanislao. 2019. Ergative is not inherent: Evidence from *ABA in suppletion and syncretism. *Glossa: a journal of general linguistics* 4(1).

Zubizarreta, Maria Luisa. 1985. The relation between morphophonology and morphosyntax: The case of Romance causatives. *Linguistic inquiry* 16:247–289.

Zyman, Erik. 2018. Quantifier float as stranding: Evidence from Janitzio P'urhepecha. *Natural Language & Linguistic Theory* 36:991–1034.

Čeremisinova, Maria. 2019. Attributiviyator əŋ/*əp štuki. [attributivizor əŋ/*əp things.]. Fieldwork handout.